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Technical Report

3rd Generation Partnership Project; Technical Specification Group Radio Access Network; Study on solutions for Ambient IoT (Internet of Things) in NR

(Release 19)



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Contents

Foreword	7
Introduction.....	8
1 Scope	9
2 References.....	9
3 Definitions of terms, symbols and abbreviations.....	20
3.1 Terms.....	20
3.2 Symbols.....	20
3.3 Abbreviations	20
4 Evaluation methodology	21
4.1 Remaining details of RAN design targets	21
4.2 Evaluation scenarios and assumptions	22
4.2.1 Evaluation scenarios	22
4.2.2 Evaluation assumptions	24
4.3 Link budget.....	27
4.3.1 Receiver sensitivity.....	27
4.3.2 Link budget template.....	28
4.4 R2D waveform generation	33
5 Ambient IoT device architectures	33
5.1 ~1 μ W devices (Device 1).....	33
5.2 $\leq$ a few hundred μ W devices (Device 2).....	34
5.2.1 External carrier wave (Device 2a)	34
5.2.1.1 Reflection amplifier	35
5.2.2 Internally-generated carrier wave (Device 2b)	36
5.2.2.1 RF envelope detector receiver.....	36
5.2.2.2 IF envelope detector receiver.....	38
5.2.2.3 ZIF receiver.....	40
5.3 Clock(s).....	41
6 Solutions for Ambient IoT	43
6.1 Physical layer	43
6.1.0 Aspects in common to R2D and D2R.....	43
6.1.0.1 General.....	43
6.1.0.2 CRC.....	43
6.1.1 R2D.....	45
6.1.1.1 R2D waveform, modulation and numerology.....	45
6.1.1.2 R2D line coding.....	47
6.1.1.3 R2D channel coding, repetition, CRC	48
6.1.1.3.1 Channel coding and CRC	48
6.1.1.3.2 Repetition	48
6.1.1.4 R2D bandwidths.....	50
6.1.1.5 PRDCH	50
6.1.1.6 R2D timing.....	51
6.1.1.6.1 R2D timing acquisition signal.....	51
6.1.1.6.2 R2D end timing	53
6.1.1.7 CFO calibration signal for device 2b.....	54
6.1.1.8 R2D multiplexing.....	54
6.1.1.9 Scheduling of R2D.....	56
6.1.1.10 R2D control information.....	56
6.1.2 D2R.....	57
6.1.2.1 Waveform and modulation.....	57
6.1.2.2 D2R line coding.....	58
6.1.2.3 D2R channel coding, repetition, CRC	58
6.1.2.3.1 Channel coding.....	58
6.1.2.3.2 CRC	60
6.1.2.3.3 Repetition	61

6.1.2.4	D2R bandwidths.....	62
6.1.2.5	PDRCH	64
6.1.2.6	D2R timing.....	65
6.1.2.6.1	D2R timing acquisition signal.....	65
6.1.2.6.2	D2R end timing	66
6.1.2.6.3	D2R midamble	66
6.1.2.6.4	Trade-offs analysis	66
6.1.2.7	D2R multiple access	70
6.1.2.7.1	Multiple access schemes	70
6.1.2.8	Scheduling of D2R.....	74
6.1.2.9	D2R control information.....	74
6.1.3	Timing relationships	75
6.1.4	Random access.....	75
6.1.5	Topology 2.....	76
6.2	Device (un)availability	77
6.2.1	Direction 1 solution details	77
6.2.2	Direction 2 solution details	99
6.3	Protocol stack and signalling procedures	137
6.3.1	General aspects and overall procedure	137
6.3.2	Protocol stack and functionality aspects.....	139
6.3.3	A-IoT paging.....	140
6.3.4	A-IoT random access procedure.....	141
6.3.5	A-IoT data transmission.....	143
6.3.6	Topology 2 aspects on the interface between UE reader and RAN.....	144
6.4	RAN architecture aspects	146
6.4.1	Support of Topology 1.....	146
6.4.2	Support of Topology 2.....	147
6.4.2.1	Solutions for Topology 2	148
6.4.2.1.0	General	148
6.4.2.1.1	Solution 1: RRC based solution	148
6.4.2.1.2	Solution 2: NAS based solution	149
6.4.2.1.3	Solution 3: UP based solution	150
6.5	Impacts on CN-RAN interface.....	151
6.5.1	Information exchanged between A-IoT CN and A-IoT RAN	151
6.5.1.1	Inventory	151
6.5.1.2	Command.....	151
6.5.1.3	A-IoT radio resource allocation in case of NAS/UP based solutions.....	151
6.5.2	Signaling and Procedures for Topology 1	152
6.5.2.1	Candidate procedures for A-IoT Inventory for Topology 1	152
6.5.2.2	Candidate procedures for A-IoT Command in Topology 1.....	153
6.5.3	Signaling and Procedures for Topology 2	154
6.5.3.1	Candidate procedures for A-IoT Inventory for Topology 2	154
6.5.3.1.1	RRC based solution.....	154
6.5.3.1.2	NAS/UP based solution.....	154
6.5.3.2	Candidate procedures for A-IoT Command for Topology 2	157
6.5.3.2.1	RRC based solution.....	157
6.5.3.2.2	NAS/UP based solution.....	157
6.6	Coexistence of ambient IoT and NR/LTE.....	158
6.6.1	Co-existence scenarios and cases	158
6.6.2	Co-existence evaluation assumptions	158
6.6.2.1	Deployment.....	158
6.6.2.2	NR BS/ A-IoT reader/ intermediate UE/ CW RF characteristics	160
6.6.2.3	NR UE/ A-IoT device RF characteristics	161
6.6.3	Co-existence simulation methodology.....	162
6.6.3.1	Coexistence evaluation methodology.....	162
6.6.3.2	SINR definition.....	162
6.6.3.3	Coupling loss	163
6.6.3.4	Impact from CW to NR.....	163
6.6.4	Co-existence evaluation results	163
6.6.5	Summary of co-existence evaluation.....	163
6.6.5.1	Scenario 1-1: D1T1-A2- NR UE only outdoors (R2D/ CW2D/ D2R: DL).....	163
6.6.5.2	Scenario 1-2: D1T1-A2- NR UE indoor (R2D/ CW2D/ D2R: DL)	164

6.6.5.3	Scenario 2-1: D1T1-B- NR UE only outdoor (R2D: DL, CW2D/ D2R: UL)	164
6.6.5.4	Scenario 2-2: D1T1-B- NR UE indoor (R2D: DL, CW2D/ D2R: UL)	165
6.6.5.5	Scenario 5-1: D2T2-A2- NR UE only outdoor (R2D/ CW2D/ D2R: UL)	165
6.6.5.6	Scenario 5-2: D2T2-A2- NR UE indoor (R2D/ CW2D/ D2R: UL)	165
6.6.5.7	Scenario 6-1: D2T2-B- NR UE only outdoor (R2D: UL, CW2D/ D2R: DL)	165
6.6.5.8	Scenario 6-2: D2T2-B- NR UE indoor (R2D: UL, CW2D/ D2R: DL)	166
6.6.5.9	Scenario 3 (Optional): D1T1-A2- NR UE indoor (R2D: UL, CW2D/ D2R: DL).....	166
6.6.5.10	General remarks on coexistence findings	167
6.7	RF requirements study.....	167
6.7.1	System parameters.....	167
6.7.1.1	Operating band.....	167
6.7.1.2	Channel bandwidth	167
6.7.1.3	Channel arrangement	168
6.7.2	Ambient IoT BS.....	168
6.7.2.1	General.....	168
6.7.2.2	Ambient IoT BS TX requirements.....	168
6.7.2.2.1	Base station output power	168
6.7.2.2.2	Output power dynamic	168
6.7.2.2.3	Transmit ON/OFF power	168
6.7.2.2.4	Transmit signal quality.....	168
6.7.2.2.5	Unwanted emissions.....	169
6.7.2.3	Ambient IoT BS Rx requirements	169
6.7.2.3.1	Reference sensitivity level.....	169
6.7.2.3.2	Dynamic range	169
6.7.2.3.3	In-band selectivity and blocking	169
6.7.2.3.4	Out-of-band blocking.....	169
6.7.2.3.5	Receiver spurious emissions	169
6.7.2.3.6	Receiver intermodulation	170
6.7.2.3.7	In-channel selectivity	170
6.7.3	Intermediate node (UE)	170
6.7.3.1	General.....	170
6.7.3.2	Intermediate UE TX requirements.....	170
6.7.3.2.1	Maximum output power	170
6.7.3.2.2	Output power dynamics.....	170
6.7.3.2.3	Transmitted signal quality.....	170
6.7.3.2.4	Output RF spectrum emissions.....	171
6.7.3.3	CW for D2T2	171
6.7.3.3.1	Output power/Output power dynamic range.....	171
6.7.3.3.2	Transmit signal quality.....	171
6.7.3.3.3	RF spectrum emission/Spurious emissions/Phase noise	171
6.7.3.4	Intermediate UE RX requirements.....	172
6.7.3.4.1	Reference sensitivity	172
6.7.3.4.2	Maximum input power	172
6.7.3.4.3	Adjacent channel selectivity.....	172
6.7.3.4.4	Blocking characteristics	172
6.7.3.4.5	Spurious response.....	173
6.7.3.4.6	Receiver intermodulation	173
6.7.3.4.7	Spurious emissions	173
6.7.4	Ambient IoT Device	173
6.7.4.1	Device 1	173
6.7.4.1.1	TX.....	173
6.7.4.1.2	RX	174
6.7.4.2	Device 2a	175
6.7.4.2.1	TX.....	175
6.7.4.2.2	RX	176
6.7.4.3	Device 2b	176
6.7.4.3.1	TX.....	176
6.7.4.3.2	RX	177
6.7.4.4	Testability	178
6.7.4.4.1	Testability of device.....	178
6.7.4.4.2	Summary of testability issues.....	179
6.7.5	CW interference cancelation study.....	179

6.7.5.1	General	179
6.7.5.2	Interference cancelation analysis	180
6.7.5.2.1	Ericsson	180
6.7.5.2.2	Qualcomm	183
6.7.5.2.3	Huawei	190
6.7.5.2.4	vivo	190
6.7.5.3	Summary	193
6.8	Characteristics of carrier-wave waveform	193
6.8.1	CW transmission	193
6.8.2	CW characteristics	195
6.9	Locating ambient IoT devices	199
6.9.1	General	199
6.9.2	Topology 1	199
6.9.3	Topology 2	199
6.9.4	Proximity determination	199
6.10	DO-A assessment	199
7	Evaluations	200
7.1	Coverage evaluations	200
7.1.1	Link budget for D1T1	200
7.1.2	Link budget for D2T2	200
7.1.3	Link performance for D2R	200
7.1.4	Link performance for R2D	200
7.2	Latency evaluations	201
7.2.1	Single device latency	201
7.2.2	Inventory completion time for multiple devices	207
8	Conclusions and recommendations	219
Annex A:	Device receiver sensitivity for <i>Budget-Alt1</i>	221
Annex B (informative):	Change history	225

Foreword

This Technical Report has been produced by the 3rd Generation Partnership Project (3GPP).

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In the present document, modal verbs have the following meanings:

- shall** indicates a mandatory requirement to do something
- shall not** indicates an interdiction (prohibition) to do something

The constructions "shall" and "shall not" are confined to the context of normative provisions, and do not appear in Technical Reports.

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- will** indicates that something is certain or expected to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document
- will not** indicates that something is certain or expected not to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document
- might** indicates a likelihood that something will happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

might not indicates a likelihood that something will not happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

In addition:

is (or any other verb in the indicative mood) indicates a statement of fact

is not (or any other negative verb in the indicative mood) indicates a statement of fact

The constructions "is" and "is not" do not indicate requirements.

Introduction

In recent years, IoT has attracted much attention in the wireless communication world. More 'things' are expected to be interconnected for improving productivity efficiency and increasing comforts of life. Further reduction of size, complexity, and power consumption of IoT devices can enable the deployment of tens or even hundreds of billions of IoT devices for various applications and provide added value across the entire value chain. It is impossible to power all the IoT devices by battery that needs to be replaced or recharged manually, which leads to high maintenance cost, serious environmental issues, and even safety hazards for some use cases, for example, wireless sensors in electrical power, and petroleum industries.

Most of the existing wireless communication devices are powered by batteries that need to be replaced or recharged manually. The automation and digitization of various industries opens numerous new markets requiring new IoT technologies of supporting batteryless devices with no energy storage capability or devices with energy storage that do not need to be replaced or recharged manually.

An example type of application is asset identification, which presently has to resort mainly to barcodes and RFID in most industries. The main advantage of these two technologies is the ultra-low complexity and small form factor of the tags. However, the limited reading range of a few meters usually requires handheld scanning which leads to labor intensive and time-consuming operations, or RFID portals/gates which leads to costly deployments. Moreover, the lack of interference management scheme results in severe interference between RFID readers and capacity problems, especially in case of dense deployment. It is hard to support a large-scale network with seamless coverage for RFID.

In contrast, this study investigates solutions for Ambient IoT, a new IoT technology to open new markets within 3GPP systems, whose number of connections and/or device density can be orders of magnitude higher than existing 3GPP IoT technologies, and which can provide complexity and power consumption orders-of-magnitude lower than existing 3GPP LPWA technologies such as NB-IoT and LTE-MTC. TSG RAN has completed a Rel-18 RAN-level SI on Ambient IoT, producing TR 38.848 which provides a terminological and scoping framework for future discussions of Ambient IoT. This has defined representative use cases, deployment scenarios, connectivity topologies, Ambient IoT devices, design targets, and required functionalities; it also conducted a preliminary feasibility assessment.

The SI reported in this present TR is now to investigate solutions in detail at RAN-WG level for Ambient IoT in 3GPP.

1 Scope

The overall objective of the SI is to study a harmonized air interface design with minimized differences (where necessary) for Ambient IoT to enable the following devices:

- $\sim 1 \mu\text{W}$ peak power consumption, has energy storage, initial sampling frequency offset (SFO) up to 10^x ppm, neither R2D nor D2R amplification in the device. The device's D2R transmission is backscattered on a carrier wave provided externally.
- $\leq$ a few hundred μW peak power consumption has energy storage, initial sampling frequency offset (SFO) up to 10^x ppm, both R2D and/or D2R amplification in the device. The device's D2R transmission may be generated internally by the device, or be backscattered on a carrier wave provided externally.

Referring to the definitions in [2, TR 38.848], this is done in the context of:

- Deployment scenario 1 (indoor-to-indoor) with Topology 1, and indoor microcell basestation.
- Deployment scenario 2 (indoor-to-outdoor) with Topology 2 and indoor UE as intermediate node under network control, and outdoor macrocell basestation.

The spectrum considered is FR1 licensed spectrum in FDD, which can be in-band to NR, in guard-band to NR/LTE, or in standalone band(s). The traffic types considered are DO-DTT and DT, focusing on indoor inventory and indoor command representative use cases. The study also assesses whether the harmonized air interface can address the DO-A use case.

Study of the design of energy harvesting signal/waveform is out of scope in Rel-19.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

- [1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".
- [2] 3GPP TR 38.848: "Study on Ambient IoT (Internet of Things) in RAN".
- [3] RP-240826: "Revised SID: Study on solutions for Ambient IoT (Internet of Things) in NR".
- [4] 3GPP TR 38.869: "Study on low-power Wake-up Signal and Receiver for NR".
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3 Definitions of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in TR 21.905 [1].

For the purposes of the study, this TR uses the following terms:

Device 1: ~ 1 μ W peak power consumption, has energy storage, initial sampling frequency offset (SFO) up to 10^x ppm, neither R2D nor D2R amplification in the device. The device's D2R transmission is backscattered on a carrier wave provided externally.

Device 2a: $\leq$ a few hundred μ W peak power consumption, has energy storage, initial sampling frequency offset (SFO) up to 10^x ppm, both R2D and/or D2R amplification in the device. The device's D2R transmission is backscattered on a carrier wave provided externally.

Device 2b: $\leq$ a few hundred μ W peak power consumption, has energy storage, initial sampling frequency offset (SFO) up to 10^x ppm, both R2D and/or D2R amplification in the device. The device's D2R transmission is generated internally by the device.

D1T1: Deployment scenario 1 with connectivity topology 1, according to TR 38.848.

D2T2: Deployment scenario 2 with connectivity topology 2, according to TR 38.848.

Inventory: The service provided by the network to discover and acquire the identifier of A-IoT device(s).

Command: The service provided by the network to send the operation instruction to the A-IoT device (e.g. read, write, etc.).

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in TR 21.905 [1].

1SB	Single sideband
2SB	Double sideband
A-IoT	Ambient IoT
A-IoT RAN	Ambient IoT Radio Access Network
BFSK	Binary frequency-shift keying
BPSK	Binary phase-shift keying
CAP	Clock-acquisition part (of R-TAS)
CFO	Carrier-frequency offset
CP	Cyclic prefix
CW	Carrier-wave
CW2D	Carrier-wave, or carrier-wave node, to device
D2R	Device to reader
DFT-s-OFDM	DFT-spread OFDM
DO-A	Device-originated autonomous
DO-DTT	Device-originated by device-terminated trigger
DT	Device-terminated
ED	Envelope detector
FAR	False alarm rate
FDR	False detection rate

FEC	Forward error-correction code
FR	Frequency Range
IF	Intermediate frequency
IoT	Internet of Things
LPWA	Low-power, wide-area
LTE-MTC	Long Term Evolution – Machine Type Communication
MCS	Modulation and coding scheme
MDR	Missed detection rate
MSK	Minimum-shift keying
NB-IoT	Narrowband IoT
OOK	On-off keying
PDRCH	Physical device-to-reader channel
PIE	Pulse interval encoding
PRDCH	Physical reader-to-device channel
R2D	Reader to device
RF	Radio frequency
RFID	Radio frequency identification
R-TAS	R2D timing acquisition signal
SER	Sample error rate
SFO	Sampling-frequency offset
ZIF	Zero IF

4 Evaluation methodology

4.1 Remaining details of RAN design targets

TR 38.848 [2] sets a number of RAN design targets. In [3], in particular three aspects of design targets beyond those in TR 38.848 are to be studied:

- Applicable maximum distance target value(s): The maximum distance targets are set separately for device 1, device 2a, device 2b, respectively. The distance target is compared to the evaluation results which assume the agreed mandatory evaluation assumptions.
 - For device 1 in D1T1: 15 m
 - For device 2a in D1T1: 25 m
 - For device 2b in D1T1: 50 m
 - For device 1 in D2T2: 10 m
 - For device 2a in D2T2: 15 m
 - For device 2b in D2T2: 40 m
- Refined definition of latency:
 - For the use case of "inventory-only":
 - The time interval between the time that the A-IoT paging is sent from reader to a A-IoT device and the time that the inventory report is successfully received at reader from the A-IoT device, i.e., for completing Step A and Step B (as per RAN2 agreements).
 - For the use case of "inventory and command":
 - The time interval between the time that the A-IoT paging is sent from reader and the time that
 - The command is successfully received at A-IoT device, i.e., for completing Step A, Step B, Step C1 (as per RAN2 agreements), if Step C2 is optional and not used (note: pending RAN2 decision on optionality of Step C2).

- The response is successfully received at Reader, i.e., for completing Step A, Step B, Step C1, and Step C2 (as per RAN2 agreements).
- See Clause 6.3.1 for descriptions of the Steps of these procedures.

Expected value of latency is calculated according a X% re-attempt probability to each attempt.

Note: The "successfully received" (i.e., decoding successfully)

Note: The latency is evaluated for a single A-IoT device.

Note: Time for energy harvesting is not included in the definition of latency.

- 2D distribution of devices: A-IoT devices are dropped uniformly distributed over the horizontal area. See Table 4.2.2-2.

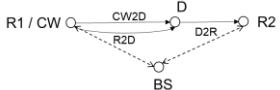
4.2 Evaluation scenarios and assumptions

4.2.1 Evaluation scenarios

The following scenarios are defined for the purpose of potential evaluation.

Table 4.2.1-1: Evaluation scenarios

Scenario	CW Inside/outside topology	Diagram of the scenario	Description of the scenario	Device 1/2a/2b	CW spectrum	D2R spectrum	R2D spectrum
D1T1-A1	CW inside topology		CW node inside topology 1 'CW' in CW2D and 'R2' in D2R are different 'CW' in CW2D and 'R1' in R2D are same 'R1' in R2D and 'R2' in D2R are different	Device 1, 2a	Case 1-1 (inside topology, DL) Case 1-2 (inside topology, UL)	Same as CW	At least DL
D1T1-A2			CW node inside topology 1 Same 'CW' and 'R' node for CW2D, D2R and R2D		Same as D1T1-A1	Same as CW	At least DL
D1T1-B	CW outside topology		CW node outside topology 1 'CW' in CW2D and 'R' in D2R are different 'CW' in CW2D and 'R' in R2D are different 'R' in R2D and		Case 1-4 (outside topology, UL)	Same as CW	At least DL

			'R' in D2R are same				
D1T1-C	No CW		No CW Node.	Device 2b	N/A	UL	At least DL
D2T2-A1	CW inside topology		CW node inside topology 2 'CW' in CW2D and 'R2' in D2R are different 'CW' in CW2D and 'R1' in R2D are same 'R1' in R2D and 'R2' in D2R are different BS communicates with R1 and R2	Device 1, 2a	Case 2-2 (inside topology, UL)	Same as CW	UL
D2T2-A2			CW node inside topology 2 Same 'CW' and 'R' node for CW2D, D2R and R2D BS communicates with R		Same as D2T2-A1	Same as CW	UL
D2T2-B			CW node outside topology 2 'CW' in CW2D and 'R' in D2R are different 'CW' in CW2D and 'R' in R2D are different 'R' in R2D and 'R' in D2R are same BS communicates with R		Case 2-3 (outside topology, DL) Case 2-4 (outside topology, UL)	Same as CW	UL
D2T2-C	No CW		No CW Node. BS communicates with R	Device 2b	N/A	At least UL	UL

Note: This table is for the case where D2R is in the same spectrum as CW2D.

4.2.2 Evaluation assumptions

The following table of coverage evaluation assumptions for link-level simulation is considered. (M) indicates a value mandatory for evaluation, (O) indicates optional for evaluation. If there are any differences between devices, they are for evaluation purposes only.

Table 4.2.2-1: Coverage evaluation assumptions for link-level simulation

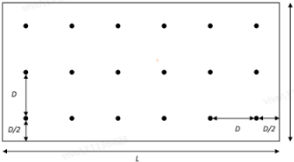
No.	Parameters		Assumptions
R2D/D2R common parameters			
[0a]	Carrier frequency		Refer to link budget template
[0b]	SCS		15 kHz as baseline
[0c]	Block structure		Blocks as agreed in 9.4.2.3, or other blocks reported by companies
[0d]	Channel model		R2D: For D2T2: BS pathloss model is reused for intermediate UE with antenna height = 1.5 m [0D]-Alt1: InF-DL NLOS, with TDL-A [0D]-Alt2: InH-Office LOS, with TDL-D For D1T1: InF-DH NLOS, with TDL-A D2R: For D2T2: BS pathloss model is reused for intermediate UE with antenna height = 1.5 m [0D]-Alt1: InF-DL NLOS, with TDL-A [0D]-Alt2: InH-Office LOS, with TDL-D For D1T1: InF-DH NLOS, with TDL-A
[0e]	Delay spread		An RMS delay spread of 30 ns (M) and [150] ns (O) is considered for TDL-A channel model. An RMS delay spread of 30 ns is considered for TDL-D channel model.
[0f]	Device velocity		3 km/h
[0g]	Number of Tx/Rx chains for Ambient IoT device		1
[0h1]	BS	Number of antenna elements	2 or 4
[0h2]		Number of TXRUs	2 or 4
[0j1]	Intermediate UE	Number of antenna elements	1 or 2
[0j2]		Number of TXRUs	1 or 2
[0m]	Reference data rate		1 kbps (M) 5 - 7 kbps (M) 48 - 60 kbps (O) 0.1 kbps for message size of 20 bits or 96 bits (O) Other data rates can be reported by companies Note 1: Companies to report the exact data rate. Note 2: The exact data rate is close to the values listed above. Note 3: The exact data rate is calculated by dividing the total message size (excluding CRC) by the total transmission time including applicable overheads(e.g., CRC, pre/mid/post-ambls if present). Note 4: The exact data rate may be related to coding scheme, repetition and etc. Note 5: All data rates considered are for evaluation purpose only
[0n]	Message size		{20 bits, 96 bits, 400 bits} are considered for message size. Note 1: companies to report the M value and chip length used for each message

		size Note 2: CRC is not included for the message size
[0p]	BLER target	1%, 10%
[0q]	Sampling frequency	Companies to report the sampling frequency (e.g., 1.92Mps or other feasible values if any) Initial SFO (Sampling Frequency Offset) (Fe): (M) Randomly select a value from the range of $[0.1 \sim 1] \cdot 10^4$ ppm for device 2, (M) Randomly select a value from the range of $[0.1 \sim 1] \cdot 10^5$ ppm for device 1, (O) Randomly select a value from the range of $[0.1 \sim 1] \cdot 10^5$ ppm for device 2, Note: For random selection, the value is randomly selected per simulation drop, according to a uniform distribution Note: Above values are only for sampling purpose. Note: Above assumptions are only for LLS evaluation purpose only for R2D and D2R. The timing drift ΔT over a time T is modelled as $\Delta T = \pm Fe \cdot T$. Note: SFO after clock calibration can be applied to Fe. CFO for device 2b: 100 ppm (M) 200 ppm (O) 1000 ppm (O, only as initial CFO) Drift rate of 0.1 or 1 ppm/s. Companies to report which value they used. Note: Above assumptions are for LLS evaluation purpose only
[0r]	Device 1/2a/2b	Options are as follows, Device 1, RF-ED Device 2a, RF-ED Device 2b, RF-ED/IF-ED/ZIF
R2D specific parameters		
[1a]	Transmission bandwidth	180 kHz as baseline. Other larger values are not precluded.
[1b]	ED bandwidth	The ED bandwidth is the bandwidth for calculating the noise/interference (if any) power: For evaluations, the value(s) of ED bandwidth is 20 MHz for RF-ED, [180] kHz for IF/ZIF receiver. Note: this does not imply that a A-IoT device supports sampling clock rate as large as RF ED bandwidth.
[1c]	BB LPF	[X]-order Butterworth/RC filter with cutoff frequency at half of R2D transmission bandwidth. Companies to report $X = \{3, 5\}$.
[1d]	Waveform	OOK waveform generated by OFDM modulator
[1e]	Modulation	OOK Companies to report, e.g., OOK-1, OOK-4 with M chips per OFDM symbol
[1f]	Line code	Companies to report, e.g., Manchester, PIE
[1g]	FEC	No FEC as baseline
[1h]	ADC bit width	1-bit for device 1 4-bit for device 2
[1j]	Detection/decoding method for Line code	Companies to report
D2R specific parameters		
[2a1]	Transmission bandwidth	[2a1]-Alt1 (M): DSB X kHz is considered for D2R transmission bandwidth. The value is for two sidebands, i.e., the total transmission bandwidth for DSB is X kHz [2a1]-Alt2: SSB X kHz is considered for D2R transmission bandwidth. The value is for one sideband, i.e., the total transmission bandwidth for SSB is X kHz. Not applicable for at least device 1 $X = \{[15 (M)], [180 (O)]\}$, other values are not precluded and reported by companies
[2a2]	[OOK/BPSK/BFSK chip rate]	Companies to report

[2a3]	Receiver bandwidth	D2R receiver bandwidth is the bandwidth used at the reader side to filter out the D2R signals for calculating noise and interference (if any) power. Assume the receiver matches the transmitter's modulation, i.e., to receiver uses SSB when transmitter uses SSB, receiver uses DSB when transmitter uses DSB. Companies to report the value, and further down-selection of the values and DSB/SSB is not precluded.
[2b]	Waveform (CW)	Companies to report waveform, e.g., unmodulated single tone, multi-tone(multiple unmodulated single tone)
[2d]	Modulation	Companies to report modulation, e.g., OOK, BPSK, BFSK
[2e]	Line code	Companies to report, e.g., Manchester encoding, FM0 encoding, Miller encoding, no line coding
[2g]	FEC	Companies to report, e.g., CC, No FEC
[2h]	ADC bit width	Companies to report, e.g., 11-bit
[2j]	D2R receiver	Companies to report, e.g., coherent receiver / non-coherent receiver
Other assumptions		
[3a]	Other assumptions	To be reported by company
[3b]	Note: Companies to report required SINR/SNR/CINR/CNR according to BLER target.	

The following layouts are used for evaluation purposes.

Table 4.2.2-2: Assumptions on layout for D1T1 and D2T2

Parameter	Assumptions for D1T1	Assumptions for D2T2	
Scenario	InF-DH	InH-office	InF-DL
Hall size	120x60 m	120 x50 m	300x150 m
Room height	10 m	3m	10 m
Sectorization	None		
BS deployment / Intermediate UE dropping	18 BSs on a square lattice with spacing D, located $D/2$ from the walls. $L=120\text{m} \times W=60\text{m}$; $D=20\text{m}$ BS height = 8 m 	$L=120\text{m} \times W=50\text{m}$; Intermediate UE height = 1.5 m Option 1: Intermediate UEs are assumed to be deployed in a grid with $D = \text{e.g. } 10 \text{ m} / 20 \text{ m}$. Option 2: Intermediate UEs are assumed to be uniformly distributed over the horizontal area. Number of UE is either 1 or 2.	$L=300\text{m} \times W=150\text{m}$; Intermediate UE height = 1.5 m Option 1: The intermediate UEs are assumed to be deployed in a grid with $D = \text{e.g. } 10 \text{ m} / 20 \text{ m}$. Option 2: Intermediate UEs are assumed to be uniformly distributed over the horizontal area. Number of UE is either 1 or 2.
Device distribution	Device Height= 1.5 m A-IoT devices drop uniformly distributed over the horizontal area	Device Height= 1.5 m A-IoT devices drop uniformly distributed over the horizontal area Option 1: The devices within the maximum distance, which is obtained by the coverage evaluation, from each intermediate UE are involved in the evaluations. Option 2: The devices within the applicable maximum distance target are involved in the evaluations.	Device Height= 1.5m A-IoT devices drop uniformly distributed over the horizontal area Option 1: The devices within the maximum distance, which is obtained by the coverage evaluation, from each intermediate UE are involved in the evaluations. Option 2: The devices within the applicable maximum distance target are involved in the evaluations.
Device mobility (horizontal plane only)	3 kph	3 kph	3 kph
NOTE: The CW distribution in D1T1-B or D2T2-B is reported by companies			

4.3 Link budget

4.3.1 Receiver sensitivity

The study uses the following definitions for receiver sensitivity.

Budget-Alt1: Receiver sensitivity is derived by a predefined threshold and no link-level simulation is needed for link budget calculation

The results rely on the received sensitivity and maximum transmit power, and directly calculate the maximum distance / pathloss based on these values and other related parameters. The link-level simulation performances, such as required SINR, can be satisfied for such case and no link-level simulation is needed for link budget calculation.

Budget-Alt1 receiver sensitivity values are summarized in Annex A.

Budget-Alt2: Receiver sensitivity is derived by required SINR which is given by LLS results

The results rely on link-level simulation results, e.g., required SINR which corresponds to detail LLS assumptions (e.g., BW, coding, data rate). And based on the required SINR, the received sensitivity can be calculated and then the maximum distance / pathloss can be derived.

Note: For noise power, a noise figure value needs to be provided.

4.3.2 Link budget template

Link budget is calculated according to the following Table 4.3.2-1. (M) denotes the value is mandatory to be evaluated. (O) denotes the value can be optionally evaluated.

Table 4.3.2-1: Link budget template

No.	Item	Reader-to-Device	Device-to-Reader
(0) System configuration			
[0A]	Scenarios	D1T1-A1/A2/B/C D2T2-A1/A2/B/C	D1T1-A1/A2/B/C D2T2-A1/A2/B/C
[0A1]	CW case	N/A	1-1/1-2/1-4/2-2/2-3/2-4
[0B]	Device 1/2a/2b	Device 1/2a/2b	Device 1/2a/2b
[0C]	Center frequency (MHz)	900MHz (M), 2GHz (O)	900MHz (M), 2GHz (O)
[0D]	Topology/Pathloss model	For D2T2: BS pathloss model is reused for intermediate UE with antenna height = 1.5 m [0D]-Alt1: InF-DL NLOS [0D]-Alt2: InH-Office LOS For D1T1: InF-DH NLOS	For D2T2: BS pathloss model is reused for intermediate UE with antenna height = 1.5 m [0D]-Alt1: InF-DL NLOS [0D]-Alt2: InH-Office LOS For D1T1: InF-DH NLOS
(1) Transmitter			
[1D]	Number of Tx antenna elements / TxRU/ Tx chains modelled in LLS	For BS: 2(M) or 4(O) antenna elements for 0.9 GHz For Intermediate UE: 1(M) or 2(O)	1
[1E]	Total Tx Power (dBm)	For BS in DL spectrum for indoor - [1E]-R2D-Alt1: 33dBm(M), - [1E]-R2D-Alt2: 38dBm(O), - [1E]-R2D-Alt3: 24dBm(M) - Companies to report if PSD constraints are imposed (company to report the condition for applying PSD constraints in Row [5A]) For UL spectrum for indoor, - [1E]-R2D-Alt4: 23dBm (M) - [1E]-R2D-Alt5: 26dBm(O)	For device 1/2a: (see note 1) - [1E]-D2R-Alt1: For scenarios 'B', the device Tx Power is calculated by CW received power which can be derived by at least CW2D distance (m) value and other related factors. - [1E]-D2R-Alt2: For scenarios 'A1' and 'A2', the device Tx Power is calculated by assuming CW2D pathloss = D2R pathloss. For device 2b: For scenarios 'C' - [1E]-D2R-Alt3: -20 dBm(M) - [1E]-D2R-Alt4: -10 dBm(O)
[1E1]	CW Tx power (dBm)	N/A	For scenario 'A1', 'A2' and 'B' - Report a value from the candidate values [1E]-R2D-Alt1 / [1E]-R2D-Alt2 / [1E]-R2D-Alt3 from [1E]-R2D if CW in DL spectrum - Report a value from the candidate values [1E]-R2D-Alt4 / [1E]-R2D-Alt5 from [1E]-R2D if CW in UL spectrum.

			Note: only applicable for device 1/2a
[1E2]	CW Tx antenna gain (dBi)	N/A	Company to report, the value equals: - UE Tx ant gain, or - BS Tx ant gain Note: Only applicable for device 1/2a
[1E3]	CW2D distance (m)	N/A	For scenarios 'B' D1T1-B: - 5m, - 10m, - 20m - CW2D distance is derived assuming CW node is located with the same position as 'R1' in 'A1' scenario. (See note 1) D2T2-B: - 5m, - 10m, For scenarios 'A1' and 'A2': Calculated (see note 1), (i.e., CW2D distance is calculated by assuming CW2D pathloss = D2R pathloss) Note 1: Only applicable for device 1/2a. Note 2: Companies to report which value(s) are evaluated.
[1E4]	CW2D pathloss (dB)	N/A	Calculated (see note1)
			Note 1: Only applicable for device 1/2a
			Note 2: For CW2D pathloss model, use the same pathloss model as used for R2D/D2R.
[1E5]	CW received power (dBm)	N/A	Calculated (see note1)
			Note: Only applicable for device 1/2a
[1F]	Transmission Bandwidth used for the evaluated channel (Hz)	180kHz(M), 360kHz(O), 1.08MHz(O)	Refer to LLS table [2a1]
[1G]	Tx antenna gain (dBi)	For BS for indoor, 6 dBi(M), 2dBi(M) For intermediate UE, 0 dBi	For A-IoT device, 0dBi
[1H]	Ambient IoT backscatter loss (dB) due to Modulation factor	N/A	OOK: 6 dB PSK: 0 dB FSK: Y dB It is applicable for device 1 and 2a.
			Companies to report and justify their assumptions for Y.
			Companies to report in row 3D if they assume any additional related loss.
[1J]	Ambient IoT on-object antenna penalty	N/A	0.9dB or 4.7dB
[1K]	Ambient IoT backscatter amplifier gain (dB)	N/A	10 dB (M) 15 dB (O) Note: Only for device 2a
[1N]	Cable, connector, combiner, body losses, etc. (dB)	For BS, X dB, X <=3 to be reported by companies with justification provided in row 5A For intermediate UE, 1 dB	N/A
[1M]	EIRP (dBm)	Calculated (see Note 1)	Calculated (see Note 1)
(2) Receiver			

[2A]	Number of receive antenna elements / TxRU / chains modelled in LLS	Same as [1D]-D2R	Same as [1D]-R2D
[2B]	Bandwidth used for the evaluated channel (Hz)	Refer to LLS table [1b] ED bandwidth	Refer to LLS table [2a3]
[2C]	Receiver antenna gain (dBi)	same as [1G]-D2R	Same as [1G]-R2D
[2X]	Cable, connector, combiner, body losses, etc. (dB)	N/A	Same as [1N]-R2D
[2D]	Receiver Noise Figure (dB)	<u>For RF-ED receiver</u> 20dB, Device 2 <u>For IF/ZIF receiver</u> 15dB, Device 2 Other values can be optionally used and reported by companies	For BS as reader: 5dB For intermediate UE as reader: 7dB
[2E]	Thermal Noise power spectrum density (dBm/Hz)	-174	-174
[2F]	Noise Power (dBm)	Calculated (see Note 1)	Calculated (see Note 1)
[2G]	Required SNR/CNR	Reported by companies for Budget-Alt2	Reported by companies for Budget-Alt2
[2H]	Ambient IoT on-object antenna penalty	0.9 dB or 4.7 dB	Not applicable
[2J]	Budget-Alt1/ Budget-Alt2	Budget-Alt1/ Budget-Alt2 (see note1)	Budget-Alt2
[2K]	CW cancellation (dB)	N/A	Companies to report for scenario A2/A1/B for BS and intermediate UE. Notes: - Only applicable for device 1/2a - The value provided is for the unmodulated single-tone CW. The value for other cases (e.g. multi-tone CW) can be reported by companies.
[2K1]	Remaining CW interference (dBm)	N/A	Calculated (see Note 1) Note: only applicable for device 1/2a
[2K2]	Receiver sensitivity loss(dB)	N/A	Calculated (see Note 1) Note: only applicable for device 1/2a
[2L]	Receiver Sensitivity (dBm)	<u>For Budget-Alt1</u> For device 1 (RF-ED), for example: {-30 dBm, -36 dBm, -40 dBm, etc} For device 2 (RF-ED), for example: {-40 dBm, -45 dBm, etc} <u>For Budget-Alt2</u> Calculated (see note1)	Calculated (see Note 1) Note 2L: the receiver sensitivity includes the receiver sensitivity loss [2K2], i.e. after CW cancellation at least if 'A2' scenario is used
(3) System margins			
[3A]	Shadow fading margin (dB)	For D1T1: 4 dB For D2T2: 3dB for InH-LOS 7.2dB for InF-DL-NLOS	For D1T1: 4 dB For D2T2: 3dB for InH-LOS 7.2dB for InF-DL-NLOS
[3B]	polarization mismatching loss (dB)	3 dB	3 dB

[3C]	BS selection/macro-diversity gain (dB)	0 dB Note: only applicable for D1T1	0 dB Note: only applicable for D1T1
[3D]	Other gains (dB) (if any please specify)	Reported by companies with justification	Reported by companies with justification
(4) MPL / distance			
[4A]	MPL (dB)	Calculated (see Note 1)	Calculated (see Note 1)
[4B]	Distance (m)	Calculated (see Note 1)	Calculated (see Note 1)
(5) Other			
[5A]	Other notes	Companies to report	Companies to report

Note 1: Calculated values are derived according to the following.

[1E3]

- For scenarios where CW2D distance is calculated by assuming CW2D pathloss = D2R pathloss, [1E3] is derived by assuming pathloss is [1E4] and use the pathloss formula as agreed.

[1E4]

- For scenarios where CW2D distance is calculated by assuming CW2D pathloss = D2R pathloss
 - $[1E4] = 0.5 * ([1E1] + [1E2] - [1N](R2D) + [2C](R2D) - [2H](R2D) - 2*[3A] - 2*[3B] + [3D](R2D) + [1K] - [1H] + [1G] - [1J] + [2C] - [2X] - [2L] + [3C] + [3D])$
 - [1K] is only for device 2a
- Otherwise
 - [1E4] is derived according to pathloss formula by assume distance is [1E3]

[1E5]

- $[1E5] = [1E1] + [1E2] - [1N](R2D) - [1E4] + [2C](R2D) - [2H](R2D) - [3A] - [3B] + [3D](R2D)$

[1E]

- $[1E] = [1E5] + [1K] - [1H]$
- [1K] is only for device 2a

[1M]:

- For R2D,
 - $[1M] = [1E] + [1G] - [1N]$
- For D2R
 - $[1M] = [1E] + [1G] - [1J]$

[2F]:

- $[2F] = [2D] + [2E] + \ln 2 dB([2B])$

[2G]

- For the R2D LLS for ED, CINR/CNR is reported, where CINR/CNR is defined as the ratio of signal power spectral density in the transmission bandwidth to the noise and interference (if any) power spectral density in the device ED channel bandwidth.
- For R2D ZIF receiver, report the same metrics (i.e., CNR/CINR, signal transmission bandwidth, ED bandwidth) as agreed for RF-ED/IF receiver.

- For the D2R LLS, the SINR/SNR is reported and it is defined as the ratio of signal power to noise and interference (if any) power in the receiver bandwidth. Receiver bandwidth is the bandwidth used at the reader side to filter the D2R signals for calculating noise and interference (if any) power.
- On/off keying backscatter loss (including DC removal loss) is not taken into account in the LLS and is included in link budget table [1H].

[2J]

- For R2D link in the coverage evaluation, for device 1
 - Budget-Alt1 is used (note: receiver architecture is RF ED)
- For R2D link in the coverage evaluation for device 2,
 - Budget-Alt1 is used if receiver architecture is RF ED
 - Budget-Alt2 is used if receiver architecture is IF/ZIF ED

Note A: this does not preclude to have LLS for device 1 and 2 R2D link with RF-ED if needed.

Note B: For device 2 R2D link with RF-ED, *Budget-Alt1* is mandatory, *Budget-Alt2* is optional.

Note C: this does not imply all M values are achievable with the sensitivity given by *Budget-Alt1* for RF ED

Note D: For device 2 with an RF ED-based receiver on the R2D link, if the receiver sensitivity derived from *Budget-Alt2*, assuming a noise figure in [2D] from the table, exceeds the receiver sensitivity based on *Budget-Alt1*, then *Budget-Alt2* is applied.

[2K1]:

- $[2K1] = [1E1] + [1E2] - [1N](R2D) + [2C] - [2X] - [2K]$

[2K2]:

- $[2K2] = \text{lin2dB} \left(1 + \frac{\text{dB2lin}([2K1])}{\text{dB2lin}([2F])} \right)$

[2L]:

- For R2D and *Budget-Alt2*,
 - $[2L] = [2G] - \text{lin2dB}([2B] / [1F]) + [2F]$

Note E: The term ' $\text{lin2dB}([2B] / [1F])$ ' is applied due to scaling from CNR/CINR to SNR/SINR.

- For D2R,
 - $[2L] = [2G] + [2F] + [2K2]$, device 1/2a
 - $[2L] = [2G] + [2F]$, device 2b

[4A]

- For R2D
 - $[4A] = [1M] + [2C] - [2H] - [2L] - [3A] - [3B] + [3C] + [3D]$
- For D2R
 - $[4A] = [1M] + [2C] - [2X] - [2L] - [3A] - [3B] + [3C] + [3D]$

[4B]

- [4B] is derived by assuming pathloss is [4A] and using the pathloss formula as agreed.

4.4 R2D waveform generation

With reference to the R2D waveform described in Clause 6.1.1.1, for evaluation purposes the waveform for DFT-s-OFDM is generated as follows:

1. The time domain OOK signal is the M chips of one OFDM symbol.
2. A chip is represented (e.g. upsampled) by L samples
 - Companies to report L
3. An N' -points DFT is performed on the samples of one OFDM symbol to obtain the frequency domain signal.
 - Companies to report N' , e.g. $N'=128$ or equal to X
4. Map the frequency domain signal obtained by N' -points DFT to the X subcarriers of $B_{\text{tx,R2D}}$.
 - Companies report how to map and report X
5. An N -points IDFT is performed to obtain the time domain signal.
 - Companies to report N , and how value was selected

Note: Companies report whether/how CP samples are added.

5 Ambient IoT device architectures

5.1 $\sim 1 \mu\text{W}$ devices (Device 1)

The architecture of such a device is summarised in Figure 5.1-1, with the blocks described as follows.

- **Antenna** could be either shared or separate for RF energy harvester and receiver/transmitter.
- **Matching network** is to match impedance between antenna and other components (including RF energy harvester and receiver related blocks).
- **RF energy harvester** can include **rectifier** performing RF signal (AC) to DC conversion.
- **Energy storage** (e.g., capacitor) stores harvested energy from RF energy harvester.
- **Power management unit (PMU)** manages storing energy to energy storage from energy harvester and supplying power to active component blocks which needs power supply.
- **Digital BB logic** includes functional blocks like encoder, decoder, controller, etc.
- **Memory** can include two types of memory: 1) Non-Volatile Memory (NVM) such as EEPROM for permanently storing device ID, etc, and 2) registers for temporarily keeping any information required for its operation only while energy is available in energy storage.
- **Clock generator** provides required clock signal(s).
- **Reception related blocks**
 - **RF BPF** for improving selectivity.
 - Depending on implementation, it may not exist. RAN4 RF requirement (if any, e.g., ACS) and peak power consumption target also need to be considered.
 - **RF Envelope Detector** converts RF signal to baseband.
 - **BB LPF** can filter out harmonics and high frequency components to improve input signal quality to comparator.
 - Depending on implementation, it may not exist. Presence of BB LPF is assumed for the study.

- **Comparator** determines high/low of input signal.
- **Transmission related blocks**
 - **Backscatter modulator** switches impedance to modulate backscattered signal with transmitted signal from BB logic.

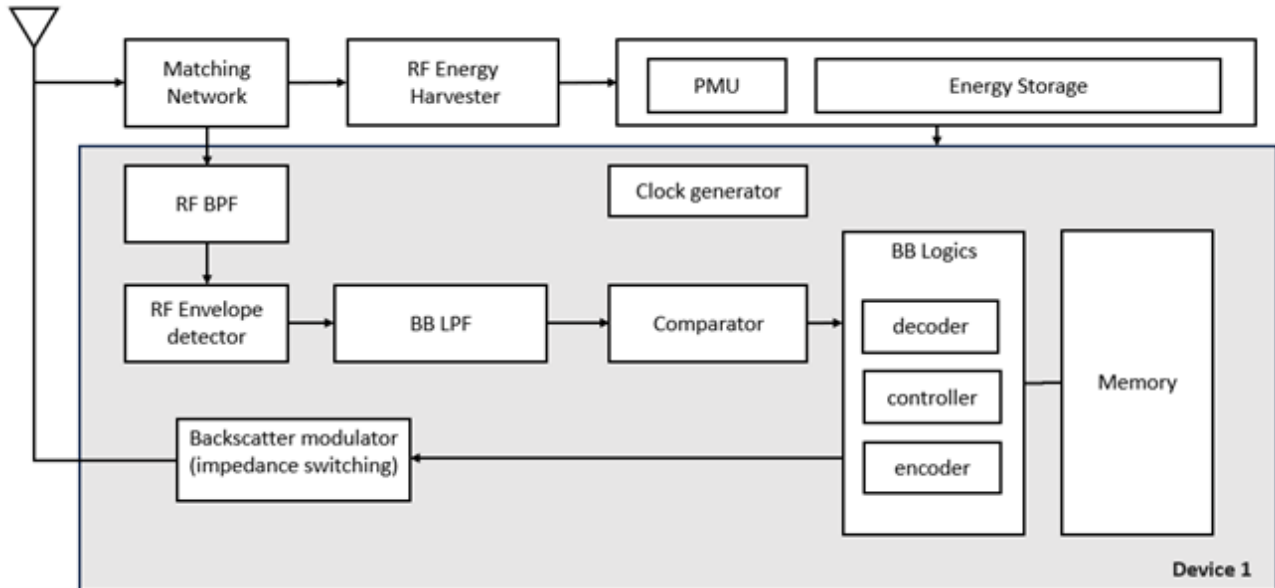


Figure 5.1-1: Architecture of device 1

5.2 $\leq$ a few hundred μ W devices (Device 2)

5.2.1 External carrier wave (Device 2a)

The architecture of device 2a is summarised in Figure 5.2.1-1, with the blocks described as follows.

- **Antenna** could be either shared or separate for RF energy harvester (if present) and receiver/transmitter.
- **Matching network** is to match impedance between antenna and other components (including RF energy harvester (if present) and receiver related blocks).
- **Energy harvester**.
- **Energy storage** (e.g., capacitor) stores harvested energy from energy harvester.
- **Power management unit (PMU)** manages storing energy to energy storage from energy harvester and supplying power to active component blocks which needs power supply.
- **Digital BB logic** includes functional blocks like encoder, decoder, controller, etc.
- **Memory** can include two types of memory: 1) Non-Volatile Memory (NVM) such as EEPROM for permanently storing device ID, etc, and 2) registers for temporarily keeping any information required for its operation only while energy is available in energy storage.
- **Clock generator** provides required clock signal(s).
- **Reflection amplifier** can amplify reflected backscattered signal.
 - At least one of R2D/CW2D and D2R could be amplified by either reflection amplifier or LNA.
- **Reception related blocks**
 - **RF BPF** filter for improving selectivity.

- Depending on implementation, it may not exist. RAN4 RF requirement (if any, e.g., ACS) and peak power consumption target also need to be considered.
- **LNA** for improving signal strength and sensitivity of receiver, if present.
 - At least one of R2D/CW2D and D2R could be amplified by either reflection amplifier or LNA.
- **RF envelope detector (RF-ED)** detects envelope from RF signal.
- **BB amplifier** amplifies BB signal to improve signal strength.
- **BB LPF** can filter out harmonics and high frequency components to improve input signal quality to comparator/ADC.
 - Depending on implementation, it may not exist.
- **Comparator** or **N-bit ADC**
- **Transmission related blocks**
 - **Backscatter modulator** switches impedance to modulate backscattered signal with tx signal from BB logics.
 - **Large Frequency shifter** (e.g., tens of MHz) for shifting backscattered signal from one frequency (e.g., FDD-DL frequency) to another frequency (e.g., FDD-UL frequency).

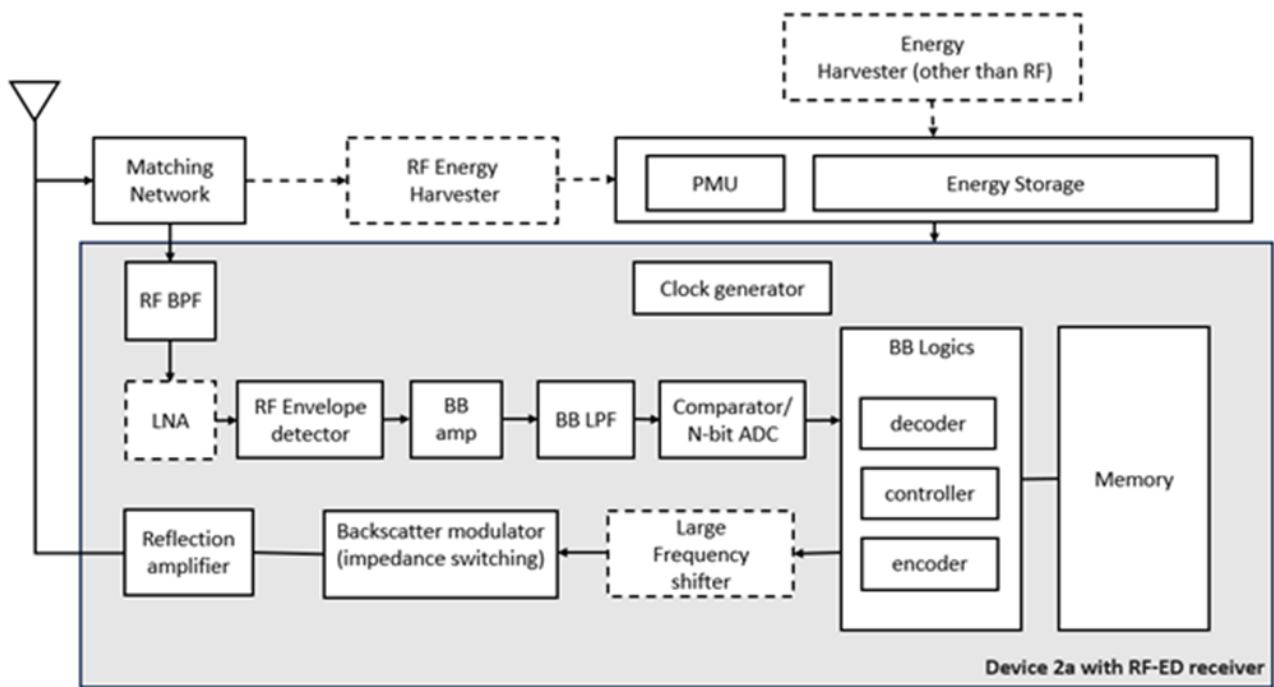


Figure 5.2.1-1: Architecture of device 2a

IF envelope detector receiver and ZIF receiver for device 2a are de-prioritized.

5.2.1.1 Reflection amplifier

For the reflection amplifier block, the following characteristics are considered for device 2a:

- Direction of amplification
 - Uni-directional reflection amplifier (baseline) can amplify backscattered signal in D2R which can improve D2R link budget.

- Bi-directional amplifier can amplify both signal in R2D and backscatter signal in D2R at least when R2D and D2R are in the same spectrum.
- Bi-directional amplifier has higher complexity, higher noise figure, and reduced isolation between tx and rx path.
- Amplification gain ranges from 10 to 20 dB.
- Power consumption of reflection amplifier is in the range of a tens of uW to 100s of uW.
- Reflection amplifier can operate in FDD frequency bands.
- Reflection amplifier bandwidth can support 10s of MHz.
- Note: reflection amplifier can get unstable when the input power exceeds a certain value, which may be frequency-dependent.

5.2.1.2 Large frequency shifter

For the large frequency shifter block, it is observed that:

- Large frequency shift can be used in shifting reflected signal in tens of MHz, e.g., from FDD DL to FDD UL frequency or vice versa.
- Large frequency shift consumes 10s of uW to 100s of uW.
- Large frequency shift is not feasible for device 1.
- Large frequency shift requires a clock for IF generation which is accurate enough to avoid large guard band and interference to adjacent channels/bands.
- Large frequency shift requires image suppression and may require harmonics suppression
- Note: details of image suppression and harmonics suppression are not discussed in RAN1
- Large frequency shift may allow the reader to avoid implementing in-band full duplex capability for scenarios e.g., D1T1-A2 and D2T2-A2.
- Large frequency shift may result in e.g., 5 kHz - 50 kHz of frequency uncertainty in target frequency for clock accuracy of e.g., 0.01% - 0.1% assuming the large frequency shift range is 50 MHz.

Large frequency shift may be technically feasible in terms of power consumption, and device complexity.

Remaining technical feasibility of large frequency shift in terms of image suppression, harmonics suppression may be relevant to device emission requirement, spectrum usage. Additional complexity and power consumption may be needed to achieve image suppression and harmonics suppression.

The necessity or effectiveness of large frequency shift may depend on spectrum for R2D/CW2D/D2R, full duplex capability of reader, whether device 1 and 2a are supported by a single reader, spectrum regulation/usage, frequency error, coverage, etc.

5.2.2 Internally-generated carrier wave (Device 2b)

5.2.2.1 RF envelope detector receiver

The architecture of device 2b with an RF envelope detector receiver is summarised in Figure 5.2.2.1-1, with the blocks described as follows.

- **Antenna** could be either shared or separate for RF energy harvester (if present) and receiver/transmitter.
- **Matching network** is to match impedance between antenna and other components (including RF energy harvester (if present) and receiver related blocks).
- **Energy harvester** for harvesting energy from e.g., RF signal, solar, vibration/movement, temperature difference, etc

- **Energy storage** (e.g., capacitor) stores harvested energy from energy harvester.
- **Power management unit (PMU)** manages storing energy to energy storage from energy harvester and supplying power to active component blocks which needs power supply.
- **Digital BB logic** includes functional blocks like encoder, decoder, controller, etc.
- **Memory** can include two types of memory: 1) Non-Volatile Memory (NVM) such as EEPROM for permanently storing device ID, etc, and 2) registers for temporarily keeping any information required for its operation only while energy is available in energy storage.
- **Clock generator** provides required clock signal(s).
- **Reception related blocks**
 - **RF BPF** filter for improving selectivity.
 - Depending on implementation, it may not exist. RAN4 RF requirement (if any, e.g., ACS) and peak power consumption target also need to be considered.
 - **LNA** for improving signal strength and sensitivity of receiver, if present
 - **RF envelope detector (RF-ED)** detects envelope from RF signal.
 - **BB amplifier** amplifies BB signal to improve signal strength.
 - **BB LPF** can filter out harmonics and high frequency components to improve input signal quality to comparator/ADC.
 - Depending on implementation, it may not exist.
 - Comparator or N-bit ADC
- **Transmission related blocks**
 - **Tx Modulator**: baseband bits are modulated according to modulation scheme. This block could be the part of BB logic.
 - **Digital to Analog Converter (DAC)** converts digital signal to analog signal.
 - **Low pass filter** for filtering out undesired signal
 - **Mixer** performs up converting baseband signal to RF range.
 - **Local oscillator (LO)** for carrier frequency generation
 - FLL(/PLL) can be used for frequency synthesis
 - Depending on implementation, FLL(/PLL) may not exist.
 - **Power amplifier (PA)** amplifies tx signal, if present
 - Details on transmitter related blocks depends on tx waveform/modulation.

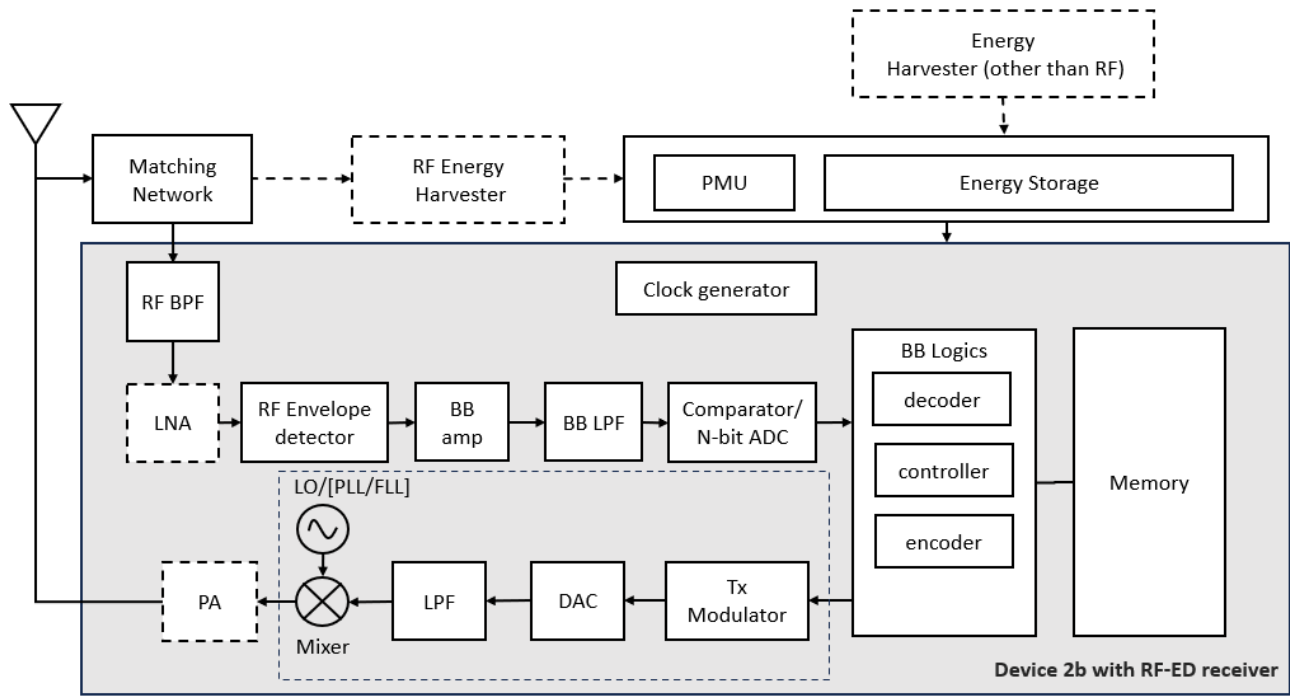


Figure 5.2.2.1-1: Architecture of device 2b with RF-ED receiver

5.2.2.2 IF envelope detector receiver

The architecture of device 2b with an IF envelope detector receiver is summarised in Figure 5.2.2.2-1, with the blocks described as follows.

- **Antenna** could be either shared or separate for RF energy harvester (if present) and receiver/transmitter
- **Matching network** is to match impedance between antenna and other components (including RF energy harvester (if present) and receiver related blocks)
- **Energy harvester** for harvesting energy from e.g., RF signal, solar, vibration/movement, temperature difference, etc.
- **Energy storage** (e.g., capacitor) stores harvested energy from energy harvester
- **Power management unit (PMU)** manages storing energy to energy storage from energy harvester and supplying power to active component blocks which needs power supply
- **Digital BB logic** includes functional blocks like encoder, decoder, controller, etc.
- **Memory** can include two types of memory: 1) Non-Volatile Memory (NVM) such as EEPROM for permanently storing device ID, etc, and 2) registers for temporarily keeping any information required for its operation only - while energy is available in energy storage

Clock generator provides required clock signal(s).

- **Local oscillator (LO)** for generating carrier frequency for Tx, or for generating carrier frequency offset by the IF for Rx
 - FLL(/PLL) can be used for frequency synthesis
 - One LO or separate LOs for Tx and Rx
- **Reception related blocks**
 - **RF BPF** filter for improving selectivity
 - Depending on implementation, it may not exist. RAN4 RF requirement (if any, e.g., ACS) and peak power consumption target also need to be considered

- **LNA** for improving signal strength and sensitivity of receiver, if present
- **Mixer** down converts RF signal to IF stage
 - Depending on implementation, there could be one or two mixers for Rx and Tx
- **IF amplifier** amplifies IF signal
- **IF filter** for filtering out unwanted RF and LO signals
- **IF envelope detector (IF-ED)** detects envelope from IF signal.
- **BB amplifier**
 - Depending on implementation, one or both of IF amplifier and BB amplifier may exist
- **BB LPF** can filter out harmonics and high frequency components to improve input signal quality to comparator/ADC
 - Depending on implementation, it may not exist
- Comparator or N-bit ADC
- Note: image rejection is required
- **Transmission related blocks**
 - **Tx Modulator**: baseband bits are modulated according to modulation scheme. This block could be the part of BB logic
 - **Digital to Analog Converter (DAC)** converts digital signal to analog signal
 - **Low pass filter** for filtering out undesired signal
 - **Mixer** performs up converting baseband signal to RF range
 - **Power amplifier (PA)** amplifies transmitted signal, if present
 - Details of transmitter related blocks depends on e.g., waveform/modulation, etc.

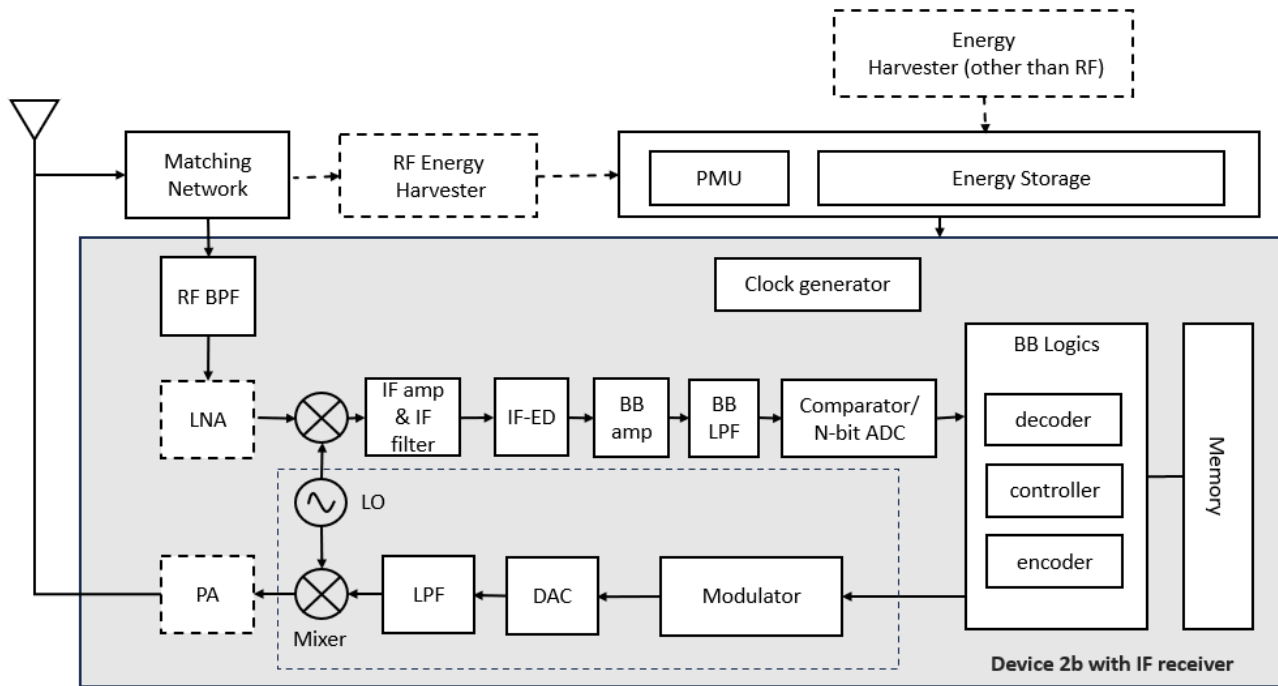


Figure 5.2.2.1-1: Architecture of device 2b with IF-ED receiver

5.2.2.3 ZIF receiver

The architecture of device 2b with a ZIF receiver is summarised in Figure 5.2.2.3-1, with the blocks described as follows.

- **Antenna** could be either shared or separate for RF energy harvester (if present) and receiver/transmitter
- **Matching network** is to match impedance between antenna and other components (including RF energy harvester (if present) and receiver related blocks)
- **Energy harvester** for harvesting energy from e.g., RF signal, solar, vibration/movement, temperature difference, etc
- **Energy storage** (e.g., capacitor) stores harvested energy from energy harvester.
- **Power management unit (PMU)** manages storing energy to energy storage from energy harvester and supplying power to active component blocks which needs power supply
- **Digital BB logic** includes functional blocks like encoder, detector, decoder, controller, etc.
- **Memory** can include two types of memory: 1) Non-Volatile Memory (NVM) such as EEPROM for permanently storing device ID, etc, and 2) registers for temporarily keeping any information required for its operation only while energy is available in energy storage
- **Clock generator** provides required clock signal(s).
- **Local oscillator (LO)** for generating carrier frequency for Tx and Rx
 - FLL(/PLL) can be used for frequency synthesis
 - One LO or separate LOs for Tx and Rx
- **Reception related blocks**
 - **RF BPF** filter for improving selectivity
 - Depending on implementation, it may not exist. RAN4 RF requirement (if any, e.g., ACS) and peak power consumption target also need to be considered

- **LNA** for improving signal strength and sensitivity of receiver, if present
- **Mixer** down converts RF signal to BB stage
 - Depending on implementation, there could be one or two mixers for Rx and Tx
- **BB amplifier** amplifies BB signal
- **BB LPF** can filter out undesired frequency components to improve input signal quality to comparator/ADC.
 - Depending on implementation, it may not exist
- Comparator or N-bit ADC
- **Transmission related blocks**
 - **Tx Modulator**: baseband bits are modulated according to modulation scheme. This block could be the part of BB logic.
 - **Digital to Analog Converter (DAC)** converts digital signal to analog signal.
 - **Low pass filter** for filtering out undesired signal
 - **Mixer** performs up converting baseband signal to RF range.
 - **Power amplifier (PA)** amplifies transmitted signal, if present
 - Details of transmitter related blocks depend on e.g., waveform/modulation, etc.

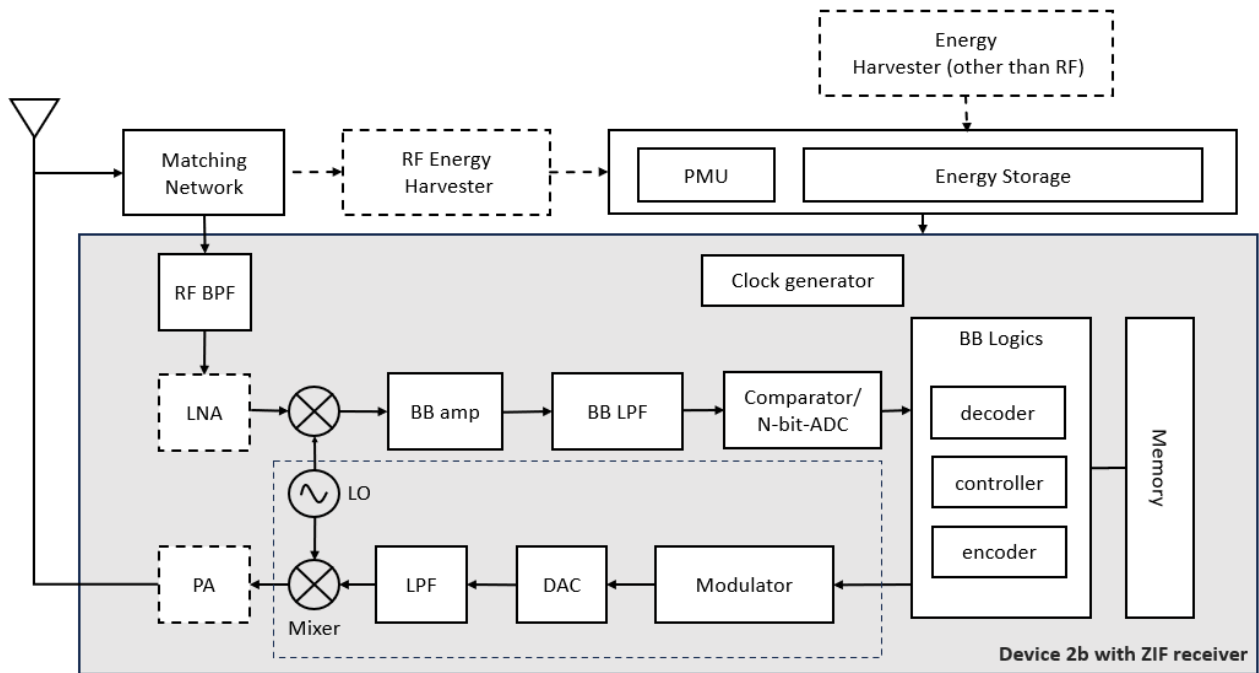


Figure 5.2.2.1-1: Architecture of device 2b with ZIF receiver

5.3 Clock(s)

For A-IoT device clock, 9 sources [7], [9], [8], [16], [10], [12], [11], [15], [17] have reported that clock calibration based on at least clock counting for a known duration is feasible for A-IoT devices.

Table 5.3-1: Descriptions of clocks/LOs

Clock purpose #	Description	Clock speed	Applicable devices	Power consumption	Initial clock Accuracy [ppm]	Accuracy after clock sync / calibration at device side [ppm]
1	Sampling	A few MHz	1 (Note 3)	<1 μ W		10^5 (Note 2)
						10^4 (Note 2) the study did not determine whether or not this accuracy can be achieved with < 1 μ W power consumption at least for device 1
			2a, 2b	<10 μ W	$10^3 - 10^4$	10^3 (Note 2)
2	Small frequency shift	A few MHz	1	<1 μ W		10^5 (Note 2)
						10^4 (Note 2) the study did not determine whether or not this accuracy can be achieved with < 1 μ W power consumption at least for device 1
			2a, 2b (if applicable)	<10 μ W		10^3 (Note 2)
3	Time counting (if supported)	e.g. tens of kHz	1 (if applicable) 2a, 2b	<0.1 μ W	$10^4 - 10^5$	Calibration may not always be feasible for this clock purpose
4	Large frequency shift (if supported for device 2a)	10s of MHz (e.g. 50 MHz)	2a	10s of μ W	$10^3 - 10^4$	10^3 (Note 2) Whether better accuracy can be achieved has not been determined.
5	LO for carrier frequency (for up/down conversion)	According to carrier frequency e.g., 900 MHz	2b	10s - 100s of μ W	$10^3 - 10^4$	Within 10s – 200 (Note 2), there is no need to determine a single value for the study
Note 1: This table does not necessarily imply that different purposes of LOs/clocks correspond to separate discrete LOs/clocks, which is up to implementation. Note 2: No drastic temperature change is assumed during calibration. Note 3: Device 2a, 2b could also use similar implementation.						

Several sources [10], [15], [13], [14], [16], [9] think that for clock purpose #1 (sampling), accuracy after clock sync / calibration at device side of 10^4 ppm is achievable with < 1 μ W power consumption for device 1, and that 10^5 ppm may not be sufficient for certain designs.

- Source [10] mentioned that device 1 can support basic calibration operation within a reasonable power consumption or complexity, achieving accuracy of 10^4 ppm.
- Source [15] mentioned that with clock calibration based on a clock acquisition part design, 10^4 ppm could be achieved after clock calibration.

- Source [13] reported that clock purpose #1 could be applied to all device types. But, considering complexity and power, device 1 and 2 could have different power, accuracy. For device1, the power consumption should be $<1 \mu\text{W}$; the timing accuracy should be $10^4 - 10^5$ ppm; the calibration can reach to 10^4 ppm.
- Source [14] proposed device 1 with accuracy of 10^4 ppm after clock sync/calibration.
- Source [16] reported that using simple clock counting based calibration method, 1% could be achieved with power consumption $<1 \mu\text{W}$. It was reported that 1% is the sweet spot of target accuracy considering performance, complexity, signal design, etc.
- Source [9] mentioned that device can use recommended clock calibration method to get high accuracy of $10^3 - 10^4$ ppm for device 1 and $10^2 - 10^3$ ppm for device 2.

6 Solutions for Ambient IoT

6.1 Physical layer

6.1.0 Aspects in common to R2D and D2R

6.1.0.1 General

The names of physical channels and signals used in this TR are for the sake of the study.

The study assumes that an A-IoT device has a single antenna for both communication (transmission/reception) and RF energy harvesting purposes.

The study defines repetition types as follows:

- Block level: All the bits received from higher layers and/or physical layer (according to what is present) after CRC attachment (if used) are blockwise repeated Rblock times
- Bit level type 1: Each bit after CRC attachment (if used) is repeated Rbit times
- Bit level type 2: Each bit after both CRC attachment (if used) and FEC (if used) is repeated Rbit times
- Chip level: Each chip after line coding (if used) or after square wave modulation (if used) is repeated Rchip time. NOTE: This is equivalent to extending the duration of each chip by Rchip times.

6.1.0.2 CRC

Sources as follows recommend that the length of the CRC used for A-IoT depends on:

- Block size or TBS-like size, according to sources [22], [23], [19], [26], [25], [28], [49], [44], [51]. Furthermore:
 - Source [22] states that specifically for short messages, the CRC length can directly impact the transmission efficiency due to the overhead caused by the CRC.
 - Source [49] state that CRC length depends on the TBS due to its impact on the transmission overhead and FAR performance. When the TBS is too small, the overhead of CRC-6 becomes quite significant.
 - Source [51] states that CRC length is fundamentally proportional to the size of the message.
- Target false alarm rate or error detection performance, according to sources [22], [23], [19], [26], [28], [41], [51]. Furthermore:
 - Source [22] evaluated that the probability of undetected errors is higher for CRC-6 as compared to CRC-16, and transmissions with CRC-16 performs better, for a given TBS=24, and source [41] say that a longer CRC would mean a better error detection performance.
- Type of message, such as a command/reply function, to determine when no CRC is used, according to sources [22], [23], [19], [26], [25], [41], [44]. Furthermore:

- Sources [22] and [23] say that similar to RFID, only important messages, decided based on their functionality, such as configuration/control/command messages, are appended with CRC to increase their reliability, while other messages are not protected by CRC to save the overhead.
- Source [19] proposes different CRC lengths for Paging-like/select/challenge messages that may or may not contain the device ID/group ID.
- Source [26] states that some messages such as a QueryRep-like command do not result in serious FAR issues if mis-detected, and hence the functionality of a message/command has to be considered.
- Overhead caused by CRC, according to sources [22], [23], [26], [49], [41], [45]
 - Sources [22], [23], [26], [41] mention that longer CRCs have high overhead but enjoy better error detection performance, affecting the transmission efficiency. In order to balance the overhead and performance, Hence, short CRC can be used for short commands to reduce the false alarm rate and make the system more robust as compared to no CRC.
 - Source [26] states that adding CRC to QueryRep-like commands which are sent repeatedly with a small payload would waste network resources, and hence the overhead should be considered.
 - Sources [49], [45] states that the overhead of CRC decreases as the TBS increases, and that CRC-16 has higher overhead than CRC-6.

For the CRC generator polynomials:

- Sources [24], [29], [27], [28], [38] recommend that the same polynomials from TS 38.212 are reused, giving justifications:
 - Sources [28], [29] state that the polynomials from TS 38.212 were already carefully and thoroughly evaluated and ensured in the NR channel coding design, so there is no need of considering other polynomials.
 - Source [27] states that the device complexity is increased when different polynomials are introduced for the D2R transmission.
 - Source [24] states that link performance is significantly impacted by the CRC lengths, and not the CRC polynomials, hence the polynomials from TS 38.212 should be used.
- On the other hand, sources [49], [45], [54] and [71] recommend that the same polynomial for CRC-6 ($g_{CRC6}(D) = [D^6 + D^5 + 1]$) from TS 38.212 is reused, but a new polynomial for CRC-16 is introduced, $g_{CRC16}(D) = [D^{16} + D^{11} + D^6 + D^5 + 1]$, incorporating the CRC-6 polynomial, giving justifications:
 - The proposed new CRC-16 polynomial nests the existing CRC-6 within its structure.
 - Using separate sets of circuits to implement the polynomials for CRC-6 and CRC-16 as per TS 38.212 requires a higher number of registers and XOR gates when compared to using the new CRC-16 polynomial.
 - FAR performance is the same for the legacy and proposed new polynomial for CRC-16, but the new polynomial reduces the complexity and power consumption of the device.

The following evaluation results of various CRC designs were provided during the study:

NOTE 1: P_{ud} = Probability of undetected error according to [75].

NOTE 2: Overhead = CRC length [bits] / (TBS [bits] + CRC length [bits]).

Table 6.1.0.2-1: CRC evaluations

TBS	CRC-6			CRC-16		
	Source	CRC overhead	P_{FA} or P_{ud}	Source	CRC overhead	P_{FA} or P_{ud}
8 bits	[22] [72]	43%	$\sim 10^{(-8)} P_{ud}$ @ 10^{-3} BER	[22] [72]	67%	$\sim 10^{(-11)} P_{ud}$ @ 10^{-3} BER
12 bits	[20] [72]	33% 33%	$\sim 10^{(-2)} P_{FA}$ $\sim 10^{(-8)} P_{ud}$ @ 10^{-3} BER	[20] [72]	57% 57%	$\sim 10^{(-5)} P_{FA}$ $\sim 10^{(-11)} P_{ud}$ @ 10^{-3} BER
16	[22]	27%		[22]	50%	

bits	[49] [72]	27% 27%	$\sim 10^{(-7)} P_{\text{Ud}} @ 10^{-3} \text{ BER}$	[49] [72]	50% 50%	$\sim 10^{(-10)} P_{\text{Ud}} @ 10^{-3} \text{ BER}$
24 bits	[22] [20] [49] [72]	20% 20% 20% 20%	$\sim 10^{(-6)} P_{\text{Ud}} @ 10\% \text{ BLER}$ and 2.7dB SNR $\sim 10^{(-2)} P_{\text{FA}}$ $\sim 10^{(-7)} P_{\text{Ud}} @ 10^{-3} \text{ BER}$	[22] [20] [72]	40% 40% 40%	$\sim 10^{(-9)} P_{\text{Ud}} @ 10\% \text{ BLER}$ and 2.8dB SNR $\sim 10^{(-5)} P_{\text{FA}}$ $\sim 10^{(-10)} P_{\text{Ud}} @ 10^{-3} \text{ BER}$
32 bits	[22] [49] [72]	16% 16% 16%	$\sim 10^{(-5)} P_{\text{Ud}} @ 10^{-3} \text{ BER}$	[22] [49] [72]	33% 33% 33%	$\sim 10^{(-10)} P_{\text{Ud}} @ 10^{-3} \text{ BER}$
40 bits	[22] [49] [72]	13% 13% 13%	$\sim 10^{(-5)} P_{\text{FA}} @ 10\% \text{ BLER}$ $\sim 10^{(-4)} P_{\text{Ud}} @ 10^{-3} \text{ BER}$	[22] [49] [72]	29% 29% 29%	$\sim 10^{(-10)} P_{\text{Ud}} @ 10^{-3} \text{ BER}$
48 bits	[20] [49] [72]	11% 11% 11%	$\sim 10^{(-2)} P_{\text{FA}}$ $\sim 10^{(-5)} P_{\text{FA}} @ 10\% \text{ BLER}$ $\sim 10^{(-4)} P_{\text{Ud}} @ 10^{-3} \text{ BER}$	[20] [49] [72]	25% 25% 25%	$\sim 10^{(-5)} P_{\text{FA}}$ $\sim 10^{(-10)} P_{\text{Ud}} @ 10^{-3} \text{ BER}$
96 bits	[22] [20] [49] [72]	6% 5% 6% 6%	$\sim 10^{(-2)} P_{\text{FA}}$ $\sim 10^{(-4)} P_{\text{Ud}} @ 10^{-3} \text{ BER}$	[22] [20] [49] [72]	14% 14% 14% 14%	$\sim 10^{(-5)} P_{\text{FA}}$ $\sim 10^{(-9)} P_{\text{Ud}} @ 10^{-3} \text{ BER}$
128 bits	[49] [72]	4% 4%	$\sim 10^{(-3)} P_{\text{Ud}} @ 10^{-3} \text{ BER}$	[49] [72]	11% 11%	$\sim 10^{(-9)} P_{\text{Ud}} @ 10^{-3} \text{ BER}$
256 bits	[22] [72]	2% 2%	$\sim 10^{(-3)} P_{\text{Ud}} @ 10^{-3} \text{ BER}$	[22] [72]	6% 6%	$\sim 10^{(-9)} P_{\text{Ud}} @ 10^{-3} \text{ BER}$

6.1.1 R2D

A dedicated physical broadcast channel, e.g. PBCH-like, and reference signals including at least DMRS, PTRS, CSI-RS/TRS, are not considered for R2D. R2D transmission without a midamble is studied as the baseline, if Manchester line coding is used; the necessity of a midamble for the R2D transmission requires further study if PIE line coding is used.

6.1.1.1 R2D waveform, modulation and numerology

An OFDM-based OOK waveform with subcarrier spacing of 15 kHz is studied for R2D, with OOK-1 for single-chip per OFDM symbol transmission, and OOK-4 for M -chip per OFDM symbol transmission, starting from the definitions in TR 38.869 [4]. For this waveform, the start of R2D transmission from the reader perspective is assumed to be aligned with the boundary of an NR OFDM symbol (including the CP) for in-band/guard-band operation. For the generation of this waveform, CP-OFDM and DFT-s-OFDM are both feasible. For $M=1$, i.e. use of OOK-1 or OOK-4 for single-chip per OFDM symbol transmission, CP-OFDM and DFT-s-OFDM are both feasible. For $M>1$, i.e. use of OOK-4 for M -chip per OFDM symbol transmission, DFT-s-OFDM is feasible. For R2D waveform generation using a DFT-s-OFDM based transmitter, from the transmitter perspective, when the generated number of chips for the R2D transmission does not fully occupy the last OFDM symbol, at least padding can be used.

R2D chip duration is a consequence of CP handling design.

For CP handling, the following candidate methods are studied, on the basis of e.g., CP impact on R2D timing acquisition, and decoding & performance of PRDCH, reader and device implementation complexities, interference between R2D and NR DL/UL if in the same NR band, spectrum efficiency.

Method Type 1: Removal of CP at device without specified transmit-side.

Method Type 2: Ensure the CP insertion of OFDM-based waveform will not introduce false rising/falling edge between the last OOK chip in OFDM symbol ($n-1$) and the first OOK chip in OFDM symbol n .

For Method Type 1, two ways that CP location/length can be determined are studied:

Alt M1-1: Device assumes same CP length for each OFDM symbol, i.e. does not distinguish exact CP length among different OFDM symbols

Alt M1-2: Duration between transition edges is utilized by device to determine CP location/length, i.e. if the duration appears to be invalid based on known chip duration

For Method Type 1, at least for Alt M1-1, device needs to be aware of or determine the boundary of an OFDM symbol (i.e. the beginning of an OFDM symbol) to determine CP location, including the start and the length of CP. Alt M1-1 and/or Alt M1-2 (if both are supported) can be up to device implementation which may depend on M values.

- Some sources [18], [21], [20], [32], [31], [42], [45], [39], [48] [23], [38], [44], [74] report device needs to be aware of or determine the boundary of an OFDM symbol (i.e. the beginning of an OFDM symbol) to determine CP location
- Some sources [25], [22], [26], [18] report the device implementation may depend on M values

For Alt M1-1, the potential impacts are discussed as follows:

- Some sources [18], [22], [23], [25], [42], [45] report that CP handling of Alt M1-1 can be used for both small and large M values for OOK-4, while [25] reports that for large M values Alt M1-1 is used in combination with Alt M1-2.
- Some sources [20], [49], [74] report that CP handling of Alt M1-1 is challenging to be used for large M values for OOK-4 considering large SFO and [25], [35] report that CP handling of Alt M1-1 may not completely remove CP samples due SFO impact.
- Among them, [22] show that the performance loss of PRDCH carrying 20 bits due CP handling is negligible at 10% BLER even for large M values (e.g. $M=24$) under large SFO (e.g. 10^4 - 10^5 ppm). Sources [25], [49] show some performance loss due CP handling for both small ($M=4$) and large M values ($M=24$) under large SFO (e.g. 10^4 - 10^5 ppm) while [49] shows [1~2 dB] loss compare to no CP case for $M<24$, and an error floor at BLER=10% for $M=24$.
- Some sources [26], [35] report that the device needs additional complexity to handle CP, while other sources [22], [42] reports that it is feasible in terms of implementation complexity based on transition edge detection.
- One source [65] reports that the device might remove the wrong portion of the CP part of the OFDM symbol due to timing error, which could introduce the false rising/falling edge for the subsequent OOK demodulation.

For Alt M1-2, the potential impacts are discussed as follows:

- Some sources [18], [21], [26], [20], [22], [23], [25], [49], [35], [42], [44], [65], [74] report that CP handling of Alt M1-2 cannot be used for large M values, e.g. $M>8$, while [25] reports that for large M values Alt M1-2 is used in combination with Alt M1-1.
- One source [39] report that CP handling of Alt M1-2 can be used for both small and large M values (e.g. $M>8$) if with the knowledge of OFDM symbol boundaries.
- Among of them, [25] show that the performance of Alt M1-2 is not applicable for large M values (e.g. $M=24$) under large SFO (e.g. 10^4 ppm).

For Method Type 2, two approaches regarding subcarrier orthogonality are studied:

Alt M2-1: Method Type 2 retains subcarrier orthogonality, i.e. CP is copied from the end of an OFDM symbol.

Alt M2-1-1: The first OOK chip(s) and the last OOK chip(s) in an OFDM symbol are the same.

Alt M2-1-2: Ensure a transition edge occurs only at the start or only at the end of the CP, and no transition edge occurs during the CP.

Alt M2-2: Method Type 2 does not retain subcarrier orthogonality.

For Method Type 2, depending on the design, the chip duration generation of OOK-4 for M -chip per OFDM symbol transmission could possibly be determined by:

- M , and the length of OFDM symbol with CP
- M , and the length of OFDM symbol without CP
- Depending on detailed solutions, chip duration may or may not be constant.
- Some sources [45], [57] report that non-constant OOK chip duration may impact performance, while some other source [49] report that non-constant OOK chip duration does not impact performance.

For Alt M2-1, the potential impacts are discussed as follows,

- Some sources [22], [26], [25], [50], [38], [28], [35], [20], [68] report that CP handling of Alt M2-1 cannot be used for large M values (e.g. $M > 8$). Source [56] reports that for $M > 8$, the CP size becomes comparable to that of the normal OOK chip, and hence it would be challenging to identify the invalid transition caused by CP. Sources [65] reports that if chip duration is comparable to CP duration, CP could not be identified as the invalid chip by the A-IoT device, e.g., $M > 8$.
- Some sources [18], [23], [45], [49] report that CP handling of Alt M2-1 can be used for both small and large M values.
 - Among of them, some sources [23], [49] show the performance of Alt M2-1 for small ($M=4$) and large M values ($M=24$) under large SFO (e.g. 10^5 ppm).
- Some sources [45], [26], [49] report that CP handling of Alt M2-1 may result in non-constant OOK chip duration around CP. Source [57] report that due to non-constant OOK chip duration around CP, Alt M2-1 has ~ 1 dB worse performance than Alt M1-1 at BLER 10% and BLER 1% when it used for small M value (e.g., $M = 6$).
- Some sources [20], [22], [28], [49], [39], [21], [44], [74] report that CP handling of Alt M2-1-1 would increase the overhead and reduce spectral efficiency.
- Some sources [42], [18], [26] report that CP handling of Alt M2-1-1 may not be completely transparent to the device thus add additional complexity.
- Source [65] report that if chip duration is significantly different from CP length, M2-1-2 would be complicated to be used for removing false transition edge occurring at the end of the CP. And M2-1-2 would require high complexity of A-IoT device implementation if it is used for the R2D preamble.

For Alt M2-2, the solutions and potential impacts are discussed as follows,

- [25], [29], [28], [38], [71] report solutions for Alt M2-2 (e.g. CP is copied from the start of OFDM symbol or do not insert CP to OFDM symbol).
- [20], [22], [23], [27], [49], [42], [45], [21], [26], [39], [44] report that CP handling of Alt M2-2 would cause interference to NR, while [25] reports single PRB guard band would be sufficient to handle interference. [71] reports the guard band would anyway be needed when SCS is different between R2D and other NR signal.
- Sources [22], [42], [45], [48], [26], [39], [44], [74] report that CP handling of Alt M2-2 would increase the transmitter complexity.

6.1.1.2 R2D line coding

The line codes studied for R2D are Manchester encoding and PIE. For R2D reception, the device can use line codes to achieve at least chip-level time tracking by using the transition(s) of the line code. A chip corresponds to one modulated symbol.

For Manchester encoding, the bit-to-chip mapping is: bit 0 $\rightarrow$ chips {10}, bit 1 $\rightarrow$ chips {01}.

It is reported by sources [22], [26], [32], [28], [49], [33], [38], [36], [48], [35], [39] that PIE has a higher ratio of high to low voltage, which enables it to provide stable instantaneous RF energy to the device. Sources [26], [38], [39], [32] and [48] consider this a justification for studying PIE. Sources [22], [28], [49], [36] and [35] acknowledge this aspect of PIE, but question its necessity.

It is stated by sources [21], [20], [25], [28], [49], [35], [44], [47] that, due to variable codeword lengths in PIE, scheduling time domain resources would be different to transmit the same amount of data due to the number of 0 and 1, as compared to the constant codeword lengths in Manchester codes.

It is stated by sources [22], [23], [28], [49], [33], [35] that, due to variable codeword lengths in PIE, the error detection of one information bit can result in error propagation for the detection of subsequent bits and therefore negatively affecting the detection of the overall data length, as compared to Manchester codes.

Source [20] says that variable codeword lengths in PIE complicates A-IoT resource management, especially in in-band deployments when a reader needs to multiplex both NR and A-IoT communication.

Source [26] says that equal codeword length can overcome the cons of variable codeword length in PIE.

According to sources [21], [20], [28], [49], [33], [38], [35], [44], transitions during Manchester code transmissions can be utilized for recovering the signal's clock.

Sources [21], [20] say that the signal does not lose information or distort when fed to a DC blocker because each Manchester encoded symbol maintains a constant DC level, i.e. always the same average power, unlike if there was a long sequence of ones or zeros.

Evaluations are reported showing that Manchester codes perform better than PIE by 1dB to 6dB @10% BLER – [22], [26], [28], [49], [47]. Source [26] claims only a 1dB gain when comparing Manchester over equal length codeword in PIE, stating that the gain is negligible since the downlink coverage bottleneck is the receiver sensitivity and not the decoding performance.

6.1.1.3 R2D channel coding, repetition, CRC

6.1.1.3.1 Channel coding and CRC

PRDCH without FEC is studied as the baseline, with evaluations performed by comparison to this baseline. The study assumes PRDCH can attach a CRC, where the baseline design is using a 6-bit or 16-bit CRC with polynomials as per TS 38.212 [5]. A baseline of no CRC attachment is also included. For the study of CRC designs, see Clause 6.1.0.2.

Sources [57], [54], [62], [59], [60], [66], [58], [63], [67] and [64] provide justifications for not having R2D FEC beyond the baseline, with the following observations:

- Sources [57], [59], [66], [63] and [67] state that FEC decoders require complicated arithmetic or logical operations which are too complicated to be implemented in device 1.
- Sources [59] and [60] state that it would be difficult for a device to implement a FEC decoder due to its low power consumption.
- Source [57] and [67] state that FEC decoder procedures such as the de-interleaving operation or route metric caching require volatile memory of a certain size with a certain reading/writing throughput, which cannot be supported by device 1.
- They also mention that the received signal power at the device can be relatively high (e.g., >-60 dBm), making the receiver sensitivity not the bottleneck of the link budget for target coverage, even for device 2b, thus questioning the necessity of R2D FEC.

Sources [56] and [72] provide the following justifications for using FEC in R2D for device 2b:

- Source [56] claim that CC with small constraint lengths (e.g., 3 or less) offer a substantial performance gain over uncoded transmission, especially in a fading environment, with reasonable complexity. CC with explicit tail-biting transmission to aid decoding may be suitable for R2D.
- Source [72] claim that even simple block code (e.g., Golay, RM) with hard decisions can significantly reduce the required SNR for achieving a target BLER e.g., 1%.

6.1.1.3.2 Repetition

Regarding R2D repetitions, it is reported by sources [28] (only for R2D control, if supported), [29], [49], [30], [38], [36], [45] and [47] that R2D repetitions should be supported. The following are observations regarding the different types of repetition that should be supported.

Bit-level repetition

Positive observations:

- Source [20] state that bit level repetition can be studied if coverage enhancement of the R2D link is required.
- Source [29] state that bit level repetition where every input bit repeated for 8 times before Manchester coding could have ~4dB gain when compared with no repetition. They claim using Manchester codes with repetitions require a simple structure and consumes extremely low power.

- Source [45] state that bit level repetitions with scrambling is required since the former would improve the link budget and the latter would add extra randomness to the information bits, providing gain by suppressing the interference. They also claim that repetitions can be used in devices that cannot soft combine the repetitions, and majority-based detection would offer gain for these devices.

Negative observations:

- Source [26] state that since envelope detection is used for R2D reception, bit level repetition may not provide expected gain for the reception.
- Source [25] state that though it may be feasible, it increases the device's processing complexity for reception, e.g., combination, repetition parameters determination.
- Source [67] state that repetition gain of a bit-level repetition, which requires additional standardization effort to define necessary control information, mainly comes from the energy accumulation of the signal, and should be similar with the achievable gain by directly lowering the chip rate/reducing the M value, which does not require this additional effort.

Block-level repetition

Positive observations:

- Source [49] state that at least for large TBs, repeatedly transmitting the TB multiple times consecutively provides time diversity gain and increases the probability that at least one of the repetitions can be successfully decoded.
- Source [59] further state that the device can perform the block-wise detection without chase combination of the repeated blocks so that block-level repetition may not need additional buffer and increase the complexity and cost.
- Source [67] state that block-level repetition can obtain a bigger repetition gain than that achieved by bit- or chip-level repetition, and can enjoy both the time diversity gain and the gain of energy accumulation.

Negative observations

- Source [25] state that considering limited capability and cost for an A-IoT device, block level repetition for R2D should be excluded.
- Source [67] state that block-level repetition additionally requires a very large volatile memory to store all received repetitions of one block.

Chip-level repetition

Positive observations:

- Source [26] state that it may be useful for R2D transmission coverage and can be considered to generate a lower data rate than 7kbps.
- Source [47] state that chip-level repetition increases the chip duration, improving the edge detection at the receiver, thereby having a ~2dB performance increase when compared to bit level repetitions.

Negative observations:

- Sources [20], [25] and [28] state that chip-level repetition is equivalent to long chip transmission, i.e., by using a smaller modulation index, and therefore, there is no need to support this option.
- Source [67] state that repetition gain of a chip-level repetition, which requires additional standardization effort to define necessary control information, mainly comes from the energy accumulation of the signal, and should be similar with the achievable gain by directly lowering the chip rate/reducing the M value, which does not require this additional effort.

On the other hand, it is reported by sources [55], [57], [58], [54] and [62] that R2D repetitions should not be supported, giving justifications:

- Source [55] mentions that the transmission power of a R2D transmission is typically much greater than its corresponding D2R transmissions, and if the R2D transmission has coverage issues, then the corresponding D2R transmission would not reach the reader. Hence it should be considered for D2R transmissions alone.
- Source [54] say that not supporting R2D repetition can be the baseline.
- Sources [58], [69] and [63] include that the decision to support R2D repetitions can be based on whether the activation threshold is a bottleneck according to the coverage evaluations.
- Sources [58], [63] and [57] also comment that from a device perspective, especially device 1 with low complexity and memory storage, it is not possible to combine multiple repetitions.

6.1.1.4 R2D bandwidths

The study defines the following bandwidths for R2D:

- Transmission bandwidth, $B_{tx,R2D}$ from a reader perspective: The frequency resources used for transmitting R2D. For an OFDM-based waveform with subcarrier spacing of 15 kHz, $B_{tx,R2D} \leq$ twelve PRBs, and would be down-selected among:
 - Alt 1: Including 180 kHz, 360 kHz
 - Alt 2: Integer multiple(s) of 180 kHz
 - Alt 3: Integer multiple(s) of the subcarrier spacing
- Occupied bandwidth, $B_{occ,R2D}$ from a reader perspective: The frequency resources used for transmitting R2D, and potential guard band.
- $B_{occ,R2D} \geq B_{tx,R2D}$.

Table 6.1.1.4-1 is a starting point for study of M values and the associated minimum $B_{tx,R2D}$ value. The reader can use any transmission bandwidth greater than or equal to the minimum $B_{tx,R2D}$ value.

Note: Depending on further study, the maximum value of M may be less than 32.

Note: The performance can be better when transmission bandwidth greater than the minimum $B_{tx,R2D}$, depending on device processing and transmit power constraint.

Table 6.1.1.4-1: Starting point for M values and the associated minimum $B_{tx,R2D}$ value

M	Minimum $B_{tx,R2D}$ # of PRBs
1	1
2	1
4	1
6	1
8	2
12	2
16	2
24	3
32	4

6.1.1.5 PRDCH

For R2D, the only physical channel is PRDCH, which carries any higher-layer payload (including system information, if defined), and L1 R2D control information, if defined. The following design options have been studied for PRDCH:

Option 1: If there is no L1 R2D control information carried by PRDCH, then PRDCH transmission with only R2D data is considered.

Note: In this option, R2D control information (if any) maybe carried by higher-layer signalling.

Option 2: If any L1 R2D control information is supported & carried by PRDCH, then for the CRC attachment (appended if there is non-zero length CRC), following two options have been studied.

Option 2-1: Joint CRC is attached to L1 R2D control information and R2D data.

Option 2-2: Separate CRC is attached to L1 R2D control information and R2D data, respectively.

PRDCH is studied via the blocks shown in Figure 6.1.1.5-1, where other clauses give their detailed descriptions.

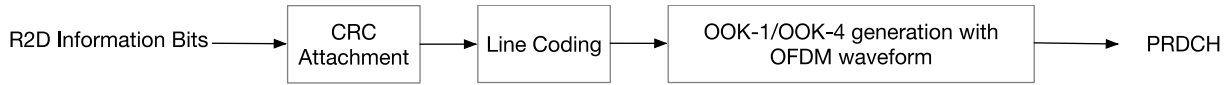


Figure 6.1.1.5-1: PRDCH generation

6.1.1.6 R2D timing

6.1.1.6.1 R2D timing acquisition signal

An R2D timing acquisition signal (R-TAS), immediately preceding the transmission of PRDCH, is included at least for timing acquisition and indicating the start of R2D transmission in the time domain. An R-TAS structure using a preamble is studied, in which a start-indicator part provides the start of the R2D transmission, and immediately precedes a clock-acquisition part which is used to determine the OOK chip duration of the subsequent PRDCH transmission. The preamble is not part of PRDCH.

The R-TAS start-indicator part is not included in T_{D2R_min} , and an ON/OFF pattern i.e., high/low voltage transmission, is applied. The following options have been studied for the start-indicator part:

- Option 1: ON-OFF transmission is considered based on energy/edge detection, and multiple alternatives have been studied including
 - Alt 1: A single ON-OFF transmission, i.e. one high-voltage transmission followed by one low-voltage transmission, where ON and OFF may have same or different durations
 - Alt 2: A multi-ON-OFF transmission, where different ON and different OFF may have same or different durations and different parts may have same or different duration
- Option 2: ON-OFF sequence-based design is considered which consists of a pre-defined sequence for detection of start-indicator part based on digital correlation
- For both the options, it is observed that a fixed duration for the start-indicator part can be considered, regardless of the value of M used for PRDCH transmissions.

MDR, FAR, and detection complexity have been evaluated for the design of the R2D start indicator part as follows:

- It is observed by one source [119] that for an FAR of ~0%, the MDR of less than 1% can be achieved with Alt 2 of Option 1 (considering 2 ON-OFF transmissions with different durations) and it is also observed that low-complexity and reduced power consumption can be achieved.
- One source [122] evaluated Alt 1 of Option 1 (considering same duration for ON and OFF) and Alt 2 of option 1 (considering multiple ON-OFF transmissions with same duration) and observed that for an FAR of ~0%, the MDR of less than 1% can be achieved and Alt 1 of Option 1 performs better than Alt 2 of Option 1.
- One source [121] observed with ON-OFF pattern, that for an FAR of ~0%, the MDR of less than 1% can be achieved with a duration of at least 1 OFDM symbol.
- One source [126] compares the performance between option 1 and option 2. It shows almost similar coverage range (SNR requirement) for target MDR of 1%. For MDR of 10%, it shows that sequence-based design provides better performance, and it is observed that during the available time, it is feasible for all devices to detect the start-indicator sequence within the power budget. It is further observed that the FAR with sequence-based design can be improved in case of interference scenarios when compared with pattern-based design.

For both the options, it may be beneficial that the start-indicator part is distinguishable at least from other parts of the R2D transmissions.

The clock-acquisition part is based on OOK without line coding, and includes rising/falling edges, including at least two rising or at least two falling edges for the device to determine the OOK chip duration. The following options for design of the clock-acquisition part are studied:

- Option 1: Duration of the clock-acquisition part is variable for different M values, i.e. the duration becomes shorter with increasing value of M .
- Option 2: Duration of the clock-acquisition part is constant for different M values based on repetition, i.e. repetition factor is increased with increasing value of M to keep the duration constant.

On impact/restriction of M values for the clock-acquisition part (CAP):

- The following 9 sources provided observations on the impact/restriction of M values for the clock-acquisition part design requirements:
 - 1 source [131] observed that increasing value of M , while retaining the same transmission duration, improves the auto-/cross- correlation properties of the sequence due to increase in sequence length and use of $M \geq 4$ provides better timing estimation accuracy even in the presence of SFO as the sequence length spans only over a shorter duration.
 - 2 sources [130], [132] observed for option 1 of the clock-acquisition part design that no restriction is required to be placed on the M values. Furthermore, 1 source [132] observed that the same 2 ON-OFF voltage (with the same duration) satisfies the FDR performance metric of less than 1% for different M values, e.g., $M = 2, 6$ and 24, where FDR is the False detection ratio, i.e. incorrectly calculating M , is the performance metric.
 - 1 source [133] observed that pattern of the clock-acquisition part is related to M chips per OFDM symbol and when M is small, the clock-acquisition part may cross multiple OFDM symbols, and the CP insertion may degrade the timing acquisition performance.
 - 1 source [134] observed that with option 2, the duration of the clock-acquisition part remains consistent across all M values, at least three OFDM symbols maybe required for clock-acquisition part and it maybe not as efficient as option.1.
 - 1 source [138] observed that among the two options studied for the clock-acquisition part, option 2 provides increased robustness, especially in case of large value of M , when compared to option 1 and potentially increase the detection performance of the clock-acquisition part.
 - 1 source [140] observed that if the chip duration is variable based on the M value used for OOK-4 waveform, the detection performance would be limited by the received SINR of the CAP with clear transition of the rising and falling edges.
 - 1 source [148] observed that the option 1 with $M > 1$ has shorter duration of clock acquisition part than $M = 1$ and worse timing acquisition accuracy. At least part of PRDCH following the clock acquisition part may need to be used to improve the timing acquisition. Furthermore, the larger M (e.g., $M > 4$) with small chip duration is more sensitive to the SFO accuracy and the restriction of M for the clock acquisition part may be needed.
 - 1 source [149] further observed that different M values may impact the chip accuracy obtained by the clock acquisition part.

On impact of CP insertion/handling on the clock-acquisition part:

- The following 10 sources observed that the CP insertion/handling may impact the design requirements of the clock-acquisition part:
 - 1 source [133] further observed that when the clock-acquisition part occupies more than one OFDM symbol, ON-OFF state transition around CP can avoid the error rising or falling edges due to the CP insertion.
 - 1 source [134] further observed that to mitigate the impact of the CP in the clock-acquisition part for large M values, it can reuse the CP handling method for PRDCH
 - 1 source [135] further observed that CP insertion/handling on the clock-acquisition part can cause false rising/falling transition and, therefore, the clock acquisition part should be designed such that it does not incur a false rising or falling edges due to CP insertion when CP-OFDM is used for OOK signal generation.
 - 1 source [137] further observed that CP insertion/handling on the clock acquisition part will impact the chip duration estimation accuracy. It is further observed that for CP handling, device may not be able to count the clock and estimate OFDM symbol duration accurately until the clock acquisition part if the start indicator only includes a single ON-OFF transmission.

- 1 source [140] further observed that the SER will be degraded due to uneven chip interval when the CP is inserted within an OFDM symbol, where SER refers to the number of samples which is mismatched for comparing to the total number of samples in a chip.
- 1 source [147] further observed if CP insertion would cause false rising/falling edges, accuracy of timing acquisition may be impacted.
- 1 source [149] further observed that the issues of chip extension, false raising/falling transition, and additional raising/falling transition caused by CP insertion/handling considering different M values will impact the chip accuracy obtained by the clock acquisition part.
- 1 source [136] further observed that the design of clock acquisition part should consider that CP insertion does not cause a false rising or falling edges and does not cause different length of multiple high / low voltages within the clock acquisition part when the clock acquisition spans multiple OFDM symbols.
- 1 source [132] observed CP insertion/handling may not impact the design requirements of the clock-acquisition part.

On purpose of SFO estimation/correction based on the clock-acquisition part

- The following 3 sources provided observations on the applicability of clock-acquisition part for frequency synchronization:
 - 1 source [131] observed that the length of preamble sequence may need to consider also the robustness against SFO
 - 1 source [140] observed that device 2a/2b may require higher synchronization accuracy for signal transmission or backscattering and therefore, the design of CAP may be required to accommodate the requirement of additional frequency synchronization and clock calibration for Device 2a/2b.
 - 1 source [148] further observed for Option 1, as the CAP duration with high M is decreased, only CAP may not be sufficient for SFO correction and for Option 2, as the CAP duration is fixed and independent from M, the CAP with long enough duration can support SFO correction.

On purpose of CFO estimation/correction based on the clock-acquisition part

- The following 2 sources provided observations on the applicability of clock-acquisition part for frequency synchronization:
 - 1 source [144] observes that the clock-acquisition part can be utilized to solve the frequency synchronization problem without impacting the time-domain sequence, for example by transmitting in some frequency resources and it can be a harmonized solution for both chip duration indication and device frequency synchronization. However, it is further observed that if the time interval between an R2D transmission and the corresponding D2R transmission and if the device loses the timing obtained from the R2D timing acquisition signal due to timing drift at the time for the D2R transmission, then an additional synchronization signal is needed
 - 1 source [148] further observed for Option 1, as the CAP duration with high M is decreased, only CAP may not be sufficient for CFO correction and for Option 2, as the CAP duration is fixed and independent from M, the CAP with long enough duration can support CFO correction.

6.1.1.6.2 R2D end timing

To determine or derive the end of PRDCH transmission, the following options are studied:

- Option 1: TBS information via implicit/explicit L1 R2D control information.
- Option 2: Postamble at the end of PRDCH.

14 sources [131], [132], [134], [133], [135], [144], [143], [146], [148], [136], [149], [150], [144], [137] provided following observations on the above two options for determining the end of PRDCH:

- 3 sources [131], [132], [134] observed that option 2 provide two benefits, namely, the variable payload length and to provide timing acquisition before the subsequent transmission of either PDRCH or PRDCH, thus improving the detectability at both reader and the device, respectively. Furthermore, 1 source [132] observed that

R2D postamble indicates the TBS with high efficiency for small packets by avoiding a large padding overhead, unlike option 1, which may require devices to perform blind detection of different PRDCH formats (if supported) and the overhead caused by the inclusion of a R2D postamble does not exceed 20% for even the smallest of message sizes and may be less than the signaling overhead caused by using a dedicated TBS indicator.

- 1 source [133] observed for option 2, that for small payload size with only a few bits, the presence of long postamble generates large resource overhead, while for large payload size with more bits, the resource overhead of postamble is smaller.
- 1 source [137] observed for option 2, that for small payload size with only a few bits, the presence of long postamble generates large resource overhead.
- 1 source [135] observed option 2 is not strictly required, however, given the possible clock drift at a device, it may be still beneficial to also attach postamble at least for the determination of the end of PRDCH at a device.
- 3 sources [143], [136], [150] observed that with option 2, the false detection may be higher for shorter postamble. Source [143] observed that in contrast to option 2, it is more reliable and efficient to indicate TBS with control information in option 1.
- 2 sources [146], [137] observed that if a message type or a command ID is included in L1 control information and implicitly indicates a known size of a fixed TB, then there is no need for either option 1 or option 2.
- 2 sources [148], [137] observed that option 1 has the advantages of avoiding blind detection of postamble and providing the power saving for non-target devices to skip the R2D detection.
- 1 source [149] observed that option 1 is feasible for the device to avoid the unnecessary reception of a TB with a specific size and thus enable power saving, e.g., when the TB has a size exceeding the allowance of the device remaining power.
- 1 source [144] observed option 2 is not strictly required if the end of PRDCH can be explicitly indicated by R2D control information, and it is subject to the miss-detection rate. It may be beneficial if a PRDCH postamble can serve as an additional timing acquisition signal prior to a PDRCH transmission.

6.1.1.7 CFO calibration signal for device 2b

For device 2b, a signal for CFO calibration should be provided to synchronize / calibrate the device clock for LO for carrier frequency (Clock purpose #5) to achieve the accuracy after clock sync / calibration at device side captured in Table 5.3-1. Frequency calibration at device 2b is beneficial at least to reduce the guard-bandwidth of D2R transmission.

For the CFO calibration signal, which is required only for device 2b to reduce the frequency offset range and the guard-bandwidth of D2R transmission, the following observations are captured in TR 38.769:

- Sources [99] report that a single-tone RF signal is used as the CFO calibration signal, it is not a part of time acquisition signal and can be transmitted as an optional R2D signal after the PRDCH transmission.
- Sources [98], [110] and [111] report that additional synchronization signal is needed.
 - [111] state the R2D timing acquisition signal may not be sufficient or may not be usable for CFO calibration since a reference frequency is needed when separate LOs are used for Tx and Rx in device 2b.
- Sources [103], [105], [114], [106] report that additional synchronization signal is needed if the synchronization for carrier frequency using R2D signal/channel does not provide required functionalities for device 2b.
- Sources [101], [115] report that it may not be possible to achieve enough frequency accuracy (0.01 ppm) even after CFO calibration based on R2D time acquisition signals for coherent detection at reader especially when the D2R data rate is low.

6.1.1.8 R2D multiplexing

For R2D, time-domain multiplexing is the baseline. Code-domain multiplexing is not considered for device 1/2a/2b. Frequency-domain multiplexing is not considered for the devices with an RF-ED receiver (see Clause 5). For device 2b with IF-ED or ZIF receivers, the study considered the following technical aspects:

Table 6.1.1.8-1: Observations on the feasibility and necessity of FDM for Device 2b

Aspects to be considered for feasibility/benefit	Observations
Inventory completion time	Sources [20], [39], [21], [29] state that FDM is beneficial to reduce the inventory completion time, especially considering more devices per reader due to the larger maximum distance for Device 2b. Source [62] state that inventory latency reduction would be limited, due to difficulty of allocating frequency resources efficiently for Msg2 by reader with uncertainty of number of successful Msg 1, and difficulty of informing an A-IoT device R2D frequency location other than Msg2.
Device implementation	Sources [35], [44], [20], [39], [29] state that channel selection may be performed by a narrowband filter (IF filter or BB filter) after the mixer for Device 2b, if the LO accuracy is sufficiently good. Source [71] states that narrowband RF filtering at device side to realize R2D FDM would be challenging considering reception performance and complexity, while such filtering would also limit the deployment scenario supported by device. Sources [51], [41], [23], [19], [22] state that it would be challenging for a device using an RF-ED receiver architecture to distinguish the different incoming signal fall into the RF BW without narrowband RF filtering which may cause increasing device implementation complexity and power consumption. Source [20] states that the larger R2D responses are harder for the devices to process in the case of TDM+FDM/TDM only for D2R/R2D, respectively.
Spectrum utilization	Source [51] state that the spectral efficiency may be impacted by the guard band across the FDMed R2D transmissions to multiple devices. Source [56] state that the spectrum utilization can still be higher for non-RF-ED based devices if FDM is used, despite guard bands. Source [62] state that spectrum efficiency improvement would be limited, due to difficulty of allocating frequency resources efficiently for Msg2 by reader with uncertainty of number of successful Msg 1, and difficulty of informing an A-IoT device R2D frequency location other than Msg2.
Coverage (in the case of single reader)	Source [22] states the R2D link budget of a reader is decreased due to the power splitting between the parallel R2D channels. Source [20] states the coverage target of Device 2b is still larger than that of Device 1 (with RF-ED architecture).
Reader implementation (in the case of single reader)	Source [22] states that additional interference suppression may be needed to deal with the intermodulation between the parallel R2D transmissions.
Harmonized design for all devices	Sources [43], [26], [36], [61] state that it is not appropriate to include FDM only for Device 2b, while Device 1 and 2a cannot support it. Source [62] state that non-harmonized resource allocation for different device types complicates the system design. Source [56] state that a deployment supporting a combination of RF-ED devices and non-RF ED devices can be harmonized by TDMA'ing different types of R2D time slots, where some time slots can support only a single frequency occasion while other time slots support multiple frequency occasions.

6.1.1.9 Scheduling of R2D

For R2D reception, the following information potentially can be explicitly/implicitly indicated to the device via the corresponding PRDCH:

- ID associated with device(s) intended for the reception of R2D, potentially including all devices (if supported)

For each information, it is for further study whether higher-layer signalling and/or L1 R2D control signalling is used - see Clause 6.1.1.10.

6.1.1.10 R2D control information

For transmission of R2D control information for R2D reception and D2R scheduling, following two options are studied and captured in the TR 38.769:

- Option 1: L1-control signaling
- Option 2: Higher-layer signaling

The following 12 sources provided following observations on the above two options for signaling R2D control information:

- 1 source [132] observed that option 2 avoids physical layer design, while at the same time offering flexibility in terms of the size of the grant. On the other hand, option 1 may require the device to perform blind detection of different PRDCH formats (if supported), while at the same time, reducing the flexibility and resource utilization efficiency. Option 1 would also require a separate CRC, which would result in breaking the pipelining of the CRC decoding due to the added latency and increases overhead.
- 3 sources [133], [134], [141] observed that for D2R scheduling information, as the PDRCH transmission only occurs when the device can successfully decode the preceding PRDCH transmission, therefore no benefit is observed for option 1 compared to option 2. Furthermore, one source [133] also observed that for R2D reception, if TBS of PRDCH is indicated by R2D control information, such R2D control information should be carried via L1 signaling as the device needs to understand the packet size of PRDCH before decoding.
- 1 source [135] observed that for D2R scheduling information indication, option 1 has an advantage as the decoding of D2R scheduling information is decoupled from that of R2D payload and the device can early start the preparation of the PDRCH. For R2D scheduling information indication, Option 1 has an advantage as the device can early detect whether the corresponding R2D transmission is addressed to the device or not, which also saves device's power consumption.
- 1 source [143] observed that D2R scheduling information can be transmitted as physical layer signalling or higher layer signalling as there is no fundamental difference between the 2 options.
- 1 source [148] observed that option 1 allows the decoding of R2D and D2R scheduling information to occur before the decoding of the R2D payload. The D2R scheduling in L1 control enables the device to provide feedback on the (un)successful detection of the R2D payload, e.g., due to insufficient energy/power or channel/interference variation. Additionally, 6 sources [134], [148], [133], [135], [143], [137] observed that the R2D scheduling information in L1 control enables early identification of target devices and power saving for non-target devices by allowing them to skip the following R2D detection.
- 1 source [136] observed that device can decode L1 control information first which can allow device to obtain the scheduling information (e.g. TBS rated information) before data part reception in option 1 and if device knows from control information that the following data is not for it, the data reception can be skipped.
- 1 source [147] observed that option 1 allows the decoding of R2D control information separately from the subsequent R2D reception, which enables devices to feedback when subsequent R2D reception is failed and also enables identification of target devices before the subsequent R2D reception.
- 1 source [150] observed that Option 1 enables the separate decoding of R2D control information before R2D data reception, which supports feedback if R2D reception fails. Additionally, L1 control can carry D2R scheduling info, allowing devices to avoid the non-intended R2D reception.

- 1 source [144] observed that with Option 1, L1 R2D control information may include important information needed to decode the data portion of the PRDCH signal. This requires L1 R2D control information part to have a fixed payload size and self-decodable with separate CRC.

6.1.2 D2R

Reference signals including DMRS, PTRS, SRS, are not considered for D2R. CSI feedback and autonomous SR are not considered for L1 D2R control information.

6.1.2.1 Waveform and modulation

For D2R by backscattering, the waveform is provided by the carrier wave, see Clause 6.7. The D2R baseband signal (as distinct from the internal or external carrier wave) is non-OFDM.

For all devices, the following D2R baseband modulations are studied:

- OOK
- Binary PSK
- Binary FSK, as MSK (and not GMSK)

OOK and BPSK for baseband modulation are feasible for D2R for all devices.

- Source [73] reports that BPSK performance with coherent receiver highly depends on the CFO value and D2R transmission duration after a D2R amble.

Sources [20], [28], [45], [33] report that MSK is feasible in some way:

- [20], [28], [53], [53] say it is feasible for all devices, for example when it is implemented with multiple impedances switching
- [45] say that it would be implemented as square-wave MSK for devices 1 and 2a, and sine-wave MSK for device 2b
- For device 1 and 2a this type of MSK does not have continuous phase
- [20] say that benefits include lower sidelobes than OOK and BPSK, and lower BER than OOK and same BER as BPSK

Sources [22], [19], [26], [24], [25], [27], [40] report that MSK is either infeasible or should be deprioritized for all devices.

- [22], [26], [24], [25], [19], [27], [40] say that MSK is less spectrally efficient than OOK and BPSK because there are issues due to poor phase accuracy in the device
- [22], [24], [19], [25], [27] say that MSK would increase reader and device complexity
- [25] say that MSK performance for device 2b would materially degrade due to CFO
- [54] say that it is difficult to modulate MSK signal using impedance switching due to the implementation complexity, including frequency mapping and phase continuation. [63] say that if multiple impedances switching are applied to maintain the phase continuity, it violates the principle of low device cost.

2SB modulation is feasible for D2R transmission for all devices. Feasibility and necessity of 1SB modulation would depend on impacts due to issues including: device-side filtering requirements (i.e. image suppression), RF resource usage / spectral efficiency, etc.

Device 2b internally generates a carrier wave with only a single tone. Device 2b does not use a '2 single tone' carrier wave. See Clause 6.8.2 relating to these terms.

6.1.2.2 D2R line coding

The line codes studied for D2R are Manchester encoding, FM0 encoding, Miller encoding, and no line coding. FM0 encoding is deprioritized. A chip corresponds to one modulated symbol at least for OOK and BPSK.

For Manchester encoding, the bit-to-chip mapping is: bit 0→chips{10}, bit 1→chips{01}. For FM0 encoding, according to Figure 6-8 and Figure 6-9 of [6]. For Miller encoding, according to Figure 6-12 of [6].

Sources [22], [26], [25], [28], [49], [38] report that Manchester codes perform better than Miller codes by 0.5dB to 5.5dB @10% BLER.

Sources [22], [26] [49] report that Manchester codes perform better than FM0 by 0.2dB to 2.5dB @10% BLER. According to sources [25], [28], Manchester codes have a similar performance as FM0.

Sources [22], [33] state that the power of Miller coding is more concentrated in the low-frequency region which is adjacent to the carrier wave frequency, as compared with FM0 code and Manchester code which leads to worse interference handling capability for Miller codes.

Source [45] state that performance of BPSK square wave is better than Miller/FM0 by ~3-4 dB @10%BLER with coherent demodulation.

6.1.2.3 D2R channel coding, repetition, CRC

6.1.2.3.1 Channel coding

For D2R, convolutional codes are studied, with comparisons to the case of no FEC. The LTE convolutional code polynomials are a reference, and other designs studied subject to:

- Constraint length, $K = 8$ or $K = 7$ or $K = 6$ or $K = 4$.
- Mother code-rate, $R = 1/6, 1/4, 1/3, 1/2$.

As shown in Table 6.1.2.3.1-1, it is observed that:

- Performance of other $\{k, m\}$ pairs decreased 0.15 – 1.4 dB when $k < 7$, with reported benefit of reduced reader and device complexity due to shorter shift registers.
- Performance of other $\{k, m\}$ pairs increased 0.22 – 1.8 dB when $k > 7$, with an increase in the number of shift register resulting in increased device complexity.
- Performance of other $\{k, m\}$ pairs decreased 0.3 – 2.5 dB when $m < 3$.
- Performance of other $\{k, m\}$ pairs increased 0.2 – 1.8 dB when $m > 3$.

where k = constraint length of the code, and $m = 1/\text{code-rate}$.

Table 6.1.2.3.1-1: Summary of study on D2R FEC

Option #	CC Design	Pros	Cons
Baseline	Constraint length 7 Code rate 1/3	[25] Decoding performance is increased by ~3dB@10% BLER, when compared to no CC but with repetitions. [25] Decoding performance is increased by ~7dB@10% BLER, when compared to no CC or repetitions. [44] Decoding performance is increased by 6.23dB@10% BLER with 2RX, when compared to no CC or repetitions. [44] Decoding performance is increased by 6.42dB@10% BLER	

		with 4RX, when compared to no CC or repetitions. [28] Decoding performance is increased by ~2dB@10% BLER, when compared to LTE CC-TBCC with code rate 1/2. [33] Decoding performance is increased by ~2.5dB@1% BER, when compared to code rate 1/2. [27] Decoding performance is increased by ~1.5dB@1% BLER, when compared to constraint length 4, code rate 1/3 [27] Decoding performance is increased by ~2.5dB@1% BLER, when compared to constraint length 6, code rate 1/3 [55] Decoding performance is increased by 3 dB@ 10% BLER with 2 RX, when compared to no CC or repetitions	
1	Constraint length 4 Code rate 1/2 – 1/4	[20] Code rate 1/2: Detection performance is increased by 3dB@10% BLER, when compared to no CC or line codes.	[26] Code rate 1/2: Decoding performance is decreased by ~0.86dB@10% BLER, when compared to constraint length 7, code rate 1/2. [49] Code rate 1/2: Decoding performance is decreased by ~1dB@10% BLER, when compared to constraint length 7, code rate 1/2. [49] Code rate 1/4: Decoding performance is decreased by ~1.4dB@10% BLER, when compared to constraint length 7, code rate 1/4. [65] Code rate 1/2, 1/3, TBCC: Decoding performance is decreased by ~1dB@10% BLER, when compared to baseline with TBCC.
2	Constraint length 6 Code rate 1/2 – 1/6	[45] Reduced reader and device complexity due to smaller number of shift registers.	[45] Code rate 1/2: Decoding performance is decreased by 0.6dB@1% BLER, when compared to baseline. [49] Code rate 1/2: Decoding performance is decreased by ~0.4dB@10% BLER, when compared to constraint length 7, code rate 1/2. [45] Code rate 1/3: Decoding performance is decreased by 0.4dB@1% BLER, when compared to baseline. [45] Code rate 1/4: Decoding performance is decreased by 0.2dB@1% BLER, when compared to baseline.

			[49] Code rate 1/4: Decoding performance is decreased by ~0.4dB@10% BLER, when compared to constraint length 7, code rate 1/4. [45] Code rate 1/6: Decoding performance is decreased by 0.15dB@1% BLER, when compared to baseline.
3	Constraint length 7 Code rate 1/2 – 1/4	[28] Code rate 1/2, TBCC: Decoding performance is increased by ~9dB@10% BLER, when compared to no CC but with Manchester line codes. [28] Code rate 1/4: Decoding performance is increased by ~1.4dB@10% BLER, when compared to baseline. [49] Code rate 1/4: Decoding performance is increased by 0.5dB@10% BLER, when compared to CC with code rate 1/2. [45] Code rate 1/4: Decoding performance is increased by 0.2dB@1% BLER, when compared to baseline.	
4	Constraint length 8 Code rate 1/2 – 1/6	[26] Code rate 1/2: Decoding performance is increased by ~0.22dB@10% BLER, when compared to constraint length 7, code rate 1/2. [49] Mother code rate 1/4, code rate 1/6: Decoding performance is increased by ~1.8@10% BLER, when compared to constraint length 8, mother code rate 1/4, code rate 1/4. [49] Mother code rate 1/3, code rate 1/6: Decoding performance is increased by ~1.8@10% BLER, when compared to constraint length 8, mother code rate 1/3, code rate 1/4. [49] Mother code rate 1/2, code rate 1/6: Decoding performance is increased by ~1.7@10% BLER, when compared to constraint length 8, mother code rate 1/2, code rate 1/4.	

6.1.2.3.2 CRC

The study assumes PDRCH can attach a CRC, where the baseline design is using a 6-bit or 16-bit CRC with polynomials as per TS 38.212 [5]. A baseline of no CRC attachment is also included. For the study of CRC designs, see Clause 6.1.0.2.

6.1.2.3.3 Repetition

For definitions of repetition types, see Clause 6.1.0. For D2R, at least block-level and bit-level repetition type 1 and type 2 are studied.

Block-level repetition

For D2R repetitions, it is reported by sources [22], [21], [19], [25], [26], [27], [28], [49], [38] that block level repetitions should be supported. The following are observations regarding the different types of repetition.

General observations

- Source [22] state that detection of D2R can obtain combining gain and time diversity gain by the block repetition
- Sources [22] and [25] state that it can enable BS to terminate reception early if it successfully decodes the block.
- Source [28] state that with small D2R information block size, memory size of A-IoT devices is comparable for bit level repetition and block level repetition, but compared to block level repetition, bit level repetition would degrade power and BLER performance of D2R transmission.
- Source [21] and [19] state that block level repetition would be a TB level repetition, while [21] goes on to state that the transmission of a PDRCH multiple times may go through multiple channels showing the diversity gain/ channel averaging impact of a specific binary data. Its implementation/specification impact would be minimum.
- Source [25], [49] state that it may require larger buffer than bit level repetition at device side, depending on device implementation.

Performance comparisons

- Source [22] state that block level repetition yields ~2.5 dB performance gain compared with bit level type 2 due to the additional time diversity gain for the combination of decoding.
- Source [27] state that block level repetition provides ~4dB performance gain @1% BLER compared with bit level type 1.
- Source [49] state that block level repetition provides ~6dB performance gain @10% BLER compared with no repetitions and the performance between block level repetition and bit level repetition type 2 is the same.
- Source [25] state that the performance difference between block level repetition and bit level repetition without CW hopping is minor, while block level repetition outperforms bit level repetition with CW hopping.
- Sources [28] and [49] state that bit level repetition and block level repetition have similar performance in the AWGN channel but block level repetition could achieve more time diversity gain than that of bit level type 2 in a fading channel.
- Source [64] state that for the no FEC case, with 3 times repetition, block level repetition provides ~5dB gain at 1% BLER when compared with bit level type 1 repetition.

Bit-level repetition

It is reported by sources [49], and [38] that bit level repetitions should be supported.

- Source [22] state that for bit level type 1, the repetition data bit is decoded after FEC detection, and hence, the digital data bit cannot be combined with combining gain.
- Source [49] state that the performance of bit level type 1 repetition is degraded by ~3.5dB @ 10% BLER under the cases of AWGN and TDL-A channel when compared to bit level type 2.
- Source [22] state that since the bit level type 2 repetition is performed after FEC at the transmitter, at the receiver side, each bit would have a log-likelihood ratio (LLR) that can be combined and provided as the input for FEC decoding, thereby providing a better performance as compared to bit level type 1.

6.1.2.4 D2R bandwidths

The following bandwidths for D2R are defined for the purpose of the study:

- Transmission bandwidth, $B_{tx,D2R}$: The frequency resources scheduled by a reader for a D2R transmission from one device.
- Occupied bandwidth, $B_{occ,D2R}$: The transmission bandwidth plus the potential associated intra A-IoT guard-bands totalling $B_{guard,D2R}$. Note: this guard band is not for coexistence with NR/LTE.
- $B_{occ,D2R} \geq B_{tx,D2R}$.

For $B_{tx,D2R}$ of the D2R transmissions associated with one/each single-tone of a carrier-wave, the following figures depict small frequency-shift, with reference to Clause 6.1.2.7.1, by:

- In Figures 6.1.2.4-1 and 6.1.2.4-3, Manchester line code with option 1, and if no D2R line-code is used (i.e., by using a square-wave corresponding to the small frequency-shift).
- In Figures 6.1.2.4-2 and 6.1.2.4-4, Manchester line code with option 2, and by Miller line code.

where:

- F_c is the carrier-wave frequency, at least when externally generated.
- FS_x depict examples of amount of small frequency shift.
- For 2SB transmission, both $+FS_x$ and $-FS_x$ parts are included in Figures 6.1.2.4-1 and 6.1.2.4-2.
- For 1SB transmission, one of the upper part or lower part applies, as appropriate, in Figures 6.1.2.4-3 and 6.1.2.4-4.

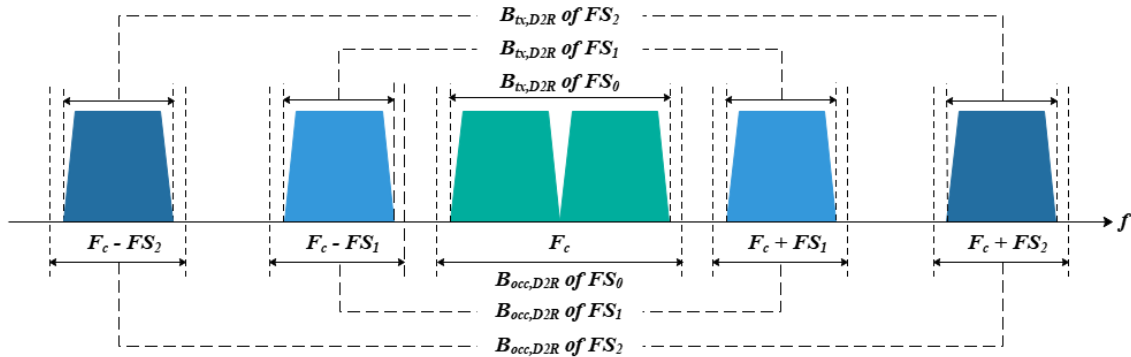


Figure 6.1.2.4-1: $B_{tx,D2R}$ and $B_{occ,D2R}$ for 2SB transmission of Manchester line code with option 1, and if no D2R line code is used (i.e., by using a square-wave corresponding to the small frequency-shift)

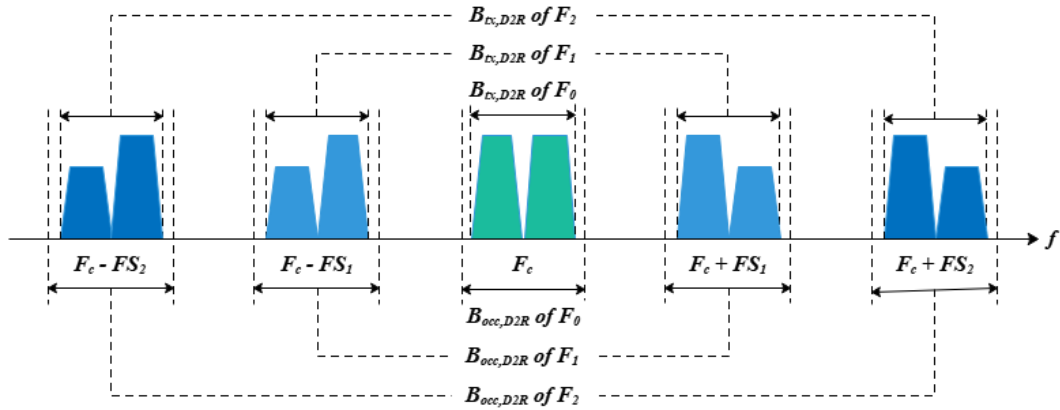


Figure 6.1.2.4-2: $B_{tx,D2R}$ and $B_{occ,D2R}$ for 2SB transmission of Manchester line code with option 2, and by Miller line code

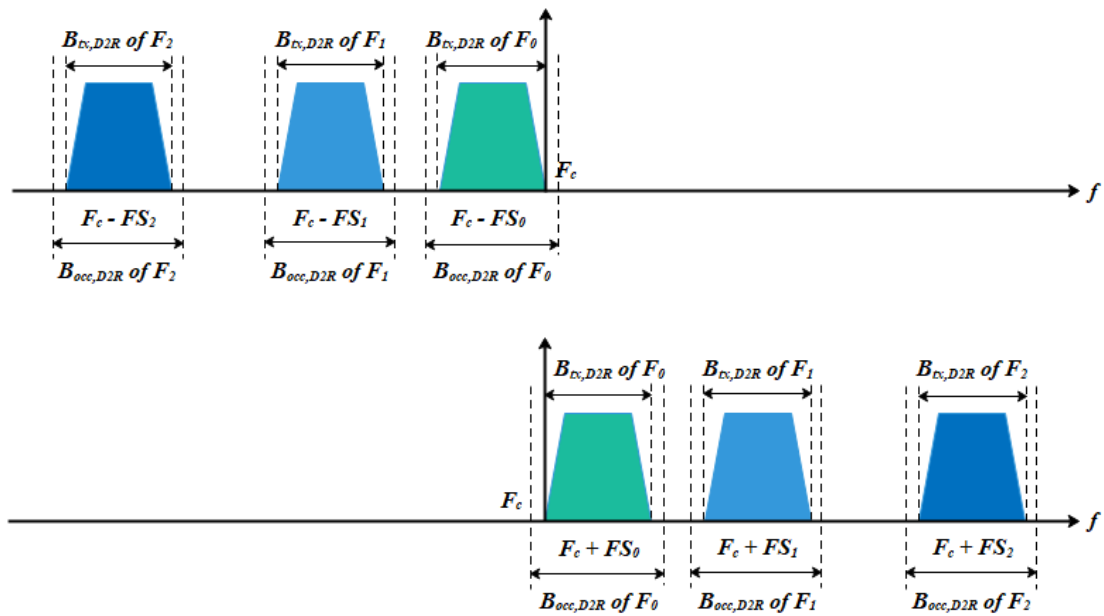


Figure 6.1.2.4-3: $B_{tx,D2R}$ and $B_{occ,D2R}$ for 1SB transmission of Manchester line code with option 1, and if no D2R line code is used (i.e., by using a square-wave corresponding to the small frequency-shift)

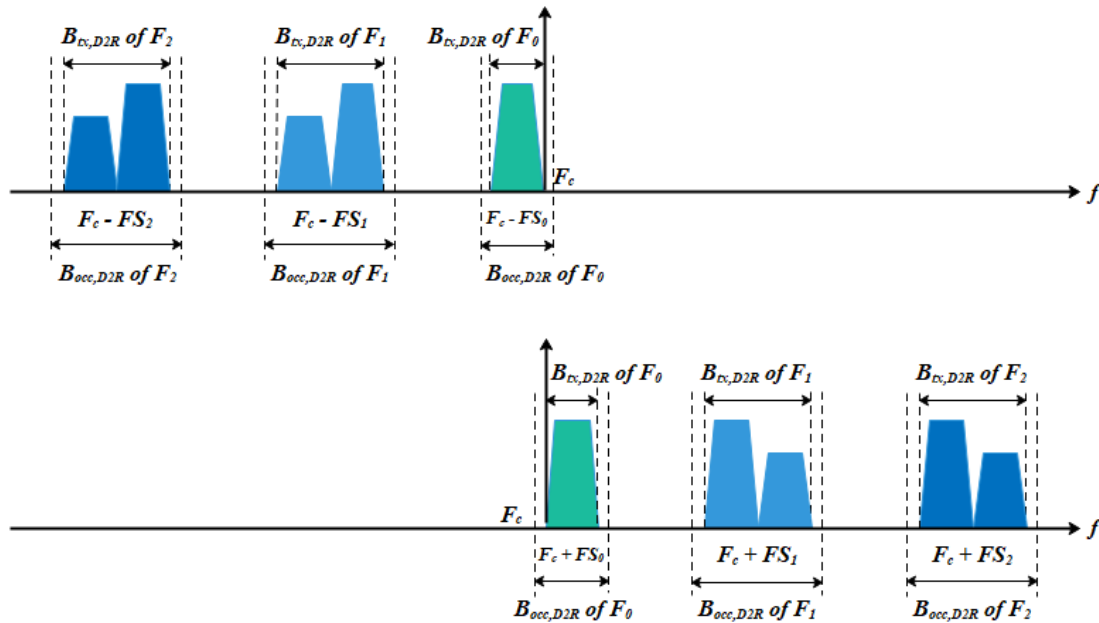


Figure 6.1.2.4-4: $B_{tx,D2R}$ and $B_{occ,D2R}$ for 1SB transmission of Manchester line code with option 2, and by Miller line code

$B_{occ,D2R}$ of the D2R transmission associated with one/each single-tone of a carrier wave would be impacted by at least the following aspects:

- A possible guard band $B_{guard,D2R}$ due to at least SFO.
- A possible guard band $B_{guard,D2R}$ due to CFO for Device 2b.

Since device 2b has internal carrier-wave generation, the 2SB and/or 1SB $B_{tx,D2R}$ and $B_{occ,D2R}$ can be affected by how it performs D2R modulation and baseband waveform generation.

6.1.2.5 PDRCH

For D2R, a physical channel PDRCH carries any higher-layer payload, the response transmitted from device to reader during the contention-based access procedure, and L1 D2R control information, if defined. PDRCH is studied via the blocks shown in Figures 6.1.2.5-1 and 6.1.2.5-2, where other clauses give their detailed descriptions and the location of the "Repetition" block depends on the repetition types described in Clause 6.1.0.1.

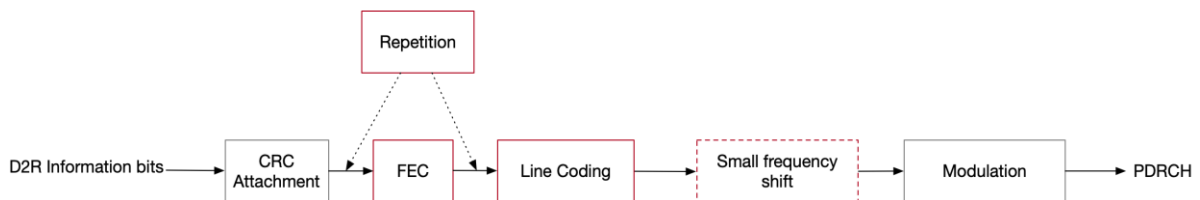


Figure 6.1.2.5-1: Single PDRCH generation at the device with line coding, for the case where FEC and repetition are present

NOTE: Presence of the block "small frequency shift" depends on the options for small frequency shift described in clause 6.1.2.7.1.

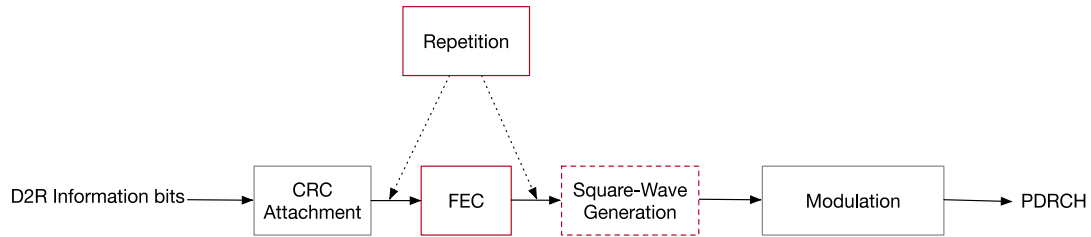


Figure 6.1.2.5-2: Single PDRCH generation at the device without line coding, for the case where FEC and repetition are present.

Scheduling information of PDRCH transmission is provided by a corresponding PRDCH.

6.1.2.6 D2R timing

6.1.2.6.1 D2R timing acquisition signal

A D2R timing acquisition signal (D-TAS), preceding each PDRCH, is included at least for timing acquisition, indicating the start of the D2R transmission in the time domain, and studied potentially for SFO estimation, CFO estimation, channel estimation, and interference estimation. A D-TAS structure using a preamble is studied, for which a binary signal is considered. The preamble is not part of PDRCH.

The following aspects have been studied for D2R preamble design:

- Autocorrelation property
 - 10 sources [131], [132], [133], [139], [140], [143], [144], [147], [148], [134] observed that the signal should have good autocorrelation properties for accurate peak detection based on the signal correlation at the reader
- Cross-correlation property
 - 7 sources [131], [133], [143], [144], [148], [134], [140] observed that the signal should have good cross-correlation properties if multiple D2R preamble sequences are considered (e.g. for multiple access schemes (if supported) for D2R transmissions).
- Line coding
 - 1 source [131] observed that line coding may impact the autocorrelation property of the sequence.
 - 1 source [132] observed that for D2R preamble, to apply backscattering, line coding can help improve the detection performance based on shifting the D2R signal's frequency location away from the carrier wave
- Sequence types (not limited to below types only)
 - M-sequence
 - 3 sources [131], [137], [139] observed that m-sequence can be considered for D2R preamble mainly owing to good correlation properties.
 - Golay sequence
 - 4 sources [133], [137], [139], [135] observed that Golay sequence can be considered for D2R preamble mainly owing to good correlation properties and availability of large number of distinct sequences and complementary pairs.
 - Walsh sequence
 - 1 source [143] observed that Walsh sequence can be considered as a candidate for D2R preamble thanks to its good auto/cross-correlation property and flexible length

There are also the following general observations:

- 1 source [132] observed can achieve 0.97% residual SFO with 98% probability under -2.5dB SNR and 0.1% MDR with $[-1/8, 1/8]$ chip timing error with 99.05% probability under -2.5dB SNR with D2R preamble

including 2-parts with clock-like sampling frequency signal and timing-acquisition signal, having 32-length '1' sequence (encoded to 64-chip Manchester code) and 32-length sequence (encoded to 64-chip Manchester code), respectively.

- 4 sources [130], [133], [134], [137] observed that for D2R preamble with binary signal, the timing synchronization performance is highly related to the sequence length of the preamble. Furthermore, 1 source [133] observed that to achieve a BLER performance at 10%, the timing synchronization error should be less than 10%. Furthermore, 1 source [134] observed that the channel estimation performance is also highly related to sequence length. 1 source [134] observed that using a 32 bits preamble provides ~8 dB, ~5 dB performance gain than using 8 bits, 16 bits preamble, respectively. And using a 64 bits preamble provides ~2.5dB performance gain than using a 32 bits preamble.
- 1 source [144] observed that for D2R preamble with binary signal, normalized SFO estimation error of less than 10% can be achieved with a training sequence length 64 or longer. The simulated D2R preamble consisting of a Golay complementary pair can tolerate SFO up to 1% (AWGN) with up to 1 dB loss in performance for a sufficiently long preamble sequence length (32 or greater).

6.1.2.6.2 D2R end timing

For the reader to acquire the end of PDRCH transmission, the following options are studied:

- Option 1: D2R postamble immediately follows the PDRCH
- Option 2: Based on control information

6.1.2.6.3 D2R midamble

The necessity of a midamble is studied at least for the purpose of performing timing/frequency tracking or channel estimation or interference estimation.

6.1.2.6.4 Trade-offs analysis

The study analysed the trade-offs among the following D2R amble(s) options:

- Option 1: D2R preamble only
- Option 2: D2R preamble + X midamble(s), where $X \geq 1$
- Option 3: D2R preamble + postamble
- Option 4: D2R preamble + Y midamble(s) + postamble, where $Y \geq 1$

For coherent detection of PDRCH with a payload of 16 bits or 20 bits with 6-bit or 16-bit CRC, using 1/2 Manchester coding and 1/3 or 1/2 convolutional code:

- Source [102] observed that with up to 10% SFO, ~1kbps data rate, when the same amble(s) overhead is maintained, Option 3 provides comparable performance results to Option 1.
- Source [103] observed that with up to 10% SFO, ~5kbps data rate, for device 1 and with up to 1% SFO for device 2, the decoding performance with/without midamble are similar
- Source [105] observed that Option 1 is sufficient to achieve 10% and 1% BLER, with no more than 8 SFO hypotheses tested at the reader side.
 - With up to 10% SFO, ~ 5kbps data rate, the SNR needed to achieve 10% and 1% BLER is similar (~ -2dB and 4 dB) for Option 1, Option 2 of D2R preamble+1midamble and Option 3.
 - With up to 1% SFO, ~ 5kbps data rate, the SNR needed to achieve 10% and 1% BLER is similar (~ -2.8dB and 3.3dB) for Option 1, Option 2 of D2R preamble+1 midamble and Option 3.
- Source [111] observed that with perfect SFO estimation, 1kbps data rate and OOK modulation, there is no noticeable performance gain from using midamble(s) and/or postamble for PDRCH transmission, regardless of the payload size.
- Source [114] observed that the required SFO estimation accuracy to achieve 1% and 10% BLER is $<5 \times 10^3$ ppm

- With up to 10% SFO, achieving the required accuracy necessitates more than 20 SFO hypotheses at the reader side for Option 1 and 10 SFO hypotheses are sufficient for Option 3 of D2R preamble + postamble. But for Option 3 reader has to store the received samples and wait for the postamble that is after the end of PDRCH for any of SFO/channel/interference estimation, demodulation, decoding, etc.
- With up to 1% SFO, 4 SFO hypotheses are sufficient for Option 1 to achieve the required accuracy.

For coherent detection of PDRCH with a payload of 96bits with 16-bit CRC (or 6-bit CRC [108]), using 1/2 Manchester coding and 1/3 or 1/2 convolutional code,

- Sources [99], [101] and [108] observed that Option 1 cannot achieve 10% BLER.
- Sources [102], [103], [105], [111] and [114] observed that Option 1 can achieve 10% BLER.
- Sources [99], [101], [102], [103], [105], [108], [117] observed that adding additional amble improves the performance.
- Source [99] observed that with up to 10% SFO,
 - Option 2 of D2R preamble+ 1 midamble achieves 10% BLER at SNR around -3dB, but cannot achieve 1% BLER.
 - Option 3 of D2R preamble+ postamble achieves 10% BLER at SNR around -4dB, and can achieve 1% BLER at SNR around 4dB.
- Source [101] observed that with up to 10% SFO, Option 3 allows reader to precisely search and detect the SFO with 0.03% residual SFO at -3dB SNR TDL-A channel, achieving 10% BLER -2.44dB SNR for ~1 kbps data rate and -2.17 dB for ~6 kbps data rate. Source [101] further observed that when the reader adopts same number of SFO hypothesis based on preamble, with 1% SFO, Option 3 can achieve 10% BLER at -4.27 dB SNR for ~1 kbps and at -4.29 dB SNR for ~6 kbps, which provides 1~2 dB performance gain when compared to 10% SFO.
- Source [108] observed that with up to 10% SFO, Option 2 of D2R preamble + 1 midamble, option 3, and option 4 of D2R preamble + 1 midamble+postamble achieve basically the same performance, the SNR for 10% BLER is 5dB for 1.25 kbps data rate.
- Source [102] observed that with up to 10% SFO, ~1kbps data rate, and the same amble(s) overhead, Option 3 can provide 1~2 dB, 5dB performance gain for 10% BLER, 1% BLER, respectively, compared to Option 1. Additionally, Option 3 can provide ~1dB, 2dB performance gain for 10% BLER, 1% BLER, respectively, compared to Option 2 of D2R preamble +1 midamble.
- Source [103] observed that for ~5kbps data rate, compared to option 1,
 - For device 1 with up to 10% SFO, Option 2 of D2R preamble + 1 midamble provides ~0.5 dB SNR gain at 10% BLER with TDL-A channel and ~0.9 dB SNR gain with TDL-D channel.
 - For device 2 with up to 1% SFO, Option 2 of D2R preamble + 1midamble provides ~1 dB SNR gain at 10% BLER with TDL-A channel and ~1.4 dB SNR gain with TDL-D channel.
 - Moreover, the additional ambles i.e., midamble(s) and/or postamble introduces additional overhead and postamble may prevent pipelined processing of the reception.
- Source [105] observed that,
 - With up to 10% SFO, ~5.5kbps data rate, Option 1 achieves 10% and 1% BLER at SNR ~0.7dB and 10dB, respectively; Additionally, maintaining the same amble overhead, Option 2 (D2R preamble + 1 midamble) and Option 3 demonstrate similar performance, achieving 10% and 1% BLER at SNR around -1.7dB and 5.2dB, respectively.
 - With up to 1% SFO, ~5.5kbps data rate, Option 1 achieves 10% and 1% BLER at SNR ~ -1.3dB and 11dB, respectively. Additionally, with the same amble overhead, the SNR difference between Option 2 (D2R preamble+1midamble) and Option 3 for 10% and 1% BLER is less than 1dB, with SNRs ~ -3.1dB to -2.5dB for 10% BLER and ~3.6dB to 4.5dB for 1% BLER.

- Source [117] observed that with up to 10% SFO, ~7.5kbps data rate, there is ~6~7dB performance gap at 10% BLER and ~10.5~11.5dB performance gap at 1% BLER between option 2 of D2R preamble+111 midambles and option 1. Note that Source [117] does not use any convolutional code.
- Source [111] observed that with perfect SFO estimation, 1kbps data rate and OOK modulation, there is no noticeable performance gain from using midamble(s) and/or postamble for PDRCH transmission, regardless of the payload size.
- Source [114] observed that the required SFO estimation accuracy to achieve 1% and 10% BLER is $<10^3$ ppm. To achieve the required accuracy,
 - For Option 1, more than 50 SFO hypotheses at reader side are necessary for device with up to 10% SFO and 6 SFO hypotheses are sufficient at reader side for device with up to 1% SFO.
 - For Option 3, 10 SFO hypotheses are sufficient for device with up to 10% SFO, but reader has to store the received samples and wait for the postamble that is after the end of PDRCH for any of SFO/channel/interference estimation, demodulation, decoding, etc.

For coherent detection of PDRCH with a payload of 400bits with 16-bit CRC, using 1/2 Manchester coding and 1/3 or 1/2 convolutional code,

- For option 1 of D2R preamble only,
 - Sources [99], [101], [102], [104], [105], [108] observed that with up to 10% SFO, 10% BLER cannot be achieved.
 - Source [111] observed that with perfect SFO estimation, 1kbps data rate and OOK modulation, there is no noticeable performance gain from using midamble(s) and/or postamble for PDRCH transmission.
- For other amble options,
 - Source [99] observed that:
 - With accurate SFO estimation, Option 2 of D2R preamble + 4 midambles can achieve 10% BLER at SNR ~ 2.7dB but cannot achieve 1% BLER.
 - With up to 10% SFO, Option 3 cannot achieve 10% BLER.
 - With up to 10% SFO, Option 4 of D2R preamble+2 midambles+postamble achieves 10% BLER at SNR of ~0.25dB; But it cannot achieve 1% BLER. Option 4 of D2R preamble+3 or 4 midambles+postamble, achieves a 10% BLER at an SNR of around -0.2 dB, and achieves 1% BLER at SNR around 9dB or 8dB, respectively.
 - Source [101] observed that with up to 10% SFO, Option 4 of D2R preamble combined with 1 to 4 midambles + postamble, achieves 10% BLER at SNR of 2.5 dB, 1 dB, 0.8 dB, or 0.5 dB, respectively, for a data rate of around 1 kbps.
 - Source [102] observed that with up to 10% SFO,
 - Option 3 can provide ~5.5 dB performance gain compared to option 2 of D2R preamble+1midamble for 10% BLER, with the same amble(s) overhead for ~1kbps data rate.
 - Option 2 of D2R preamble+1midamble cannot achieve 1% BLER for ~1kbps data rate.
 - Option 4 of the D2R preamble+1 or 2 midamble(s)+postamble, has similar performance, it can achieves a 10% BLER at SNR of -1dB and achieves a 1% BLER at SNR of 6dB and 5dB respectively for ~1kbps data rate.
 - Source [104] observed that with up to 10% SFO,
 - Option 3 of D2R preamble+ postamble cannot achieve 10% BLER for ~7kbps.
 - Option 4 of D2R preamble + 1 midamble + postamble can achieve 10% BLER and 1% BLER at SNR around -6dB and 0 dB, respectively for ~7kbps data rate.
 - Source [105] observed that

- With up to 10% SFO, maintaining the sameamble overhead, both Option 2 of D2R preamble+5 midambles and Option 4 of D2R preamble+4 midambles+1 postamble demonstrate similar performance, achieving 10% BLER at SNR ~0.2dB. While for 1% BLER, the SNR for Option 2 and Option 4 is ~9.2dB and 12.8dB, respectively for ~5.5kbps data rate.
- With up to 1% SFO, maintaining the sameamble overhead, both Option 2 of D2R preamble+5 midambles and Option 4 of D2R preamble+4 midambles+1postamble demonstrate similar performance, achieving 10% BLER at SNR around -1.2dB. While for 1% BLER, the SNR for Option 2 and Option 4 is ~7.8dB and 9.1dB, respectively for ~5.5kbps data rate.
- Source [108] observed that with up to 10% SFO, Option 2 of D2R preamble+3 midambles and Option 4 of D2R preamble+3 midambles+postamble can achieve 10% BLER when the SNR is within the range of [15, 25] dB for 1.25 kbps data rate.
- Source [114] observed that the required SFO estimation accuracy to achieve 1% and 10% BLER is much smaller than 10^3 ppm. To achieve the required accuracy,
 - For Option 2 of D2R preamble+X midamble(s) where midamble inserted per every certain number of PDRCH bits (e.g., 192 bits),
 - For SFO estimation using each amble for the subsequent PDRCH bits (e.g., 192 bits), with up to 10% SFO, more than 50 SFO hypotheses are necessary at the reader side and with up to 1% SFO, 6 SFO hypotheses are sufficient at the reader side.
 - For SFO estimation based on the time gap between preamble and midamble, with up to 10% SFO, 10 SFO hypotheses are used, but reader has to store the received samples and wait for the midamble to start SFO/channel/interference estimation, demodulation, decoding, etc.
 - For Option 3 of D2R preamble+postamble, SFO estimation is based on the time gap between preamble and postamble, with up to 10% device SFO, 10 SFO hypotheses are used for reader, but reader has to store the received samples and wait for the postamble that is after the end of PDRCH for any of SFO/channel/interference estimation, demodulation, decoding, etc.
- Source [103] observes that the additional ambles i.e., midamble(s) and/or postamble introduces additional overhead and postamble may prevent pipelined processing of the reception.

For the synchronization and timing tracking of D2R transmission,

- Source [101] report that with up to 10% SFO, option 1 is not sufficient for D2R reception since the residual SFO at reader side is larger than 1%. While with option 3, the reader can precisely search and detect the SFO with a residual SFO of 0.03% at -3dB SNR TDL-A channel.
- Source [108] report that
 - For packet size of 96bits, when the SNR is increased from -4dB to 20dB, the ratio of device residual SFO over 100ppm decreases to 6% for Option 2, 3 and 4, but remains at 95% for Option 1.
 - For packet size of 400bits, when the SNR is increased from -4dB to 20dB, the ratio of device residual SFO larger than 10ppm decreases to 5% for Option 2, 3, and 4, but is still 99.6% for Option 1.
- Sources [105], [109] report that SFO estimation based on D2R preamble can achieve accurate estimation without additional ambles (midamble or postamble).
- Source [105], [103] observed that for non-coherent detection of PDRCH, the number of SFO hypotheses and the SNR needed for 10% and 1% BLER cannot significantly be reduced for option 2, 3 and 4 compared to the option 1. Moreover, the additional ambles i.e., midamble(s) and/or postamble introduces additional overhead and postamble may prevent pipelined processing of the reception.
- Source [109] observed that
 - The coarse estimation of SFO based on the D2R preamble indicates that the SFO estimation error is less than 1% with a probability of 99.3%, and less than 0.1% with a probability of 49.9%.
 - The fine estimation of SFO based on the D2R preamble shows that the SFO estimation error is less than 1% with a probability of 99.5%, and less than 0.1% with a probability of 90.8%.

- Reader/gNB can achieve a probability of not less than 99.5% for SFO estimation error below 1%, and 90.8% for SFO estimation error below 0.1% by receiving D2R preamble signals.
- Source [114] report that for D2R with coherent demodulation at reader, the reader needs to estimate the device clock frequency with the accuracy of 0.5% (5×10^3 ppm) or lower for a short message (e.g., 72 bits after CRC/coding) and 0.1% (10^3 ppm) or lower for a long message (e.g., 224 bits after CRC/coding). The source further reports that design of D2R preamble(s) (e.g., overhead) and the correspondingly required number of SFO hypothesis for the estimation depend on the sampling clock accuracy that the device uses for D2R.
- Source [115] reports that transmitting 96-bit packet size with 16-bit CRC requires residue SFO after reader compensation to be 1000 ppm, and transmitting 1000-bit packet size with 16-bit CRC requires residue SFO after reader compensation to be 100 ppm.

Note: In the observations above where coherent detection is used, sources that evaluated option 3 and option 4 assumed that the postamble is used at least for time/frequency tracking and for channel estimation.

6.1.2.7 D2R multiple access

6.1.2.7.1 Multiple access schemes

Time-domain multiple access, and frequency domain multiple access at least by using a small frequency shift in baseband are studied. For device 2b, FDMA in D2R can be achieved by direct modulation of its internally generated carrier wave at the desired frequency. Whether code-domain multiple access is feasible and necessary for all devices would require further analysis which was not carried out in this study.

Time-domain multiple access is the baseline. Sources [26], [28], [20], [18], [55], state that the benefit of TDMA is the low implementation complexity for both device and reader, while the inventory efficiency may be relatively low for TDMA only, and sources [44], [39], [41], [65], [55], [72], state that the guard interval, if supported, between consecutive D2R transmissions from different devices depends on the SFO after clock calibration.

According to sources [20], [26], [28], [52], [44], [18], [22], [42], [55], the potential benefit of frequency-domain multiple access is to increase the transmission efficiency and reduce collisions, while the cons include more complicated frequency resource management and reception processing at reader according to source [26], and potentially increased power consumption for devices according to sources [28], [34], [29], [55]. It is observed that the performance of FDMA may be impacted by the following aspects.

- Large SFO of device
 - Sources [45], [49] state that large SFO (e.g. up to 10^5 ppm) produces higher BLER degradation due to inter-device interference than a smaller (e.g. up to 10^4 ppm) or the ideal case of zero SFO. Source [72] states that under the case of the large SFO (e.g. up to 10^5 ppm), two among four devices using small frequency shifts have BLER floor and cannot achieve BLER 1%. Source [49] state that under the case of the large SFO (e.g. up to 10^5 ppm), two FDMA-ed devices induce about 0.6~1dB performance loss compared to single device.
 - Source [22] state that the large SFO (e.g. up to 10^5 ppm) has little impact (e.g., ≤ 1 dB) on the performance of FDMA between multiple devices.
 - Sources [25] think that sufficient gap between D2R transmissions should be reserved to accommodate frequency error caused by SFO/CFO
 - Sources [34] think that the required guard band size increases for higher switching frequencies for passive devices.
 - Source [58] observed that the performance gap among 10% SFO and 1% SFO (residual) is similar compared FDMA with no FDMA, e.g., about 0.5dB @ 10% BLER, i.e., the performance gap among 10% SFO and 1% SFO (residual) is irrelevant to whether FDMA scheme is used or not.
 - Source [57] state that large SFO (e.g. up to 10^5 ppm) has little impact on the link performance of FDMA, as it is observed that the performance gap between 10% and 1% SFO is negligible in the case of FDMA among 4 devices.
- Timing offset between devices

- Sources [42] state that timing offset results in a degradation of up to ~4 dB and the loss varies for different devices depending on the level of experienced interference
- Maximum range of small frequency shift
 - Sources [41] think that the frequency gap among devices will impact the interference, which is highly depends on the small frequency shift capability, i.e., how large the frequency shift can be via small frequency shift.
- Harmonics in the backscattered signal
 - Sources [25] state that FDMA is feasible by proper frequency resource allocation not using odd harmonic frequency of FDMed D2R transmissions.
- Potential intermodulation spectral leakage in the backscattered signal
- Frequency resource collision
 - Source [68] thinks that if the guard band size between D2R transmissions is fixed, allocating passive devices with large SFO to frequency shifts closer to the A-IoT carrier frequency and either (1) passive devices with smaller SFO or (2) active devices to frequency shifts further from the A-IoT carrier frequency reduces frequency resource collision.
- Number of multiplexed devices
 - Source [49] reports that performance loss increases with the increase of device number. Besides, for FDMA detection at reader side, there is about 1.5 - 3dB performance loss from 6 FDMA-ed devices compared to single device.
 - Source [68] thinks that the potential number of multiplexed devices depends in the maximum rate of frequency switching.
 - Source [68] thinks that if the guard band size depends on the SFO capability of the device, the number of multiplexed devices can be increased if passive devices with large SFO are allocated frequency shifts closer to the A-IoT carrier frequency and passive / active devices with smaller SFO are allocated frequency shifts further from the A-IoT carrier frequency.

According to sources [28], [IIT], [29], [49], [55], CDMA can improve the resource utilization efficiency without increasing the device complexity significantly. Sources [44] thinks CDMA would help the multiplexing among readers and it can alleviate the cross-link interference. Source [29] thinks CDMA is also beneficial for the latency reduction and success rate improvement of access procedure. Source [65] thinks that the CDMA scheme is mostly used for the signals without carrying information. However, sources [35], [43], [20], [18], [22], [23], [25], [55] show concerns on the necessity and feasibility of CDMA, especially considering the limited capability (e.g., large SFO/CFO) of Ambient IoT devices and the cost (additional memories to store a set of codewords at device) versus benefits. In detail, the observations are as follows.

Impact of SFO on the performance of CDMA

In the case of large SFO (e.g., 10^5 ppm):

- Source [23] think that the orthogonality between different codes/sequences will be severely disrupted, as the large SFO will accumulate an additional sampling error of 10 points of 100 points.
- Source [25] think that the increased inter-device interference would materially degrade D2R performance, e.g., increased false alarm rate and miss detection probability, which in turn reduce spectrum efficiency even lower than the case of simple TDMA.
- Source [18] think that the accurate timing and power control required by CDMA are far-fetched for Ambient IoT devices, referring to the IS-95 CDMA system.
- Source [49] state that CDM-ed MA has comparable or better performance than FDM-ed MA under different SFO assumptions (i.e., $0/10^3/10^4/10^5$) and device numbers (i.e., $1/2/3$). Source [49] state that CDMA by mapping Manchester encoded bit or convolutional encoded bit with 4-length orthogonal code is feasible for D2R transmission carrying 20 information bits.

- Source [59] state that convolutional codes and BPSK modulation for the CDMA improve the D2R transmission performance with multiple multiplexing devices. Source [59] state that for D2R transmission with TBS16+CRC0, the performance difference between the CDMA scheme using 4-length orthogonal code for 4 devices and the repetition scheme for single device at 1%BLER is less than 0.5dB. While for D2R transmission with TBS96+CRC16, the performance difference between the CDMA scheme with 4-length orthogonal code for 4 devices and the repetition scheme for single device at 1%BLER is also less than 0.5dB.
- Source [72] state that CDMA for msg-1 enables multiplexing large number of msg-1 sequences even when power variation is [-9dB, +9dB], while quite large number of SFO hypotheses is necessary at reader to achieve reasonable false-alarm and miss detection probabilities with the SFO of $[0.1 - 1] * 10^5$ ppm.
- Source [22] state that correlation properties of sequences are severely damaged with the SFO of 10^5 ppm. Besides, D2R receiver fails to estimate the SFO of each of the parallel D2R transmissions for the SFO of 10^5 ppm.
- Source [29] state that CDM of RACH preambles using either m-sequences or Gold sequences of length 63 is feasible and preambles from multiple devices can be clearly detected by the reader, even in challenging conditions (SFO = 5%, SNR = 0dB). For 1% missed-detection rate, simulation results showed that m-sequences and Gold sequences are able to achieve this performance level when SNR is about -24dB and -23dB, respectively.

In the case of relatively smaller SFO (e.g., 10^4 ppm),

- Source [72] state that CDMA for msg-1 enables multiplexing large number of msg-1 sequences even when power variation is [-12dB, +12dB] for the SFO of $[0.1 - 1] * 10^4$ ppm with reasonable number of SFO hypotheses at reader.

Impact of CFO on the performance of CDMA for Device 2b

- Source [22] state that codeword detection at D2R receiver fails considering non-coherent demodulation has to be used due to the quick phase rotation caused by the residual CFO of e.g. 10s or 100s of Hz after CFO estimation and correction.

Impact of timing offset between CDMed D2R transmissions on the performance of CDMA for Device 2b

- Source [48] state that binary modulated orthogonal sequence such as Golay sequence can tolerate timing error by selecting a suitable cyclic shift spacing.
- Source [29] state that it is possible to detect multiple transmitters with timing difference and power difference among devices.
- Source [41] think that poor synchronization performance in time and frequency domain of device would degrade the code orthogonality and thus results in a bad cross-correlation performance.
- Source [38] think that the different propagation delays from devices may also degrades decoding performance.
- Source [49] state that the negative impact of asynchronization can be mitigated with some enhancements, e.g. enhanced synchronization sequence and enhanced detection method at reader/BS side, e.g., sliding window based detection and setting constraints on the start of D2R transmission.

The impact of power variation between devices on the performance of CDMA is studied as follows.

- Source [72] state that the larger power variation of at least up to [-9dB, +9dB] can be addressed by reader receiver and CDMA can achieve significant throughput gain for msg-1.
- Source [49] state that the greater the disparity in received power among multiple devices, the better performance will be obtained by SIC receiver with CDM-based multiple access scheme.

Except the impact of SFO/CFO of devices, whether CDMA provides benefit is also studied as depending on the length of the orthogonal or pseudo-orthogonal code and the number of available codes for parallel D2R transmissions:

- Source [20] think that using spreading sequence can lead to transmitting a larger number of bits which can be extremely inefficient considering that the devices are extremely power inefficient.

- Source [20] think that CDMA might be too complex to implement in A-IoT devices, which might involve complexities with generating orthogonal sequences.
- Source [41] think that a large device density (e.g., 150 devices per 100 m² for indoor scenarios per TR) requires a long code sequence, which is challenging for the device with limited buffer size.
- Source [19] think that CDMA leads to higher power consumption and lower data rate.
- Source [25] think that the usable number of binary sequences would be much smaller due to impairment such as timing/frequency error and interference.
- Source [29] state that in comparison to RN16, when Msg. 1 is transmitted using RACH preamble m-sequences or Gold sequences, the number of usable binary sequences that can be used is large since the base sequence design from LTE and NR can be reused.
- Source [29] state RN16 cannot tolerate collision for any one of its bits. Once collided, the bit sequence is changed and became non-detectable. On the other hand, m-sequences and Gold sequences are able to tolerate transmission overlap.
- Source [49] state that CDMA by mapping Manchester encoded bit or convolutional encoded bit with 16-length or 64-length orthogonal code improve the D2R transmission performance and multiplexing capacity compared with using 4-length orthogonal code for mapping. Source [59] state that for D2R transmission with TBS16+CRC0 under the assumption of SFO being 10⁴~10⁵, the performance difference between the CDMA scheme with 64-length orthogonal code for 4 devices and the repetition scheme for single device at 1%BLER is about 1dB.
- Source [49] state that BPSK modulation and convolutional code for the CDMA further improve the D2R transmission performance with multiple multiplexed devices compared with OOK based modulation. Source [59] state that for D2R transmission with TBS16+CRC0 under the assumption of SFO being 10⁴-10⁵ and using convolutional codes and BPSK modulation, the performance difference between the CDMA scheme using 16-length or 64-length orthogonal code for 6 devices and the repetition scheme for single device at 1%BLER is less than 1dB.

6.1.2.7.1 Small frequency shifts

For OOK and BPSK, small frequency shifts are studied:

- For applying with Manchester line codes with a repetition number $R \geq 1$:
 - Option 1: Each Manchester codeword is repeated by a codeword repetition number R , within the same time duration T_b corresponding to an information bit, where $R = T_b / (2 \times \text{chip length})$, such that the amount of small frequency shift in Hz is $R/T_b = 1/(2 \times \text{chip length})$.
 - Option 2: By multiplying the Manchester codeword with a square wave corresponding to the small frequency shift, the time duration T_b corresponding to each information bit includes R number of square wave periods, where $R = T_b / (2 \times \text{chip length})$, such that the amount of small frequency shift in Hz is $R/T_b = 1/(2 \times \text{chip length})$. The multiplication operation is performed as either an XOR or XNOR operation between a Manchester codeword corresponding to the information bit and the square wave for the small frequency shift.
- For applying with Miller line codes, according to Figure 6-13 of [6].
- For FM0, small frequency shift is not defined
- If no D2R line code is used, by using a square-wave corresponding to the small frequency-shift, the time duration T_b corresponding to each information bit includes R number of square wave periods generated by 2R OOK chips [0, 1, 0, 1 ...]/[1, 0, 1, 0 ..] or BPSK chips [-1, +1, -1, +1, ...]/[+1, -1, +1, -1, ...], such that the amount of small frequency shift in Hz is R/T_b .
- Potential purposes include:
 - FDMA of D2R, if supported
 - CW interference avoidance, if supported

Note: Small frequency shifts for D2R are studied for the same potential purposes for MSK.

For study purposes, for BPSK and OOK, the small-frequency-shift factor is defined as the ratio of bit-length to two times the chip length. The value of small-frequency-shift factor depends on details of design, which according to proposals considered could be, e.g.:

- The Manchester codeword repetition number R within duration T_b .
- The M value of Miller in Figure 6-12 of UHF RFID standard.
- The number of square-wave periods within T_b .

Sources [18], [25], [45], [57] state that Manchester codeword repetitions within the same time duration corresponding to an information bit is equivalent to bit-level repetitions within the same duration prior to Manchester encoding. Sources [58] and [62] state that option 1 has a more concentrated spectrum, and requires lesser bandwidth as compared to Option 2. Source [62] further states that while Option 1 and option 2 show similar BLER performance for single device case, Option 1 outperforms Option 2 with FDMA, especially with presence of 10^5 ppm SFO. Option 1 can achieve additional gain for coverage evaluation due to lower effective noise power. Source [70] observed that while Option 1 has a narrower main lobe, it has higher sidelobes than Option 2.

Sources [25], [49] and [30] state that the output waveform for Manchester line codes by Option 2 introduces a phase reversal of the output waveform in the middle of the time duration corresponding to an information bit as compared to Option 1.

Sources [18], [20] [26] [45] [42] [39] state that the output waveform using Manchester line codes by Option 2, and no line codes, is equivalent to using Manchester line codes by repetition of the codewords within the same time duration corresponding to an information bit.

Source [49] shows that small frequency shift by Manchester Option 1 is more sensitive to SFO and the time difference between devices than small frequency shift by Manchester Option 2, and small frequency shift via Miller code has the worst performance compared with by Manchester Option 1 and Option 2.

6.1.2.8 Scheduling of D2R

For D2R scheduling, the following information potentially can be explicitly/implicitly indicated to the device via the corresponding PRDCH:

- Time domain resources
- Frequency domain resources
- MCS-like information
- Chip duration
- ID associated with device(s)
- Repetitions
- Midamble (if supported) related information

For each information, it is for further study whether higher-layer signalling and/or L1 R2D control signalling is used - see Clause 6.1.1.10.

6.1.2.9 D2R control information

The following 14 sources studied feedback information regarding the (un)successful reception of the R2D command at the device and provided following observations:

- 5 sources [132], [141], [134], [139], [145] observed that L1-based ACK/NACK is unnecessary or infeasible for ambient IoT, since all D2R transmission are controlled and activated by the reader. Additionally, 1 source [134] observed that it is infeasible for devices to report NACK since it cannot distinguish whether the R2D transmission is dedicated for it and the resource allocation for D2R NACK if the R2D transmission is not successfully detected. And ACK can be transmitted in higher layer signaling.
- 11 sources [135], [136], [143], [144], [147], [148], [131], [149], [150], [139], [138] observed that ACK/NACK feedback improves transmission reliability, when the BS or intermediate node sends R2D transmission to the

device in the command procedure and if the device does not respond ACK/NACK, the BS or intermediate node cannot determine whether the transmission is successful. 1 source [142] observed that in the use cases of command where some device actions are required and no D2R message needs to be fed back, only D2R ACK feedback in response to the PRDCH is beneficial for the reader to know that the reception of the PRDCH is successful and the required actions have been done by the device.

6.1.3 Timing relationships

A-IoT processing time aspects are studied in terms of the following timing relationships:

T_{R2D_min} :	Minimum time between a R2D transmission and the corresponding D2R transmission following it.
T_{D2R_min} :	Minimum time between a D2R transmission and the corresponding R2D transmission following it.
T_{D2R_max} :	Maximum time between the D2R transmission and the corresponding R2D transmission following it, so that the R2D transmission timing is expected to be within $[T_{D2R_min}, T_{D2R_max}]$, when a R2D transmission in response to a D2R transmission is expected for A-IoT Msg2 response to A-IoT Msg1 for the A-IoT device. See clause 6.3 for message descriptions.
$T_{R2D_R2D_min}$:	Minimum time between two different consecutive R2D transmissions to the same A-IoT device.
$T_{D2R_D2R_min}$:	Minimum time between two different consecutive D2R transmissions from the same A-IoT device.

For the time interval between a R2D transmission and the corresponding D2R transmission following it, there are two options studied:

- Option 1: Define a maximum time T_{R2D_max} between a R2D transmission and the corresponding D2R transmission following it, so that the device transmits D2R transmission within $[T_{R2D_min}, T_{R2D_max}]$.
- Option 2: The corresponding D2R transmission timing T_{R2D} following a R2D transmission is determined based on the control information in the R2D transmission, where $T_{R2D} \geq T_{R2D_min}$.

6.1.4 Random access

From the perspective of the physical layer, at least when a response is expected from multiple devices that are intended to be identified, an A-IoT contention-based access procedure initiated by the reader is used, for which at least slotted-ALOHA based access and FDMA, are studied. The study of FDMA includes how the frequency domain resources for Msg1 are allocated, and how a device determines that frequency-domain resource allocation.

The response transmitted from the device to the reader during this procedure is transmitted on PDRCH.

A R2D transmission triggering random access determines X time domain resource(s) for D2R transmission(s) for Msg1, where each D2R transmission for Msg1 occurs in one time domain resource of the X time domain resource(s).

The study includes:

- $X=1$ and $X>1$ and $X \geq 1$. The maximum value of $X>1$ should be set considering the device implementation complexity, device power consumption, the resource usage efficiency affected at least by SFO, and inventory latency.
- Size(s) for resource allocation in the time domain.
- Determination of the X time domain resource(s) by the device.
- Addressing timing errors for adjacent time domain resources due to residual SFO of the device.

For Msg2 transmission in response to multiple Msg1 transmissions initiated by a R2D transmission triggering random access, the following options have been identified:

Option 1: A PRDCH for Msg2 transmission corresponds to an A-IoT Msg1 received from one device.

Option a: the starting time for Msg2 monitoring is separate for each Msg1 resource.

Option b: the starting time for Msg2 monitoring is common for multiple Msg1 resources.

Option 2: A PRDCH for Msg2 transmission corresponds to multiple A-IoT Msg1 received from different devices, the starting time for Msg2 monitoring is common for multiple Msg1 resources.

To determine the time duration for monitoring Msg2 in response to multiple Msg1 transmissions initiated by a R2D transmission triggering random access, the following options have been identified:

Option 1: Predefined

Option 2: Indicated in the R2D transmission triggering random access

Note: Option 1 and Option 2 may not be mutually exclusive.

For a device to determine the reference time for the starting time for monitoring Msg2 in response to multiple Msg1 transmissions initiated by a R2D transmission triggering random access, the following options have been identified:

Option 1: End of time domain resource where the device transmitted the Msg1

Option 2: End of the last of X time domain resource(s) determined for Msg1 transmission(s)

Option 3: End of the R2D transmission triggering random access

Option 4: End of a R2D transmission carrying Msg2

Note: The reference time is used to determine the starting time for Msg2 monitoring, it does not mean that the starting time always equals the reference time.

For Msg3 from multiple devices in response to a given set of one or multiple Msg2 transmission(s) during access procedure, FDMA and/or TDMA of D2R transmissions are studied, including how the frequency and time domain resources for Msg3 are allocated.

6.1.5 Topology 2

For resource allocation and/or controlling of intermediate UE in topology 2 for R2D and D2R transmissions via the Uu interface between the BS and intermediate UE, following two options have been studied:

- Option 1: Higher-layer signalling is used.
- Option 2: L1 and higher-layer signalling are used.

From RAN1 perspective, detailed aspects on the two options are left up to work item phase.

The following observations related to the two options have been made:

- Four sources [129], [124], [120], [125] observed that Option 1 can save overhead compared to Option 2.
- One source [119] observed that the timing gap between R2D message and D2R message is not extended, and inventory rate is comparable with UHF RFID and furthermore, it is observed that with Option 1, the standard impact includes no or less impact on L1 signalling for resource allocation and controlling of the intermediate UE.
- Six sources [129], [126], [123], [118], [128], [127] observed that Option 2 with L1 signalling provides better resource utilization compared to Option 1.
- 1 source [137] observed that if UE determines the resource used for A-IoT transmission within A-IoT resources configured by the network, option 1 provides better resource efficiency and option 2 enables faster A-IoT resource adaptation compared to option 1. And it is further observed that if the resource allocation from NW is for each R2D and/or D2R transmission and is indicated by a L1 signaling of option2, option 2 may affect the A-IoT communication timing relations and causes substantial overhead between NW and UE reader and may lead to resource inefficiency as NW lacks direct information of channel condition between A-IoT device and UE reader.
- 1 source [139] observed that option 1 can save signaling overhead b/w gNB and intermediate UE compared to option 2 but may cause resource waste issue. For resource allocation of intermediate UE, Option 2 can provide better resource efficiency compared to Option 1 but requires more signaling overhead b/w gNB and intermediate UE.

- 1 source [140] compared option 1 and option 2 in terms of signaling overhead, signaling latency, flexibility and resource utilization. It is observed that signaling overhead is higher with option 2, but latency is higher with option 1. Option 2 may provide higher flexibility and better resource utilization compared to option 1.
- 1 source [142] observed that option 1 only requires the standardization workload for adding new higher layer signaling and does not impact the current DCI design for UEs, but the resource utilization efficiency of option 1 may be lower than that of option 2. Option 2 can provide better resource usage efficiency than that of option 1, but Option 2 requires standardization effort on renewing DCI design for intermedia nodes.
- 1 source [148] observed that option 2 is too costly in terms of overhead and latency if using DCI-based scheme to trigger each R2D or D2R, but it is beneficial for flexible resource sharing between AIoT communications as well as the coexistence with the legacy transmission.
- 1 source [149] observed that Option 2 would increase the latency of A-IoT transmission. Furthermore, it is observed that the two options are not necessarily exclusive.
- 1 source [136] observed that the signaling overhead of option 2 is higher and the spec impact in RAN1 is larger. If each R2D or D2R transmission resource is scheduled by DCI, option 2 will bring high transmission latency and low transmission efficiency.
- 1 source [134] observed that Option 2 has larger standardization impacts, for example the DCI design. And Option 2 may affect the A-IoT timing definitions.

6.2 Device (un)availability

The following directions, not for down-selection, are studied regarding the potential impact of device unavailability due to energy harvesting:

- Direction 1: Reader does not provide information to a device regarding when the device may become available/unavailable.
- Direction 2: Reader can provide information to a device based on which the device may become available/unavailable.

6.2.1 Direction 1 solution details

The following solution descriptions, observations or analysis or evaluations, and specification impacts if any, were provided during the study.

Table 6.2.1-1: Details from Source [76]

Source	Details
Source [76]	Solution description By implementation, the reader can provide the device enough time to charge. In addition, the reader can arrange the sequence of messages so that a device can perform its communications / operations spread out over time if possible to allow a device to remain available. Observations or Analysis or Evaluations N/A Specification impacts, if any Additional benefits in availability of "ID associated with device(s) intended for the reception of R2D" when it is located at the front of the PRDCH, using a L1 R2D control information.

Table 6.2.1-2: Details from Source [77]

Source	Details
Source [77]	Solution description The reader does not provide information to a device regarding when the device may become available/unavailable and the device behavior will entirely depend on the available energy in its energy storage and be highly implementation dependent

	Observations or Analysis or Evaluations Observation #1: RF energy harvesting is a suitable way to deploy Device 1 with capacitor size that is sufficient to complete at least an inventory round. Observation #3: RF energy harvesting is not a suitable way to deploy Device 2. Other ambient sources, e.g., indoor lighting, vibration, etc., which are pervasive and continuous, and provide higher energy conversion, should be used. Observation#4: It is expected that Direction 1 has no or less specification impact from both reader and device perspective. Specification impacts, if any There are no direct specification impact unless some mechanisms need to be specified to mitigate the potential drawbacks of Direction 1.
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Table 6.2.1-3: Details from Source [97]

Source	Details
Source [97]	Solution description The reader may be able to request the A-IoT devices to provide information on the current energy status and/or the ability on energy harvesting such as charging rate and capacity, which could be different depending on the device types. If needed, the reader can predict the device availability, but it does not mean the reader provide information on device availability to a specific device. Observations or Analysis or Evaluations Basically, the A-IoT device availability/unavailability cannot be controlled by the reader, which is up to its current energy status at the device. The A-IoT-devices may have the following operation state. - ON state: transmission, reception, and a clock running - OFF state: energy harvesting only - SLEEP state: energy harvesting, monitoring, maintaining memory, and a rough clock running The A-IoT devices may switch the state depending on its energy harvesting status. Once it receives a triggering signal, its status may be switched from OFF to ON, but it may not be controllable by the network at least for the passive devices. If the reader wants information relevant to energy harvesting, the device needs to provide information. Specification impacts, if any The spec needs to support necessary procedures and signaling such as indication on the A-IoT device to report the energy status. How to utilize the information is up to the reader implementation.

Table 6.2.1-4: Details from Source [78]

Source	Details
Source [78]	Solution description The reader by implementation maintains the maximum time between the Paging messages as $\leq T_{\text{discharge}}/2$, ensuring that the device can switch to its ON duration after charging completely, receive the Paging message and complete the entire inventory procedure before running out of charge. The device is expected to be fully charged and then switch to an ON duration to be able to receive the Paging message from the reader. The interval of the Paging messages would ensure that the device receives at least one Paging message before it runs out of charge. Once the device completes the inventory procedure, the device maintains a state flag using an ultra-low power latch during the OFF duration such that the device does not perform inventory repeatedly after it completes a charging cycle. If the device runs out of charge during the inventory procedure, or misses out on Paging messages during its OFF or charging duration, the device is simply expected to charge fully and then monitor for the next Paging message.

	Any failed reception of Paging messages during the device charging time is handled by the higher layer re-access procedure. The multi device inventory latency for direction 1 is 18.3 seconds, to reach 100% of the devices. Observations or Analysis or Evaluations The leakage current for a given capacitor size has to be much lesser than the output of the energy harvester in order to achieve satisfactory charging. For capacitor sizes larger than 10 μF, the leakage current is in the order of micro amperes, resulting in a slow charging or continuous discharging using RF energy and hence is an unsuitable energy source for such capacitors. For device 1, the discharging time is in the order of a few seconds, providing the devices with adequate time to complete an inventory round. For device 2, the discharging time is in the order of milli seconds, which is too short a duration for the device to complete an entire inventory procedure. For device 2 to be able to increase its discharge times, capacitors with larger capacitance, in the order of 1000 μF can be used, however, it would be infeasible to use RF energy as a source due to the high leakage current. RF energy harvesting is not a suitable way to deploy devices with short discharge times, e.g. device 2, due to the following reasons: - The discharge time is in the order of milli seconds, which is not enough for the device to complete an entire inventory round. - Increasing the capacitor size results in a high leakage current, resulting in a slow charging or continuous discharging, deeming RF energy as an unsuitable energy source. - Other ambient sources, e.g. indoor lighting, vibration, etc. which are pervasive and continuous, and provide higher energy conversion, should be used. RF energy harvesting is a suitable way to deploy devices with long discharge times, and the reader's scheduling implementation is able to appropriately handle the duration of an inventory round. For calculating the multi device inventory latency, the charging time taken by the first set of devices that receive the first Paging message and are to participate in the first inventory round is taken into account when calculating the latency. Specification impacts, if any No specification impact. Possible with proper implementation of devices with adequate charge/discharge times and capacitor size, as well as of the reader with the required time gap between the Paging messages, there is no specification impact.
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Table 6.2.1-5: Details from Source [78]

Source	Details
Source [78]	Solution description The device provides information to the reader pertaining to e.g. the remaining charge or charge status, such that the reader can optimally transmit the Paging messages to ensure that all devices are inventoried in the shortest amount of time. The reader is expected to transmit a request message to the device, responding to which the device would provide the reader with a remaining charge status report. The reader would then group the devices with similar amount of charge left and based on this grouping, the reader can then transmit the Paging message accordingly for each of these groups to participate in different inventory rounds. For devices that did not have enough charge remaining to complete an inventory round, the reader would wait for a favorable report such that it can be inventoried in subsequent rounds. Observations or Analysis or Evaluations Solution 2: The device cannot transmit remaining charge-related information to the reader without the reader requesting for such a report.

	Transmission of the request message and corresponding report from each of the devices multiple times to the reader would cause a substantial overhead. Additional hardware is required for the device to be able to determine the amount of charge remaining in its capacitor. Specification impacts, if any Solution 2: Significant specification impact. Request message and report might require new signals or channels. Definitions of these messages are required. Changes to the architecture to include additional hardware for the device to measure the remaining charge.
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Table 6.2.1-6: Details from Source [104]

Source	Details
Source [104]	Solution description The energy harvest and transition between states (ON and OFF) are achieved by device implementation The device indicates to the reader before entering the OFF state. Observations or Analysis or Evaluations For device 1, reader can assume that device 1 with 10uF capacitor is always available during an inventory or command procedure, if the inventory/command procedure is immediately after the device is powered on. For device 2, by implementing a larger capacitor, device 2 can be assumed available during an inventory or command procedure, if the inventory/command procedure is immediately after the device is powered on. RF energy harvesting is not enough for larger capacitors (e.g., 100uF), other energy sources should be considered. Before entering the OFF state, the device may indicate to the reader which may be beneficial for the reader to perform the subsequent scheduling or transmission. Specification impacts, if any no specification impact is needed for energy harvest and transition between states. Indication design (e.g., from device to reader)

Table 6.2.1-7: Details from Source [95]

Source	Details
Source [95]	Solution description: • Signaling of energy status reporting • Energy aware segmentation Observation: • RAN1 is still discussing whether device 1 can support sleep state in addition to the ON and OFF state, however, it is clear that the device 2b can support ON; OFF and sleep state. • The RF based energy harvesting eliminates the negative consequence of device unavailability due to smaller capacitor size, • Even though, reader does not provide any information related to the device availability or unavailability as part of direction#1, reporting of coarse energy status helps reader schedule or not schedule subsequent R2D transmission to the device. • For 96 bits payload, the transmission time increases from 14.9ms to 89.6ms when the data rate decreases from 6kbps to 1kbps. ○ With duty-cycle and energy harvesting, the device can be sustainably operated in case of 6kbps with storage smaller of less than 0.5uF, while it requires at least 1.5uF storage to be sustainably operate in case of 1kbps.

- **With always ON condition, the device can be sustainably operated in case of 6kbps with storage smaller of less than 2uF, while it requires at least 10uF storage to be sustainably operate in case of 1kbps.**
- **For 400 bits payload, the transmission time increases from 69.3ms to 388ms when the data rate decreases from 6kbps to 1kbps.**
 - **With duty-cycle and energy harvesting, the device can be sustainably operated in case of 6kbps with storage smaller of less than 1.5uF, while it requires at least 6uF storage to be sustainably operate in case of 1kbps.**
 - **With always ON condition, the device can be sustainably operated in case of 6kbps with storage smaller of less than 6uF, while it requires very high storage capacity storage to be sustainably operate in case of 1kbps.**

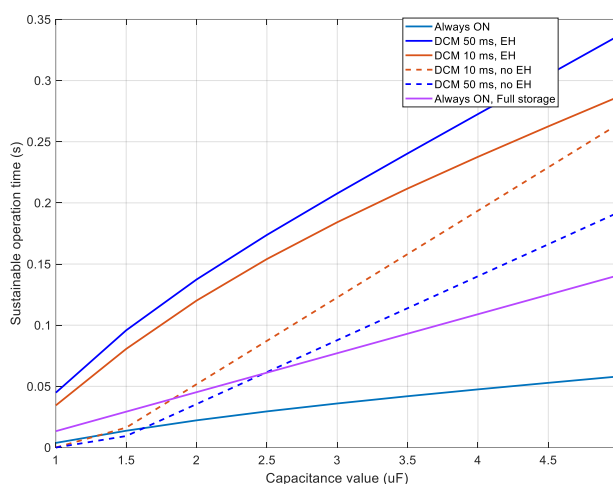


Figure 1: Sustainable operation time of device with duty cycle and w/o duty cycle, distance = 12m

- Since device may run out of charge during the transmission of larger payload size, retransmitting the entire packet requires more charging time. Hence, device can transmit part of the packet using segmentation based on the available energy while indicating to the reader about the subsequent packets to be transmitted after charging.

Specification impact:

- Device signaling its coarse energy status as part of the Msg-1 or Msg-3 transmission to the reader. Reader can schedule or not schedule subsequent R2D transmission to the device. Device behavior when the expected R2D transmission is not received.
- RAN2 can discuss the energy-based segmentation.

Table 6.2.1-8: Details from Source [94]

Source	Details
Source [94]	Solution description A-IoT device is assumed to have two states: ON, OFF. - ON state supports transmission, reception for communication and does not support energy harvesting considering device has a single antenna for both communication and RF energy harvesting in TDMed manner. - OFF state supports at least energy harvesting and does not support at least transmission, reception for communication, maintaining a timer. There are two schemes for a device switching between ON and OFF states: - Scheme 1 (OFF when fully discharged): The device switches to OFF when fully discharged and turns ON when there is stored energy and the incoming RF power is above the activation threshold. - Scheme 2 (OFF when energy < X%): The device switches to OFF when the stored energy falls below a threshold X, e.g., X=90% and turns ON when the stored energy exceeds that threshold and the incoming RF power is above the activation threshold.

	Scheme 2 has advantages over scheme 1, such as shorter device downtime for energy harvesting and sufficient energy for communication once the target PRDCH is received. However, its implementation complexity and cost may be higher. There are two options for a reader to transmit the PRDCH e.g., A-IoT paging triggering the random access (RA). • Option 1 (P-R2D Tx): Reader transmits the PRDCH periodically. • Option 2 (A-R2D Tx): Reader transmits the PRDCH multiple times on demand. Option 1, where the paging triggering the RA occurs periodically, remains transparent to the device as the reader cannot control or be aware of the device's state. Additionally, the reader can use option 2 to decrease the likelihood that the device will miss the paging when it is in the OFF state. Observations or Analysis or Evaluations Evaluations and observations can be found in Table 6.2.2-1 [94]. Specification impacts, if any For both Schemes, it is expected to have no or minimal specification impacts.
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Table 6.2.1-9: Details from Source [79]

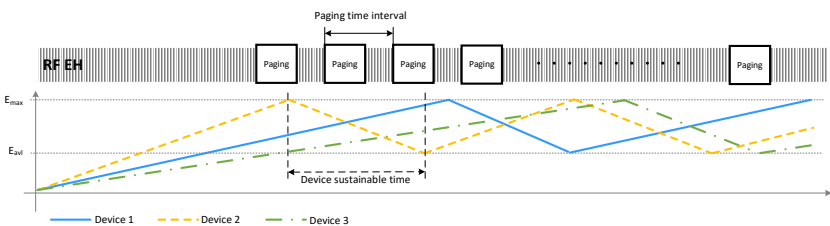
Source	Details
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Solution description
In Direction 1, the reader does not provide information to a device nor be aware of a device regarding when the device may become available/unavailable. The device is assumed to have two states:

- ON state: Device can at least perform transmission and reception, run MHz sampling clock, and retain memory.
- OFF state: Device cannot perform transmission and reception, no clock running. Device can harvest energy. Device can retain memory of being inventoried.

From reader perspective, it transmits paging messages with time interval shorter than the device sustainable time. To avoid devices running out of energy before successfully completing the inventory, the paging time interval can be comparable to half of the device sustainable time.

From device perspective, it becomes available when it is fully charged, and then keeps performing Tx/Rx procedure until it runs out of its available energy. Once a device finishes the inventory, it needs to retain a memory of being inventoried so that it does not re-enter the inventory procedure.



Observations or Analysis or Evaluations

A. Results for Device 1

Figure 3.2-2 and Figure 3.2-3 show the evaluation results of percentage of successfully inventoried devices with respect to the inventory time for Device 1, where Figure 3.2-2 provides results assuming {R2D, D2R} data rate of {56, 160} kbps and Figure 3.2-3 provides results assuming {R2D, D2R} data rate of {14, 40} kbps. The inventory completion time considering 99% and 100% of devices successfully complete inventory are summarized in Table 3.2-1. As the sustainable time of Device 1 is 4.75 s, and therefore two paging time interval of 2 s and 4 s are considered in the evaluation.

Source
[79]

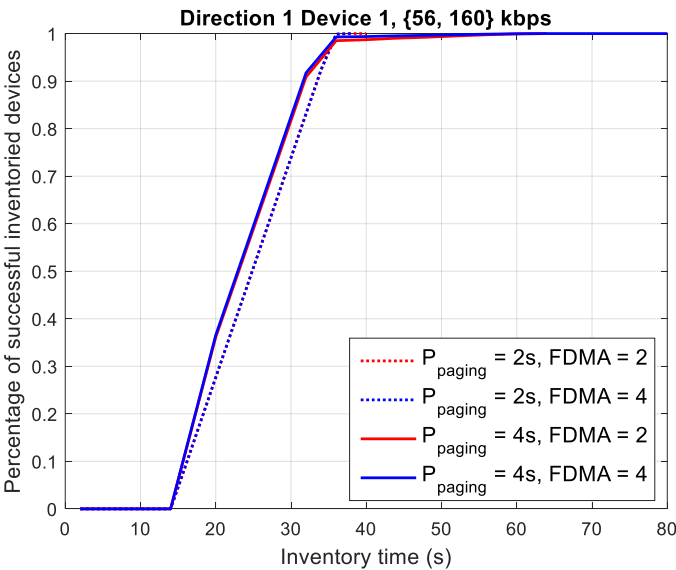


Figure 3.2-2: Percentage of successful inventoried devices with respect to inventory time for Direction 1 Device 1 in case of {R2D, D2R} data rate of {56, 160} kbps.

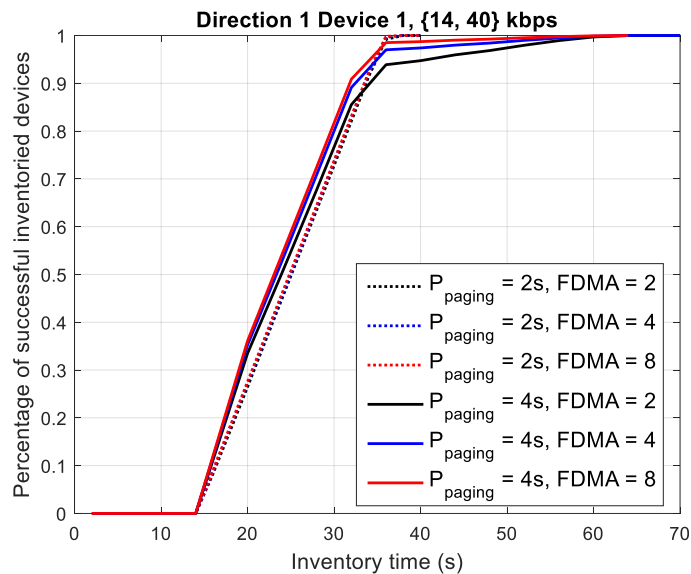


Figure 3.2-3: Percentage of successful inventoried devices with respect to inventory time for Direction 1 Device 1 in case of {R2D, D2R} data rate of {14, 40} kbps.

Table 3.2-1: Inventory completion time for Direction 1 Device 1

Successful rate	{R2D, D2R} Data rate	FDMA resources	Inventory completion time	
			Paging time interval = 2 s	Paging time interval = 4 s
99%	{56,160} kbps	2	36.40 s	46.40 s
		4	36.40 s	48.40 s
	{14,40} kbps	2	36.40 s	62.40 s
		4	36.40 s	58.40 s
		8	36.40 s	46.40 s
100%	{56,160} kbps	2	39.42 s	64.02 s
		4	39.17 s	59.64 s
	{14,40} kbps	2	40.34 s	69.63 s
		4	39.83 s	67.03 s
		8	39.43 s	63.96

B. Results for Device 2

Figure 3.2-4 and Figure 3.2-5 show the evaluation results of percentage of successfully inventoried devices with respect to the inventory time for Device 2, where Figure 3.2-4 provides results assuming {R2D, D2R} data rate of {56, 160} kbps and Figure 3.2-5 provides results assuming {R2D, D2R} data rate of {14, 40} kbps. The inventory completion time considering 99% and 100% of devices successfully complete inventory are summarized in Table 3.2-2. Two paging time interval of 20 ms and 40 ms are considered considering that the sustainable time of Device 2 is only 47.5 ms.

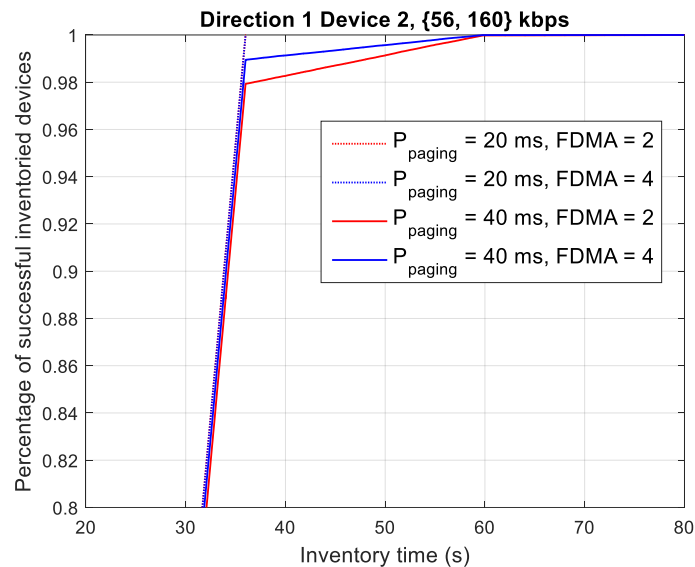


Figure 3.2-4: Percentage of successful inventoried devices with respect to inventory time for Direction 1 Device 2 in case of {R2D, D2R} data rate of {56, 160} kbps.

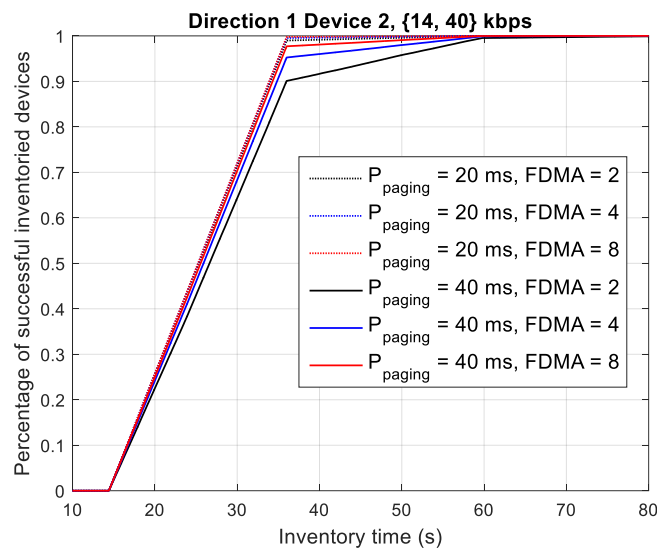


Figure 3.2-5: Percentage of successful inventoried devices with respect to inventory time for Direction 1 Device 2 in case of {R2D, D2R} data rate of {14, 40} kbps.

Table 3.2-2: Inventory completion time for Direction 1 Device 2

Successful rate	{R2D, D2R} Data rate	FDMA resources	Inventory completion time	
			Paging time interval = 20 ms	Paging time interval = 40 ms
99%	{56,160} kbps	2	35.80 s	48.44 s
		4	35.80 s	37.16 s
	{14,40} kbps	2	36.34 s	58.36 s
		4	35.90 s	55.32 s

		8	35.84 s	49.32 s
100%	{56,160} kbps	2	35.99 s	57.38 s
		4	35.99 s	53.55 s
	{14,40} kbps	2	55.57 s	72.66 s
		4	51.18 s	62.32 s
		8	46.35 s	57.34 s

Observation 3: For Direction 1, when the paging time interval is comparable to half of the sustainable time, at least 99% of devices can be successfully inventoried before running out of energy and no device goes through recharging.

Observation 4: For Direction 1, with the increase of paging time interval, when the paging time interval is comparable to the sustainable time, there are devices cannot complete inventory before running out of energy and therefore the inventory completion time sufficiently increases.

Observation 5: For Direction 1, as the sustainable time of Device 1 can be up to several seconds, it is practical for a reader to transmit paging messages with a time interval comparable to half of device sustainable time, e.g., in the level of seconds.

Observation 6: For Direction 1, as the sustainable time of Device 2 can only last for tens of ms, a reader has to transmit paging messages very densely, e.g., in the level of tens of ms.

Specification impacts, if any
For Direction 1, no specification impact is expected at reader side nor device side.

Table 6.2.1-10: Details from Source [80]

Source	Details
Source [80]	Solution description The reader can set a proper value of the interval in-between two paging messages, to ensure all the devices can complete at least one entire inventory procedure. Observations or Analysis or Evaluations The reader may or may not have the prior information on each device's corresponding discharging time, which may rely on the deployment; The R2D signalling overhead may also need to be considered. Specification impacts, if any No

Table 6.2.1-11: Details from Source [81]

Source	Details
Source [81]	Solution description In Direction 1, an A-IoT device would become available after fully charged and become unavailable at the time of charging or the energy in the storage running out. A-IoT devices can perform transmission and reception for communication in available state and perform energy harvesting in unavailable state. The switching between device availability and device unavailability is based on implementation without specification impact. In this case, carrier wave should always exist to provide energy for A-IoT devices. Observations or Analysis or Evaluations Observation 3: In Direction 1, carrier wave should always exist on to provide energy for A-IoT devices and is not guaranteed in the A-IoT field deployments for all A-IoT devices. The availability/unavailability of A-IoT devices are not fully controlled by the network during A-IoT operation due to the limitation of device within the carrier wave coverage. Specification impacts, if any In Direction 1, the switching between device availability and device unavailability is based on

	implementation without specification impact.
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Table 6.2.1-12: Details from Source [82]

Source	Details
Source [82]	Solution description Method 1: The Reader sends continuous CW before paging message to ensure that all devices achieve a high energy storage state. After the continuous CW, the reader transmits the paging message. Devices can harvest energy from the CW while in unavailable state. Upon the end of the continuous CW through power detection, devices switch to the available state to receive the subsequent paging message and enter the random access process. Figure 8 illustrates the transmission and reception between the reader and the device in Method 1. Reader CW Paging message Device Unavailable ON Figure 8 Transmission and reception between reader and device in Method 1 Observations or Analysis or Evaluations Method 1: Before random access, Method 1 based on Direction 1 can ensure that devices receive the paging message at a high power level and enter the random access process Specification impacts, if any Method 1: The energy node transmits the CW before paging message. The device is required to detect the end of continuous CW and receive subsequent paging message.

Table 6.2.1-13: Details from Source [82], [102]

Source	Details
Source [82], [102]	Solution description Method 2: - The reader can transmit multiple paging messages in one access round. Wherein, each paging message includes an indicator that indicates whether the paging message is the initial paging message of the access round. - During one access round, the initial paging message indicates the total number of occasions that the access round contains and other paging messages indicate the remaining number of occasions after them. - If a device receives the initial paging message, it performs random access based on the number of occasions indicated by the initial paging message.

- If a device receives a paging message for the first time, and this paging message is not the initial paging message, it will perform random access based on the number of occasions indicated by this paging message.
- If a device receives a paging message that is neither the first paging message it has received nor the initial paging message, it will decrease its stored occasion counter.

To facilitate the comparison between the existing slotted-ALOHA and Method 2, Figures 9 and 10 respectively illustrate the function of paging message and R2D trigger (i.e. counter decrement command) of the two methods. It can be observed that Option 2 replaces some R2D triggers with paging messages in one access round.

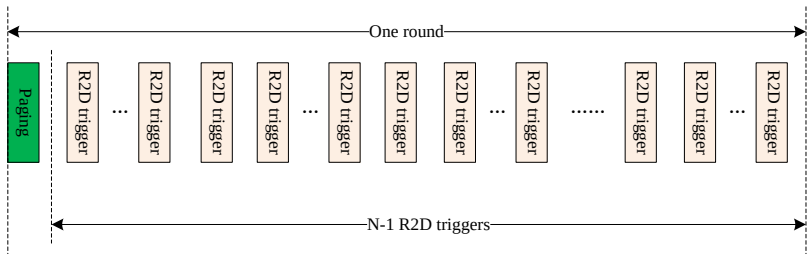


Figure 9 Function of paging message and R2D triggers in slotted-ALOHA process

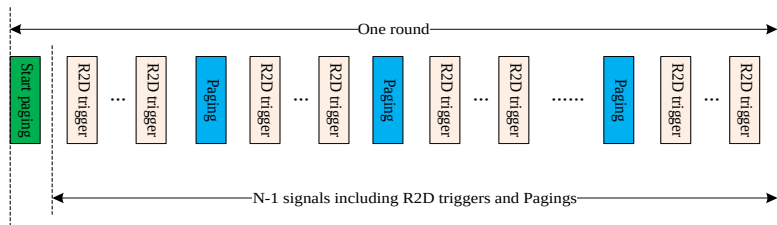
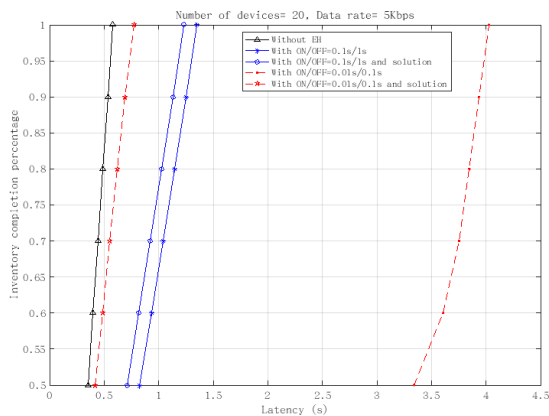


Figure 10 Function of paging messages and R2D triggers in Method 2

Observations or Analysis or Evaluations

Method 2:



- Before random access, Method 2 based on Direction 1 can increase the probability of devices receiving the paging message without incurring additional R2D overhead.
- For the inventory process of 20 devices, device (un)availability leads to an increase in random access latency. The Method 2 based on Direction 1 can significantly mitigate the latency impact caused by device (un)availability.

Specification impacts, if any

Method 2:

	Define the paging message transmission mechanism and relevant signaling indication. Specify the device's behavior after receiving a paging message.
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Table 6.2.1-14: Details from Source [83]

Source	Details
Source [83]	Solution description Since the reader does not provide any information (e.g., scheduling or coordination) that can be used by a device for switching from one operating state to another (e.g., for the purpose of power saving or energy harvesting), the state transition in a device is expecting to be based on its implementation choice. For example, - For a device that supports only 2 states (ON / OFF), the state transition is expecting to be based on device energy storage level (i.e., switch to the ON state when the storage level is charged above a X% and switch to the OFF state when the storage level is dropped below a Y%, where X could be 50/80/100 and Y could be 20/10/0 depending on device implementation choice). That is, a device could be power-ON only when it is fully charged and power-OFF only when it is fully dis-charged. Therefore, it is not possible / infeasible for a device to perform energy re-charging in the middle of an inventory or command process, and as such, the device should complete an inventory or command process as soon as power-ON and before it is fully dis-charged. - For a device that supports 3 states (ON / OFF / SLEEP), the timing/determination of state transition between ON and SLEEP will be also solely base on its implementation choice (since the reader does not provide any information). It is mentioned in the past, a device may employ a duty-cycle based ON / SLEEP state switching/transition. For example, for every duty-cycle period (which is to be determined by the device implementation), the device switches to the ON state for a small amount of time to monitor/receive any paging message from the reader and switches back to the SLEEP state for a longer duration for energy re-charging. The device repeats this duty-cycle state switching operation to keep its energy storage level high, until it detects during a ON period a paging message that triggers an inventory or command process, then it remains in the ON state to complete the process. In order to cater for the above possible state transition behaviours, it is mentioned in one solution that the reader periodically transmits a paging message triggering an inventory or command process with a periodicity equal to $\frac{1}{2}$ of the expected sustainable operating time (ON duration) of a device (where the triggered process also finishes within the periodicity). If the device power-ON or switches to the ON state with a remaining energy level that is or near 100% full, even if the next paging message is transmitted $\frac{1}{2}$ of the expected sustainable operating time later, the remaining energy level should still be able to sustain device operation for the entire process. Observations or Analysis or Evaluations For A-IoT devices that support only 2 states (ON / OFF), the reader should frequently transmit a paging message triggering an inventory or command process in order to ensure a device does not spend a lot of energy monitoring the channel for a paging message. Although the above-described solution is in-principle workable from the view point of a device sustainable operating time, however, this does not take into account when the number of devices trying to access the same process is more than the capacity that can be handled within a process, then TX collision will occur and some devices would not complete the process. Subsequently, these devices may re-attempt in the next period / triggered process, take up time and frequency resources, but not be able to finish the process. Furthermore, these device with failed attempts will take 10's of seconds to be re-charged to 100%, wake-up and try a new process again until they complete one round of inventory or command process. As seen, this type of mechanism is prone / exposed to TX collisions and long latency delays. For A-IoT devices support 3 operating states (ON / OFF / SLEEP), the reader does not need to frequently transmit the paging message as devices are always in a high energy storage level. But the drawback would be the inventory / command process completion time for multiple devices. Even to make an inventory of just one device, the reader may need to transmit the

	paging message multiple times, as the device ON timing and duration is not influence by any information from the reader. Reader transmission of a paging message that falls within the device ON period will be purely coincident. To make an inventory of a large number of devices, at the beginning the paging transmission may easily coincide with one or multiple devices, which is fine. But when only a few devices remained, the coincident may not happen often. Based on the above solution description and analysis, the following observations are reached for Direction 1: - When the device supports only two operating states (ON / OFF), - the reader should frequently transmit a paging message triggering an inventory or command process in order to ensure a device does not spend a lot of energy monitoring the channel for a paging message, and - for a mechanism where a paging trigger is periodically transmitted and the periodicity is $\frac{1}{2}$ of the expected sustainable operating time of the device, this type of mechanism may be prone / exposed to TX collisions and long latency delays to complete a round of inventory or command process with the reader. - When the device supports three operating states (ON / OFF / SLEEP), - when a device employs a duty-cycle based ON / SLEEP state transition scheme, device energy storage could always remain in a high level, but the inventory / command process completion time for multiple devices may be long, since the timing of a paging message transmission that falls within the ON duration of a device is based on coincident. Specification impacts, if any Behaviour(s) and restriction(s) for autonomous state transition in a device may need to be specified (e.g., TS 38.213 or 214) in order to ensure the overall operation of an inventory or command process (A-IoT communication) is not negatively impacted. Detailed behaviour(s) and restriction(s) should be further studied and discussed.
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Table 6.2.1-15: Details from Source [84]

Source	Details
Source [84]	Solution description Reader scheme: Transmits PRDCH multiple times. Device scheme: Either continuous On until fully discharged or continuous On until discharged to a threshold. Observations or Analysis or Evaluations Reader does not control the device state Specification impacts, if any no specification impact

Table 6.2.1-16: Details from Source [86]

Source	Details
Source [86]	Solution description The device is assumed to always have sufficient energy to receive commands from the reader at any time. Observations or Analysis or Evaluations The power consumption of the device must be less than the achievable rate of energy harvesting. The range from device to CWE is limited by the power consumption of the device. CWE power consumption and interference are greater for this direction than for direction 2. Specification impacts, if any Specification impacts are minimal.

Table 6.2.1-17: Details from Source [89]

Source	Details
Source [89]	Solution description When CW and communication is same frequency, - To use only this direction can work. When CW and communication are different frequencies, - After one transaction (inventory or command) or multiple of transactions, reader is not required to call the successful communication device again. In such situation, the reader does not provide information to a device regarding when the device may become available/unavailable as next inventory or command are sufficiently future like 1 hour or 1 days later. - On the other hand, during one transaction or multiple of transactions, when communication is possible needs to be aligned between the reader and the devices. If not, the reader is not able to initiate the communication. This operation is controlled by the timer between ON and RETENTION. Observations or Analysis or Evaluations If the device is able to know no more transaction is expected after successful communication, the device can go for OFF. This may be realized by "no intention for the more paging for now" kind of the message by the reader. Or other implicit means. If the device is not sure no more transaction or not even after successful transaction, the device needs to repeat the transition between ON and RETENTION by the timer for possible future paging when CW and communication are different frequencies. Specification impacts, if any What information needs to be kept in volatile Memory and what timers are required to be run during RETENTION. When CW and communication is same frequency, how the device distinguish them using the start-indicator part. When CW and communication is different frequency, in what condition, the device is not required to repeat the transition between ON and RETENTION.

Table 6.2.1-18: Details from Source [90]

Source	Details
Source [90]	Solution description Define a minimum and a maximum time between two PRDCHs providing random access triggering messages such that the device is not required to monitor a potential R2D transmission during the minimum time, if the device does not participate the corresponding random access round. During the minimum time, the device may transition into the PS state. After the minimum time, the device is require to monitor a potential R2D transmission at least until the maximum time from the preceding PRDCH reception. Define a minimum time for a device, who successfully finished the random access procedure, such that the device is not required to monitor a potential R2D transmission during the defined minimum time. The minimum time may be dependent on the count of the current random access round. Define a minimum time for a device to continuously monitor a potential R2D transmission, whenever the device becomes active after unavailable periods from energy harvesting, such that a reachability from a reader to the device is guaranteed. Observations or Analysis or Evaluations N/A Specification impacts, if any N/A

Table 6.2.1-19: Details from Source [114]

Source	Details
Source [114]	Solution description Applicable primarily to Device 1 (ultra-low-power devices) Relies on the device's long availability duration (e.g., >10 seconds) Reuse simple slotted-ALOHA based access mechanism R2D includes end timing information for scheduled D2R or R2D transmission Devices make autonomous decisions based on their remaining energy and the scheduled transmission duration Observations or Analysis or Evaluations Device 1 operates at ultra-low power (~1μW) and can rely solely on RF energy harvesting. Long availability duration (>10 seconds) is possible with a 10 μF capacitor for Device 1. This approach is suitable when the device's available duration is long enough to cover the period of slotted-ALOHA. Devices can estimate their remaining energy using the formula $P1 = (M1 - N1)/M1$, where M1 is the minimum available duration and N1 is the counted slot number after entering the available state. By providing end timing information, devices can autonomously decide whether to participate in scheduled transmissions based on their energy status, potentially avoiding D2R or R2D failures due to energy shortage. Specification impacts, if any Device: Definition and test requirement on M1, counting behavior for N1, skip and go unavailable if identifying energy shortage for upcoming scheduling, reporting behavior of remaining energy information Reader: Command periodicity for initial contact with the device based on M1 value, end timing information in R2D transmissions for scheduled D2R or R2D transmissions

Table 6.2.1-20: Details from Source [91]

Source	Details
Source [91]	Solution description RAN1 assumes the device has enough energy for at least for one message (e.g., Msg1) then leave it to RAN2 to add signaling from the device to indicate to the reader if it does not have enough energy for follow up messages (e.g., energy status report in Msg1). Observations or Analysis or Evaluations RAN1 can assume the device has enough energy, with a fallback energy status report. Specification impacts, if any RAN2 specification impact, if RAN2 decides to add an energy status report.

Table 6.2.1-21: Details from Source [93]

Source	Details
Source [93]	Solution description Each device reduces a duration of contiguous unavailable time and reader transmits paging frequently Observations or Analysis or Evaluations This direction would mean that each device wakes up and starts monitoring paging just when available energy is sufficient. If unavailable time of a device is long, inventory completion time will be degraded. For example, a paging is transmitted right after a device becomes long unavailable state for energy harvesting. In this case, the inventory session cannot be completed until the device wakes up, notices a re-paging after a lot of re-paging, and detects

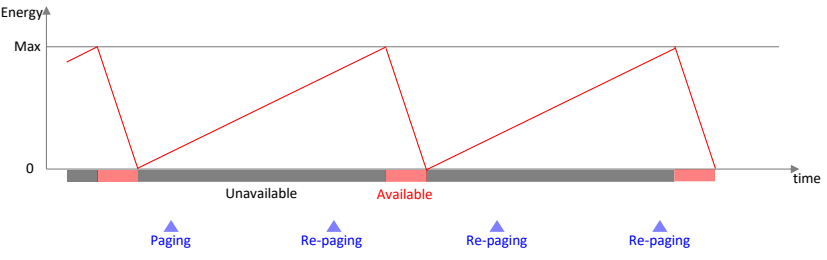
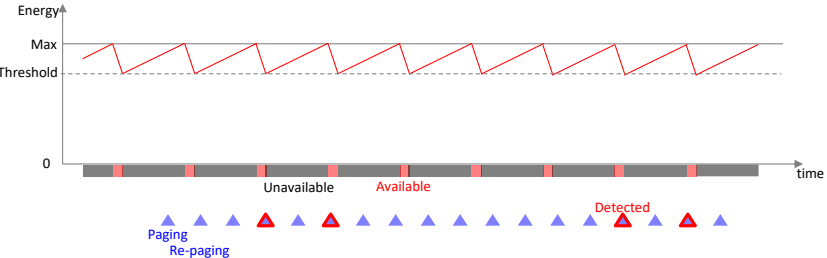
	satisfaction of response conditions.  Fig: An example of Direction 1. To avoid this kind of situation, a possible way is that each device reduces a duration of contiguous unavailable time and reader transmits paging frequently so that device notices existence of an on-going inventory session earlier. For spec impact, such a kind of device behavior is defined with a threshold of energy storage, and it may be better to define device behavior after a paging/re-paging is detected in consideration of missing paging/re-paging a lot of times.  Fig: A possible way of Direction 1. Specification impacts, if any E.g., a threshold of energy storage is defined to do state transition for energy harvesting. E.g., device behavior to monitor paging/re-paging and to perform Msg1 TX is defined in consideration of missing a lot of paging/re-paging.
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Table 6.2.1-22: Details from Source [96]

Source	Details
Source [96]	Solution 1 description Reader continues to try reaching A-IoT device until it becomes available Observations or Analysis or Evaluations The simple solution for a reader to communicate with an unavailable A-IoT device is to wait until the device is available after charging. Without the knowledge/controllability of device availability/unavailability, the reader has to repeat/retry the communications over time. For example, suppose the case where a reader wants to do inventory for A-IoT devices. The reader should repeat msg-0 over time so that whenever the target device is available, it can receive the msg-0. If there is no msg-0 received during an available time duration, the device is not aware of the inventory procedure from the reader. Therefore, the reader should send msg-0 at least once per device available time duration. The reader should send msg-0 at least once per these time durations. Unavailable time duration for charging depends on Rx power, RF EH efficiency, capacitor size, etc. Therefore, the reader is not able to identify when and how long each device becomes available. The best thing that a reader can do is to continue the tries as much as possible and as long as possible.

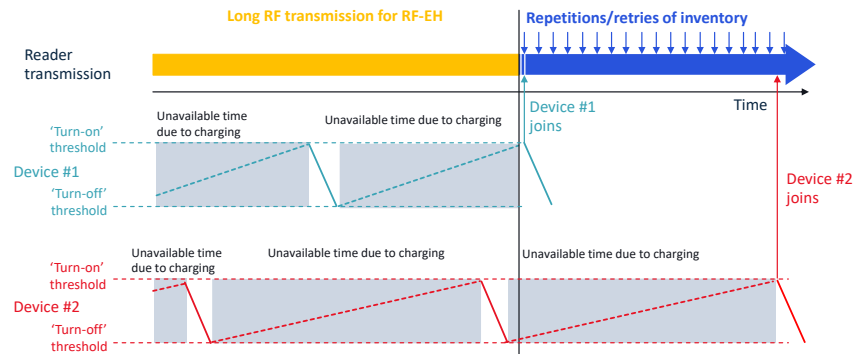


Fig. A1 Direction 1: a reader has no idea and cannot control when a device is available/unavailable

A typical inventory procedure is a slotted-ALOHA based contention-access procedure. If there are a large number of devices, a number of devices compete with each other in the slotted-ALOHA msg-1 transmissions. In such case, the inventory procedure for a device may not be able to be completed within the available time duration. Once the device becomes unavailable, the device will not be available for tens or hundreds of seconds due to charging. As such, congestion of the contention-access procedure would further increase the time duration where a reader has to try.

Observation A1:

- Direction 1 is the case where a reader has no idea and cannot control when a device is available/unavailable due to the amount of energy in the storage.
- As a solution, a reader should try to reach the target device in blind manner.
- Take an inventory procedure as an example, reader should transmit msg-0 multiple times until msg-1 is received, where the msg-0 transmission should be at least once per device available time duration.
- If an A-IoT communication activity (e.g., message exchanges for contention-based access procedure) cannot be completed within an available time duration of a device for some reason (e.g., congestion of slotted-ALOHA), the device becomes not accessible (unavailable) due to charging again, which leads additional delay for the reader.

Solution 2 description

Improved energy harvesting efficiency by deployment/implementation

Observations or Analysis or Evaluations

The other solution is to improve energy harvesting efficiency by deployment/implementation. For example, network can deploy energizers densely such that all the devices are always fully energized and available. Or, device can implement energy harvesting sources such as solar, vibration, etc, which increases the form factor of the A-IoT devices.

Note that large capacitor size for a device does not resolve the issue. With a large capacitor size, the device available time duration gets longer. However, it also prolongs the device unavailable time duration for charging. There must be cases where some devices has (almost) zero available energy when a reader starts repeat/retry of inventory procedure, and such devices needs longer charging time until it becomes first time available after the start of the reader's inventory.

Observation A2:

- Another solution for direction 1 to improve energy harvesting efficiency by deployment/implementation.
- A network can deploy energizers densely such that all the devices are always fully energized and available.
- Devices can implement energy harvesting sources such as solar, vibration, etc, which increases the form factor of the A-IoT devices.
- Using large capacitor size cannot be a solution for direction 1

	- In direction 1, reader cannot make sure all the devices are fully charged when the reader starts the overall inventory procedure Specification impacts, if any No specification impact is envisioned for direction 1.
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Table 6.2.1-23: Details from Source [92]

Source	Details
Source [92]	Solution description When the energy storage is full or reaches a certain level, or the device receives a RF power larger than an activation threshold, the device wakes up to detect the R2D signal and then transmit the D2R signal. When the procedure of inventory/command is completed or the energy storage is not enough to process the following communication, the device switches to OFF and only performs the energy harvesting. Observations or Analysis or Evaluations N/A Specification impacts, if any How and when to transmit CW for energy harvesting, for example, controlled by the reader Energy harvesting in Topology 2, for example, when the RF energy source and the UE are the same node, the impacts of different RRC states (IDLE/Inactive/Connected mode) Transmitting power of RF energy source

Table 6.2.1-24: Details from Source [112]

Source	Details
Source [112]	Solution description - Device states are not defined in A-IoT system. - The energy harvesting can be handled by device implementation. That is, if the energy is insufficient, the device power is turned off, and device performs energy harvesting. After that, if the energy is charged, the device power is turned on, and device performs TX/RX. - Reader can provide inventory round triggering command (e.g., Query command) frequently, allowing devices with relatively short available time to perform inventory round. - For example, reader can set a small Q value (which determines the total number of slots) for each inventory round so that each inventory round can be terminated quickly. - When the reader provides the inventory round triggering command, available devices will participate in the inventory round. - In other words, the device determines whether it can remain available during the corresponding inventory round. Observations or Analysis or Evaluations - Small specification impact. - R2D resource overhead may increase because the reader needs to transmit inventory round triggering command frequently. - Device available time is relatively short because the energy harvesting is not considered until the device is turned off. Specification impacts, if any

	- Potential information feedback on the energy state (e.g., remaining energy) of the device.
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Table 6.2.1-25: Details from Source [117]

Source	Details												
Source [117]	Solution description for direction 1												
	State description												
	<table><tr><th>State</th><th>Description</th><th>Transition event</th><th>Device</th></tr><tr><td>ON</td><td>Transmission and reception running a clock for timing maintaining the memory</td><td>Transit to 'ON' state: • the charging energy is higher than an energy threshold or a certain percentage, and the incoming RF power is higher than an activation threshold Transit from 'ON' state: • the charging energy is lower than an energy threshold or a certain percentage, or fully discharged • Receives an indication from Reader to switch to the other state.</td><td>Device 1 Device 2a</td></tr><tr><td>OFF</td><td>Energy harvesting</td><td>Transit to 'OFF' state: • the charging energy is lower than an energy threshold or a certain percentage or fully discharged • Receives an indication from Reader to switch to 'OFF' state. Transit from 'OFF' state: • the charging energy is higher than an energy threshold or a certain percentage</td><td>Device 1 Device 2a</td></tr></table>	State	Description	Transition event	Device	ON	Transmission and reception running a clock for timing maintaining the memory	Transit to 'ON' state: • the charging energy is higher than an energy threshold or a certain percentage, and the incoming RF power is higher than an activation threshold Transit from 'ON' state: • the charging energy is lower than an energy threshold or a certain percentage, or fully discharged • Receives an indication from Reader to switch to the other state.	Device 1 Device 2a	OFF	Energy harvesting	Transit to 'OFF' state: • the charging energy is lower than an energy threshold or a certain percentage or fully discharged • Receives an indication from Reader to switch to 'OFF' state. Transit from 'OFF' state: • the charging energy is higher than an energy threshold or a certain percentage	Device 1 Device 2a
	State	Description	Transition event	Device									
	ON	Transmission and reception running a clock for timing maintaining the memory	Transit to 'ON' state: • the charging energy is higher than an energy threshold or a certain percentage, and the incoming RF power is higher than an activation threshold Transit from 'ON' state: • the charging energy is lower than an energy threshold or a certain percentage, or fully discharged • Receives an indication from Reader to switch to the other state.	Device 1 Device 2a									
	OFF	Energy harvesting	Transit to 'OFF' state: • the charging energy is lower than an energy threshold or a certain percentage or fully discharged • Receives an indication from Reader to switch to 'OFF' state. Transit from 'OFF' state: • the charging energy is higher than an energy threshold or a certain percentage	Device 1 Device 2a									
	Scheme:												
	- Reader transmits the control information based on the PRDCH to the devices. ➤ Option 1: Reader transmits the scheduling information on the PRDCH periodically. ➤ Option 2: Reader transmits the scheduling information on the PRDCH multiple attempts based on the reader's on-demand scheduling. - The device switches to ON when the charging energy is higher than an energy threshold and the incoming RF power is higher than an activation threshold, then the device receives the reader's control information in a certain period or one attempt of the multiple attempts. - Reader stops its attempt or period transmission when a response has been received by the reader side or a max attempt amount has been achieved.												
	Observations												
	- High communication delay due to multiple transmission attempts. - Low communication efficiency.												
Specification impacts, if any													
No or minimal specification impacts.													

Table 6.2.1-26: Details from Source [116]

Source	Details
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Source [116]	Solution description The reader does not provide information to a device nor be aware of a device regarding when the device may become available/unavailable. The device can report the energy status to the reader. In this case, the device is assumed to have two states i.e. ON and OFF state. Observations or Analysis or Evaluations Reduce the time interval between the paging messages based on the device capacitor size and the threshold value of the device The device can report its energy status during the random access Scheduling for R2D/D2R can be based on the reported energy status Specification impacts, if any Define the paging message transmission mechanism and relevant signaling indication. Device behavior for e.g. reporting of energy status Content of R2D/D2R control information
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6.2.2 Direction 2 solution details

The following solution descriptions, observations or analysis or evaluations, and specification impacts if any, were provided during the study.

Table 6.2.2-1: Details from Source [76]

Source	Details
Source [76]	Solution description A device is (nearly) fully charged when it first becomes available. A device monitors R2D transmissions to determine whether it should participate in a round of communications. If a device is not among the device or a group of devices scheduled to participate, it can use a next scheduled opportunity field in the R2D transmission to determine when it can/should be available. This field indicates the next round of communications. A device then may become unavailable and can harvest energy. When the device determines it is near the next scheduled opportunity, and it has sufficient energy for a round of communications, a device can become available to process the PRDCH. If a device is (nearly) fully charged, it may become available at any time. Observations or Analysis or Evaluations N/A Specification impacts, if any Availability of "ID associated with device(s) intended for the reception of R2D" is located at the front of the PRDCH, using a L1 R2D control information. Availability of a next scheduled opportunity for another round of communications, using a L1 R2D control information. The next scheduled opportunity field can represent a number of events, such as R2D transmissions.

Table 6.2.2-2: Details from Source [77]

Source	Details
Source [77]	Solution description Support for wake-up signal in the SLEEP state Observations or Analysis or Evaluations Observation #1: There is a need to study a solution for extending the sustainable operation

	time for devices with short discharge times and long charge time by harvesting only RF energy. Observation #2: The state transitions to or from the OFF states can be based on an energy level of the device, which can be on implementation issue. On the other hand, the state transitions between ON and SLEEP states need to be studied as it involves a monitoring behavior of a device, which may be a potential scope of specification. Observation #3: Following benefits are achievable by reader controlling the device's state transition between the ON and SLEEP: - Avoiding very long unavailable time duration after the reader starts inventory: Since the consumed energy in the energy storage in each available time duration is quite limited, the time duration that a device is unavailable for energy harvesting is short. - Keeping large amount of energy in the storage all the time: Whenever a device joins inventory, the energy storage is (almost) full. This avoids the case where a device joins inventory but the remaining amount of energy is not sufficient. Specification impacts, if any At least a wake-up signal needs to be specified. The wake-up signal support should be specified in a way so that it is transparent for (and can be ignored by) devices that do not support it.
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Table 6.2.2-3: Details from Source [78]

Source	Details
Source [78]	Solution description The following states and the corresponding functions of the device for each of the states are considered, where their duration is determined by device implementation.: - ON state – Device is fully operational – it can transmit and receive communication messages, run a clock, maintain memory. - OFF state – Device is not operational – it cannot transmit or receive any messages, nor run a clock. In this state, the device performs energy harvesting and maintains an inventory flag. - SLEEP state – Device is semi-operational – it cannot transmit or receive communication messages, but it can run a clock and perform energy harvesting. It may also maintain memory. The objective of this solution is not to force all the devices to adhere to a single configuration to periodically shift between the ON and SLEEP states, but rather to ensure that all devices shift to ON state when the Paging messages are being transmitted. To achieve this, we propose an on-demand shifting of states where the reader controls the device to switch to the ON state from the SLEEP state using an indicator which would inform the device of the duration it has to remain in the SLEEP state. The device is expected to perform energy harvesting and charge itself during the OFF state, and when its fully charged, it would shift to the ON state and listen for the indicator to be transmitted by the reader. The indicator would indicate the time left for the inventory procedure to start and for the device to receive a Paging or Query message. The reader transmits the indicator in an aperiodic manner, controlled by reader implementation, such that the device has to be in an ON state for only a short time, enabling it to switch to the SLEEP state until the next Select or Query message is expected to be received to conserve energy. At the indicated time, the device will shift to the ON state and perform the inventory procedure, ensuring that the device would have enough energy to be able to complete the procedure during the ON state. Within an inventory round, if the device failed to receive a Query message, the device can remain in ON state until it receives the indicator, after which it can go to SLEEP state until the next expected Query message, based on the indicated duration. The following are the possible transitions that the device would be expected to perform.

	- OFF to ON state – Transition based on when the device is fully charged. - ON to OFF state – Transition based on when the device runs out of charge, or is discharged. - ON to SLEEP state – Transition based on when the device receives the indicator. - SLEEP to ON state – Transition based on when the device completes the time indicated. - SLEEP to OFF state - Transition based on when the device runs out of charge, or is discharged. The multi device inventory latency for direction 2 is 18.3 seconds. Observations or Analysis or Evaluations During the SLEEP state, it has to be ensured that the power consumption of the device while running the clock is lower than the charging rate of the capacitor. The on-demand shifting of states does not require the device to maintain any fixed duration of the states or a periodicity to shift between states, thus avoiding any potential need for the device to be preconfigured with any information. Specification impacts, if any Minor specification impact. The indicator needs to be specified to indicate the time left for the inventory procedure to start, which can be a higher layer message carried on the PRDCH.
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Table 6.2.2-4: Details from Source [104]

Source	Details
Source [104]	Solution description Duty cycle-like solution (with sleep state) The device indicates to the reader before entering the OFF state. Observations or Analysis or Evaluations Reader can provide an indication to the device to perform sleep state transition. Energy threshold based transition can be up to implementation. Timer or counter based transition can be considered Before entering the OFF state, the device may indicate to the reader which may be beneficial for the reader to perform the subsequent scheduling or transmission. Specification impacts, if any Sleep-state and state transition conditions Indication or timer/counter design for state transition

Table 6.2.2-5: Details from Source [95], [100]

Source	Details
Source [95], [100]	Solution description: • Wake up signal for the device or configuring device monitoring periods • Configuring the duty cycles and aligning duty cycle among group of devices • Energy aware scheduling and selection of subset of device in each access round within the inventory Observation: • Reader can control the power states of device 2b - ON; OFF and sleep and help to extend the operational time of devices. • The device can be periodically configured to transition from sleep to the ON state as part of the duty cycle for monitoring the trigger from the reader. Contrary to it, wake up signaling indicating an identifier associated to a device or a group of device wakes up selected device for the access round.

- Contrary to the traditional Q based device scheduling implemented in UHF RFID, devices can be selected for each access occasion where each access occasion reserve resources for multiple devices.
- Energy aware selection of device for each access round decrease the storage size as shown in our analysis.
- For 96 bits payload, the transmission time increases from 14.9ms to 89.6ms when the data rate decreases from 6kbps to 1kbps.
 - With duty-cycle and energy harvesting, the device can be sustainably operated in case of 6kbps with storage smaller of less than 0.5uF, while it requires at least 1.5uF storage to be sustainably operate in case of 1kbps.
 - With always ON condition, the device can be sustainably operated in case of 6kbps with storage smaller of less than 2uF, while it requires at least 10uF storage to be sustainably operate in case of 1kbps.
- For 400 bits payload, the transmission time increases from 69.3ms to 388ms when the data rate decreases from 6kbps to 1kbps.
 - With duty-cycle and energy harvesting, the device can be sustainably operated in case of 6kbps with storage smaller of less than 1.5uF, while it requires at least 6uF storage to be sustainably operate in case of 1kbps.
 - With always ON condition, the device can be sustainably operated in case of 6kbps with storage smaller of less than 6uF, while it requires very high storage capacity storage to be sustainably operate in case of 1kbps.

Evaluation:

- Comparing the sustainable operation time of device supporting always ON with that of duty cycle and energy harvesting is provided in Fig. 3.

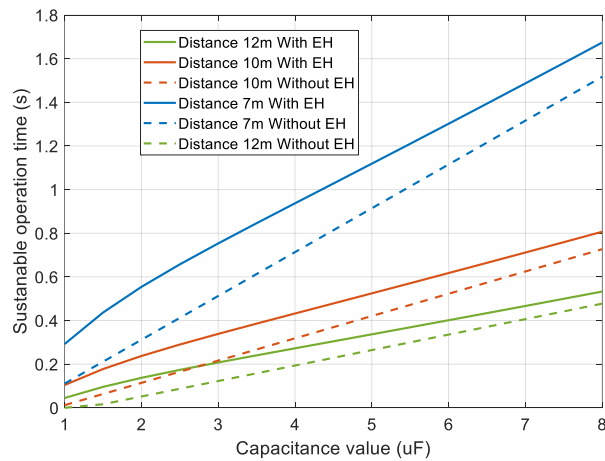


Figure 3: Comparison of operation time of Ambient IoT devices

- Energy aware scheduling benefits illustrated in Fig. 6. As it reduces the required capacitance value significantly

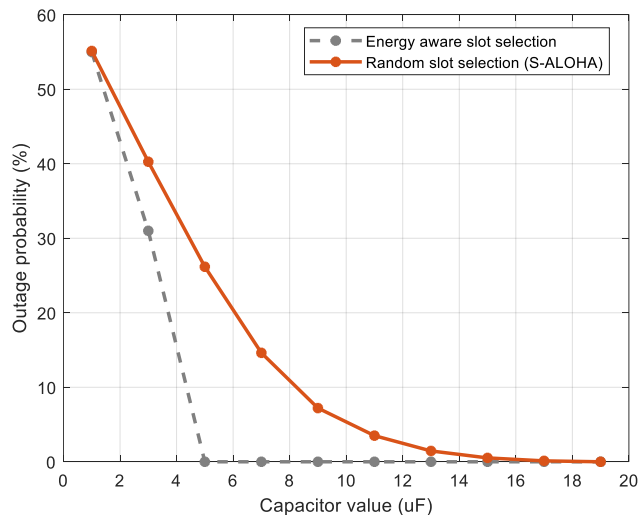


Figure 6 Comparison of outage for different scheduling mechanism for Ambient IoT

	devices Specification impact: - Wake up signal design and content otherwise device monitoring periodicity - Duty cycle configuration for a group of devices Energy aware scheduling using selecting a subset of devices for each access round.
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Table 6.2.2-6: Details from Source [94]

Source	Details																			
Source [94]	Solution description A-IoT device is assumed to have three states: ON, OFF, SLEEP. - SLEEP state supports maintaining a timer, maintaining a memory content from ON state, energy harvesting and does not support transmission, reception for communication. - The functions for ON and OFF states supported and unsupported are the same as those captured for direction 1. Under direction 2, one scheme for a device switching between ON and SLEEP states: - Scheme 3 (SLEEP w R2D control): Device switches between ON and SLEEP states under reader's control. The reader can provide the time duration for ON or SLEEP and their periodicity for the device to perform ON and SLEEP state transition. Thus, only periodic paging transmission, i.e., Option 1 (P-R2D Tx) is used for reader to imitates RA. Observations or Analysis or Evaluations The detailed evaluation methods and simulation assumptions listed in Table 3-3 can be found in section 3.3 of [94]. Case 1: For 2 RA resources and paging period of 6ms and 16ms																			
	Paging period = 6ms Paging period = 16ms																			
	Figure 1. Time required to inventory 100% of devices for 2 RA resources and 6/16ms paging period																			
	Table 1: Time required to inventory 100% of devices using 2 RA resources per paging message																			
	<table><tr><th rowspan="2">Schemes</th><th colspan="2">Totally inventory time [s]</th></tr><tr><th>Paging period: 6ms</th><th>Paging period: 16ms</th></tr><tr><td>OFF when fully discharged w P-R2D Tx</td><td>21.16</td><td>55.69</td></tr><tr><td>OFF when energy <90% w P-R2D Tx</td><td>17.918</td><td>42.019</td></tr><tr><td>SLEEP w R2D control w P-R2D Tx</td><td>6.309</td><td>12.833</td></tr><tr><td>OFF when fully discharged w A-R2D Tx</td><td>19.677</td><td>44.179</td></tr><tr><td>OFF when energy <90% w A-R2D Tx</td><td>13.30</td><td>23.747</td></tr></table>	Schemes	Totally inventory time [s]		Paging period: 6ms	Paging period: 16ms	OFF when fully discharged w P-R2D Tx	21.16	55.69	OFF when energy <90% w P-R2D Tx	17.918	42.019	SLEEP w R2D control w P-R2D Tx	6.309	12.833	OFF when fully discharged w A-R2D Tx	19.677	44.179	OFF when energy <90% w A-R2D Tx	13.30
Schemes	Totally inventory time [s]																			
	Paging period: 6ms	Paging period: 16ms																		
OFF when fully discharged w P-R2D Tx	21.16	55.69																		
OFF when energy <90% w P-R2D Tx	17.918	42.019																		
SLEEP w R2D control w P-R2D Tx	6.309	12.833																		
OFF when fully discharged w A-R2D Tx	19.677	44.179																		
OFF when energy <90% w A-R2D Tx	13.30	23.747																		

Case 2: For 4 RA resources and paging period of 6ms and 16ms

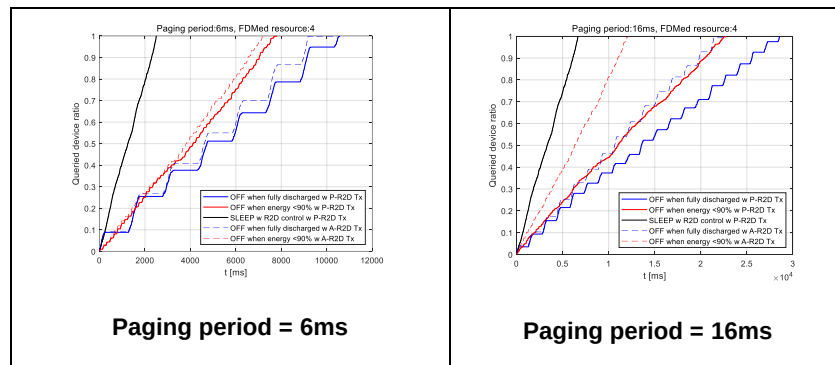


Figure 2. Time required to inventory 100% of devices for 4 RA resources and 6/16ms paging period

Table 2: Time required to inventory 100% of devices using 4 RA resources per paging message

Schemes	Totally inventory time [s]	
	Paging period: 6ms	Paging period: 16ms
OFF when fully discharged w P-R2D Tx	10.567	28.538
OFF when energy <90% w P-R2D Tx	7.805	22.732
SLEEP w R2D control w P-R2D Tx	2.50	6.67
OFF when fully discharged w A-R2D Tx	10.441	22.684
OFF when energy <90% w A-R2D Tx	7.284	12.081

Observations based on above Case 1 and Case 2

Observation 1:

- Decreasing the paging period reduces total inventory latency across all schemes.
- For 2 RA resources scheduled by a single paging message, when the paging period is reduced from 16ms to 6ms,
 - Schemes 1 and 2 reduce the time to inventory all devices by about 62% and 57% for periodic R2D transmission, and by about 55% and 44% for aperiodic R2D transmission.
 - Scheme 3 achieves about 51% reduction for periodic R2D transmission.
- For 4 RA resources scheduled by a single paging message, when the paging period is reduced from 16ms to 6ms,
 - Schemes 1 and 2 show about 63% and 66% time reduction for periodic R2D transmission, and 54% and 40% for aperiodic R2D transmission.
 - Scheme 3 shows about 62% latency reduction for periodic R2D transmission.

Observation 2:

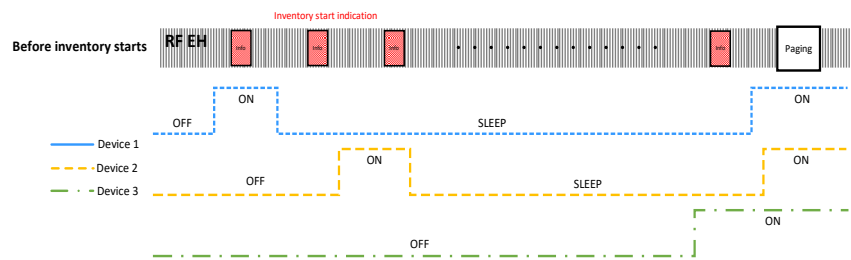
- Increasing the random access (RA) resources for D2R transmission scheduled by a single A-IoT paging reduces total inventory latency significantly across all schemes.
- For 6ms paging period, when RA resources is increased from 2 to 4,
 - Scheme 1 and 2 reduce the time to inventory all devices by about 50% and 56% for periodic R2D transmission, and 47% and 45% for aperiodic R2D transmission.
 - Scheme 3 show about 60% latency reduction for periodic R2D transmission with the reduced paging period from 16ms to 6ms.

	- For 16ms paging period, when RA resources is increased from 2 to 4, - Schemes 1 and 2 show about 49% and 46% time reduction for periodic R2D transmission, and about 48% and 49% for aperiodic R2D transmission. - Scheme 3 shows about 48% latency reduction for periodic R2D transmission. Observation 3: - For Scheme 1 and Scheme 2, aperiodic R2D transmission can further reduce the inventory time, especially for long paging period, although it results in more paging overhead than the periodic R2D transmission. - For 2 RA resources, compared to 6ms/16ms-period R2D transmissions, Scheme 1 and 2 further reduce the time to inventory all devices by about 7%/21% and 26%/43% with additional aperiodic R2D transmission - For 4 RA resources, compared to 6ms/16ms-period R2D transmissions, Scheme 1 and 2 further reduce the time to inventory all devices by about 1.2%/20% and 6.7%/49% with additional aperiodic R2D transmission Observation 4: - Devices under Scheme 1 needs to recharge more often compared to Scheme 2 and Scheme 3. - Scheme 3 generally offers the shortest inventory time, while Scheme 1 has the longest. - For scheme 1 and scheme 2 that reader cannot control or be aware of the device state, short paging period and/or aperiodic R2D transmission can reduce inventory latency. For scheme 3, to achieve the tradeoff between the inventory latency and R2D overhead, it is advisable to set the paging period and the ON to SLEEP time ratio to allow device harvest enough energy. Specification impacts, if any For scheme 3, it may have following specification impacts: - Define SLEEP, ON, and OFF states for the A-IoT device with their functions. - Include signaling from the reader to the device for transitioning between ON and SLEEP states. - Detail the device's monitoring behavior, transmission, and reception procedures.
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Table 6.2.2-7: Details from Source [79]

Source	Details
Source [79]	Solution description In Direction 2, solutions to extend device availability for both stages before the inventory starts and during the inventory procedure are proposed. The device is assumed to have three states: - ON state: Device can at least perform transmission and reception, run MHz sampling clock, and retain memory. - OFF state: Device cannot perform transmission and reception, no clock running. Device can harvest energy. Device can retain memory of being inventoried. - SLEEP state: Device cannot perform transmission and reception. Device can run a low accuracy clock, retain memory, may or may not harvest energy (depends on the relationship of the consumed power and the input power). Solution before inventory starts From reader perspective, it provides indication of the potential starting time or remaining time of the future paging message. By implementation, the reader can provide the indication in a shorter time interval, so that a device does not need to drain its energy to monitor the indication. From device perspective, it becomes available when it is fully charged and monitors the indication of the transmission time of the upcoming paging message. The device transits to

SLEEP state to save or harvest energy, if necessary. During SLEEP state, the device needs to retain a clock for time counting.



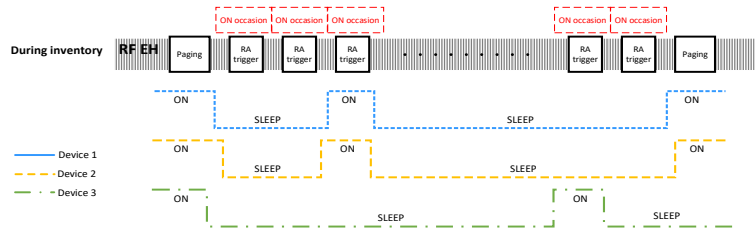
Solutions during inventory procedure

B1: Non-slot-ALOHA based solution

This solution is a proposed new random access solution for Ambient IoT, and is not based on the slot-ALOHA solution specified in UHF RFID.

From reader perspective, it provides duty cycle information in paging message, which includes at least number and duration of ON occasions.

From device perspective, upon receiving a paging message, it randomly selects an ON occasion for random access, and transits to SLEEP state, if necessary, to conserve and harvest energy. During SLEEP state, the device needs to retain memory regarding its ON occasion information and runs a clock for time counting. The device transits to ON state till the selected ON occasion is coming. Within the ON occasion, the device performs random access upon receiving triggering message. If collision occurs, devices transit to SLEEP state and wait for the next paging message. If a device does not successfully access before running out of available energy, it recharges until fully charged again.



B2: Slot-ALOHA based solution

This solution is an optimization based on the slot-ALOHA solution specified in UHF RFID.

From reader perspective, it additionally sends a slot decrement indication to inform the decrement progress to devices.

From device perspective, upon receiving a paging message and acquires the Q value, it first randomly selects an access slot within $[0, 2Q-1]$. The device transits to SLEEP state before the selected access slot, if necessary. During SLEEP state, the device needs to retain memory regarding its access slot information and runs a clock for time counting with an assuming slot duration. The device transits to ON state when it counts to the selected slot. The device monitors the decrement indication to obtain the actual slot decrement progress and determines whether it re-transits to SLEEP state or keeps monitoring to its upcoming access slot. If a device finds out that its access slot has passed, it responds immediately in the follow-up slot. If collision occurs, devices transit to SLEEP state and wait for the next paging message. If a device does not successfully access before running out of available energy, it recharges until fully charged again.

Observations or Analysis or Evaluations

Necessity of Direction 2 for Device 1

Figure 3.3-4 to Figure 3.3-7 show the comparison results between Direction 1 and Direction 2 for Device 1.

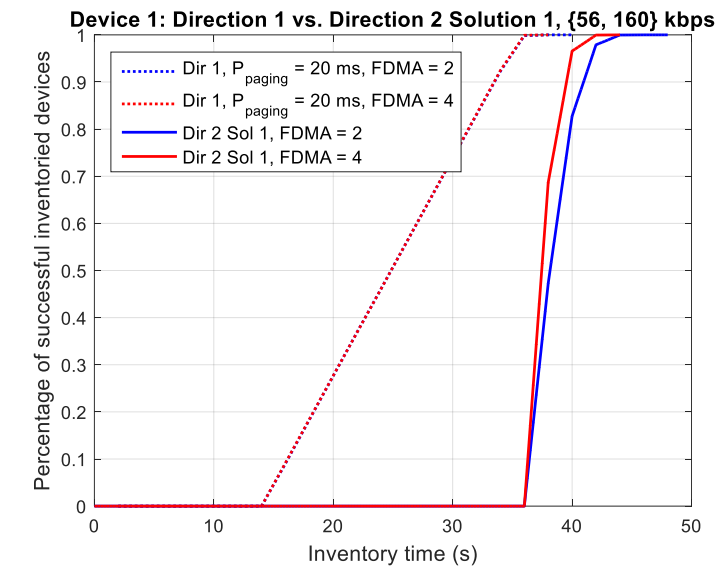


Figure 3.3-4: Comparison of percentage of successful inventoried devices with respect to inventory time for Direction 2 Device 1 Solution 1 versus Direction 1 Device 1 in case of {R2D, D2R} data rate of {56, 160} kbps.

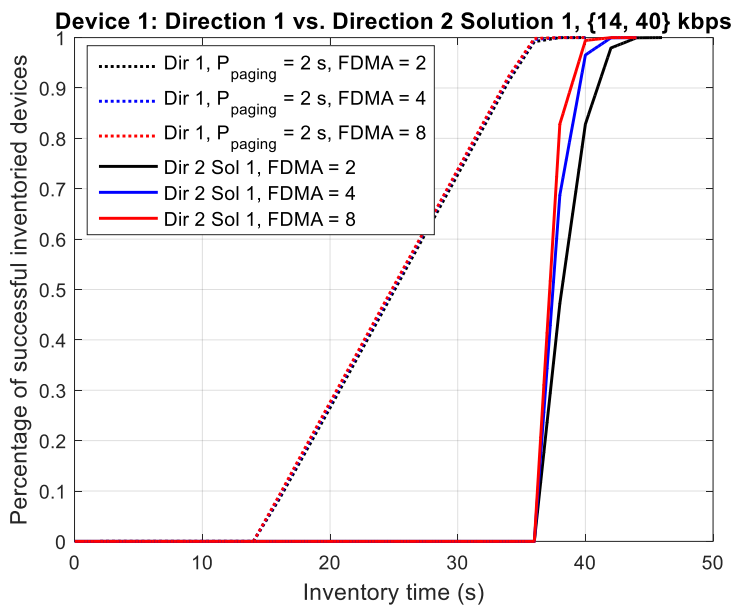


Figure 3.3-5: Comparison of percentage of successful inventoried devices with respect to inventory time for Direction 2 Device 1 Solution 1 versus Direction 1 Device 1 in case of {R2D, D2R} data rate of {14, 440} kbps.

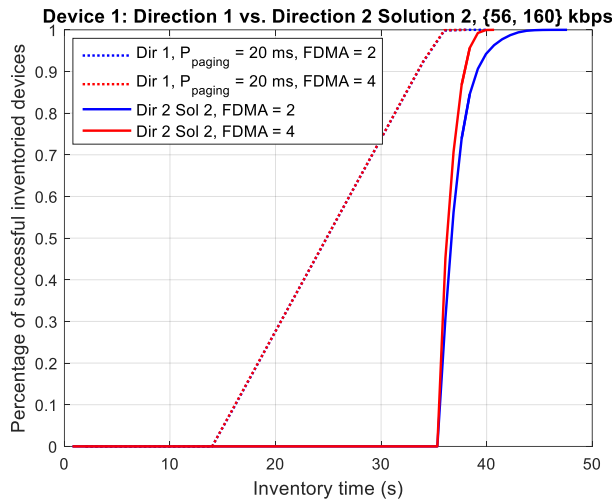


Figure 3.3-6: Comparison of percentage of successful inventoried devices with respect to inventory time for Direction 2 Device 1 Solution 2 versus Direction 1 Device 1 in case of {R2D, D2R} data rate of {56, 160} kbps.

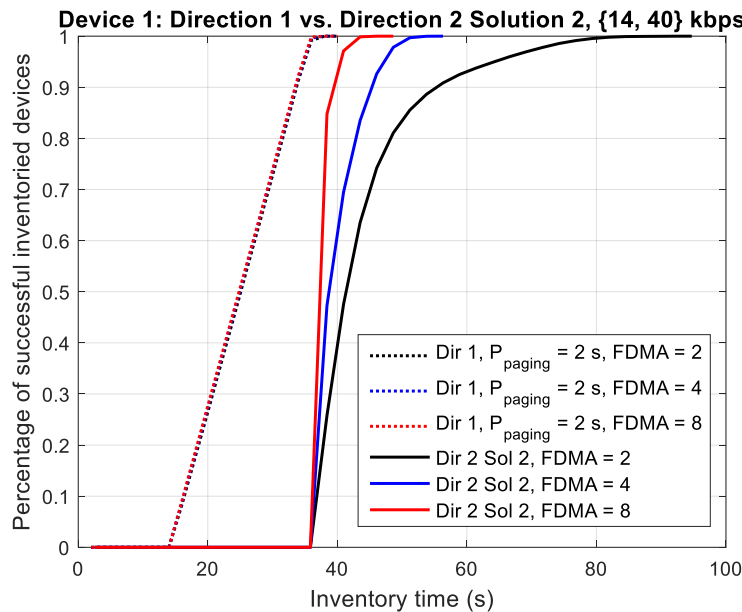


Figure 3.3-7: Comparison of percentage of successful inventoried devices with respect to inventory time for Direction 2 Device 1 Solution 2 versus Direction 1 Device 1 in case of {R2D, D2R} data rate of {14, 440} kbps.

Observation 7: For Device 1, the inventory completion time for Direction 1 is shorter than Direction 2 for all evaluation cases.

Observation 8: At least for Device 1 implemented with large capacitor size (e.g., 10 uF), it seems not necessary to study Direction 2.

Results for Direction 2 Device 2

B1. Results for Solution 1 (Solution A + Solution B1)

Figure 3.3-8 and Figure 3.3-9 show the evaluation results of percentage of successfully inventoried devices with respect to the inventory time for Solution 1, where Figure 3.3-8 provides results assuming {R2D, D2R} data rate of {56, 160} kbps and Figure 3.3-9 provides results assuming {R2D, D2R} data rate of {14, 40} kbps. The inventory completion time considering 99% and 100% of devices successfully complete inventory are summarized in Table 3.3-1. In addition, we also provide comparison between Direction 1 and Direction 2 Solution 1, as shown in Figure 3.3-10 and Figure 3.3-11.

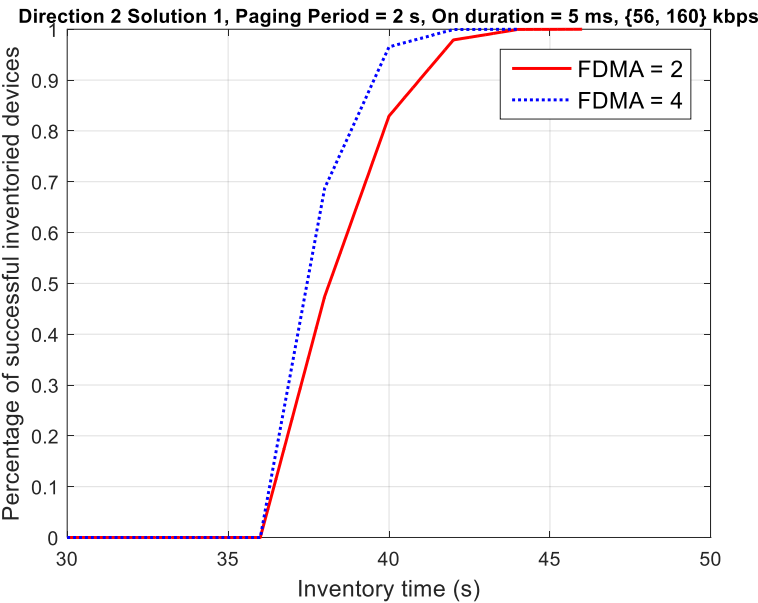


Figure 3.3-8: Percentage of successful inventoried devices with respect to inventory time for Direction 2 Device 2 Solution 1 in case of {R2D, D2R} data rate of {56, 160} kbps.

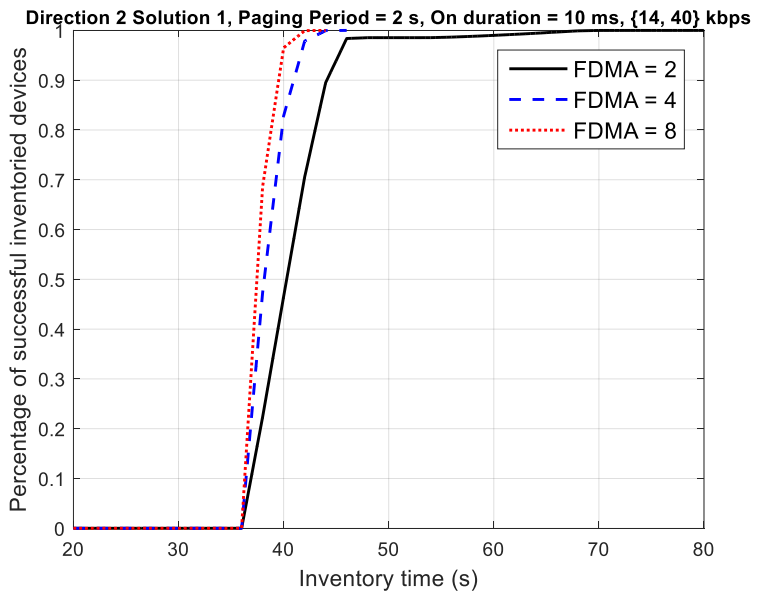


Figure 3.3-9: Percentage of successful inventoried devices with respect to inventory time for Direction 2 Device 2 Solution 1 in case of {R2D, D2R} data rate of {14, 40} kbps.

Table 3.3-1: Inventory completion time for Direction 2 Solution 1

Successful rate	{R2D, D2R} Data rate	FDMA resources	Inventory completion time
99%	{56,160} kbps	2	43.08 s
		4	41.44 s
	{14,40} kbps	2	59.97 s
		4	43.09 s
		8	41.46 s

100%	{56,160} kbps	2	43.96 s
		4	42.09 s
	{14,40} kbps	2	67.11 s
		4	43.69 s
		8	42.10 s

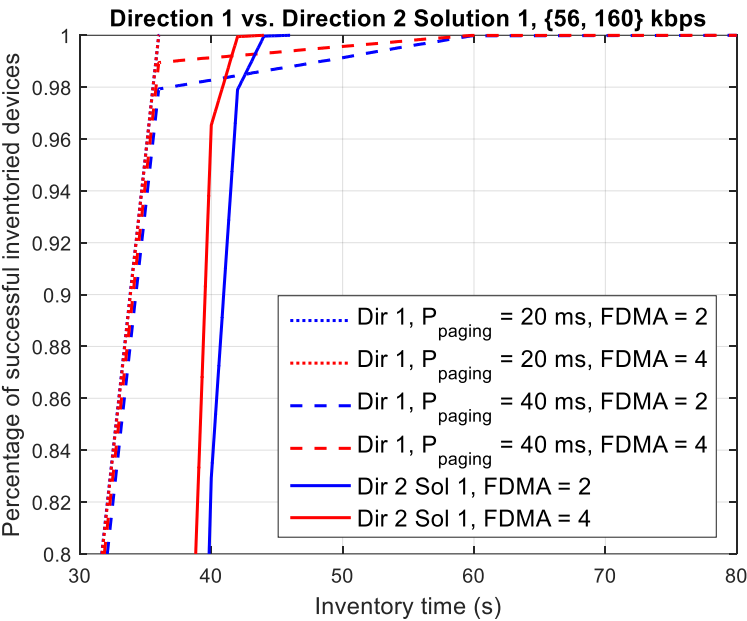


Figure 3.3-10: Comparison of percentage of successful inventoried devices with respect to inventory time for Direction 2 Device 2 Solution 1 versus Direction 1 Device 2 in case of {R2D, D2R} data rate of {56, 160} kbps.

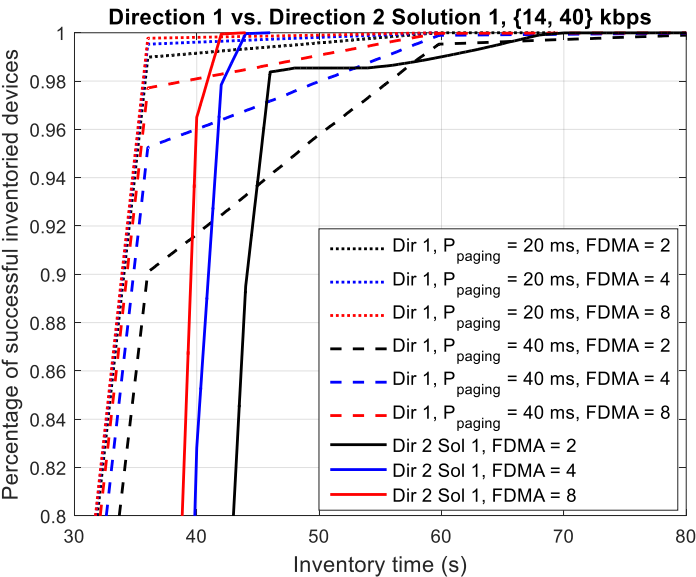


Figure 3.3-11: Comparison of percentage of successful inventoried devices with respect to inventory time for Direction 2 Device 2 Solution 1 versus Direction 1 Device 2 in case of {R2D, D2R} data rate of {14, 40} kbps.

B2. Results for Solution 2 (Solution A + Solution B2)

Figure 3.3-12 and Figure 3.3-13 show the evaluation results of percentage of successfully inventoried devices with respect to the inventory time for Solution 2, where Figure 3.3-12 provides results assuming {R2D, D2R} data rate of {56, 160} kbps and Figure 3.3-13 provides

results assuming {R2D, D2R} data rate of {14, 40} kbps. The inventory completion time considering 99% and 100% of devices successfully complete inventory are summarized in Table 3.3-2. In addition, we also provide comparison between Direction 1 and Direction 2 Solution 2, as shown in Figure 3.3-14 and Figure 3.3-15.

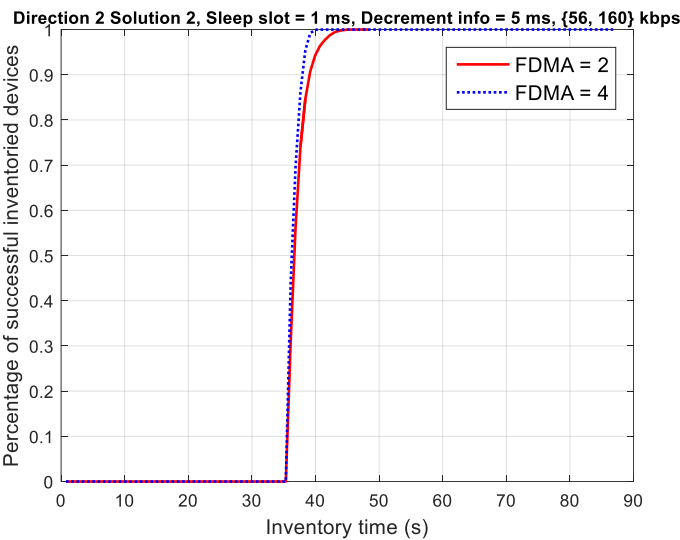


Figure 3.3-12: Percentage of successful inventoried devices with respect to inventory time for Direction 2 Device 2 Solution 2 in case of {R2D, D2R} data rate of {56, 160} kbps.

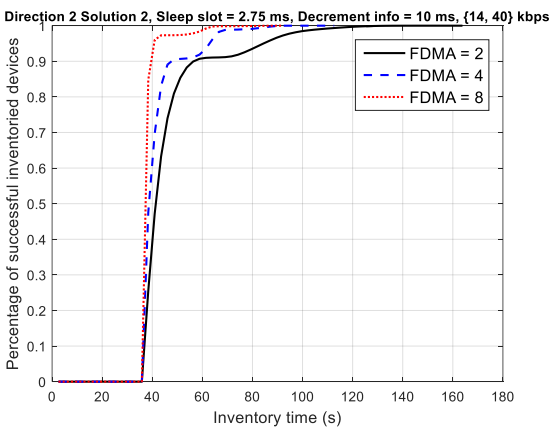


Figure 3.3-13: Percentage of successful inventoried devices with respect to inventory time for Direction 2 Device 2 Solution 2 in case of {R2D, D2R} data rate of {14, 40} kbps.

Table 3.3-2: Inventory completion time for Direction 2 Solution 2

Successful rate	{R2D, D2R} Data rate	FDMA resources	Inventory completion time
99%	{56,160} kbps	2	43.68 s
		4	39.84 s
	{14,40} kbps	2	107.68 s
		4	76.96 s
		8	61.60 s
100%	{56,160} kbps	2	44.06 s
		4	41.05 s

	{14,40} kbps	2	107.68 s
		4	87.73 s
		8	71.87 s

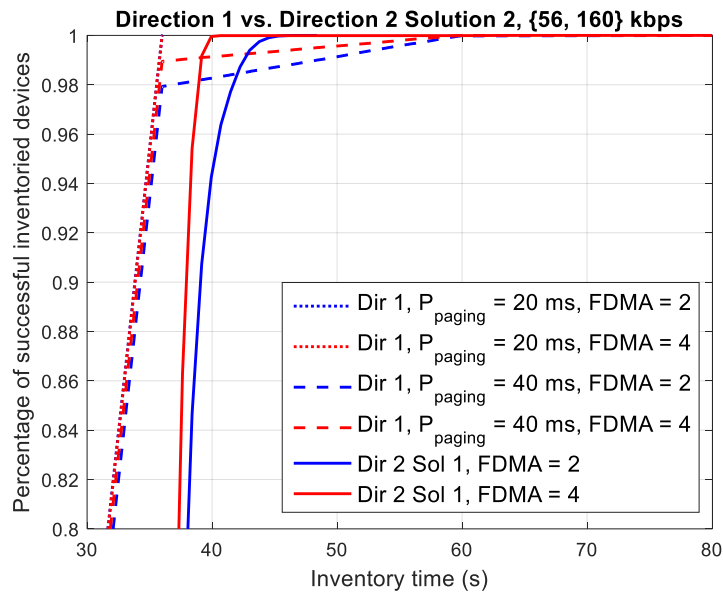


Figure 3.3-14: Comparison of percentage of successful inventoried devices with respect to inventory time for Direction 2 Device 2 Solution 2 versus Direction 1 Device 2 in case of {R2D, D2R} data rate of {56, 160} kbps.

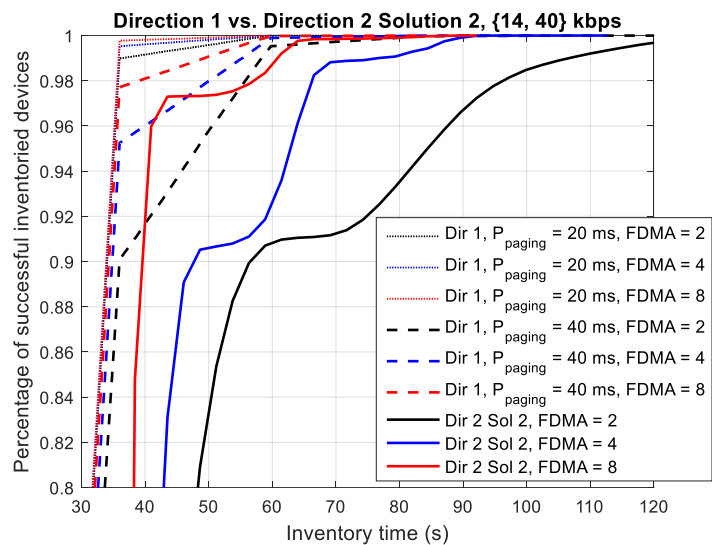


Figure 3.3-15: Comparison of percentage of successful inventoried devices with respect to inventory time for Direction 2 Device 2 Solution 2 versus Direction 1 Device 2 in case of {R2D, D2R} data rate of {14, 40} kbps.

- Observation 9: For Direction 2, and for both Solution 1 and Solution 2, when considering higher {R2D, D2R} data rate of {56, 160} kbps, at least 99% of devices can be successfully inventoried before running out of energy and no device goes through recharging.
- Observation 10: For Direction 2, and for both Solution 1 and Solution 2, when considering lower {R2D, D2R} data rate of {14, 40} kbps, there are devices cannot complete inventory before running out of energy and therefore the inventory completion time sufficiently increases.
- Observation 11: For Direction 2, and for both Solution 1 and Solution 2, when compared to Direction 1 assuming paging time interval is comparable to half of the device sustainable time:

- In case of higher {R2D, D2R} data rate of {56, 160} kbps, Direction 2 consumes longer inventory completion time when at least 99% of device successfully complete inventory, since the first paging message is sent at the time when the farthest device finish energy harvesting in Direction 2.
- In case of lower {R2D, D2R} data rate of {14, 40} kbps, Direction 2 consumes longer inventory completion time when at least 99% of device successfully complete inventory, since there are devices goes through recharging for Direction 2 but there is no device goes through recharging for Direction 1.

Observation 12: For Direction 2 Solution 1, when compared to Direction 1 assuming paging time interval is comparable to the device sustainable time, Solution 1 has benefits in the following cases:

- Considering 99% devices successfully complete inventory, in case of higher {R2D, D2R} data rate of {56, 160} kbps and 2 FDMA resources; in case of lower {R2D, D2R} data rate of {14, 40} kbps and 4/8 FDMA resources.
- Considering 100% devices successfully complete inventory, in case of higher {R2D, D2R} data rate of {56, 160} kbps and 2/4 FDMA resources; in case of lower {R2D, D2R} data rate of {14, 40} kbps and 2/4/8 FDMA resources.

Observation 13: For Direction 2 Solution 2, when compared to Direction 1 assuming paging time interval is comparable to the device sustainable time, Solution 2 has benefits in the following cases:

- Considering 99% devices successfully complete inventory, in case of higher {R2D, D2R} data rate of {56, 160} kbps and 2/4 FDMA resources.
- Considering 100% devices successfully complete inventory, in case of higher {R2D, D2R} data rate of {56, 160} kbps and 2/4 FDMA resources.

Specification impacts, if any

For Direction 2, potential specification impact is expected at both reader side and device side.

From reader perspective,

- Before the inventory starts, the reader provides indication of the potential starting time or remaining time of the future paging message.
- During the inventory procedure, for Solution 1, the reader provides duty cycle information at least including number and duration of duty cycle ON occasions.
- During the inventory procedure, for Solution 2, the reader provides slot decrement indication to inform the slot decrement progress.

From device perspective,

- Before the inventory starts, the available device monitors the starting time indication of the paging message and can transits to SLEEP state, if necessary.
- During the inventory procedure, for Solution 1, the device randomly selects a duty cycle ON occasion for random access. During SLEEP state, the device retains memory regarding its ON occasion information and runs a clock for time counting.

During the inventory procedure, for Solution 2, the device randomly selects an access slot and can transits to SLEEP state, if necessary. During SLEEP state, the device retains memory regarding its access slot information and runs a clock for time counting with an assuming slot duration. The device transits to ON state when it counts to the selected slot, monitors the decrement indication to obtain the actual slot decrement progress and determines whether it can re-enter SLEEP state or keeps monitoring to its upcoming access slot.

Table 6.2.2-8: Details from Source [80]

Source	Details
Source [80]	Solution description The device can switch to "SLEEP" state to recharge the power and wake up before the selected access occasion starting time. If a device has selected a later access occasion based on the indicated "Q" value. Observations or Analysis or Evaluations The SLEEP state may need to be defined in which the device can maintain a timer and the memory content from ON state; The device's SFO may need to be taken into account. Specification impacts, if any Designing a timer to control the time length of "SLEEP" state and/or additional information carried in the R2D command like "QueryRep" [RAN2].

Table 6.2.2-9: Details from Source [81]

Source	Details
Source [81]	Solution description In Direction 2, A-IoT devices perform transmission and reception for communication in available state. The devices would become unavailable during charging or after running out the stored energy. While for A-IoT devices in unavailable state, devices may perform energy harvesting or just be in a fully charged state without the requirement of transmission and reception for communication. The unavailable devices do not become available to perform transmission and reception for communication until receiving the status switching information provided by the reader. Direction 2 has more specification impacts, e.g. specify the content and the form of the information. In addition, carrier wave should not always exist to provide energy for A-IoT devices since the devices may be fully charged in unavailable state. Observations or Analysis or Evaluations Observation 5: In Direction 2, carrier wave should not always exist on to provide energy for A-IoT devices. Observation 6: In Direction 2, A-IoT devices are unavailable when they are in a fully charged state without the requirement of transmission and reception for communication. Specification impacts, if any In Direction 2, at least the content and the form of the information used for switching A-IoT devices to become available needs to be specified.

Table 6.2.2-10: Details from Source [83]

Source	Details
Source [83]	Solution description In Direction 2, since a reader provides assistance or coordination information to assist a device to determine whether/when to become available or unavailable, it is assumed the device should support 3 operating states (ON, OFF, SLEEP), such that the device can switch from SLEEP-to-ON based on an internal clock (time counting). Switching from OFF-to-ON is, in general, based on device implementation (e.g., whenever the energy storage level has reached X % full, where X=70/80/100). To switch between ON/SLEEP for energy harvesting (e.g., based on time counting via an internal clock), the main intention is to prolong/extend the sustainable operating time of a device such that the device has sufficient energy to complete at least a round of inventory or command process. Energy harvesting for a device should focus on two stages: before and after an inventory or command process is triggered. Before an inventory or command process is triggered: - A device power-ON once it is sufficiently charged and receives information from the reader relating to the timing/periodicity of future inventory/command processes to be triggered. - Device selects one of the future periodic inventory / command processes, switches to SLEEP state for power saving and energy harvesting and performs time counting

	using an internal clock. To minimize the congestion / collisions of every device trying to wake-up at the same time and attempt to access the same inventor process together, it is beneficial for the reader to indicate multiple future processes and a device selects one of the future processes (e.g., randomly), so that all devices are spread out across the multiple processes. - Before start of the selected inventory / command process, device switches to ON state to receive paging message from the reader. After an inventory or command process is triggered: - After receiving the paging message that triggers an inventory or command process and selecting an access slot (q) as part of slotted-ALOHA, the device switches to SLEEP state for power saving and energy harvesting/re-charging. - The device counts the time (via internal clock) or detects number of paging repeat messages or timing acquisition signals / R2D preambles, and switches to ON state before the start of the selected access slot (q) to carry out a random-access procedure with the reader. Observations or Analysis or Evaluations Based on the solution descriptions, the goal / intention is for a device to be in the ON state only when it is necessary, i.e., for the selected process (before the inventory / command process is triggered) and the selected access slot (after the inventory / command is triggered). In other time, the device switches / remains in the SLEEP state for energy harvesting. Hence, the energy storage level in the device remains in a high level and the sustainable operating time is maximized to ensure the device is able to complete an inventory/command process with the reader. Specification impacts, if any To ensure the energy storage level in a device is maximized for prolonging the sustainable operating time and to ensure a smooth operation of running an inventory or command process, a unified state transition behaviour(s) based on reader signaling or assistance/coordination information should be specified.
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Table 6.2.2-11: Details from Source [84]

Source	Details
Source [84]	Solution description Solution 1: Reader scheme: Transmits indications multi times to notify the next access occasion(s). Device scheme: Random On to receive the indication and select the access occasion by device. Observations or Analysis or Evaluations Solution 1: Reader does not control the device state Specification impacts, if any Solution 1: minor specification impact.

Table 6.2.2-12: Details from Source [84]

Source	Details
Source [84]	Solution description Solution 2: Reader scheme: Periodically transmits PRDCH. Device scheme: Periodically On to access according to the reader configuration. Observations or Analysis or Evaluations Solution 2: Reader controls the device state Specification impacts, if any Solution 2: has specification impact for both reader and device.

Table 6.2.2-13: Details from Source [85]

Source	Details
Source [85]	Solution description Each device can be associated with a number of "unavailable time windows", within which the device is not available for communications e.g. due to charging by energy harvesting. The reader can implicitly or explicitly indicate the "unavailable time windows" (or how to determine them) to a device. For example, if a device successfully receives an R2D transmission carrying A-IoT paging message intended for a specific device group NOT including the device, it can be a good indication that the device will not miss anything even if it stops transmissions/receptions for a period of time right after the the R2D transmission, and that period of time can be a "unavailable time window" for the device. Precision in definition of a "unavailable time window" can take the SFO of a device into account. Observations or Analysis or Evaluations "Unavailable time windows" for A-IoT devices can be exploited for energy harvesting, minimizing the negative impacts to communication delays due to charging by energy harvesting. Specification impacts, if any The reader needs to indicate "unavailable time windows" (or how to determine them) to a device, implicitly or explicitly.

Table 6.2.2-14: Details from Source [86]

Source	Details
Source [86]	Solution description The device operates on a duty-cycled basis for channel monitoring and reception. The reader indicates the times at which the device should be ON and when the device may sleep. The reader can only communicate with the device when the device is ON. The device indicates to the reader the maximum ratio of ON to OFF time, based on the power consumption and energy harvesting rate of the device. Observations or Analysis or Evaluations The range from CWE to device is not limited by the power consumption of the device. The range is limited by the threshold at which energy can be harvested. Devices that are further from the CWE operate with a longer duty cycle. Devices that are close to the CWE can operate in "direction 1 mode". CWE power consumption and interference are less than for direction 1. The reader needs to signal the duty cycle timing to the device. The device needs to signal its supported duty cycles to the reader, based on current energy harvesting conditions. Specification impacts, if any Procedures for duty cycle operation need to be defined. Reader needs to be signal the ON / OFF duty cycle to the device. The device needs to signal its desired ON / OFF duty cycle to the reader.

Table 6.2.2-15: Details from Source [87]

Source	Details
Source [87]	Solution description The reader can group the devices in the same "available duration" for scheduling. Reader indicates to the devices the time for activity cycle allowing the devices to monitor R2D transmissions during the same available duration. To adjust timing drift of devices due to clock inaccuracy, the reader transmits a "Sync Align" outside the activity cycle to help devices to align with the same available duration. Optionally, the Reader can indicate a minimum applicable duty cycle (ratio of available to total time) for the device in the Sync Align message. If a device can determine its duty cycle, it follows the indication if its duty cycle is higher than the indicated minimum. This enables grouping devices by their duty cycle. Observations or Analysis or Evaluations By having a common "available duration" among different devices, the reader can schedule multiple devices in the same duration. Broadcast/groupcast message can be then transmitted

	during the common available duration. The solution is suitable for all device types and can achieve better coexistence between device type 1 and type 2a/2b. Specification impacts, if any Introducing R2D broadcast message to indicate information related to activity cycle.
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Table 6.2.2-16: Details from Source [88]

Source	Details
Source [88]	Solution description Define a duty-cycle based paging procedure where each cycle include 'monitoring period' and 'nonActive period', where reader transmits paging command within 'monitoring' period to trigger inventory procedure. An A-IOT device maintains a running clock to count the timing location of 'monitoring period' and 'nonActive period'. An A-IOT Device moves to 'ON' state if it wants to involve a paging procedure; Otherwise, it moves to 'Sleep' state for RF energy harvesting. This is one solution in Direction 2. Observations or Analysis or Evaluations N/A Specification impacts, if any Define a duty-cycle based paging procedure where each cycle include 'monitoring period' and 'nonActive period', where reader transmits paging command within 'monitoring' period to trigger inventory procedure. Introduce a signal to indicate the duty-cycle configuration.

Table 6.2.2-17: Details from Source [89]

Source	Details
Source [89]	Solution description When CW and communication is same frequency, - The start-indicator part can be used to switch CW and communication. There is no need to indicate it in advance. When CW and communication are different frequencies, - During one transaction (inventory or command) or multiple of transactions, the reader would inform or specify "when Msg 0 and/or 2 and/or 4" are to be sent. This situation is "reader provide the information to a device on which the device become available. Observations or Analysis or Evaluations Without such some level of (rough) aligned timing operation, reader and device are not able to communicate well. Specification impacts, if any When CW and communication is different frequency,the timer value related to "when Msg 0 and/or 2 and/or 4" are to be sent.

Table 6.2.2-18: Details from Source [90]

Source	Details
Source [90]	Solution description R2D signaling indicating a device to go dormancy and the allowed time duration. R2D wake-up signal to instruct a device to transition into an active state and monitor a potential R2D transmission.

	Direct indication of the one or more timers described under Direction 1. Observations or Analysis or Evaluations N/A Specification impacts, if any N/A
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Table 6.2.2-19: Details from Source [114]

Source	Details
Source [114]	Solution description Applicable to both Device 1 and Device 2 (higher power devices) Reader provides counter information in R2D transmission for slotted-ALOHA mechanism Includes periodicity information of potential Reader command occasions (or expected device monitoring occasions) Remaining energy estimation and report of the information by device Scheduling commands provide end timing information for R2D reception or D2R transmission (utilization of remaining energy information) Observations or Analysis or Evaluations Necessary for resolving unavailability impact on slotted-ALOHA mechanism. Helps devices align to periodic occasions of potential Reader commands. Allows devices to estimate remaining energy and adjust behavior accordingly. Enables Reader to optimize communication based on device energy information. Examples to improve successful R2D or D2R transmissions, Small payload: Higher priority for devices with low remaining energy Large payload: Higher priority for devices with high remaining energy Addresses challenges when devices are scattered in a wide region with unaligned availability. Specification impacts, if any Define format for counter information in R2D transmission. Specify periodicity information format for potential Reader command occasions. Define end timing information format in scheduling commands. Explicit: Indication of transmission time/length Implicit: Derived from TBS and data rate indicated Establish protocol for devices to report remaining energy information to Reader. Design of initial R2D periodicity to ensure efficient initial contact with unaligned devices.

Table 6.2.2-20: Details from Source [113]

Source	Details
Source [113]	Solution description Reader provides information on when paging/re-paging are potentially transmitted, and device wakes up based on the information. Observations or Analysis or Evaluations This direction would intend to control device availability/unavailability by reader/NW side. When reader transmits paging/re-paging can be announced to devices, and each device can wake up at the announced timing. In this case, unlikely to direction 1, device can detect paging/re-paging a lot of times or device can sleep until re-paging to which the device can respond. Necessary rule between reader and devices will be what kind of information is provided and how/when, and detailed conditions to wake-up in consideration of energy storage status and the indicated information. In addition, how to achieve accurate synchronization is

	also important to make the information announcement meaningful. In our view, temporarily periodic R2D sync signal is a straightforward solution for this sync issue. Fig: A possible way of Direction 2. Specification impacts, if any E.g., what kind of information is provided and how/when. E.g., detailed conditions to wake-up in consideration of energy storage status and the indicated information. E.g., device monitors temporarily periodic R2D sync signals.
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Table 6.2.2-21: Details from Source [114]

Source	Details
Source [114]	Solution description Device duty-cycle monitoring (DCM) Observations or Analysis or Evaluations We believe a feasible solution for all A-IoT devices is <u>device duty-cycle monitoring (DCM)</u>. The essence of DCM is quite simple: <u>A device shortens/controls the available time duration</u>. Outside the available time duration, the device is unavailable, and harvests energy from RF without communication activities. A benefit of device DCM is that the device unavailable time duration becomes shorter. Figure B1 illustrates an image of how inventory procedure with DCM looks like. Similar to the case of direction 1, different devices may become available at different timings due to different energies. However, by shortening/controlling device available duration, DCM enables much shorter available/unavailable cycles for a device. In case of inventory, devices can receive msg-0 from the reader at very early stage of the inventory procedure that the reader tries. Fig. B1 Duty-cycle monitoring (DCM) Even after a device joins the inventory procedure running by a reader, the DCM can still work for longer sustainable operation. For example, in an inventory procedure that has multiple contention-access

procedures, if a device has received msg-0 and transmitted msg-1 but failed to receive the corresponding msg-2, the device can consider the current contention-based access has failed and expect another msg-0 reception in future for the retry. If the device knows the reader's msg-0 periodicity/frequency, or the expected available/unavailable cycle for the device, the device can behave such that it can make use of DCM to save the energy consumption as an example being illustrated in Fig. B2.

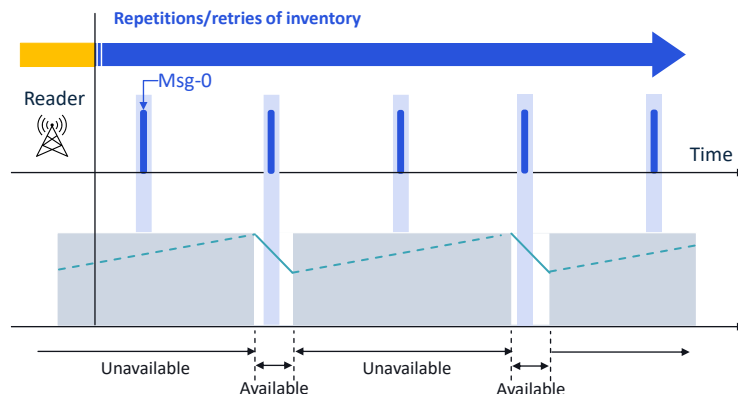


Fig. B2 DCM example after a device joins inventory procedure from a reader

DCM can be realized by a device by counting the time after it becomes available and switches the device state from available state (ON state) to an unavailable state. As for the unavailable state, two options can be considered. The first option is to define a SLEEP state. During the SLEEP state, the device may retain a certain memory and run a low-power sleep clock (e.g., 30kHz) while harvesting energy from RF. With the SLEEP state, the device is able to identify when it should be available (switch to ON state) again.

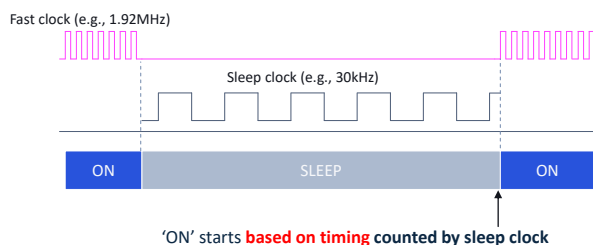


Fig. B3 Switch between ON state and SLEEP state

The other option to realize DCM is to use OFF state as the unavailable state. In the OFF state, the device may not retain a memory and may not run a clock. Therefore, the device is not able to control the unavailable time duration. The unavailable time duration is determined by the time to charge the energy storage until the 'Turn-on' threshold, which depends on Rx power, RF-EH efficiency, and the amount of energy consumed during available time duration.

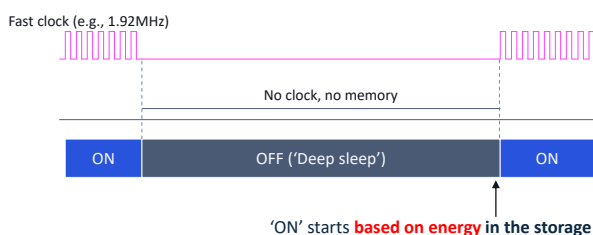


Fig. B4 Switch between ON state and OFF state

Observation B1:

- Direction 2 is the case where a reader has some ideas on when/how a device is available/unavailable, and possibly is able to control the device availability/unavailability.
- As a solution, device duty-cycle monitoring (DCM) is promising.
 - For DCM, a device shortens/controls the available time duration, and for outside the available time duration, the device is unavailable and harvests energy from RF without communication activities.
 - ✧ The device counts the time after it becomes available and switches the device state from available state (ON state) to an unavailable state.
 - ✧ As for unavailable state, two options can be considered:
 - Opt.1: SLEEP state
 - In the SLEEP state, the device may retain a certain memory and run a low-power sleep clock (e.g., 30kHz) while harvesting energy from RF
 - In the SLEEP state, the device is able to identify when it should be available (switch to ON state) again
 - Opt.2: OFF state
 - In the OFF state, the device may not retain a memory and may not run a clock while harvesting energy from RF
 - In the OFF state, the device is not able to identify when it should be available (switch to ON state) again
 - The device is switched from OFF state to ON state once the amount of energy in the storage reaches the 'Turn-on' threshold
 - ✧ An appropriate DCM operation enables efficient energy usage of a device and reduced unavailable time duration for charging

Let us consider the following readers transmission behavior: (1) before starting inventory, the reader transmits RF for energy harvesting as the pre-inventory charging phase; (2) then the reader starts inventory which contains multiple contention-based access procedures, each of which is triggered by a msg-0.



Fig. B5 Inventory model

In each round, reader transmits a msg-0 triggering of up to M msg-1 transmissions from devices. The contention-based access procedure is assumed to contain following steps:

- Msg-0: Reader transmits a msg-0 that triggers msg-1 transmissions from devices
 - Msg-0 indicates a set of msg-1 resources
- Msg-1: A-IoT device that has received a msg-0 selects one of the msg-1 resources and transmit msg-1
 - Multiple A-IoT devices may select a same resource, in which case collision occurs
- Msg-2: Reader transmits a msg-2 as a response to each msg-1
 - In case of msg-1 collision, msg-2 reception fails; then the devices does not proceed this round
- Msg-3: A-IoT device who has received the msg-2 transmits a msg-3 as a response to the msg-2
- Msg-4: Reader transmits a msg-4 as a response to the devices that were successfully identified

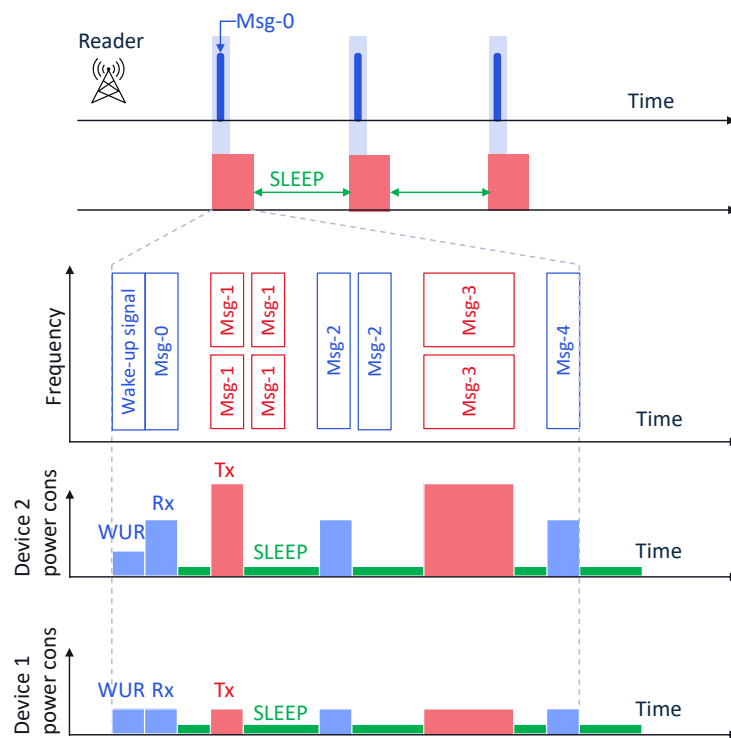


Fig. B6 The model of random-access procedure for each msg-0

Other assumptions/parameters are listed in the table below. For simplicity, we assume following:

- All devices are assumed to have same Rx power, -28dBm or -32dBm, same RF-EH efficiency
- Separate evaluations for device 1 and device 2 (no mixed scenario)

The baseline is no DCM. For DCM, we evaluated two cases.

- Case 1: SLEEP-based msg-0 monitoring before the device joins inventory
 - Device switches ON state and SLEEP state until it detects msg-0 first time
 - Device uses SLEEP state during contention-based access procedures after it detects the first msg-0 whenever possible
 - This enables controlled duty-cycle monitoring all the time
 - However, SLEEP state consumes a certain amount of energy for sleep clock/memory, leading to long unavailable time duration when device cannot sustain SLEEP state due to low Rx power
- Case 2: OFF-based msg-0 monitoring before the device joins inventory
 - Device switches ON state and OFF state until it detects msg-0 first time
 - Device uses SLEEP state during contention-based access procedures after it detects the first msg-0 whenever possible
 - This enables controlled duty-cycle monitoring after the first msg-0 detected
 - Until a device detects msg-0 first time, there is no/negligible power consumption during OFF state

Note that in both cases, if a device consumes all the available energy during ON or SLEEP state, the device is in OFF state until it is fully charged.

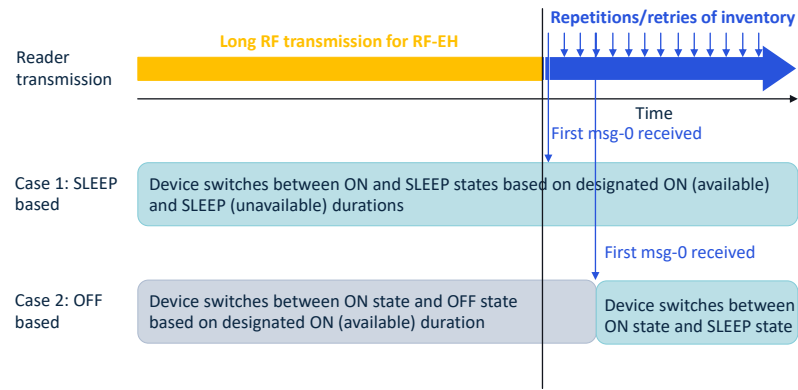


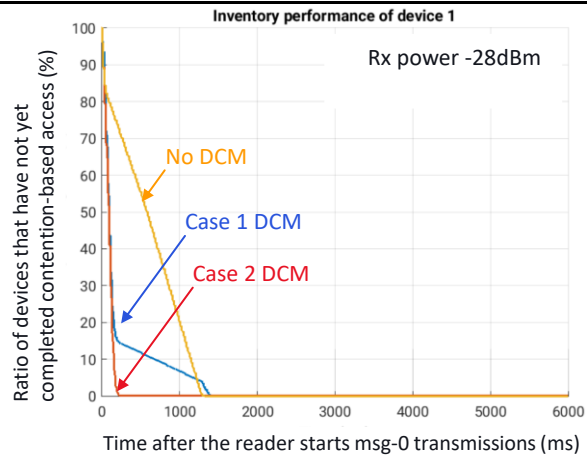
Fig. B7 Case 1 and Case 2 of DCM

Table B1 Evaluation parameters/assumptions

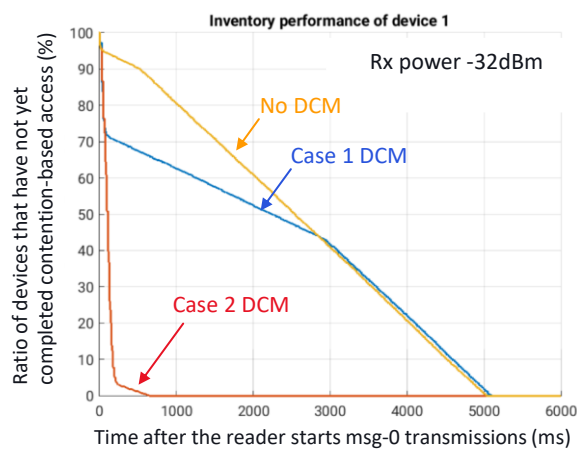
Parameters [⚡]			Device 1 [⚡]	Device 2 [⚡]
Number of A-IoT devices [⚡]			20 (all assumed to have same Rx power) [⚡]	
Rx power [⚡]			-28dBm, -32dBm [⚡]	
RF energy harvesting [⚡]			0.2uW/sec @ Rx power -28dBm [⚡] 0.05uW/sec @ Rx power -32dBm [⚡]	
Available energy in storage [⚡]			0.25uJ (1uF capacitor) [⚡]	2.5uJ (10uF capacitor) [⚡]
Power consumption [⚡]	ON state [⚡]	Msg-0 monitor [⚡]	1uW [⚡]	
		Rx [⚡]	1uW [⚡]	50uW [⚡]
		Tx [⚡]	1uW [⚡]	100uW [⚡]
	SLEEP state [⚡]		0.1uW [⚡]	
	OFF state [⚡]		Negligible [⚡]	
Inventory msg duration [⚡]		Msg-0 [⚡]	1ms [⚡]	
		Msg-1 [⚡]	0.5ms [⚡]	
		Msg-2 [⚡]	0.5ms [⚡]	
		Msg-3 [⚡]	3ms [⚡]	
		Msg-4 [⚡]	0.5ms [⚡]	
Number of resources for msg-1 [⚡]			8 resources per msg-0 [⚡]	16 resources per msg-0 [⚡]
DCM periodicity before the first msg-0 [⚡]		No DCM [⚡]	N/A [⚡]	
		Case 1 DCM [⚡]	20ms [⚡]	50ms [⚡]
		Case 2 DCM [⚡]	Energy amount dependent [⚡]	
Msg-0 periodicity [⚡]			20ms [⚡]	50ms [⚡]
Device available duration before the first msg-0 [⚡]		No DCM [⚡]	250ms [⚡]	2500ms [⚡]
		Case 1 DCM [⚡]	2ms [⚡]	1ms [⚡]
		Case 2 DCM [⚡]	30ms [⚡]	75ms [⚡]

In Fig. B8, $X = 0$ is the time that the reader starts inventory (the timing that the first msg-0 is transmitted by the reader), and Y-axis indicates the ratio (%) of A-IoT devices that are not yet completed the contention-based access procedure. Fig. 9 shows that in case no DCM, the ratio of non-inventorized devices linearly decreases as the time from the start of the inventory increases. This is because available time of different devices are distributed in time and hence in each msg-0 timing, some devices are available. However, this implies that when the reader starts inventory, there can be devices that has almost zero available energy. Such devices are accessible only after they are charged until 'Turn-on' threshold. This is observed as the bottleneck, i.e., inventory completion time from the reader perspective cannot be better than the charging time for these unavailable devices.

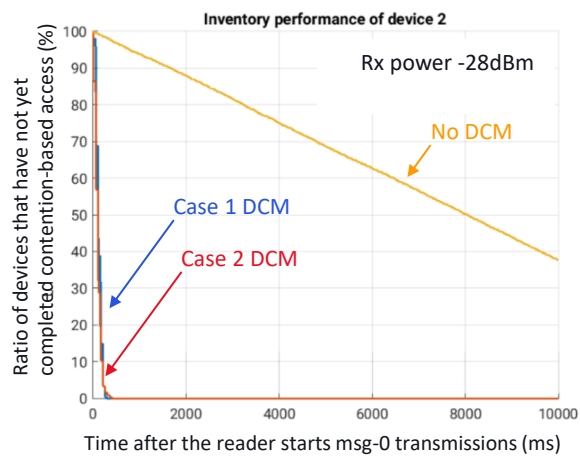
On the other hand, DCM achieves much shorter inventory completion time. This is because the device unavailable time is much shorter than the case of no DCM. Between two cases of DCM, case 1 DCM is sensitive to the Rx power; it can achieve much shorter inventory completion time under relatively high Rx power (e.g., $> -28\text{dBm}$) compared to no DCM, while it requires similar completion time as no DCM with low Rx power (e.g., $< -30\text{dBm}$). This is because SLEEP state is assumed to consume a certain power. The actual gain of Case 1 DCM highly depends on power consumption of SLEEP state. On the other hand, case 2 DCM offers significant performance benefit for all the evaluated cases.



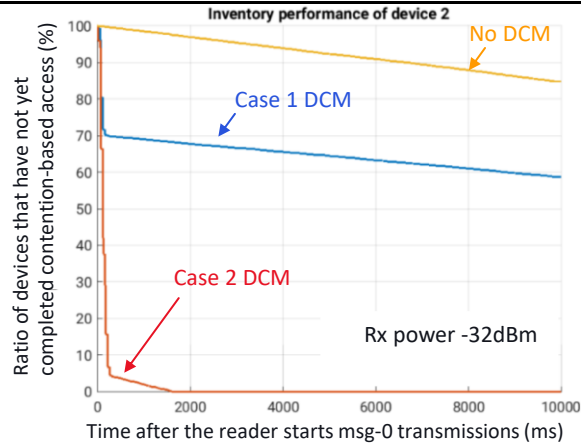
(a) Device 1, Rx power = -28dBm



(b) Device 1, Rx power = -32dBm



(c) Device 2, Rx power = -28dBm



(d) Device 2, Rx power = -32dBm

Fig. B8 Performance evaluation of inventory with and without duty-cycle monitoring

Observation B2:

- Simulation results indicate that duty-cycle monitoring is essential for A-IoT to complete inventory in a reasonable time duration.
 - Case 1 DCM is sensitive to Rx power (or RF EH efficiency) since SLEEP state consumes a certain power; it works if Rx power (or RF EH efficiency) is relatively high.
 - Case 2 DCM works with various range of Rx power (or RF EH efficiency).
 - With no DCM, inventory completion time is limited by device charging time for large amount of energy (from 'Turn-off' threshold to 'Turn-on' threshold)

Further, the analysis of inventory completion time for 600 devices is conducted. Based on the previous results, we focus on 'Case 2 DCM' as the DCM scheme.

- Method 1: Access probability control
 - For each msg-0, there are a number of msg-1 resources. The reader monitors the occupancy of msg-1 resources associated with the i -th msg-0 and indicates the access probability $P(i+1)$ in the $(i+1)$ -th msg-0. The indication of access probability $P(i+1)$ is determined such that the occupancy of msg-1 resources observed by the reader is around 36.8%.
 - A device that receives the $(i+1)$ -th msg-0 select one of the msg-1 resources associated with the msg-0 and transmits the msg-1 with the access probability $P(i+1)$.
 - Note: method 1 is similar to the Q value indication in Query command of UHF RFID in a sense that the probability of msg-1 transmission from a device is controlled by the msg-0 indication.
 - Note: method 1 is applicable to both 'No DCM' and 'Case 2 DCM'.
- Method 2: Device grouping for time-domain distribution of DCM
 - Reader transmits msg-0 with the periodicity P_t from $t = 0$.
 - A device, after the first msg-0 it receives, monitors msg-0 with periodicity $N \times P_t$ based on the reader's instruction, where N is a non-zero integer value. For example, when $N = 2$, a device that receives $(2n+1)$ -th msg-0 from the reader as the first msg-0 monitors $(2n+3)$ -th, $(2n+5)$ -th, ..., msg-0s from the reader using DCM, while another device that receives $(2n+2)$ -th msg-0 from the reader as the first msg-0 monitors $(2n+4)$ -th, $(2n+6)$ -th, ..., msg-0s from the reader using DCM, as illustrated in Fig. B9.
 - Note: method 2 can be realized in direction 2 where the reader can provide information to a device based on which the device may become available/unavailable.
 - Note: method 2 is not applicable to 'No DCM'.

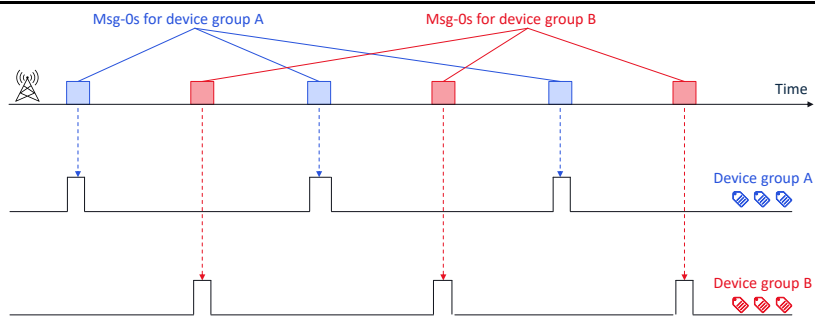


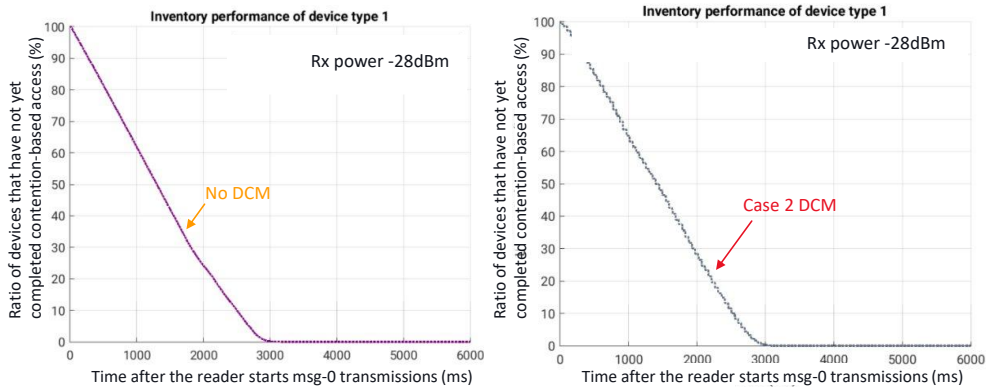
Fig. B9 Device grouping for time-domain distribution of duty-cycle monitoring

To accelerate the inventory for such a large number of devices, msg-0 transmission for the case 'No-DCM' is assumed to be aperiodic, i.e., the reader transmits msg-0 whenever possible and as much as possible. For 'Case 2 DCM', msg-0 periodicity is set to 12ms. In this additional analysis, we added Rx power -36dBm with device energy harvesting efficiency 5% as an additional case. Overall, the assumptions different from Table B1 is summarized in Table B2 below.

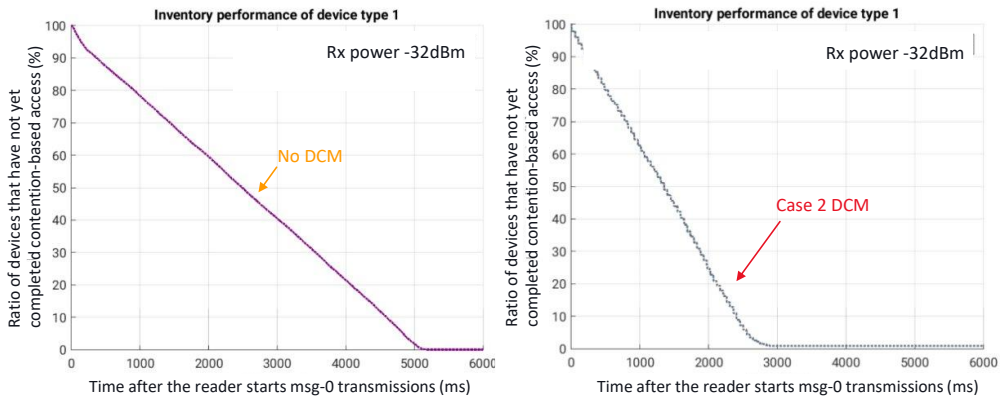
Table B2 Parameters/assumptions of inventory performance analysis

Parameters			Device 1
Number of A-IoT devices			600 (all assumed to have same Rx power)
Rx power			-28dBm, -32dBm, -36dBm
RF energy harvesting			0.2uW/sec @ Rx power - 28dBm 0.05uW/sec @ Rx power - 32dBm 0.025uW/sec @ Rx power - 36dBm
Available energy in storage			0.25uJ (1uF capacitor)
Power consumption	ON state	Msg-0 monitor	1uW
		Rx	1uW
		Tx	1uW
	SLEEP state		0.1uW
	OFF state		Negligible
Inventory msg duration		Msg-0	1ms
		Msg-1	0.5ms
		Msg-2	0.5ms
		Msg-3	3ms
Number of resources for msg-1			8 resources per msg-0
DCM periodicity before the first msg-0		No DCM	N/A
		Case 2 DCM	Energy amount dependent
Msg-0 periodicity		No DCM	Aperiodic (msg-0s as many as possible)

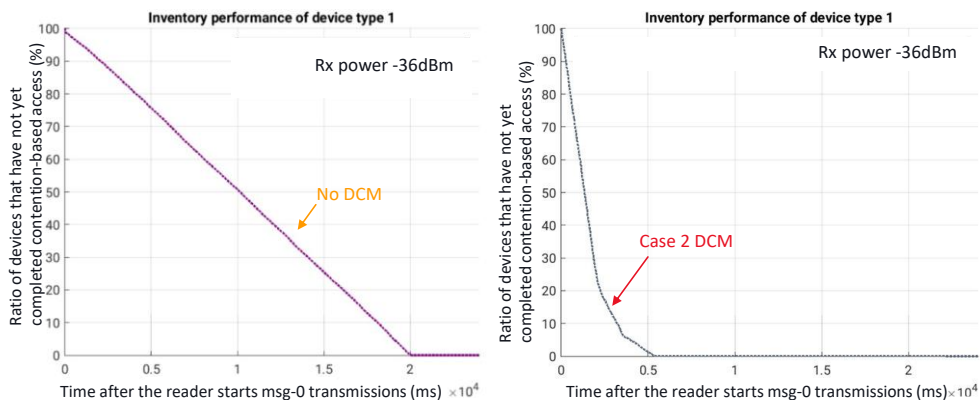
	Case 2 DCM	Periodic (12 ms)
Device available duration before the first msg-0	No DCM	250ms
	Case 2 DCM	30ms



(a) Rx power -28dBm, 'No DCM' and 'Case 2 DCM'



(b) Rx power -32dBm, 'No DCM' and 'Case 2 DCM'



(c) Rx power -36dBm, 'No DCM' and 'Case 2 DCM'

Fig. B10 Inventory completion time for 600 devices

With Rx power = -28dBm, inventory completion time is not much different between 'No DCM' and 'Case 2 DCM'. This is because the charging time from 'turn-off' threshold to 'turn-on' threshold is shorter than the necessary time to complete random access for 600 devices and hence the charging time in 'No DCM' is not visible. With lower Rx power such as -32dBm and -36dBm, the inventory completion time for 'No DCM' significantly increases since some devices need long time to charge from 'turn-off' threshold to 'turn-on' threshold when the reader starts inventory, which takes around 5 seconds with Rx power = -32dBm and 20 seconds with Rx power = -36dBm, respectively. On the other hand, 'Case 2 DCM' guarantees all the

devices have enough energy when the reader starts inventory, regardless of Rx power for each device. It is observed that the inventory completion time of 'Case 2 DCM' with Rx power = -32dBm is almost same as that with Rx power = -28dBm. With Rx power = -36dBm, 'Case 2 DCM' can complete inventory for 600 devices in around 5.5sec, whereas 'No DCM' needs 20sec.

Observation B3:

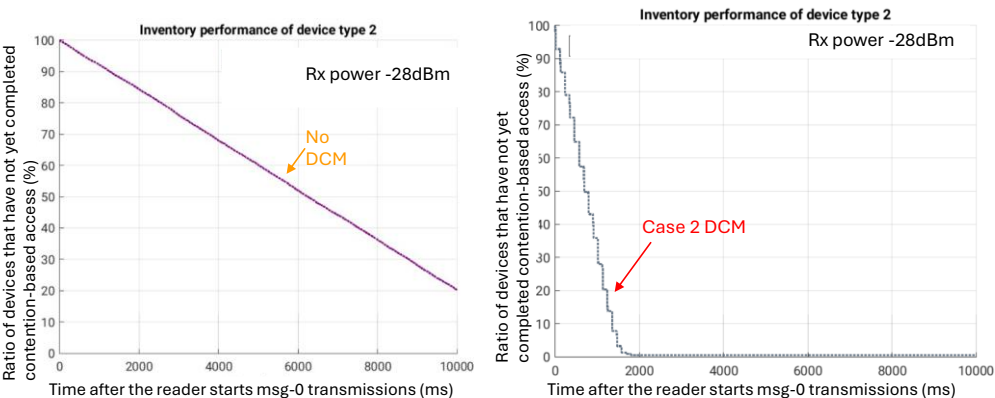
- Simulation results indicate that duty-cycle monitoring is essential for A-IoT to complete inventory in a short time duration even for 600 devices.
 - Case 2 DCM can keep the inventory completion time unchanged between Rx powers -28dBm and -32dBm.
 - Case 2 DCM can complete inventory with around 5.5 sec, whereas 'No DCM' requires around 20 sec.

We have also evaluated inventory completion time of 600 devices assuming device 2 capacitor size and power consumption. The assumptions are captured in Table B3. Except for available energy in storage and power consumption models, most of the parameters are same as those for inventory completion time evaluation for 600 device 1 above.

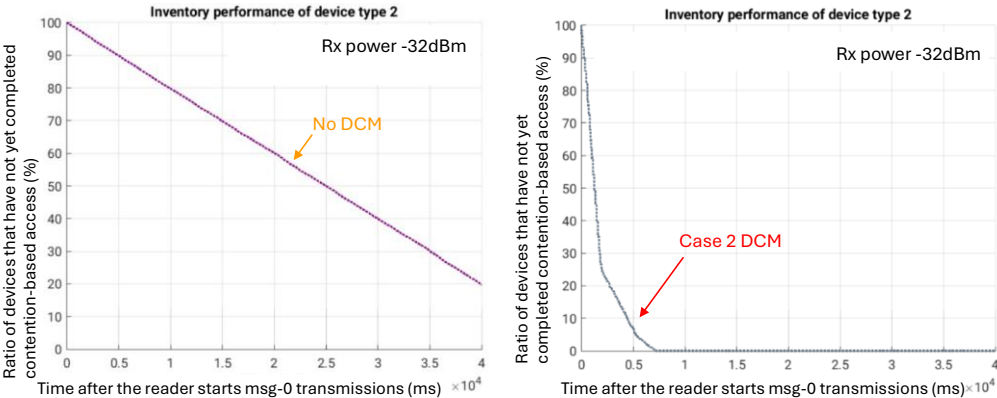
Table B3 Parameters/assumptions of inventory performance analysis

Parameters			Device 2
Number of A-IoT devices			600 (all assumed to have same Rx power)
Rx power			-28dBm, -32dBm, -36dBm
RF energy harvesting			0.2uW/sec @ Rx power - 28dBm 0.05uW/sec @ Rx power - 32dBm 0.025uW/sec @ Rx power - 36dBm
Available energy in storage			2.5uJ (10uF capacitor)
Power consumption	ON state	Msg-0 monitor	1uW (only before the device detects the first msg-0)
		Rx	50uW
		Tx	200uW
	SLEEP state		0.1uW
	OFF state		Negligible
Inventory msg duration		Msg-0	1ms
		Msg-1	1ms
		Msg-2	0.5ms
		Msg-3	3ms
Number of resources for msg-1			16 resources per msg-0
DCM periodicity before the first msg-0		No DCM	N/A
		Case 2 DCM	Energy amount dependent
Msg-0 periodicity		No DCM	Aperiodic (msg-0s as many as possible)
		Case 2 DCM	Periodic (14 ms)
		No DCM	2500ms

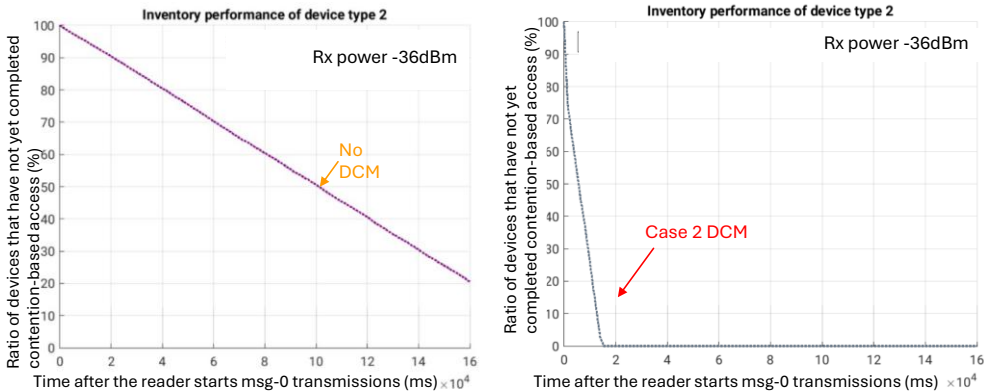
Device available duration before the first msg-0	Case 2 DCM	75ms
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(a) Rx power -28dBm, 'No DCM' and 'Case 2 DCM'



(b) Rx power -32dBm, 'No DCM' and 'Case 2 DCM'



(c) Rx power -36dBm, 'No DCM' and 'Case 2 DCM'

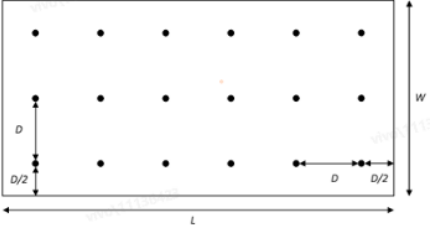
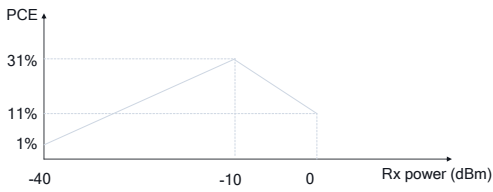
Fig. B11 Inventory completion time for 600 devices (device 2)

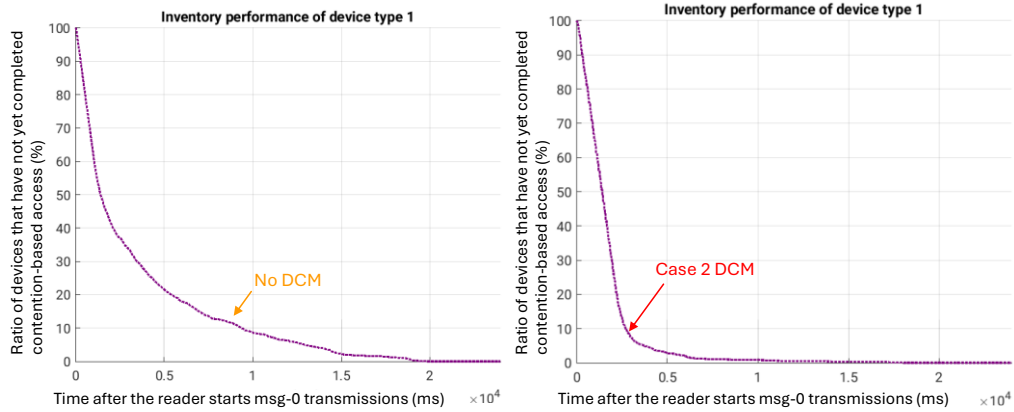
Observation B4:

- Simulation results indicate that duty-cycle monitoring is essential for A-IoT to complete inventory in a short time duration even for 600 devices (device 2).
 - Case 2 DCM can achieve inventory completion for the given device Rx power roughly within 1/10 of the time that is necessary for the case no DCM for the worst case where all devices have Rx power of -36dBm.

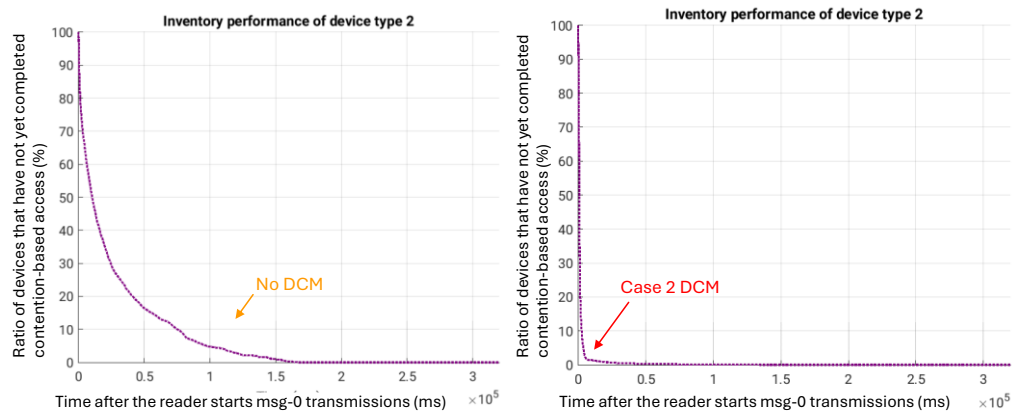
Finally, the results with the device Rx power distribution based on geometry are provided. Device Rx power distribution is assumed to be based on D1T1 as summarized in Table B4. RF energy harvesting efficiency for a given Rx power is modelled as in the table.

Table B4 Assumptions/parameters for additional performance analysis of device duty-cycle monitoring (DCM) for 600 devices with Rx power distribution

Parameter	Value
Scenario	InF-DH
Hall size	120 x 60m
Room height	10m
Sectorization	None
BS deployment	18 BSs on a square lattice with spacing D, located $D/2$ from the walls. $L=120\text{m} \times W=60\text{m}$; $D=20\text{m}$ BS height = 8 m 
Device distribution	Device Height= 1.5 m A-IoT devices drop uniformly distributed over the horizontal area
Device mobility	3kph
Device receiver sensitivity	-36dBm
RF energy harvesting	- Rx power (dBm) + 41 for Rx power (dBm) < -10 dBm - $2 * \text{Rx power (dBm)} + 11$ for Rx power (dBm) $\geq -10\text{dBm}$ 
Other parameters/assumptions are same as in Tables 2.3.2a-1 and 2.3.2b-1	



(a) Device 1



(b) Device 2

Fig. B12 Inventory completion time for 600 devices with geometry-based Rx power distribution

Even with the Rx power distribution among 600 devices, we see a significant gain of DCM over no-DCM. For device 1, inventory completion time for 99% devices is about 18 seconds with "No-DCM", while it is about 8 seconds with "Case 2 DCM". For device 2, inventory completion time for 99% devices is about 150 seconds with "No-DCM", while it is less than 25 seconds with "Case 2 DCM".

Observation B5:

- Simulation results indicate that duty-cycle monitoring is essential for A-IoT to complete inventory in a short time duration even for 600 devices with geometry-based Rx power distribution.
 - For 600 of device 1, case 2 DCM can complete inventory with less than 1/2 of the time that is necessary for the case no DCM.
 - For 600 of device 2, case 2 DCM can complete inventory with less than 1/6 of the time that is necessary for the case no DCM.

Specification impacts, if any

For direction 2, specification impacts can be considered for reader and/or device perspective. For example, if msg-0 transmission periodicity/frequency for contention-access procedure is specified (or is designed as configurable), the device DCM operation can be designed to fit with that. On the other hand, if device DCM related behaviors, such as the duration of ON state or duty-cycle periodicity, are specified (or are designed as configurable), the reader's inventory/communication procedure can be designed to fit with that.

Further, accuracies of device ON state duration and/or SLEEP state duration would need to be understood. If the device clock error is assumed to be quite large, e.g., +/- 10%, for DCM operation, the uncertainty of timings/durations may need to be taken into account in the specifications related to resource/timing.

Observation B4:

- For direction 2, impacts on specifications and designs are envisioned.
 - If msg-0 transmission periodicity/frequency for contention-access procedure is specified (or is designed as configurable), the device DCM operation can be designed to fit with that.

	➤ If device DCM related behaviors, such as the duration of ON state or duty-cycle periodicity, are specified (or are designed as configurable), the reader's inventory/communication procedure can be designed to fit with that. - Accuracies of device ON state duration and/or SLEEP state duration would need to be discussed. ➤ If the device clock error is assumed to be quite large, e.g., +/- 10%, for DCM operation, the uncertainty of timings/durations may need to be taken into account in the specifications related to resource/timing. ➤ SFO correction (clock calibration) framework would offer better clock accuracy such as clock error of < +/-10% for direction 2.
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Table 6.2.2-22: Details from Source [92]

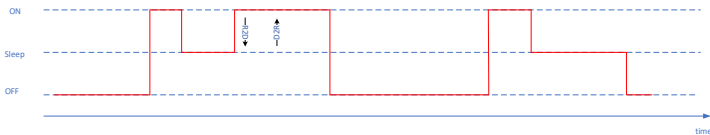
Source	Details
Source [92]	Solution description For direction 2, the reader can control the time of ON and Sleep and the device will switch the states. As shown in Figure 1, when the energy storage is full or reaches a certain level, or the device receives a RF power larger than an activation threshold, the device switches to ON to detect the R2D signal and transmit the D2R signal. By explicit or implicit way, for example, indicated by the reader, the device switches to Sleep state from ON state. After the sleep time, the device switches to ON. Additionally, it is possible that the device switches to OFF state from ON/Sleep state if the energy storage is not enough.  Figure 1 An illustration of device's states
	Observations or Analysis or Evaluations N/A
	Specification impacts, if any The device's states transition, including the device behavior in each state, the explicit or implicit indication, transition time, etc.
	The energy status reporting from the device Impacts of random access procedure If the RF energy source is applied, How and when to transmit CW for energy harvesting, for example, controlled by the reader Energy harvesting in Topology 2, for example, when the RF energy source and the UE are the same node, the impacts of different RRC states (IDLE/Inactive/Connected mode) Transmitting power of RF energy source.

Table 6.2.2-23: Details from Source [82], [102]

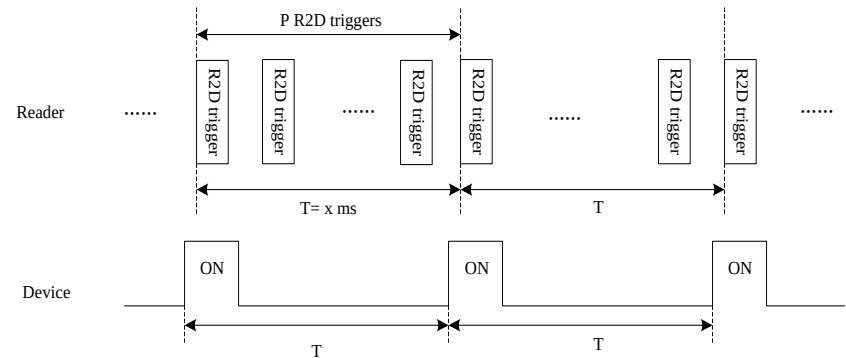
Source	Details
Source [82], [102]	Solution description The reader transmits counter decrement commands in one access round. Wherein, some counter decrement commands are necessarily transmitted on specific periodic time points. The period is equal to an absolute time length, e.g. x ms.
	If the stored occasion count is larger than or equal to P, the device is required to only detect the periodic decrement commands. Each time the device detects the periodic decrement

command, it reduces the stored occasion counter by P, where P is the number of decrement commands transmitted in a period. The value of P could be fixed in one round or be indicated in the periodic decrement commands.

After receiving the periodic decrement command each time, if the stored occasion counter is larger than or equal to P, the device can switch to sleep state to conserve power.

If the stored occasion counter is less than P, the device maintains ON state and detects the decrement commands sequentially. Each time the device detects a decrement command, it reduces the stored occasion counter by 1.

When the stored occasion counter decreases to 0, the device transmits Msg1.



Observations or Analysis or Evaluations

Table 4 Potential device availability time without the solution

Parameters	Values			
Device type	1		2	
Capacitance (uF)	10		1000	
High operating voltage U_{high} (V)	1	2	1	2
Low operating voltage U_{low} (V)	0.5	1	0.5	1
Available energy in storage (uJ) $C(U_{high}^2 - U_{low}^2)/2$	3.75	15	375	1500
Rx power consumption (uW)	1		150	
Available time for Rx	3.75 s	15 s	1.5 s	10 s

Table 5 Available time with the solution for device 1

Parameters	Values							
Available energy (uJ)	3.75				15			
Rx power consumption during ON state (uW)	1							
Power consumption during SLEEP state (uW)	0.2							
Ratio of ON and SLEEP duration	1:3	1:5	1:10	1:25	1:3	1:5	1:10	1:25

	Available time for Rx (s)	9.375	11.25	13.75	16.25	37.5	45	55	65
	Table 6 Available time with the solution for device 2								
	Parameters	Values							
	Available energy (uJ)	375				1500			
	Rx power consumption during ON state (uW)	150							
	Power consumption during SLEEP state (uW)	1							
	Ratio of ON and SLEEP duration	1:3	1:5	1:10	1:25	1:3	1:5	1:10	1:25
	Available time for Rx (s)	9.8	14.5	25.8	55.7	39	58	103	222
	Observation 1: During random access process, the above method based on Direction 2 can reduce the number of counter decrement commands that the device needs to detect and conserve device's power.								
	Observation 2: The solution in Table 6.2.2-23 of TR38.769 can increase the available time for R2D signal reception to at least 2.5 times for device 1 and 3.9 times for device 2.								
Specification impacts, if any									
Define the transmission period of the periodic decrement commands.									
Define the number of decrement commands transmitted in a period.									
Specify the device's behaviour after receiving a decrement command.									

Table 6.2.2-24: Details from Source [112]

Source	Details
Source [112]	Solution description - Device states are defined in A-IoT system. - Three states (ON/OFF/SLEEP) or two states (ON/OFF) can be considered. - Reader needs to provide necessary information to enable the device to transition between states (ON/OFF/SLEEP or ON/OFF). - For example, the reader can provide the information that the current slot is for the device that selected slot number counter X (i.e., $0 \leq X \leq 2^Q - 1$, Q value can be provided in inventory round triggering command). - For another example, the reader can provide the information that when the next inventory round would be started (or when the next inventory round triggering command would be sent). Observations or Analysis or Evaluations - Device available time is relatively long because the information provided by the reader allows the device to perform state transition and energy harvesting.

	- R2D resource overhead may not increase because the reader doesn't need to transmit inventory round triggering command frequently. - Large specification impact. Specification impacts, if any - Potential information feedback on the energy state (e.g., remaining energy) of the device. - State definition (ON/OFF/SLEEP or ON/OFF). - State transition mechanism (e.g., timing, threshold). - Definition of additional information to support direction 2.
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Table 6.2.2-25: Details from Source [107]

Source	Details
Source [107]	Solution description ON to OFF: Support state transition from ON to OFF directly for 3-state device by indication from Reader. ON to Sleep: Support state transition from ON to Sleep immediately for 3-state device by indication from Reader. Observations or Analysis or Evaluations ON to OFF: ● In some cases, device state transit from ON to OFF directly is necessary. One typical scenario would be Reader envisions no more inventory/command task in a short period for this device after current round of inventory/command task. In this case, regardless of device's power level, state transition directly to OFF state is more power efficient than putting it to Sleep. To achieve this, disabling of Sleep state can be indicated from Reader to guide device state transition to OFF directly. The necessity of this indication is that ON state can transit to either Sleep state or OFF state depends on its power level. ON to Sleep: ● In some cases, device state transit from ON to Sleep instantly is necessary for better power saving of device. One common scenario would be when device finish the inventory/command, device can go to Sleep state immediately for power saving. To achieve this, an end indication can be provided so that device can immediately change to Sleep state upon successful reception of that PRDCH. Compared to existing solutions, an additional monitoring of PRDCH in an extra time window can be saved, thereby further saving the latency of state transition as well as device power consumption for extra monitoring. Specification impacts, if any ON to OFF: At least a disabling Sleep indication needs to be specified for 3-state devices. ON to Sleep: At least an explicit end indication on inventory/command needs to be specified for 3-state devices.

Table 6.2.2-26: Details from Source [117]

Source	Details
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Source [117]	Solution description for direction 2			
	State description			
	State	Description	Transition event	Device
	ON	Transmission and reception running a clock for timing maintaining the memory	Transit to 'ON' state: - the charging energy is higher than an energy threshold or a certain percentage, and the incoming RF power is higher than an activation threshold 'SLEEP' state duration time has been achieved. Transit from 'ON' state: - the charging energy is lower than an energy threshold or a certain percentage, or fully discharged - the 'ON' state duration time has been achieved. - Receives an indication from Reader to switch to the other state.	Device 1 Device 2a
	OFF	Energy harvesting	Transit to 'OFF' state: - the charging energy is lower than an energy threshold or a certain percentage or fully discharged - Receives an indication from Reader to switch to 'OFF' state. Transit from 'OFF' state: - the charging energy is higher than an energy threshold or a certain percentage	Device 1 Device 2a
	SLEEP	Energy harvesting running a clock for timing maintaining the memory	Transit to 'SLEEP' state: - the charging energy is lower than an energy threshold or a certain percentage - the 'ON' state duration time has been achieved. - Receives an indication from Reader to switch to the other state. Transit from 'SLEEP' state: - the 'SLEEP' duration time has been achieved. - the charging energy is lower than an energy threshold or a certain percentage or fully discharged	Device 2a
	Scheme: - Reader transmits the control information of the corresponding energy harvesting process based on the PRDCH to the devices, which may include the duty-cycle periodicity and duty-cycle duration related information.¹ - Case 1: baseline case, the device can receive the reader's first command successfully. - The device switches to ON when the charging energy is higher than an energy threshold and the incoming RF power is higher than an activation threshold, then the device receives the reader's control information and acquires the duty-cycle related information. - The device transits to 'SLEEP' state to save power or 'OFF' state to complete this communication process or ON' state again to continue the communication process based on the time duration configuration or the reader's indication. - Case 2: enhancement case, the device misses the reader's first command. - The device misses the first command and does not receive the periodicity reader-assist timing indication, and switches to 'OFF' or 'SLEEP' state when the energy is lower than a threshold. - The device switches to ON when the charging energy is higher than an energy threshold and the incoming RF power is higher than an activation threshold, then the device receives the reader's assist information in one periodicity or one attempt of the multiple attempts.			

	◇ The device transits to the 'SLEEP' state to save power and will transit to the 'ON' duration when the next reader command arrives based on the time duration obtained by the reader assist information. - The device executes the corresponding behavior based on the reader's command or indication.² Note 1: For enhancement case, reader will start to transmit the assist information in a certain periodicity or multiple attempts, containing the gap information between the current occasion and the next command reception occasion. Note 2: For enhancement case, reader will stop the attempt or period transmission of its assist information when a response has been received by the reader side or a max attempt amount has been achieved. Observations • Low communication delay and high communication efficiency. • More resource allocation and more signaling overhead due to configuration or scheduling. • Low power consumption and low energy harvesting requirement due to the introduction of 'SLEEP' state. Specification impacts, if any Much more specification impacts including the configuration method or the device behavior definition.
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Table 6.2.2-27: Details from Source [116]

Source	Details
Source [116]	Solution description
	The reader provides control information to assist a device to determine whether/when to become available or unavailable. The device is assumed to have three states i.e. ON, OFF, and sleep state.
	Observations or Analysis or Evaluations
	Design the duty cycle for Tx and Rx for the device
	Increase the operational time for the device
	The time interval between the paging message can be defined based on the device energy storage
	Reader can provide the information related to switching between the states by the device
	Specification impacts, if any
	Scheduling can be based on the duty cycle
	Designing of the timer of sleep state
	Device behavior for monitoring the paging message
	Signaling indication for switching the states
	Impacts on random access procedure
	Potential information feedback on the energy state

6.3 Protocol stack and signalling procedures

6.3.1 General aspects and overall procedure

The study aims that the design on the A-IoT radio interface between reader and A-IoT device is common for Topology 1 and Topology 2. The difference of topologies is transparent to the A-IoT devices and has no impact on A-IoT devices. Unless explicitly stated, the descriptions in clause 6.3 apply to all A-IoT device types and both Topology 1 and Topology 2.

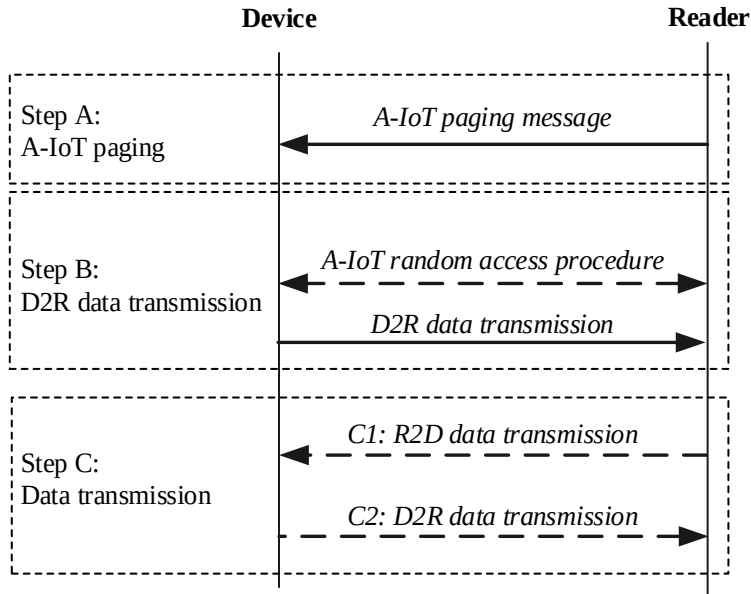


Figure 6.3.1-1 Overall AS procedures between A-IoT device and reader

The overall AS procedures can be formulated as:

- Step A: A-IoT paging. Based on the service request, the reader sends the A-IoT paging message indicating device(s) that need to respond. See clause 6.3.3.
- Step B: D2R data (device ID) transmission. Triggered A-IoT device(s) perform the device ID transmission via the A-IoT random access procedure or without using the A-IoT random access procedure. See clause 6.3.4 (and 6.3.5)..
- Step C1: Possible R2D data transmission (e.g., for sending the command). See clause 6.3.5.
- Step C2: Possible D2R data transmission (e.g., the corresponding response to command). See clause 6.3.5.

Then, above AS procedure can support indoor inventory and indoor command use cases by the following manners:

- For the detailed use case of "inventory-only", it is supported by the procedure with Step A and Step B as baseline.
- For the detailed use case of "inventory and command", it is supported by the procedure with Step A, Step B, Step C1 and Step C2, as baseline.

NOTE 1: For the use case of "inventory and command", it does not imply that the A-IoT paging message includes both the inventory and command and it does not imply the inventory and command are received by the reader at the same time from upper layer.

- For the detailed use case of "command-only":
 - It can be also supported by the baseline procedure with Step A, Step B, Step C1 and Step C2.
 - In addition, another candidate to support this use case is following, whose feasibility still depends on the conclusion from [151] and [152]:
 - Step A': A-IoT paging. Based on the service request, the reader sends the A-IoT paging message including the command, indicating device(s) to process/respond the command.
 - Step C2: Possible D2R data transmission (e.g., the device ID and/or the corresponding response to command), via the A-IoT random access procedure or without using the A-IoT random access procedure.

The following information are considered useful to be visible to the reader from CN:

NOTE 2: It needs to be further discussed on whether following information is mandatory or optional.

- The A-IoT service type (e.g., inventory, command)

NOTE 3: It needs to be further discussed if more information on command type (e.g. read/write/disable) is useful. The reader needs to know whether a D2R response is expected in D2R direction and, if so, the expected D2R message size. It needs to be further discussed on the details on how the reader gets that information (related to the "expected D2R message size" discussion in clause 6.3.5). It needs to be further discussed on whether the command type, if needed, is explicit or inferred from the "expected D2R message size" if available.

- Whether the service is targeted for one or more than one A-IoT devices
- The approximate number of target A-IoT devices of this service

6.3.2 Protocol stack and functionality aspects

For A-IoT, it is assumed the commands (e.g., read/write/disable) and/or inventory information are carried over A-IoT radio interface as the upper layer data. As to the protocol stack for A-IoT radio interface between A-IoT device and reader, it is assumed:

- RRC layer is not supported
- SDAP layer is not supported
- PDCP layer is not supported
- RLC layer is not supported
- A-IoT MAC layer is supported
- A-IoT physical layer is supported
- There is no control plane and user plane differentiation

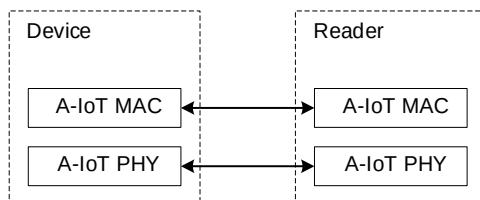


Figure 6.3.2-1 Protocol stack for A-IoT radio interface between A-IoT device and reader

As to the A-IoT required functionalities, the following functionalities are supported:

- A-IoT paging (see clause 6.3.3)
- A-IoT random access procedure (see clause 6.3.4)
- A-IoT data transmission (see clause 6.3.5)

As to the A-IoT required functionalities, at least the following functionalities are NOT supported (see TS 38.300 [153] for references for any legacy NR functionality):

- AS security (The AS layer design assumes no support of AS security, unless the study in [152] further concludes differently.)
- RRC states
- RRC connection management
- RRM L3 measurement reporting
- Mobility

- ASN.1 encoding/decoding
- Periodical system information and MIB
- Tracking/RAN area update procedure
- Per-packet QoS and per-QoS flow at AS level
- HARQ
- RLC ARQ/AM
- AS-layer (above physical layer) RLC-like/ARQ-like retransmission
- AS-layer (above physical layer) repetition

NOTE 1: It is not precluded that the reader and A-IoT device send the "payload" again as new transmission from A-IoT MAC perspective.

- Multiple A-IoT logical channels for upper layer data
- NR SR, as in TS 38.300 [153]
- NR BSR, as in TS 38.300 [153]

6.3.3 A-IoT paging

In A-IoT AS layer, the A-IoT paging functionality is to use A-IoT paging message to indicate device(s) that need to respond.

As to the A-IoT paging message, the identifier may be required to identify the device/group of devices in this trigger message (e.g., for the case of reaching a single or a group of devices). Following cases are studied:

- The A-IoT paging message containing an identifier of a single A-IoT device
- The A-IoT paging message containing a group ID that maps to multiple A-IoT devices
- The A-IoT paging message that does not contain any identifier, i.e., indicating all A-IoT devices that can receive the A-IoT paging message need to respond
- The A-IoT paging message containing multiple identifiers of A-IoT devices. The need for this use case is still to be confirmed/dependent according to the conclusion in [151]. From RAN2 perspective, it is feasible to support paging multiple identifiers of A-IoT devices, pending on TB size and multiplexing design of A-IoT paging message.

NOTE 1: The details of the above identifier and group ID and also the use case/scenario are studied in [151].

As to the A-IoT paging message, it can additionally indicate the information from which the device(s) can determine the resource(s) to be used for D2R response message(s). It can be further considered on more details for the discussion in clause 6.1.

For A-IoT device paging functionality, it is understood that the legacy paging message, legacy paging occasion and legacy DRX from NR are not supported (See TS 38.300 [153] for references for any legacy NR functionality). From RAN2 perspective, it is assumed that the A-IoT device can receive as long as there is enough energy (it can be further considered on the discussion in clause 6.1 and 6.2).

It is supported that the reader can send multiple (subsequent) A-IoT paging messages that are associated with the same service request from the CN. The duplicated response from devices for the same service request should be avoided. The A-IoT paging message can include information to avoid this duplicated response from the device to a reader. It needs to be further discussed on how to design this information in A-IoT paging message (e.g., including stage-3 details and considering the related aspects from other WGs). Then, based on this information, the device determines whether to skip sending the response to A-IoT paging message or not (if the device had successfully responded the same service before). This information should be short and simple. This information is one ID, while it needs to be further discussed

whether the ID is generated by the reader or by the core network. It needs to be further discussed on the size of this information.

It needs to be further discussed for the scenario that different readers may send A-IoT paging messages, which are associated with the same service request from the CN, to the same device for response. If this scenario is in the scope, it needs to be further discussed, by considering the progress from all the WGs.

It needs to be further discussed on whether other information is included in A-IoT paging message to indicate service type/command type or to indicate whether there will be subsequent messages.

6.3.4 A-IoT random access procedure

A-IoT random access procedure is used for the Ambient IoT device(s) to access the network for data transmission.

The A-IoT random access procedure is triggered by the reader, including triggering the access for a single A-IoT device, group of A-IoT devices, or all A-IoT devices under the coverage of the reader.

The slotted-ALOHA is the baseline for A-IoT random access procedure.

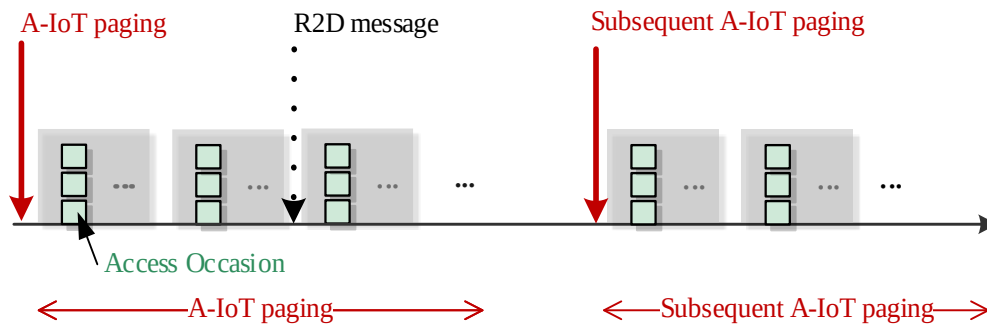


Figure 6.3.4-1 General framework of slotted-ALOHA for A-IoT random access procedure

Access occasion: An opportunity of time-frequency resource for A-IoT device(s) to perform access (e.g., transmitting the A-IoT Msg1 by the device). A set of access occasion(s) for different A-IoT device(s) is scheduled via the R2D message (referring to the "R2D transmission triggering random access" in clause 6.1.4) by the reader.

When the A-IoT device is selected to respond in accordance to the clause 6.3.3, the A-IoT device performs the following procedure:

- **Step 1:** Random access type (i.e., contention-free or contention-based) and access occasion/resource determination:
 - The A-IoT device determines the random access type from the A-IoT paging message, in accordance to clause 6.3.3. The reader can configure either contention-free access or contention-based random access (and the corresponding configurations). It needs to be further discussed on whether it is explicitly or implicitly.
 - If the random access is contention-free access:
 - Selects the indicated D2R occasion/resource;
 - Skips the contention resolution in Step 2 and performs the data transmission in accordance to clause 6.3.5.
 - If the random access is contention-based random access:
 - Performs access occasion/resource selection: as the baseline for CBRA, at least for TDMA case, the device can randomly select one access occasion for A-IoT Msg1 within the access occasions provided/assigned by the reader. It needs to be further discussed if this is applicable to FDMA case. Further enhancement option(s) can be also considered after more physical layer detailed design on TDMA and FDMA;
 - Performs the Step 2 for contention resolution.
- **Step 2:** Contention resolution of contention-based random access:

- There are three candidate solutions being studied for the contention resolution, as below (it needs to be further discussed on the down selection and/or the unified design):
 - **Solution 1: A-IoT Msg1 without data**
 - A-IoT Msg1: When the A-IoT device identifies the start of its own access occasion, it sends one 16-bit random ID generated by the A-IoT device to the reader.
 - A-IoT Msg2: The reader responds with the successfully received random ID.

If the A-IoT device receives the A-IoT Msg2 including a random ID, which is the same as the previously transmitted one in A-IoT Msg1, it considers the contention resolution as successful.
 - **Solution 2: A-IoT Msg1 with data**
 - A-IoT Msg1: When the A-IoT device identifies the start of its own access occasion, it sends the A-IoT Msg1 including the upper layer data, which can be the device ID and/or any other upper layer data, in addition to one 16-bit random ID generated by the A-IoT device to the reader.
 - A-IoT Msg2: The reader may respond with the successfully received random ID.

If the A-IoT device receives the A-IoT Msg2 including a random ID, which is the same as the previously transmitted one in A-IoT Msg1, it considers the contention resolution as successful. If the A-IoT Msg2 is not received by the device, the re-access is not autonomously performed i.e., the re-access is always controlled by the reader.
 - **Solution 3: A-IoT Msg1 optionally includes data (unified solution supporting both Solution 1 and 2)**
 - A-IoT Msg1: When the A-IoT device identifies the start of its own access occasion, it sends one 16-bit random ID generated by the A-IoT device to the reader. In addition, it is controlled by the reader on whether to also include the upper layer data, which can be the device ID and/or any other upper layer data.
 - A-IoT Msg2: The reader responds with the successfully received random ID.

If the A-IoT device receives the A-IoT Msg2 including a random ID, which is the same as the previously transmitted one in A-IoT Msg1, it considers the contention resolution as successful.

NOTE 1: The random ID is randomly generated by the A-IoT device.

NOTE 2: The A-IoT Msg2 is used for contention resolution, since it is assumed that the size of random ID in A-IoT Msg1 should be sufficient for contention resolution purpose. It is assumed that, with the sufficient range for random ID, the probability of multiple A-IoT devices selecting the same access occasion and the same random ID for A-IoT Msg1 transmission will be sufficiently low.

NOTE 3: It is up to the reader implementation whether A-IoT Msg2 is sent in random access *Solution 2*.

NOTE 4: Further information may be included in A-IoT Msg2 by considering the discussion in clause 6.1.

NOTE 5: For the advantage and disadvantage of the above solutions, it is understood that: *Solution 1* can support relatively large coverage for A-IoT Msg1 than *Solution 2*. *Solution 2* has the message number efficiency in case of low probability of A-IoT Msg1 collision (e.g. it can reduce one step for data transmission interaction in case of success).

After the A-IoT device considers the contention resolution as successful if the contention-based random access is used, or if the contention-free access is used, it may perform the upper layer data transmission with the reader, which can be the device ID and/or any other upper layer data, if any, in accordance to clause 6.3.5.

It is also studied for the possible unification between the above contention-free access and contention-based random access, from the device perspective. In this solution, when the A-IoT device identifies the start of its own access occasion, it sends at least one 16-bit random ID generated by the A-IoT device to the reader (regardless of the contention-free or contention-based access occasion). The motivation and disadvantage to include the random ID in contention-free access are still to be further discussed. For instance, one disadvantage is that the reader may not know whether the intended device is responding on the given resources.

It is supported for the A-IoT device to re-access in another opportunity controlled/provided by the reader (i.e., to retry the random access above), in case of D2R data transmission failure and contention resolution failure of contention-based random access.

The A-IoT device is not expected to autonomously re-access. The re-access is always controlled by reader. It is supported for reader to use the optional explicit R2D failure/success feedback indication to determine the re-access of device:

- This indication can be used at least to determine the re-access for addressing the transmission failure of the first D2R message, which contains the device ID and/or any other upper layer data (i.e. "Msg3");
- This indication can be also used for the following D2R data (as described in clause 6.3.5), to determine the re-access for addressing the transmission failure.

The R2D message is used by the reader to provide access occasion(s), which can be used for re-access purpose. It needs to be further discussed if additional information is needed in this R2D message to differentiate the re-access purpose.

- A-IoT paging message is one of the options for this R2D message (e.g., see the "subsequent A-IoT paging message" in Figure 6.3.4-1).
- Another option can be some R2D messages between A-IoT paging (e.g., see the other "R2D message" in Figure 6.3.4-1).

6.3.5 A-IoT data transmission

According to the protocol stack and functionalities in clause 6.3.2, more detailed data transmission functionalities are studied in this clause.

The handling of data transmission failure has been studied. It is understood that the subsequent R2D data transmission after the D2R data transmission does not need to be always sent. In case of D2R data transmission failure, the A-IoT device follows the reader subsequent R2D instruction, if any.

- For instance, the reader can repeat the R2D upper layer "command" to trigger the A-IoT device to re-send the same D2R upper layer "response" (i.e., the A-IoT device just follows the received R2D to transmit following D2R).
- For instance, the reader can re-send A-IoT Msg2 to specific device(s) to echo the random ID(s), in case of failure reception of corresponding D2R data transmission (i.e. "Msg3") after the initial A-IoT Msg2, together with the corresponding D2R resource scheduling. This can trigger the A-IoT device to re-send the same D2R data transmission (i.e. "Msg3"). If it is supported to include the echoed random IDs for multiple devices, the re-sent A-IoT Msg2 only includes the random IDs of the devices, whose "Msg3" is not successfully received.

It is studied on the functionality of segmentation. It is understood that segmentation and reassembly would add complexity. Therefore, the simplified segmentation in the A-IoT MAC layer is studied, considering following studies in physical layer:

- In both R2D and D2R directions, there is no lower bound on the minimum TB size, and a maximum TB size of around 1000 bits can be supported.
- How large TB that can be transported at a given time depends on target coverage/data rate, energy consumption/device availability, etc.

The segmentation study focuses on D2R direction (it can be further considered on R2D direction). For the candidate solution for segmentation:

- The sequence number, the segment number and the number of segments are not supported;
- An indication is used to indicate to the reader on whether the data is segmented and whether the MAC PDU is the last segment. It needs to be further discussed on the size of this indication (one or two bits) and the corresponding further details;
- It is beneficial for the reader to be able to trigger a re-transmission of a segment;

- It is assumed that the A-IoT device will not support AS layer buffering for A-IoT segmentation functionalities, i.e., all segment(s) are stored in upper layer(s).

The use of the following assistance information is studied (the need of each is still to be decided):

- The energy status report from the A-IoT device to the reader, which indicates that the A-IoT device, if the A-IoT device can determine this, does not have sufficient energy to perform the remaining/follow-up procedure.
- The A-IoT device may report a 1-bit energy status indication to the reader in a D2R message. It needs to be further discussed if and how the reader control this indication of such report.
- The reader may take this indication into account for the remaining/follow-up procedure (up to the reader implementation), e.g., the reader may choose not to transmit the subsequent messages for some time, or the reader may choose to not do anything in particular, etc.
- It is beneficial for the reader to know an estimate of expected D2R message size. This D2R message size information would be beneficial but it is not essential. Two options are studied:
 - Option 1: The estimate of expected following D2R message size from the CN to the reader;
 - Advantages/disadvantages of option 1: If this information is available from CN, it saves the overhead/latency caused by Option 2 in AS layer. It can be also used by the reader as early as for the first D2R message, if needed.
 - Option 2: The simple message size indication from the device to the reader.
 - Advantages/disadvantages of option 2: It is useful in case the CN is not able to provide or does not provide the expected D2R message size. Limited granularity is supported considering the AS layer overhead. It requires D2R message to carry this information in advance, before the reader can actually use this size information for scheduling.

From higher layer perspective, it is assumed that "AS ID" (if defined according to the design in clause 6.1) is used at least for purpose of D2R scheduling and R2D reception. From higher layer perspective, it is assumed that this "AS ID" should be a short AS layer ID, rather than the full upper layer device ID. It needs to be further discussed if this "AS ID" can be based on partial upper layer device ID. It needs to be further discussed on the length of this "AS ID". From higher layer perspective, following options are possible for this "AS ID" (it is aimed to define one common design for all access procedures in clause 6.3.4, if technically possible):

- Option 1: a random ID (if used in first D2R message) can be reused;
- Option 2: the reader assigns this "AS ID" to the device. It needs to be further discussed via which R2D message.
- Option 3: It is up to the reader whether to reuse the random ID (if used in first D2R message) as the "AS ID" or to assign a new "AS ID". It needs to be further discussed via which R2D message.

NOTE 1: The further down-selection for "AS ID" needs to consider the involved message and whether the reader needs to handle the collision.

6.3.6 Topology 2 aspects on the interface between UE reader and RAN

For Topology 2, the architecture/protocol stack options in [151] are studied (also corresponding to the studies in clause 6.4.2.1), while no new AS layer architecture/protocol stack options will be studied:

- RRC based solution: A-IoT upper layer information is explicitly forwarded via UE reader's NR Uu RRC message.
- NAS based solution: A-IoT upper layer information is transmitted over UE reader's NAS message.
- UP based solution: A-IoT upper layer information is transmitted as UE reader's user plane data.

It is assumed that the intermediate UE authorization is performed by upper layers, according to [151] and [152]. It is not studied, from RAN2 signalling perspective, the scenario to support bistatic mode of operation (e.g., D2T2-A1).

The radio resources used by A-IoT radio interface between the A-IoT device(s) and UE reader are controlled by the network. The radio resources, which are dedicated for a UE reader, are only configured to the UE reader via dedicated signalling. The mechanisms for shared resource pool amongst UE readers are not considered in this release. The UE reader in coverage of BS scenario is supported. The UE reader may perform the A-IoT procedure on A-IoT radio interface between the reader and the device(s), only if the radio resource configuration is valid in the cell, which is under network control. The radio resource validity across multiple cells is not supported in this release. Hence, for all options below, the resources remain valid only in the same cell in which the resources were configured.

For the case when the radio resources are allocated semi-statically by the network, the radio resource validities are studied in the following scenarios with candidate options (down-selection can be decided in the WI phase):

- Scenario 1: In RRC_CONNECTED, the UE reader can perform A-IoT operations using the allocated A-IoT radio resource for A-IoT radio interface:
 - Option 1: UE reader receives the resources configuration in NR Uu dedicated RRC signalling. The radio resources remain valid until the network releases them explicitly. This would be supported by the NR Uu RRC reconfiguration; and/or
 - Option 2: UE reader receives the resources configuration in NR Uu dedicated RRC signalling, which configures a time period in which the corresponding resource can be used. The UE reader considers that the resources remain valid for that time period, unless the resource configuration is explicitly released by the network by the NR Uu RRC reconfiguration.
- Scenario 2: In RRC_CONNECTED, the UE reader can perform A-IoT operations using the allocated A-IoT radio resource for A-IoT radio interface, upon its temporary out of connection scenarios (e.g., RLF and handover cases):
 - Option 1: The UE reader considers the resources as temporarily invalid during the temporary out of connection condition. The resource may become valid again after the UE recovers from the condition.
 - Option 2: The UE reader considers that the resources remain valid for a time period. It needs to be further discussed on whether this time period belongs to the one in Option 2 for Scenario 1 or is a different time period. It needs to be further discussed on whether the duration of this time period is related to the ongoing A-IoT operations.

NOTE 1: The A-IoT radio resource can be (re)configured by UE reader's RRC reconfiguration (including during handover procedure, after re-establishment of the UE reader, etc.)

- Scenario 3: In RRC_INACTIVE, the UE reader can perform A-IoT operations using the allocated A-IoT radio resource for A-IoT radio interface:
 - Option 1: UE receives the resources configuration in NR Uu dedicated RRC signaling, before moving to RRC_INACTIVE. The radio resources remain valid until the network releases them explicitly. This would be supported by the NR Uu RRC reconfiguration (the UE may need to resume or be paged by the network back to RRC_CONNECTED before that).
 - Option 2: UE reader receives the resources configuration in NR Uu dedicated RRC signalling, before moving to RRC_INACTIVE, which configures a time period in which the corresponding resource can be used. The UE reader can consider that the resources remain valid for that time period.
- Scenario 4: In RRC_IDLE, the UE reader can perform A-IoT operations using the allocated A-IoT radio resource for A-IoT radio interface.
 - If this scenario 4 is supported, the options of scenario 3 can be the candidates.
 - The support of scenario 4 (UE reader performing A-IoT operations in RRC_IDLE) is down-prioritized. It can be considered in WI phase, if feasible with minimum specification impact.

For the radio resources allocation request, following alternatives are studied:

- Alternative 1: BS configures/allocates a set A-IoT radio resource to the UE reader, in response to a request from the UE reader; and/or
- Alternative 2: BS configures/allocates a set A-IoT radio resource to the UE reader, based on the service request from CN.

NOTE 2: The above two alternatives can co-exist. The configuration for radio resource allocation may also include the validity criteria (if applicable, see the time period in previous option 2 of those scenarios)

For the UE reader selection, it is up to RAN3 and SA2 on whether BS or CN selects the UE reader(s) and what information is shared between BS and CN. From RAN2 perspective, it is assumed, at least for RRC based solution, there may be radio related reasons for the BS to be involved in the UE reader selection.

6.4 RAN architecture aspects

This clause attempts to identify and describe architectural elements necessary to define a RAN architecture for support of Ambient IoT embedded in the overall 5G system architecture in support of topology 1 and topology 2 (as defined in TR 38.848 [2]).

This clause also attempts to identify a functional split between RAN and CN.

The logical system architecture for A-IoT consists of the following architectural elements:

A-IoT device: Equipment with characteristics outlined e.g., in TS 22.369 [154] and TR 38.848 [2].

A-IoT RAN: Hosts certain functions for A-IoT as part of the functional in RAN.

A-IoT radio: Radio interface between A-IoT device and A-IoT RAN node in topology 1 and between A-IoT device and A-IoT-enabled UE in topology 2.

A-IoT CN: Hosts certain functions for A-IoT as of the functional in CN.

NOTE 1: the details of A-IoT CN are subject to SA2.

NOTE 2: Further details regarding A-IoT functions hosted in the A-IoT CN and in the A-IoT RAN and the respective functional split to be decided by RAN2, RAN3 and SA2.

XX interface: Interface between the A-IoT RAN/A-IoT-enabled gNB and the A-IoT CN on which certain A-IoT specific functions are performed.

XX interface is NG interface, NGAP is used between A-IoT RAN and A-IoT CN.

NOTE 3: How the functions represented by the XX interfaces are defined in various solutions and A-IoT CN topologies (direct/indirect connection between A-IoT RAN and A-IoT CN).

Common reader function: A function that communicates with the A-IoT device by means of A-IoT radio.

A-IoT RAN node function: A function that contains e.g., the control of the A-IoT radio resources used towards the A-IoT device.

NOTE 4: Aspects concerning coordination of the Upper Layer functions (e.g., Inventory, Command) e.g., in case these functions have to be performed over a multitude of instances of the Common Reader Function, need further discussion.

6.4.1 Support of Topology 1

Figure 6.4.1-1 depicts a logical system architecture for topology 1, where the Common reader function and A-IoT RAN node function are deployed within an A-IoT RAN.

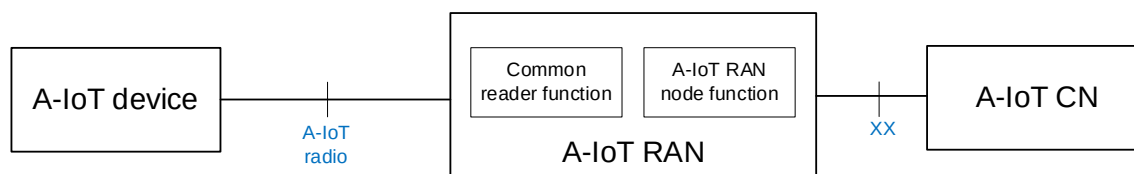


Figure 6.4.1-1 Logical system architecture for topology 1

For Topology 1, architecture and protocol aspects of split RAN architecture are not studied.

In Topology 1, the XX interface is the NG-C interface.

Figure 6.4.1-2 shows the Protocol stack for Topology 1:

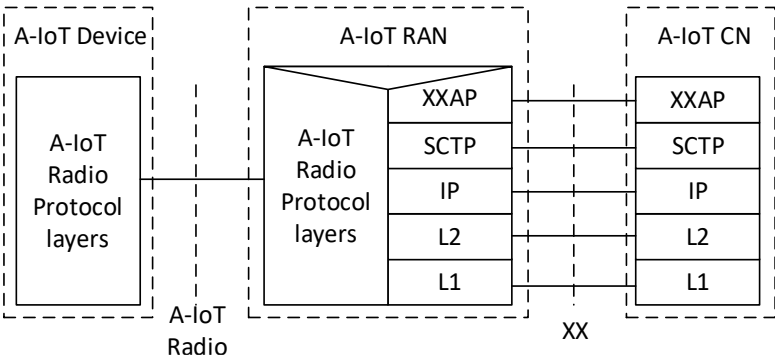


Figure 6.4.1-2. Protocol Stack for Topology 1

NOTE 1: The protocol stack in Figure 6.4.1-2 does not illustrate how A-IoT upper layer information, if any, is transported over XXAP.

For topology 1, the XXAP is terminated at an A-IoT RAN node.

The signalling transport for XXAP at the A-IoT RAN node is SCTP/IP. Other options of signalling transport for XXAP at the A-IoT RAN node (e.g., HTTP/2/TLS/TCP) were discussed, but will not be pursued.

NOTE 2: Whether "XXAP" is realised by including AIoTF information in NGAP or by carrying a new protocol layer over NGAP, needs further discussion and decision.

NOTE 3: The A-IoT CN may include AMF and A-IoTF which is up to SA2 decision.

In Topology 1, an A-IoT RAN node may serve one or more readers.

The A-IoT RAN node should enable the coordination of the usage of the A-IoT radio resources among readers.

NOTE 4: How to perform coordination of the usage of the A-IoT radio resources among readers needs further discussion.

Reader selection may need coordination between A-IoT RAN node and A-IoT CN.

NOTE 5: How to perform reader selection needs further discussions.

NOTE 6: In Topology 1, whether the AIoTF needs to be aware of the readers served by the A-IoT RAN node and their location and how this information is provided to the AIoTF needs further discussion.

6.4.2 Support of Topology 2

Figure 6.4.2-1 depicts a logical system architecture for topology 2, where the Common reader function is located at an A-IoT-enabled UE, and the A-IoT RAN node function is located at an A-IoT-enabled gNB.

The following definitions apply:

A-IoT-enabled gNB: A gNB supporting A-IoT RAN node function in topology 2, which is able to communicate with the A-IoT-enabled UE via NR Uu interface.

A-IoT-enabled UE: A UE supporting Common reader function, which is able to communicate with the A-IoTdevice via the A-IoT radio interface.

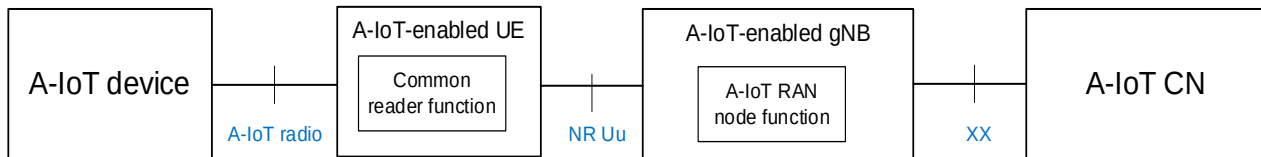


Figure 6.4.2-1 Logical system architecture for topology 2

NOTE 1: Figure 6.4.2-1 doesn't illustrate the protocol between A-IoT enabled UE and A-IoT CN, if needed.

NOTE 2: The A-IoT CN could include AMF, UPF and A-IoTF, which is up to SA2 decision.

NOTE 3: Details of the radio resource management performed by the A-IoT enabled gNB are subject to RAN1 and RAN2 mechanisms.

For Topology 2, architecture and protocol aspects of split RAN architecture are not studied.

In Topology 2, XX interface is NG interface.

In Topology 2, the RAN architecture should enable the coordination of the usage of the A-IoT radio resources among readers.

NOTE 4: How to perform coordination of the usage of the A-IoT radio resources among readers needs further discussion.

An A-IoT-enabled gNB could support both topology 1 and topology 2, this is an implementation matter.

6.4.2.1 Solutions for Topology 2

6.4.2.1.0 General

To support Topology 2, the following solutions are to be studied for conveying A-IoT upper layer information:

- **RRC based solution.** With this solution, A-IoT CN applies A-IoT upper layer information explicitly over XXAP signaling. A-IoT upper layer information is then relayed explicitly to/from the A-IoT-enabled UE via NR Uu RRC.
- **NAS based solution.** With this solution, there is no explicit termination of A-IoT upper layer information at A-IoT-enabled gNB. A-IoT upper layer information is transmitted over A-IoT enabled UE's NAS.
- **UP based solution.** With this solution, there is no explicit termination of A-IoT upper layer information at A-IoT-enabled gNB. A-IoT upper layer information is transmitted as A-IoT-enabled UE's user plane data.

NOTE 1: The protocol stack for each solution option does not illustrate A-IoT CN internal architecture and how A-IoT upper layer information is transported, if any.

No evaluation and no down selection for T2 solutions has been performed in RAN3.

Only if the UE is authorized to perform the A-IoT service, the UE can communicate with the A-IoT device and be configured with A-IoT radio resources controlled by the A-IoT enabled gNB.

NOTE 2: The NGAP protocol functions for the A-IoT-enabled UE authorization needs further discussions.

In RRC based solution, Reader selection may need coordination between A-IoT-enabled gNB and A-IoT CN.

NOTE 3: How to perform reader selection needs further discussions.

NOTE 4: In NAS/UP based solution, whether A-IoT CN selects the A-IoT-enabled UE for the selection needs further discussion.

6.4.2.1.1 Solution 1: RRC based solution

Upon receiving XXAP: A-IoT related message from A-IoT CN, the A-IoT-enabled gNB transmits the related information towards the A-IoT-enabled UE via NR Uu RRC, and vice versa.

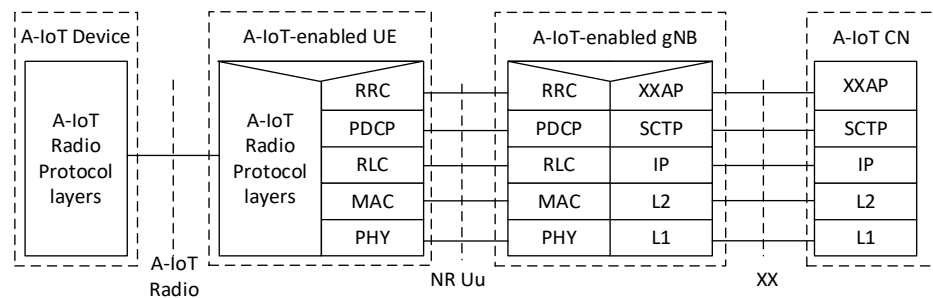


Figure 6.4.2.1.1-1: RRC based solution of Topology 2

In this solution, XX interface is NG-C interface.

NOTE: Whether "XXAP" is realised by including AIoTF information in NGAP or by carrying a new protocol layer over NGAP, needs further discussion and decision.

6.4.2.1.2 Solution 2: NAS based solution

The A-IoT related messages between the AIoTF and the A-IoT-enabled UE are carried via A-IoT-enabled UE's DL/UL NAS packets, the A-IoT-enabled gNB handles the A-IoT-enabled UE's NAS packets as legacy, i.e., using DL NAS Transport and UL NAS Transport procedures over NGAP.

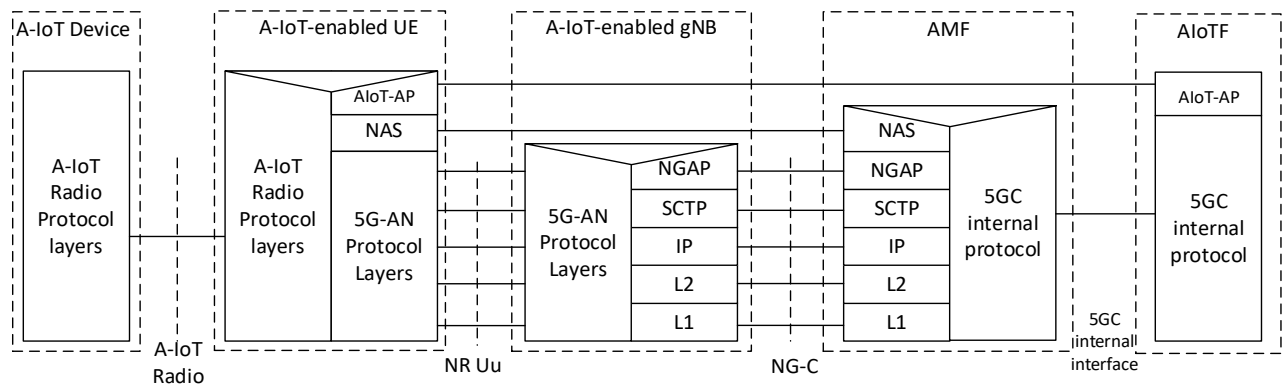


Figure 6.4.2.1.2-1: NAS based solution of Topology 2

NOTE 1: The communication between AIoTF and the A-IoT-enabled UE via A-IoT related messages and the presence of AIoT-AP protocol are being discussed in SA2. If agreed by SA2, the AIoT-AP may carry same/similar information as defined for other solutions over XXAP, this requires coordination between RAN3 and SA2.

NOTE 2: The definition and description of AIoTF and AIoT-AP refers to TR 23.700-13 [151].

There are different ways discussed to achieve A-IoT radio resource coordination and allocation as described in 6.5.1.3. Figure 6.4.2.1.2-2 illustrates one candidate solution to support AIoT session resource control for the NAS based solution by utilising XXAP between A-IoT CN and A-IoT enabled gNB.

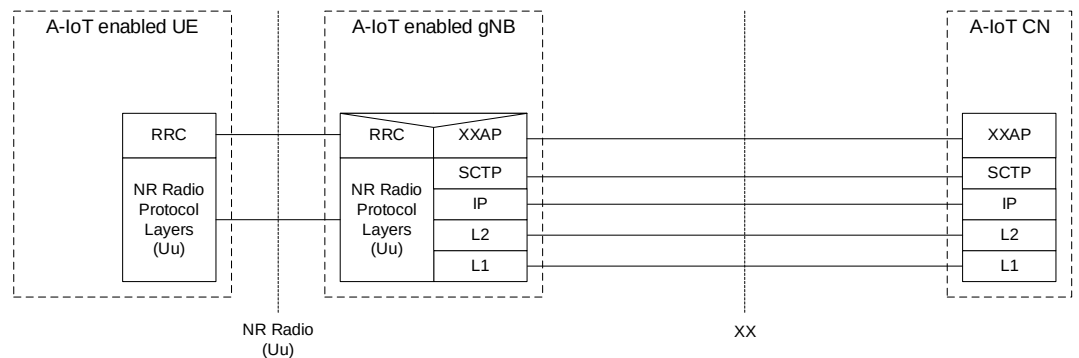


Figure 6.4.2.1.2-2: one candidate solution of Resource control for NAS based solution of Topology 2

For the NAS-based solution there are other candidate solutions discussed (e.g., A-IoT enabled UE request-based resource allocation) described in clause 6.5.1.3 which might not need this additional protocol layer (XXAP) for resource allocation, refer to clause 6.5.3.1.2.

6.4.2.1.3 Solution 3: UP based solution

The A-IoT related messages between the A-IoTF and the A-IoT-enabled UE are carried via A-IoT-enabled UE's PDU Session, the A-IoT-enabled gNB handles the A-IoT-enabled UE's user plane data as legacy, i.e., over NG-U GTP-U tunnels.

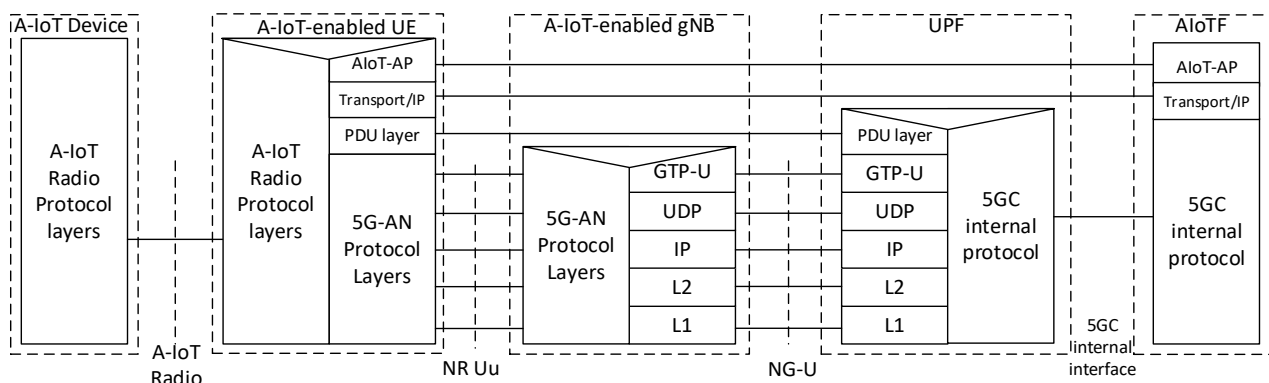


Figure 6.4.2.1.3-1: UP based solution of Topology 2

NOTE 1: The communication between A-IoTF and the A-IoT-enabled UE via A-IoT related messages and the presence of A-IoT-AP protocol is being discussed in SA2. As the A-IoT-AP may carry same/similar information as defined for other solutions over XXAP, this requires coordination between RAN3 and SA2.

NOTE 2: The definition and description of A-IoTF and A-IoT-AP/Transport/IP refers to TR 23.700-13 [151].

There are different ways discussed to achieve A-IoT radio resource coordination and allocation as described in 6.5.1.3. Figure 6.4.2.1.3-2 illustrates one candidate solution to support A-IoT session resource control for the UP based solution by utilising XXAP between A-IoT CN and A-IoT enabled gNB for resource control.

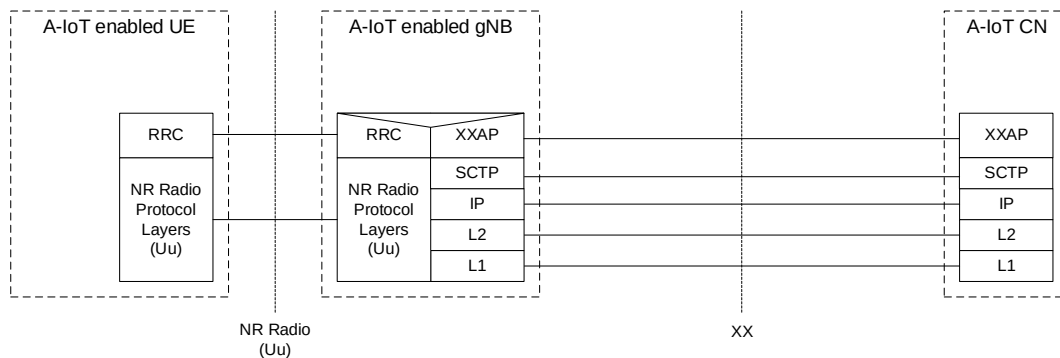


Figure 6.4.2.1.3-2: one candidate solution of Resource control for UP based solution of Topology 2

For the UP based solution there are other candidate solutions discussed (e.g., A-IoT enabled UE request-based resource allocation) described in clause 6.5.1.3 which might not need this additional protocol layer protocol layer (XXAP) for resource allocation, refer to clause 6.5.3.1.2.

6.5 Impacts on CN-RAN interface

6.5.1 Information exchanged between A-IoT CN and A-IoT RAN

6.5.1.1 Inventory

Inventory can be sent by the A-IoT CN for a single device, or a group of devices, or all devices.

The Inventory Request from the A-IoT CN to the A-IoT RAN, may include the following:

- (1) A-IoT Device Identification (to find a single device, a group of devices, or all devices)

NOTE 1: The definition of this identification is out of RAN3 scope.

For topology 1, A-IoT RAN needs to store the A-IoT Device Identification received from the inventory request.

NOTE 2: Whether to interpret and process A-IoT Device Identification received from the inventory request needs further discussion.

- (2) Scope of inventory request (e.g., a certain area in which the inventory is to be triggered)

Multiple individual A-IoT Device IDs (one ID per device) can be provided to the A-IoT CN via a single Inventory Report.

NOTE 3: For topology 1, whether device ID received from the device is sent transparently to the A-IoT CN by the A-IoT RAN depends on RAN2 and SA2.

NOTE 4: Whether the description in this clause apply to A-IoT enabled gNB in topology 2 needs further discussion and is dependent on SA2 and RAN2.

6.5.1.2 Command

Command can be sent by the A-IoT CN for a single device.

For topology 1, A-IoT RAN node should be able to differentiate between command and inventory.

NOTE 1: Whether command can be performed for a group of devices depends on RAN2, SA2 and SA3.

NOTE 2: Whether the A-IoT-enabled-gNB should be able to differentiate between command and inventory needs further discussion, which depends on SA2 and RAN2.

NOTE 3: Whether the description in this clause apply to A-IoT enabled gNB in topology 2 needs further discussion and is dependent on SA2 and RAN2.

6.5.1.3 A-IoT radio resource allocation in case of NAS/UP based solutions

In NAS/UP based solutions, there are different ways to trigger A-IoT session resource allocation, upon CN request or upon UE request.

A-IoT session resources can be requested by the A-IoT CN in advance or in parallel to the NAS/UP based communication with the A-IoT device.

NOTE 1: In case of UE request, whether and how the A-IoT enabled gNB is informed about the A-IoT session when A-IoT enabled UE requests for A-IoT radio resources and the interaction with resource allocation needs further discussion.

NOTE 2: There were also discussions about requesting A-IoT session resources along with the NAS/UP based communication. Details would need further discussions.

6.5.2 Signaling and Procedures for Topology 1

NOTE: XX communication depicted in the following clauses uses protocol elements (messages and information elements) detailed in clause 6.5.1 and are not repeated unless additional description is necessary.

6.5.2.1 Candidate procedures for A-IoT Inventory for Topology 1

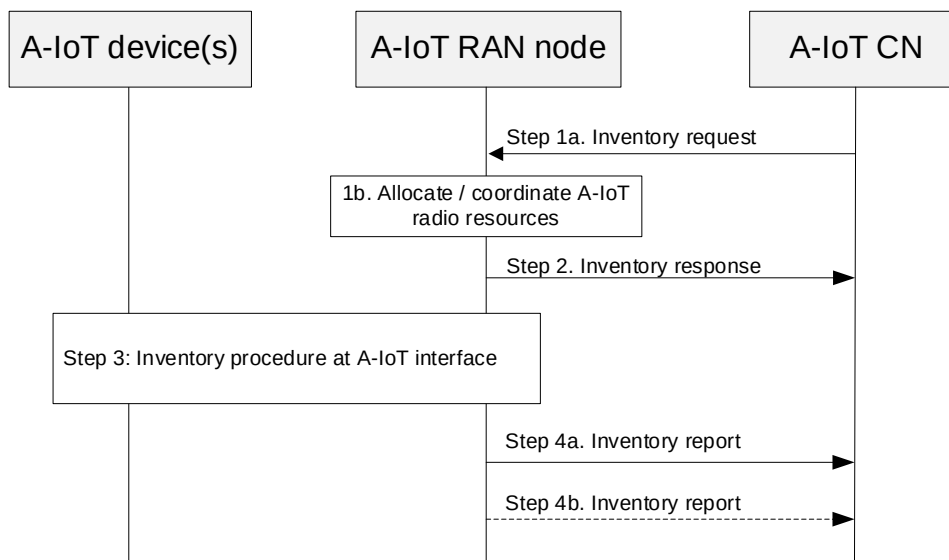


Figure 6.5.2.1-1: Message flow for A-IoT Inventory in Topology 1

1a. The A-IoT CN sends an Inventory request message to the A-IoT RAN node, taking into account, among others, the A-IoT transaction scope.

1b. The A-IoT RAN node allocates and coordinates the usage of A-IoT radio resources.

2. The A-IoT RAN node sends an Inventory response message to the A-IoT CN.

NOTE 1: In step 2, the A-IoT RAN node may instead send an Inventory failure message to the A-IoT CN indicating that the inventory procedure could not be initiated towards the A-IoT device(s).

3. The A-IoT RAN node performs the inventory procedure towards the A-IoT device(s) over the A-IoT radio interface.

4a/4b. After receiving the inventory result from the A-IoT device(s), the A-IoT RAN node may send one or multiple Inventory reports towards the A-IoT CN including the received inventory result.

NOTE 2: Steps 4a/4b may happen in parallel with Step 3 for different A-IoT devices.

6.5.2.2 Candidate procedures for A-IoT Command in Topology 1

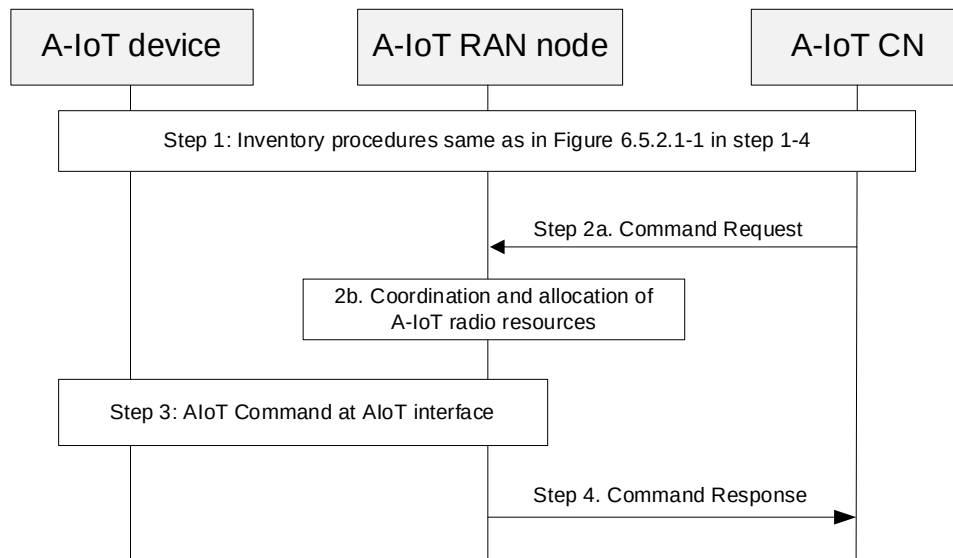


Figure 6.5.2.2-1: Message flow for A-IoT Command in Topology 1

1. The Inventory procedures are performed in step 1, which are the same as in Figure 6.5.2.1-1 in step 1-4.

NOTE 1: Step 1 is performed for the "A-IoT Inventory and Command" case. Whether and under which conditions step 1 may be skipped in case of "Command-only" and which consequences this would have for the overall message flow depends on discussions led by SA2, SA3 and RAN2.

NOTE 2: Whether and which way steps 2a and 2b may be part of step 1 is dependent on discussions led by SA2, SA3 and RAN2.

2a. The A-IoT CN node sends a Command Request message to the A-IoT RAN.

2b. The A-IoT RAN node coordinates the usage of A-IoT radio resources and allocates A-IoT radio resources for the A-IoT command session.

3. The A-IoT RAN node performs A-IoT command procedures towards the A-IoT device over the A-IoT radio interface.

4. The A-IoT RAN node sends a Command Response message to the A-IoT CN, if any command result is received from A-IoT device, the A-IoT RAN node may include the command result in the Command response message.

NOTE 3: In step 4, the A-IoT RAN node may instead send a Command Failure message to the A-IoT CN indicating that the command procedure has failed.

6.5.3 Signaling and Procedures for Topology 2

6.5.3.1 Candidate procedures for A-IoT Inventory for Topology 2

6.5.3.1.1 RRC based solution

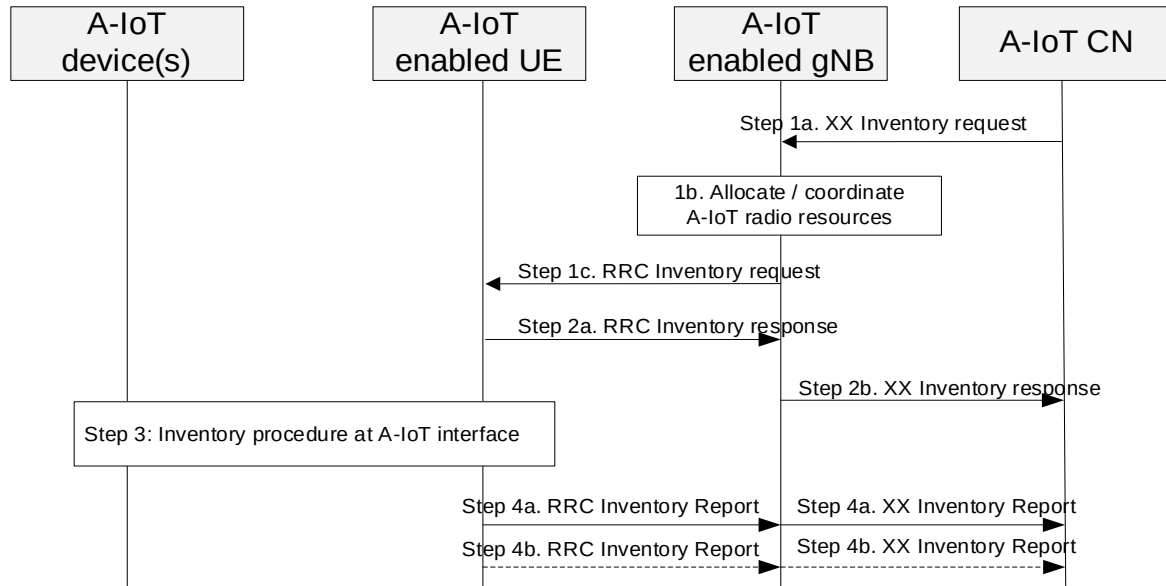


Figure 6.5.3.1.1-1: Message flow for A-IoT Inventory in Topology 2 - RRC-based solution

1a. The A-IoT CN sends an Inventory request message to the A-IoT enabled gNB

1b. The A-IoT enabled gNB allocates and coordinates usage of A-IoT radio resources.

1c/2a RRC communication with the A-IoT enabled UE takes place.

NOTE 1: RRC based communication is only depicted schematically, details subject to RAN2.

2. The A-IoT-enabled gNB sends an Inventory response message to the A-IoT CN.

NOTE 2: In step 2, the A-IoT-enabled gNB may instead send an Inventory failure message to the A-IoT CN indicating that the inventory procedure could not be initiated towards the A-IoT device(s).

3. The A-IoT-enabled gNB requests the A-IoT-enabled UE(s) to trigger inventory procedure towards the A-IoT device(s).

4a/4b. After receiving inventory result reported from the A-IoT enabled UEs, the A-IoT-enabled gNB may send one or multiple Inventory reports towards the A-IoT CN including the received inventory result.

NOTE 3: Steps 4a/4b may happen in parallel with Step 3 for different A-IoT devices.

NOTE 4: Step 4a/4b between A-IoT-enabled UE and A-IoT-enabled gNB subject to RAN2.

6.5.3.1.2 NAS/UP based solution

Note: Whether and which of the following options on CN triggered A-IoT session resource allocation and UE triggered A-IoT session resource allocation will be finally considered would need further discussion involving other WGs as needed. This was not done in this study.

NAS/UP based solution – A-IoT CN triggered A-IoT session resource allocation

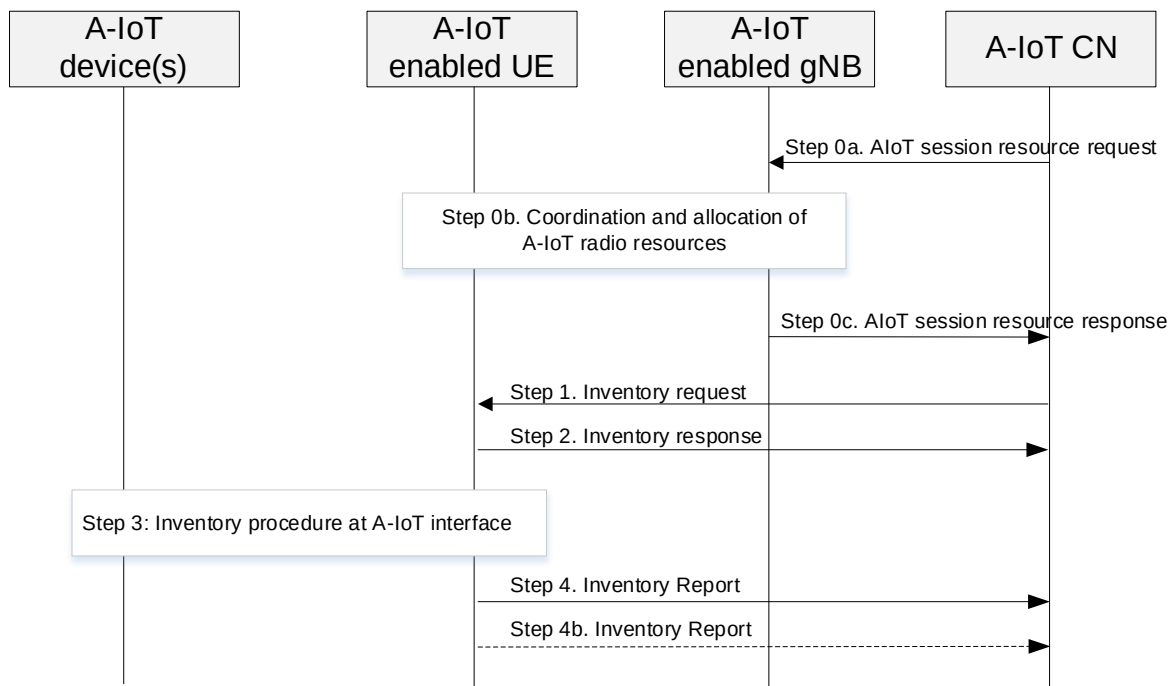


Figure 6.5.3.1.2-1: Message flow A-IoT Inventory in Topology 2 - NAS/UP based solution – A-IoT CN triggered A-IoT session resource allocation

This option is characterised by the A-IoT enabled gNB receiving the request to establish A-IoT session resources by the A-IoT CN.

0a. The A-IoT CN requests A-IoT session resources.

0b. The A-IoT enabled gNB coordinates A-IoT radio resources and allocates A-IoT radio resources to the A-IoT enabled UE accordingly.

0c. The A-IoT enabled gNB confirms the request for A-IoT session resources.

NOTE 1: In step 0c, the A-IoT-enabled gNB can reject the request for A-IoT session resource.

1. The A-IoT CN sends an Inventory request message to the A-IoT-enabled UE.

2. The A-IoT-enabled UE(s) sends an Inventory response message to the A-IoT CN.

NOTE 2: In step 2, the A-IoT-enabled UE may instead fail the Inventory request.

3. The A-IoT-enabled UE performs the inventory procedure towards the A-IoT device(s).

4/4b. The A-IoT-enabled UE may send one or multiple Inventory reports towards the A-IoT CN including the received inventory result.

NOTE 3: Steps 4/4b may happen in parallel with Step 3 for different A-IoT devices.

NAS/UP based solution – A-IoT enabled UE triggered A-IoT session resource allocation

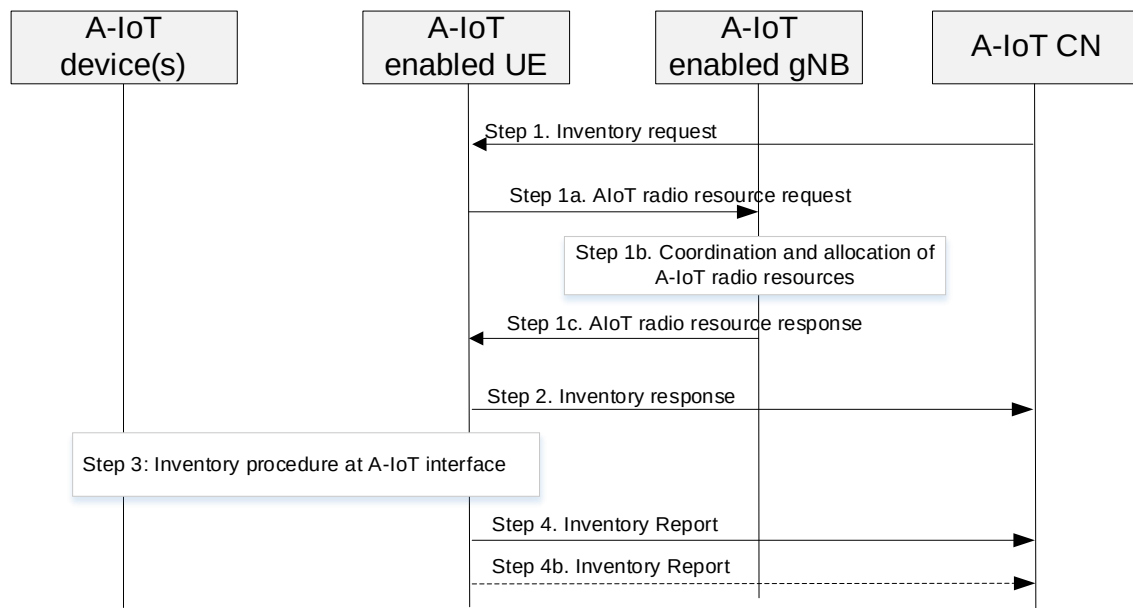


Figure 6.5.3.1.2-2: Message flow for A-IoT Inventory in Topology 2 (if - NAS/UP based solution is used) – A-IoT enabled UE triggered A-IoT session resource allocation

This option is characterised by the A-IoT enabled gNB receiving the request to establish A-IoT session resources by the A-IoT enabled UE.

Additional means to enable the A-IoT CN to provide A-IoT session related information directly to the A-IoT enabled gNB, not shown in Figure 6.5.3.1.2-2, may be needed.

1. The A-IoT CN sends an Inventory request message to the A-IoT-enabled UE.
- 1a. The A-IoT enabled UE request A-IoT radio resources from the A-IoT enabled gNB.
- 1b. The A-IoT enabled gNB coordinates A-IoT radio resources and allocates A-IoT radio resources to the A-IoT enabled UE accordingly.
- 1c. The A-IoT enabled gNB responds the A-IoT radio resources to the A-IoT-enabled UE.

NOTE 4: In step 1c, the A-IoT-enabled gNB can reject the A-IoT radio resource request. Details w.r.t. steps 1a and 1c are subject to RAN2 discussions.

2. The A-IoT-enabled UE sends an Inventory response message to the A-IoT CN.

NOTE 5: In step 2, the A-IoT enabled UE may instead send an Inventory failure message to the A-IoT CN indicating that the inventory procedure could not be initiated towards the A-IoT device(s), and the procedure ends.

3. The A-IoT-enabled UE(s) performs the inventory procedure towards the A-IoT device(s) over the A-IoT radio interface.

- 4/4b. After receiving inventory result reported from the A-IoT device(s), the A-IoT-enabled UE may send one or multiple Inventory reports towards the A-IoT CN including the received inventory result.

NOTE 6: Steps 4/4b may happen in parallel with Step 3 for different A-IoT devices.

6.5.3.2 Candidate procedures for A-IoT Command for Topology 2

6.5.3.2.1 RRC based solution

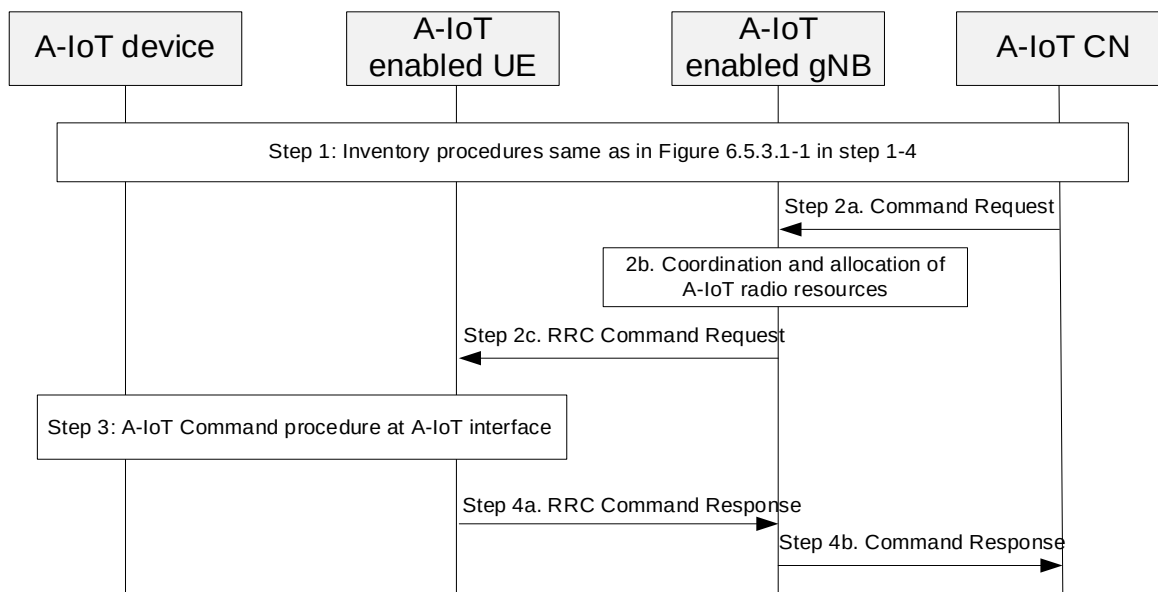


Figure 6.5.3.2-1: Message flow for A-IoT Command in Topology 2 (RRC-based solution)

1. The Inventory procedures are performed in step 1, which are the same as in Figure 6.5.3.1.1-1 in step 1-4.

NOTE 1: Step 1 is performed for the "A-IoT Inventory and Command" case, Whether and under which conditions step 1 may be skipped in case of "Command-only" and which consequences this would have for the overall message flow depends on discussions led by SA2, SA3 and RAN2.

NOTE 2: Whether and which way steps 2a, 2b and 2c may be part of step 1 is dependent on discussions led by SA2, SA3 and RAN2.

2a. The A-IoT CN sends a Command Request message to the A-IoT enabled gNB.

2b. The A-IoT enabled gNB coordinates A-IoT radio resources and allocates A-IoT radio resources to the A-IoT enabled UE accordingly.

2c. The A-IoT enabled gNB sends a RRC Command Request message to the A-IoT enabled UE.

NOTE 3: RRC based communication, i.e., performing the A-IoT Command procedure and allocation of A-IoT radio resources, is only depicted schematically, details subject to RAN2.

3. The A-IoT-enabled UE performs A-IoT command procedures at A-IoT interface towards the A-IoT device over the A-IoT radio interface.

4a. A-IoT enabled UE sends a Command Response message to A-IoT enabled gNB, if the command result is received from A-IoT device, the A-IoT enabled UE may include the command result in the Command Response message.

4b. The A-IoT enabled gNB sends a Command Response message to the A-IoT CN, it may include the command result in the Command Response message, if any.

NOTE 4: In step 4b, the A-IoT enabled gNB may instead send a Command Failure message to the A-IoT CN indicating that the command procedure is failed towards the A-IoT device.

6.5.3.2.2 NAS/UP based solution

The A-IoT Command procedure also applies to the NAS/UP based solutions, and the A-IoT Command procedure signalling are exchanged between the A-IoT enabled UE and the A-IoT CN over UE's NAS/PDU session. The A-IoT session resource allocation steps described in 6.5.3.1.2 also applies to this procedure.

6.6 Coexistence of ambient IoT and NR/LTE

6.6.1 Co-existence scenarios and cases

The coexistence evaluation is conducted considering the different scenarios listed in Table 6.6.1-1

Table 6.6.1-1: Co-existence scenarios

Deployment scenario No. (Case No.)	Topology	Spectrum
1-1(a/b/c/d)	D1T1-A2- NR UE only outdoor	R2D: DL CW2D and D2R: DL
1-2(a/b/c/d)	D1T1-A2- NR UE indoor	R2D: DL CW2D and D2R: DL
2-1(e/f/c/d)	D1T1-B- NR UE only outdoor	R2D: DL CW2D and D2R: UL
2-2(e/f/c/d)	D1T1-B- NR UE indoor	R2D: DL CW2D and D2R: UL
3 (Optional)(a/b/g/h)	D1T1-A2- NR UE indoor	R2D: UL CW2D and D2R: UL
5-1(e/f/g/h)	D2T2-A2- NR UE only outdoor	R2D: UL CW2D and D2R: UL
5-2(e/f/g/h)	D2T2-A2- NR UE indoor	R2D: UL CW2D and D2R: UL
6-1(Optional)(a/b/g/h)	D2T2-B- NR UE only outdoor	R2D: UL CW2D and D2R: DL
6-2(Optional)(a/b/g/h)	D2T2-B- NR UE indoor	R2D: UL CW2D and D2R: DL

The main co-existence scenario considered for ambient IoT is D1T1 and D2T2. The deployment parameters are described in Table 6.6.2.1-1.

The co-existence evaluation captures cases where NR and A-IoT (Reader/device) are both victim and aggressor networks. It is to evaluate impact on legacy NR networks if A-IoT is introduced in indoor scenario, also to understand impact of the legacy NR network on A-IoT system. The co-existence cases are listed in Table 6.6.1-2.

Table 6.6.1-2: Co-existence cases

Case No.	Aggressor	Victim	note
a	Device	NR DL	D2R
b	NR DL	reader	D2R
c	Reader	NR DL	R2D
d	NR DL	device	R2D
e	Device	NR UL	D2R
f	NR UL	reader	D2R
g	Reader	NR UL	R2D
h	NR UL	device	R2D

6.6.2 Co-existence evaluation assumptions

6.6.2.1 Deployment

Simulation assumptions related to network layout is captured for D1T1 and D2T2 in Table 6.6.2.1-1.

Table 6.6.2.1-1. Deployment parameters for D1T1 and D2T2

Parameter	D1T1	D2T2
Carrier frequency	900 MHz (Band n8)	

BW for NR	10MHz with 15KHz SCS	
BW for A-IOT system	180kHz, for 15kHz SCS	
Waveform (R2D)	OOK waveform generated by OFDM modulator	
Waveform (CW)	Unmodulated single tone	
A-IoT DL power control	No	
A-IoT UL power control	No	
Traffic model	Full buffer	
Frequency reuse	1	
NR BS deployment (outdoor), i.e. scenario 1-1 and 1-2	Hexagonal grid, 19 macro sites, 3 sectors per site with wrap around, 1 A-IOT indoor scenario per sector the minimum 2D distance between macro BS and indoor factory center is set as 100m. Inter-NR BS distance: 750m NR BS height: 25m	
NR BS deployment (indoor), i.e. scenario 2-1 and 2-2	NR indoor BS deployed co-site with A-IoT indoor reader; NR indoor BS ISD as 20m; NR indoor Min BS-UE distance: 0m; NR indoor UE uniformly distributed. NR indoor BS Tx power assumed [24] dBm Antenna pattern of A-IoT reader is reused. The self-interference cancellation of R2D in UL for co-located scenario can be reported by interested company.	
A-IoT reader deployment / Intermediate UE dropping	For D1T1-A2 and D1T1-B: 18 A-IoT readers on a square lattice with spacing D, located D/2 from the walls. L=120m x W=60m; D=20m Reader height = 8 m Room height = 10m 2 A-IoT readers are activated in one drop as baseline. Minimum distance between active readers: 60m as baseline	For D2T2-A2 and D2T2-B: The intermediate UEs selected from the fixed positions. L=120m x W=50m; D=20m Intermediate UE height = 1.5 m Room height = 3m Number of intermediate UE for simulation: 2 UE at one drop. Minimum distance between intermediate UEs: 60m as baseline
CW deployment	For D1T1-A2 and D2T2-A2, CW and A-IoT reader are collocated. For D1T1-B and D2T2-B, CW topology layout is the same as A-IoT reader. For each device, the nearest CW node will be activated during the simulation, and CW node is not co-located with any activated reader	
Device distribution	Device Height= 1.5 m A-IoT devices drop uniformly distributed over the horizontal area Number of A-IoTs = Total area × activated density (1.5 A-IOT devices/m ²) 1 active A-IoT device under one reader at one drop Minimum distance between reader and device is 1m	
NR UE dropping	For NR UE only outdoor, uniformly distributed outdoor. For NR UE indoor, uniformly distributed, Option1: 10% indoor, 90% outdoor, Option2: 100% indoor UE number: - DL active UE: 1 UE per cell - UL active UE: 3 UE per cell	
NR BS Inter-site distance	MCL of 70 dB	
Pathloss model	$PL = PL_b + PL_{tw} + PL_{in} + N(0, \sigma_p^2)$ - $PL_{in} = 0.5 * d_{2D-in}$ where d2D-in is the distance to nearest factory/office boundary on the line between Tx and Rx point. Set maximum value of d2D-in as [25m] as optional	
	PL_b : NLOS and LOS in TR38.901 NR BS – NR UE: Uma Outdoor NR UE – A-IoT reader/ device: Umi Device – A-IoT reader: InF-DH Indoor NR UE – device: InH-Office Indoor NR UE –A-IoT reader: InF-DH	PL_b : NLOS and LOS in TR38.901 NR BS – NR UE: Uma Outdoor NR UE – intermediate UE/ device: Umi Device – Intermediate UE: InH-Office Indoor NR UE – device: InH-Office Indoor NR UE – intermediate UE: InH-Office
O2I penetration loss	High penetration loss as in TR 38.901	

6.6.2.2 NR BS/ A-IoT reader/ intermediate UE/ CW RF characteristics

The NR BS and A-IoT reader are defined for two different antenna configurations as illustrated in Table 6.6.2.2-1 and Table 6.6.2.2-2. Assumptions related to CW RF characteristics relevant for different deployment scenarios are captured in Table 6.6.2.2-4.

Table 6.6.2.2-1: NR BS RF parameters

NR BS Parameter	Values for evaluation
Macro-BS Tx power (dBm)	46
BS antenna gain (dBi) and antenna pattern	Antenna Array Geometry: BS point at fixed beam direction: vertical: $\theta_{\text{tilt}} + 90^\circ$, horizontal: 0, 120, 240 ° Antenna pattern (horizontal) (For 3-sector cell sites with fixed antenna patterns): $A_H(\phi) = -\min \left[12 \left(\frac{\phi}{\phi_{3dB}} \right)^2, A_m \right] \quad \phi_{3dB} = 65 \text{ degrees}, A_m = 25 \text{ dB}$ Antenna pattern (vertical) (For 3-sector cell sites with fixed antenna patterns): $A_V(\theta) = -\min \left[12 \left(\frac{\theta - 90^\circ - \theta_{\text{etilt}}}{\theta_{3dB}} \right)^2, SLA_v \right] \quad \theta_{3dB} = 10 \text{ degrees}, SLA_v = 25 \text{ dB}, \theta_{\text{etilt}} = 9 \text{ degrees}$ Combining method in 3D antenna pattern: $A(\varphi, \theta) = -\min \{ -[A_H(\varphi) + A_V(\theta)], A_m \}$ BS antenna gain (dBi) (including feeder loss): 15
Height of macro NR BS (m)	25
NR Macro-BS Noise Figure (dB)	5
Network location	outdoor
ACLR	Option 1: 30dBc, Option 2: 17dBc

Table 6.6.2.2-2: A-IoT reader RF parameters

A-IoT reader parameters	Values for evaluation
A-IoT reader total Tx power	33 dBm
A-IoT reader receiver Noise Figure (dB)	10
A-IoT reader antenna gain (dBi)	6
ACLR	ACLR of legacy NB -IoT gNB (i.e. ACLR1:40dB, ACLR2:50dB)
ACS	Same as legacy NR BS
Antenna configuration	Antenna Array Geometry: equals to omni-directional antenna pattern in GCG in horizontal Antenna pattern (horizontal): $A_H(\phi) = -\min \left[12 \left(\frac{\phi}{\phi_{3dB}} \right)^2, A_m \right], \quad \phi_{3dB} = 90^\circ, A_m = 15 \text{ dB}$ Antenna pattern (horizontal): $A_V(\theta) = -\min \left[12 \left(\frac{\theta - 90^\circ}{\theta_{3dB}} \right)^2, SLA_v \right], \quad \theta_{3dB} = 90^\circ, SLA_v = 15 \text{ dB}$ Combining method in 3D antenna pattern: $A(\varphi, \theta) = -\min \{ -[A_H(\varphi) + A_V(\theta)], A_m \}$ BS antenna gain (dBi) (including feeder loss): 6

Table 6.6.2.2-3: Intermediate UE RF parameters

intermediate UE parameters	Values for evaluation
Intermediate UE total Tx power (dBm)	Baseline: 23 Optional: Tx power control is allowed
Gain of antenna intermediate UE (dBi)	0
Intermediate UE receiver Noise Figure (dB)	9
Antenna configuration	Omni directional antenna

Table 6.6.2.2-4: CW RF parameters

CW parameters	D1T1	D2T2
Tx power (dBm)	If UL spectrum is used, UE Tx power is assumed, i.e. 23dB If DL spectrum is used, A-IoT reader Tx power is assumed, i.e. 33 dBm	Intermediate UE Tx power is assumed.
Antenna gain	Same as A-IoT reader	Same as intermediate UE

6.6.2.3 NR UE/ A-IoT device RF characteristics

Assumptions relevant for modelling the NR UE and A-IoT device RF characteristics are captured in Table 6.6.2.3-1 and Table 6.6.2.3-2.

Table 6.6.2.3-1: NR UE RF parameters

NR UE Parameter	Values for evaluation
UE TX power in dBm	-40 to 23
NR UE Antenna gain (dBi)	0
Height of UE antenna (m)	1.5
NR UE ACLR (dB)	30
NR UE Noise Figure (dB)	9
Antenna configuration	Omni direction antenna

Table 6.6.2.3-2: A-IoT device RF parameters

A-IoT device parameters	Device 1	Device 2a
A-IoT device effective antenna gain per Tx or Rx branch (dBi)	0	0
A-IoT device reflection (backscatter) loss (dB)	OOK: -6 dB	OOK: -6 dB
A-IoT device power gain of reflection amplifier (dB)	N/A	10(M)
A-IoT Device receiver sensitivity (dBm) Use this value to determine whether device can camp on the cell.	-36	-45
A-IoT device noise figure (dB)	24	20
Guard band	0PRB	0PRB

6.6.3 Co-existence simulation methodology

6.6.3.1 Coexistence evaluation methodology

The coexistence evaluation methodology can be summarized as:

- 1) Aggressor and victim network are generated. NR UEs and A-IoT devices are distributed as described by parameter assumptions.
- 2) UEs are associated to BS based on coupling loss, and A-IoT devices are associated to A-IoT reader or intermediate UE based on coupling.

NOTE: For A-IoT devices are associated to A-IoT reader or intermediate UE, there are two ways applied by companies:

- a) The device access to the reader is randomly selected from candidate devices whose R2D receiving power/coupling from one reader is higher than the other. The receiving power of device meets the device receiver sensitivity.
 - b) The device can access the reader with best receiving power among of 12 or 18 total readers in one building. The receiving power of device meets the device receiver sensitivity.
- 3) Once association is done, round robin scheduling is used.
 - 4) For inter-system interference (between A-IoT and NR):
 - If SINR degradation is smaller than and equal to 1 dB, it can be considered that inter-system interference is negligible.
 - Note: For SINR degradation, SINR refers to the 5% and 50% CDF SINR

6.6.3.2 SINR definition

For D1T1-A1and -B /D2T2-A1and -B

SINR definition for D2R:

- SINR includes CW interference is used as the baseline reference for co-existence evaluation for CW reader.

$$\text{baseline SINR} = \frac{\text{received wanted signal power}}{(\text{noise} + \text{intra_system interference}) + (\text{residual CW interference after cancellation})}$$

- SINR is calculated as total power ratio:

$$\text{SINR} = \frac{\text{received wanted signal power}}{(\text{noise} + \text{intra_system interference} + \text{inter_system interference}) + (\text{residual CW interference after cancellation})}$$

For the CW outside scenarios (D1T1-B and D2T2-B), the impact to SINR due to phase noise are also evaluated. The baseline D2R SNR for CW outside can be calculated by

$$\text{baseline SINR for CW outside} = \frac{\text{received wanted signal power}}{(\text{noise} + \text{intra_system interference} + \text{phase noise})_{\text{within total receiver bandwidth}} + (\text{residual CW interference after cancellation})_{\text{in linear scale}}}$$

Phase noise can be modelled as XdBc/Hz, e.g., -112dBc/Hz compared to the received power for D2R.

The noise and intra-system interference are within total receiver bandwidth, and the residual CW interference after cancellation is in linear scale.

SINR definition for R2D:

- Signal power of device to the noise and interference within 10MHz is baseline assumption
- Consider 180KHz noise and interference bandwidth after BB LPF as optional

6.6.3.3 Coupling loss

The Coupling Loss (CL) is defined as the loss in signal between NR BS-to- NR UE, A-IoT reader -to- A-IoT device and intermediate UE -to- A-IoT device. CL is defined as the loss including propagation loss and antenna gains.

6.6.3.4 Impact from CW to NR

In 6.6.3.2, the SINR definition includes the impact from CW to D2R which is treat as intra-system interference. However, Whether and how the CW will influence the NR system also needs to be studied. In [156], the impact of CW signal under different assumptions are investigated and the evaluation results are shown in figure 6.6.3.4-1.

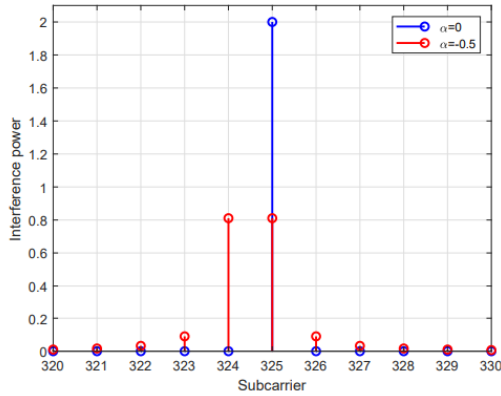


Fig. 2. Interference power distribution after the FFT when the OFDM system is subjected to one CW signal (closest to subcarrier $m = 325$), with separation $\alpha = 0$ and $\alpha = -0.5$ from the subcarrier 325.

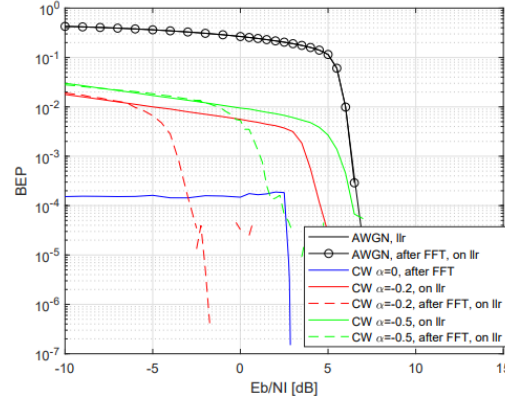


Fig. 6. Resulting BEP as a function of E_b/N_1 when limiters are used to mitigate interference consisting of AWGN or one CW signal. Limiters are placed after FFT or on the demodulator soft output LLR.

Figure 6.6.3.4-1: Impact from CW to NR system [156]

The results shows that when the CW is orthogonal with NR subcarrier, i.e., $\alpha=0$, the influence of interference is much lower and also easier to be removed after FFT, only if the power of CW interference does not exceed the dynamic range of receiver.

Based on the analysis above, the impact from CW to NR is not modelled in the co-existence evaluation for simplicity.

6.6.4 Co-existence evaluation results

The simulation results submitted for the ambient IoT and NR co-existence study from all contributing companies in RAN4 were summarized in [155] attached to this report.

6.6.5 Summary of co-existence evaluation

The clause captures the co-existence simulation results for ambient IoT categorized by scenarios and cases. The conclusions below are derived from the coexistence results.

6.6.5.1 Scenario 1-1: D1T1-A2- NR UE only outdoors (R2D/ CW2D/ D2R: DL)

Scenario 1-1 considers D1T1 topology, all the NR UE outdoors and Readers/CW collocated, with DL spectrum applying to reader-to-device link. The conclusions of each cases are listed per scenario in Table 6.5.6.1-1.

Table 6.6.5.1-1: Scenario 1-1 co-existence conclusions

Case No.	Aggressor	Victim	Co-existence conclusion
a	Device	NR DL	Negligible NR DL throughput degradation for both average throughput and cell edge throughput is observed
b	NR DL	reader	SINR degradation is less than 1dB under the assumption of 30dBc inband leakage for legacy NR gNB. Inter-system interference is negligible. 2 companies observed some SINR degradation under the assumption of 17dBc inband leakage for legacy NR gNB.
c	Reader	NR DL	Negligible NR DL throughput degradation for both average throughput and cell edge throughput is observed
d	NR DL	device	SINR degradation is less than 1dB. Inter-system interference is negligible.

6.6.5.2 Scenario 1-2: D1T1-A2- NR UE indoor (R2D/ CW2D/ D2R: DL)

Scenario 1-2 considers D1T1 topology, 10% of the NR UE indoors and Reader/CW collocated, with DL spectrum applying to reader-to-device link. The conclusions of each cases are listed per scenario in Table 6.5.6.2-1.

Table 6.6.5.2-1: Scenario 1-2 co-existence conclusions

Case No.	Aggressor	Victim	Co-existence conclusion
a	Device	NR DL	Negligible NR DL throughput degradation for both average throughput and cell edge throughput is observed
b	NR DL	reader	SINR degradation is less than 1dB under the assumption of 30dBc inband leakage for legacy NR gNB. Inter-system interference is negligible. 1 company observed some SINR degradation under the assumption of 17dBc inband leakage for legacy NR gNB.
c	Reader	NR DL	Under the assumption of 10% NR UE indoor, NR DL throughput degradation is observed. Under the assumption of 100% NR UE indoor, significant NR DL throughput degradation is observed.
d	NR DL	device	SINR degradation is less than 1dB. Inter-system interference is negligible.

6.6.5.3 Scenario 2-1: D1T1-B- NR UE only outdoor (R2D: DL, CW2D/ D2R: UL)

Scenario 2-1 considers D1T1 topology, all the NR UE outdoors and Readers/CW non-collocated, with DL spectrum applying to reader-to-device link. The conclusions of each cases are listed per scenario in Table 6.5.6.3-1.

Table 6.6.5.3-1: Scenario 2-1 co-existence conclusions

Case No.	Aggressor	Victim	Co-existence conclusion
e	Device	NR UL	Negligible NR UL throughput degradation for both average throughput and cell edge throughput is observed.
f	NR UL	reader	SINR degradation is less than 1dB. Inter-system interference is negligible.
c	Reader	NR DL	Negligible NR DL throughput degradation for both average throughput and cell edge throughput is observed.
d	NR DL	device	SINR degradation is less than 1dB. Inter-system interference is negligible.

6.6.5.4 Scenario 2-2: D1T1-B- NR UE indoor (R2D: DL, CW2D/ D2R: UL)

Scenario 2-2 considers D1T1 topology, 10% of the NR UE indoors and Readers/CW non-collocated, with DL spectrum applying to reader-to-device link. The conclusions of each cases are listed per scenario in Table 6.5.6.4-1.

Table 6.6.5.4-1: Scenario 2-2 co-existence conclusions

Case No.	Aggressor	Victim	Co-existence conclusion
e	Device	NR UL	Negligible NR UL throughput degradation for both average throughput and cell edge throughput is observed.
f	NR UL	reader	Under the assumption of 10% and 100% NR UE indoor, significant SINR degradation is observed
c	Reader	NR DL	Under the assumption of 10% and 100% NR UE indoor, NR DL throughput degradation is observed.
d	NR DL	device	SINR degradation is less than 1dB. Inter-system interference is negligible.

6.6.5.5 Scenario 5-1: D2T2-A2- NR UE only outdoor (R2D/ CW2D/ D2R: UL)

Scenario 5-1 considers D2T2 topology, all the NR UE outdoors and Readers/CW collocated, with UL spectrum applying to reader-to-device link. The conclusions of each cases are listed per scenario in Table 6.5.6.5-1.

Table 6.6.5.5-1: Scenario 5-1 co-existence conclusions

Case No.	Aggressor	Victim	Co-existence conclusion
e	Device	NR UL	Negligible NR UL throughput degradation for both average throughput and cell edge throughput is observed
f	NR UL	reader	SINR degradation is less than 1dB. Inter-system interference is negligible.
g	Reader	NR UL	NR UL throughput degradation is observed
h	NR UL	device	SINR degradation is less than 1dB. Inter-system interference is negligible.

6.6.5.6 Scenario 5-2: D2T2-A2- NR UE indoor (R2D/ CW2D/ D2R: UL)

Scenario 5-2 considers D2T2 topology, 10% of the NR UE indoors and Readers/CW collocated, with UL spectrum applying to reader-to-device link. The conclusions of each cases are listed per scenario in Table 6.5.6.6-1.

Table 6.6.5.6-1: Scenario 5-2 co-existence conclusions

Case No.	Aggressor	Victim	Co-existence conclusion
e	Device	NR UL	Negligible NR UL throughput degradation for both average throughput and cell edge throughput is observed
f	NR UL	reader	Significant SINR degradation is observed for both 10% NR indoor UE and 100% NR indoor UE assumption.
g	Reader	NR UL	NR UL throughput degradation is observed
h	NR UL	device	Under the assumption of 10% NR UE indoor, SINR degradation is larger than 1dB. Under the assumption of 100% NR UE indoor, significant SINR degradation is observed.

6.6.5.7 Scenario 6-1: D2T2-B- NR UE only outdoor (R2D: UL, CW2D/ D2R: DL)

Scenario 6-1 considers D2T2 topology, all the NR UE outdoors and Readers/CW non-collocated, with UL spectrum applying to reader-to-device link. The conclusions of each cases are listed per scenario in Table 6.5.6.7-1.

Table 6.6.5.7-1: Scenario 6-1 co-existence conclusions

Case No.	Aggressor	Victim	Co-existence conclusion
a	Device	NR DL	Negligible NR DL throughput degradation for both average throughput and cell edge throughput is observed.
b	NR DL	reader	SINR degradation is less than 1dB under the assumption of 30dBc inband leakage for legacy NR gNB. Inter-system interference is negligible.
g	Reader	NR UL	NR UL throughput degradation is observed.
h	NR UL	device	SINR degradation is less than 1dB. Inter-system interference is negligible.

6.6.5.8 Scenario 6-2: D2T2-B- NR UE indoor (R2D: UL, CW2D/ D2R: DL)

Scenario 6-2 considers D2T2 topology, 10% of the NR UE indoors and Readers/CW non-collocated, with UL spectrum applying to reader-to-device link. The conclusions of each cases are listed per scenario in Table 6.5.6.8-1.

Table 6.6.5.8-1: Scenario 6-2 co-existence conclusions

Case No.	Aggressor	Victim	Co-existence conclusion
a	Device	NR DL	Negligible NR DL throughput degradation for both average throughput and cell edge throughput is observed
b	NR DL	reader	SINR degradation is less than 1dB under the assumption of 30dBc inband leakage for legacy NR gNB. Inter-system interference is negligible. 1 company observed some SINR degradation under the assumption of 17dBc inband leakage for legacy NR gNB
g	Reader	NR UL	NR UL throughput degradation is observed.
h	NR UL	device	Under the assumption of 10% NR UE indoor, SINR degradation is larger than 1dB. Under the assumption of 100% NR UE indoor, significant SINR degradation is observed.

6.6.5.9 Scenario 3 (Optional): D1T1-A2- NR UE indoor (R2D: UL, CW2D/ D2R: DL)

Scenario 3 optionally considers D1T1 topology, 10% of the NR UE indoors and Readers/CW collocated, with UL spectrum applying to reader-to-device link. The conclusions of each cases are listed per scenario in Table 6.5.6.9-1.

Table 6.6.5.9-1: Scenario 3 co-existence conclusions

Case No.	Aggressor	Victim	Co-existence conclusion
a	Device	NR DL	Negligible NR DL throughput degradation for both average throughput and cell edge throughput is observed
b	NR DL	reader	1 company provides simulation results based on assumption of 30dBc NR gNB inband leakage and observed less than 1dB SINR degradation. 1 company provides simulation results based on assumption of 17dBc NR gNB inband leakage and observed larger than 1 dB SINR degradation.
g	Reader	NR UL	NR UL throughput degradation is observed.
h	NR UL	device	1 company provides simulation results based on assumption of 10% NR UE indoor and observed less than 1dB SINR degradation 1 company provides simulation results based on assumption of 100% NR UE indoor and observed larger than 1 dB SINR degradation.

6.6.5.10 General remarks on coexistence findings

For D1T1 deployment scenario 1-1 (R2D, CW2D/D2R: DL) and 2-1(R2D: DL, CW2D/D2R: UL), which has no legacy NR UE indoor. Negligible performance degradation for NR system is observed. Less than 1 dB SINR degradation due to inter-system interference from NR to ambient IOT system is observed. Co-existence between ambient IOT and NR is feasible based on performance metric of NR throughput degradation and ambient IOT SINR degradation, no additional coexistence measures are required. Ambient IOT intra-system performance evaluated by link level simulation is also one of the key aspects affecting ambient IOT system performance, which will have impact to the outage of ambient IOT system.

For the deployment scenarios with legacy NR UE indoor (scenario 1-2, 2-2, 5-2, 6-2), which either significant NR throughput degradation or significant SINR degradation for ambient IOT system is observed, some existing mechanisms based on gNB implementation can be considered to avoid/mitigate the interference. For example, for RRC idle mode, NR UEs can camp on a frequency different from ambient IOT system based on frequency priority configured by network; for RRC connected mode, network can handover NR UEs to a different cell (e.g. indoor gNB on a different frequency) based on existing measurement report triggering and event. Besides, other interference management solutions to reduce the inter-system interference from ambient IOT to proximity NR UE and vice versa can also be considered.

For the D2T2 deployment scenarios (scenario 5-1, 5-2, 6-1, 6-2), NR UL throughput degradation is observed. The mechanisms to mitigate the interference can be considered. For example, enhanced emission requirements for intermediate UE or power control for intermediate UE [157], [158].

Additionally, RAN4 has also considered using guard band to mitigate the interference between ambient system and NR system.

6.7 RF requirements study

6.7.1 System parameters

6.7.1.1 Operating band

According to the SID (RP-240826), Ambient IoT uses FR1 licensed spectrum in FDD. NR band n8 can be used as an example band.

Table 6.7.1.1-1 summarized the A-IoT system spectrum usage from co-existence scenario. It can be observed that The spectrum usage for A-IoT operation can be flexible, and it depends on several factors, such as whether the reader is A-IoT BS or intermediate UE, and whether CW transmission is deployed inside or outside topology. For the case which different spectrum is used for D2R and R2D, the FDD band definition can be kept. As CW transmission may interfere the R2D reception, the R2D and D2R should be TDM and therefore, for FDD band definition, it should be half duplex mode.

Table 6.7.1.1-1: Spectrum usage for Ambient IoT

	Spectrum for R2D	Spectrum for D2R	Spectrum for CW2D transmission
A-IoT BS	DL spectrum	DL spectrum	DL spectrum
	DL spectrum	UL spectrum	UL spectrum
	UL spectrum	DL spectrum	DL spectrum
	UL spectrum	UL spectrum	UL spectrum
Intermediate UE as reader	UL spectrum	UL spectrum	UL spectrum
	UL spectrum	DL spectrum	DL spectrum

6.7.1.2 Channel bandwidth

The R2D waveform is agreed to be OFDM-based waveform, therefore the channel bandwidth definition can follow the channel bandwidth definition in NR. The R2D channel bandwidth can be defined as one or multiple PRBs with guardband if necessary. The value for this can be determined during the WI phase.

6.7.1.3 Channel arrangement

The channel spacing may be specified for D2R FDM operation for device 1/2a.

For Ambient IoT BS and intermediate UE, channel spacing and channel raster may be needed for deployment purpose, for example, to co-deploy with existing NR network.

6.7.2 Ambient IoT BS

6.7.2.1 General

For Release 19, the RF requirement discussion focus on BS type 1-C for Ambient IoT indoor scenarios. The RF requirement for ambient IoT BS is studied based on the assumption that ambient IoT BS only transmits and receives ambient IoT signal.

Current the coexisting scope focuses on device 1 and device 2a, it includes the scenarios and cases which describes the different factors to be considered for the ambient IoT BS. These factors are listed in Table 6.7.2.1-1 below. CW signal can be transmitted by A-IoT BS if manufacture declares this capability. CW can be transmitted by a separate node, therefore CW related RF requirements can be specified separately for ambient IoT BS or a separate CW node.

Table 6.7.2.1-1: Factors impacting the ambient IoT BS operation to device 1 and device 2a

	Spectrum for R2D	Spectrum for D2R	Spectrum for CW transmission	A-IoT BS transmitting the CW signal
A-IoT BS	DL spectrum	DL spectrum	DL spectrum	Yes
	DL spectrum	DL spectrum	DL spectrum	No
	DL spectrum	UL spectrum	UL spectrum	Yes
	DL spectrum	UL spectrum	UL spectrum	No

For the scenarios only considering device 1 and device 2a, simultaneously CW and R2D transmission will interfere the R2D reception, as such the R2D and D2R transmission should be operated in TDM for device 1 and device 2a.

6.7.2.2 Ambient IoT BS TX requirements

6.7.2.2.1 Base station output power

Base station output power requirement is potentially needed, RAN4 agrees to use current requirements of Micro NR BS type 1-C as the starting point.

6.7.2.2.2 Output power dynamic

Output power dynamic requirement usually includes RE power control dynamic range requirement and total power dynamic range requirement. Power boosting requirement is not needed in R19 with the assumption that ambient IoT BS only transmits and receives AIOT signal.

6.7.2.2.3 Transmit ON/OFF power

Regarding the transmit on/off requirement, it usually includes of transmitter off power requirement and transmitter transient period requirement, RAN4 agrees to further discuss in the WI phase.

6.7.2.2.4 Transmit signal quality

Frequency error and modulation quality (EVM) are potentially needed. TAE is only applicable for BS transmitting from multiple antennas via MIMO, CA, or combination of them. A-IoT BS doesn't support these features, so TAE requirement is not needed.

6.7.2.2.5 Unwanted emissions

Unwanted emission requirement should be defined because it is the requirement to limit the emission to other channel or system to make this equipment can co-exist with other equipment. OBW requirement, ACLR requirement, OBUE requirement, transmitter spurious emission requirement is potentially needed.

6.7.2.3 Ambient IoT BS Rx requirements

6.7.2.3.1 Reference sensitivity level

Regarding the reference sensitivity level for A-IoT BS, RAN4 reached the following consensus:

For BS type 1-C,

$$P_{\text{REFSENS}}(\text{dBm}) = -174\text{dBm} + 10 \times \log_{10}(\text{BW}) + N_F + I_M + \text{SNR} + [\text{desens target}]$$

- BW is the transmission bandwidth for D2R.
- N_F is the noise figure of A-IoT BS, equal to 10 dB for Medium Range BS.
- I_M is the implementation margin, equal to 2dB.
- SNR is the SNR value for D2R decoding, the detailed metrics would be discussed in the WI phase.
- Desens target is considered for CW signal interference cancellation in D1T1 scenario. The desens level for CW signal cancellation might be different for D1T1-A1/D1T1-A2 and D1T1-B. For D1T1-A2, CW signal interference is coming from self-interference and also potential CW interference from other nodes transmitting CW signal. For D1T1-A1, the CW signal interference is coming from other nodes transmitting CW inside the topology. For D1T1-B, the CW signal interference is coming from CW nodes transmitting outside topology.

6.7.2.3.2 Dynamic range

For NR and LTE, receiver dynamic range requirement is to ensure that the base station can receive high throughput also in the presence of increased interference levels. The interference level is derived from simulation of co-channel interference from the same operator. It measures the effects of base station receiver impairments. For Ambient IoT indoor deployment, the co-channel interference comes from co-channel device transmission and CW interference. The necessity and the value can be studied at WI phase.

6.7.2.3.3 In-band selectivity and blocking

Regarding ACS requirement and in-band blocking requirement, RAN4 reached the following consensus:

ACS requirement and the interference level shall be determined by RAN4 co-existence study and wanted signal configuration and its corresponding configuration could be postponed to WI phase.

In-band blocking requirement and the interference level shall be determined by RAN4 co-existence study, the wanted signal configuration and its corresponding configuration could be postponed to WI phase.

6.7.2.3.4 Out-of-band blocking

Regarding Out-of-band blocking requirement, the existing out-of-band blocking requirement may be still applicable for A-IoT BS except for the wanted signal level.

6.7.2.3.5 Receiver spurious emissions

Regarding the receiver spurious emission requirement, this requirement might differ as following:

- For D1T1-A1 and D1T1-B, the receiver spurious emission requirement is still needed due to is half duplex operation mode.
- For D1TA-A2, due to the simultaneous transmission of CW signal and D2R reception, if the spurious emission requirement of CW signal is exceeding the receiver spurious emission requirement, then the receiver spurious

requirement is not needed similar as FDD BS otherwise this requirement is still needed to regulate the receiver RF performance.

6.7.2.3.6 Receiver intermodulation

Regarding the general intermodulation requirement, it is agreed to postpone to WI phase due to the potential relationship between D2R operation assumption e.g. single tone or multiple tone D2R assumption.

Regarding the narrowband intermodulation requirement, it is potentially necessary due to the potential Narrow band blocking at the operating frequency range, however its detailed requirement could be postponed to WI phase.

6.7.2.3.7 In-channel selectivity

Regarding the receiver in-channel selectivity requirement, this is agreed to be discussed in WI phase.

6.7.3 Intermediate node (UE)

6.7.3.1 General

The intermediate UE, under NW control, can send Ambient IoT data and/or signalling to the device, and receive the backscatter signal from the device. The intermediate UE can also operate as a legacy NR UE. For simplicity, non-concurrent operation of Ambient IoT and NR for intermediate UE is assumed, and focus on the case where the intermediate UE acts as an A-IoT Reader.

Considering the intermediate UE need to support both A-IoT and NR operation and if the requirements in NR and A-IoT is the same, it is possible to verify only on the NR side.

6.7.3.2 Intermediate UE TX requirements

6.7.3.2.1 Maximum output power

The goal of the maximum output power is to achieve the DL coverage. To reuse of the legacy NR UE hardware, PC3 NR UE can serve as a baseline for intermediate UE using FDD UL spectrum as R2D spectrum. High power UE is not precluded in the future release for better coverage.

6.7.3.2.2 Output power dynamics

Power control is not designed for A-IoT R2D link currently, and intermediate UE transmit at maximum output power for better coverage, making a minimum output power and power control accuracy requirements unnecessary. However, during the co-existence study, some companies propose that implementing the power control can help intermediate UE to mitigate the interference to NR system, so this requirement will be further discussed in WI phase depending on RAN1 specification.

Transmit OFF power is defined to ensure the UE will not affect the reception of other UE or BS when the transmitter is inactive. Since for A-IoT standalone operation, the HD-FDD mode could be one option for outside topology and the requirement can be further discussed in WI phase.

For the transmit ON/OFF time mask, refer to the RFID specification, a settling time requirement is used to assess signal ripple characteristics. It can be determined whether or not this settling time is necessary in WI phase. For the intermediate node that has capability to transmit CW signal, the transient time between R2D to CW transmission also needs to be further discussed.

6.7.3.2.3 Transmitted signal quality

Frequency error mainly depends on the timing and synchronization performance and PLL performance within transceiver chain. A-IoT intermediate UE is expected to reuse the existing timing and synchronization network directly. Meanwhile, similar PLL performance of A-IoT intermediate UE as NR UE is also expected as they are operating at the same frequency range. With the above considerations, frequency error requirement of NR UE can be considered reusing for A-IoT intermediate UE.

For legacy NR UE, transmit modulation quality includes EVM, carrier leakage and In-band emissions. These requirements may still be needed as R2D waveform is based on OFDM transmitter.

The carrier leakage requirement intends to reduce the DC offset which will distort the constellation for QAM signal. For OOK signal, the DC offset will increase the amplitude of waveform and the carrier leakage should also be low enough to ensure the received signal still within the receiver dynamic range. Whether and how to define this requirement can be further discussed in WI phase

The in-band emission in NR can limit the interference between UEs, but when intermediate UE perform as A-IoT reader, this requirement is unnecessary if all RB within the channel bandwidth always allocated for A-IoT transmission.

6.7.3.2.4 Output RF spectrum emissions

Output RF spectrum emissions for legacy NR UE includes occupied bandwidth, out of band emission (ACLR), spurious emissions and transmit intermodulation.

The occupied bandwidth is the width of a frequency band such that, below the lower and above the upper frequency limits, the mean powers emitted are each equal to a specified percentage $\beta/2$ ($\approx 0.5\%$) of the total mean transmitted power. Occupied bandwidth: Potentially needed for intermediate UE. The specific value is dependent on the physical layer design of R2D, which has not yet been finalized at this stage.

ACLR can be defined based on the co-existence study. During SI co-existence evaluation, the IBE of legacy NR is used as emission leakage model, and performance degradation is observed in the results, so the ACLR better than IBE may be required, and the details can be determined in WI phase.

Existing spurious emission requirement for legacy NR UE can be applicable, since this is coming from the ITU-R recommendation SM.329.

The transmit intermodulation in NR is the capability of the transmitter to inhibit the generation of signals in its non-linear elements caused by presence of the wanted signal and an interfering signal reaching the transmitter via the antenna. this requirement can be further discussed in WI.

6.7.3.3 CW for D2T2

6.7.3.3.1 Output power/Output power dynamic range

The output power of CW is needed to ensure the link quality for D2R. For the intermediate node that has capability to transmit CW signal, the MPR/A-MPR needs to be studied as the CW waveform defined with unmodulated sinusoid waveform is new waveform.

Considering the CW and R2D cannot transmitting simultaneously, the output power dynamic related requirements, e.g., transmit OFF power, can be further discussed in WI phase.

6.7.3.3.2 Transmit signal quality

Since the D2R signal transmission for device1/2a closely rely on the CW, and the signal quality may need to be considered e.g., frequency error which needs more discussion.

6.7.3.3.3 RF spectrum emission/Spurious emissions/Phase noise

The emission related requirement needs to be specified. Meanwhile, close-in phase noise may drown the D2R signal. CW cancellation can provide help in the CW inside topology scenario, but the phase noise might play a critical role for CW outside topology. In addition, the regulation is also needed to be considered. Therefore, the details of how to specify RF spectrum emission, and whether and how to specify spurious emission and phase noise can be further discussed in WI phase.

6.7.3.4 Intermediate UE RX requirements

6.7.3.4.1 Reference sensitivity

The reference sensitivity power level REFSENS is the minimum mean power applied to each one of the intermediate UE antenna ports at certain performance metric. Distinguished from the legacy NR UE, Ambient IoT intermediate UE may suffer from residual noise from CW interference, So it should be considered when defining receiver sensitivity. Reference sensitivity requirement for intermediate UE could be derived based on the following equation:

$$P_{\text{REFSENS}}(\text{dBm}) = -174 \text{ dBm} + 10 \cdot \log_{10} \text{BW} + \text{NF} + \text{IM} + I_{\text{CW}} + \text{SNR}$$

Where:

- BW is the maximum transmission bandwidth
- NF is noise figure of intermediate node
- IM is the implementation margin, equal to 2dB.
- For inside topology, I_{CW} is IoT due the residual CW interference, For outside topology, I_{CW} would be less and in this case, receiver to decode the backscattered signal without desensitization of receiver caused by CW leakage. value will be defined in the WI.
- SNR is the SNR value for which we reach 10% BLER target. Each company provided simulation results, and average will be done for each BW.
- The REFSENS requirements of an intermediate UE may be affected when it simultaneously transmits uu UL and A-IoT signals.

For inside topology, the receiver needs to tolerate the high leakage CW power from the transmitter when intermediate node can transmit CW signal and desensitization of receiver may be allowed with a CW cancellation function enabled in such case. For outside topology, the received CW power would be less and in this case, receiver to decode the backscattered signal without desensitization of receiver caused by CW leakage. Therefore, for REFSENS requirements, different requirements can be defined, corresponding to inside topology and outside topology respectively.

6.7.3.4.2 Maximum input power

Maximum input level for legacy NR UE is defined as the maximum mean power received at the UE antenna port. Considering the intermediate UE may need to implement the receiver at UL spectrum when CW/D2R are also located in UL, which is different from legacy NR UE, this requirement can be further discussed in WI phase.

6.7.3.4.3 Adjacent channel selectivity

Adjacent channel selectivity (ACS) is a measure of a receiver's ability to receive an NR signal at its assigned channel frequency in the presence of an adjacent channel signal at a given frequency offset from the center frequency of the assigned channel.

ACS requirement and the interference level depend on co-existence study.

6.7.3.4.4 Blocking characteristics

The blocking characteristic is a measure of the receiver's ability to receive a wanted signal at its assigned channel frequency in the presence of an unwanted interferer on frequencies other than those of the spurious response or the adjacent channels, without this unwanted input signal causing a degradation of the performance of the receiver beyond a specified limit.

In-band blocking requirement and the interference level depend on co-existence study.

The existing out-of-band blocking requirement for legacy NR UE is applicable for intermediate UE.

Narrow band blocking can be discussed in WI phase.

6.7.3.4.5 Spurious response

Spurious response is a measure of the ability of the receiver to receive a wanted signal on its assigned channel frequency without exceeding a given degradation due to the presence of an unwanted CW interfering signal at any other frequency for which a response is obtained. The existing spurious emission requirement for legacy NR UE is applicable for intermediate UE.

6.7.3.4.6 Receiver intermodulation

Intermodulation response rejection is a measure of the receiver's capability to receive a wanted signal on its assigned channel frequency in the presence of two or more interfering signals with a specific frequency relationship to the wanted signal. It depends on the RAN1 design. If multiple-tone carrier wave is supported, intermodulation requirements potentially needed.

6.7.3.4.7 Spurious emissions

The spurious emissions power is the power of emissions generated or amplified in a receiver that appear at the UE antenna connector. The legacy NR UE receiver spurious emission requirement could be applicable.

6.7.4 Ambient IoT Device

6.7.4.1 Device 1

6.7.4.1.1 TX

Transmitter power

For device 1, D2R backscattering is transmitted on a carrier wave provided externally, backscattering power and power backscattering loss are potentially needed to be defined to ensure the coverage range between reader and device. Maximum output power and other transmit output power requirements are potentially not needed.

The minimum backscattered modulated power can be specified for device 1. The backscatter power is the amount of power reflected back from the device when a successful command sequence is completed at the device at a specific sensitivity level.

The backscatter loss due to the modulation can be defined as scattering cross-section or modulation efficiency in equation (4). From equation (4), it can be observed that the modulation efficiency is defined as the ratio of the power radiated by the device to the power density incident on its antenna.

Assume the power density of the incident radio wave on the device antenna is S

$$S = \frac{P_t \cdot G_t}{4\pi r^2} \quad (1)$$

Where the P_t is transmission power of a CW signal and G_t is the transmitter antenna gain, r is the distance between test equipment to the device.

The received power on device antenna is P_a

$$P_a = \frac{\lambda^2 \cdot G_d}{4\pi} \cdot S \quad (2)$$

Where the λ is the radio wavelength and G_d is the device antenna gain.

The power backscattered back in the direction of the transmitter is $P_{backscatter}$

$$P_{backscatter} = K \cdot P_a \cdot G_d = \frac{4 \cdot R_a^2}{|Z_a + Z_{c1,c2}|^2} \cdot \frac{\lambda^2 \cdot G_d^2}{4\pi} \cdot S \quad (3)$$

Where the $Z_a = R_a + jX_a$ is complex antenna impedance and $Z_{c1,c2} = R_{c1,c2} + jX_{c1,c2}$ is complex chip load in different modulation status.

The backscatter loss or cross section of the device 1 can be defined as

$$\sigma = \frac{P_{backscatter}}{S} = \frac{R_a^2}{|Z_a + Z_{c1,c2}|^2} \cdot \frac{\lambda^2 \cdot G_d^2}{\pi} \quad (4)$$

The received backscattered signal at the test device:

$$P_{received} = \frac{P_{backscatter} \cdot G_r}{4\pi r^2} = \frac{P_t \cdot G_t}{4\pi r^2} \cdot \frac{G_r}{4\pi r^2} \cdot \sigma \quad (5)$$

Output power dynamics

Output power dynamics includes minimum output power, transmit OFF power, transmit ON/OFF time mask and power control.

Power control is not designed for A-IoT device 1 D2R link, and the transmit power of device 1 comes from CW node, making Output power dynamics requirements unnecessary.

Transmitted signal quality

Transmitted signal quality includes frequency error, carrier leakage, in-band emission and EVM.

As D2R backscattering is transmitted on a carrier wave provided externally, the requirements for frequency error, carrier leakage and in-band emission are potentially unnecessary.

The waveform for D2R could be OOK or BPSK. OOK signal is not mapped by the legacy constellation. EVM may not be suitable for D2R OOK waveform, and whether EVM testing is feasible for BPSK waveform needs to be discussed. Requirement to guarantee the modulation quality is potentially needed. How to define modulation quality requirements will be discussed in WI.

Output RF spectrum emissions

Output RF spectrum emissions includes occupied bandwidth, out of band emission (SEM, ACLR), spurious emissions and transmit intermodulation.

Occupied bandwidth is defined as the bandwidth containing 99 % of the total integrated mean power of the transmitted spectrum on the assigned channel. Occupied bandwidth requirement is potentially needed. The specific value is dependent on the physical layer design of D2R, which has not yet been finalized at this stage.

SEM and spurious emissions are related to co-existence study. Whether to define these requirements need to be discussed in WI. Based on the co-existence study, ACLR is not needed.

The transmit intermodulation performance is a measure of the capability of the transmitter to inhibit the generation of signals in its non-linear elements caused by presence of the wanted signal and an interfering signal reaching the transmitter via the antenna. However, device1 is low complexity and low cost, and has no active components at the transmitter. SEM, spurious emissions may cover this function and requirement. Transmit intermodulation requirement is unnecessary.

6.7.4.1.2 RX

Reference sensitivity

Reference sensitivity is strongly related to the coverage of R2D. Reference sensitivity is potentially needed.

Maximum input level

The received power level for device 1 is up to the Reader and the distance between Reader and device. No power control for device. Whether maximum input level to be defined needs to be discussed in WI.

Adjacent channel selectivity(ACS)

Device1 is RF-ED architecture with low complexity and cost, apart from the simple LPF and matching network, other components cannot suppress the interference. Hence, potentially, there is no need to define ACS.

Adjacent Subcarrier selectivity(ASCS)

Same as ACS, there is no need to define ASCS.

In-band blocking

Same as ACS, potentially, there is no need to define in-band blocking.

Out-of-band blocking

Same as ACS, potentially, there is no need to define in-band blocking.

Spurious emissions

The spurious emissions power is the power of emissions generated or amplified in a receiver that appear at the UE antenna connector. The received power comes from R2D signal and device 1 has no LNA. Hence, potentially, there is no need to define spurious emissions.

Spurious response

Same as ACS, potentially, there is no need to define spurious response.

Intermodulation Characteristic

Same as ACS, potentially, there is no need to define spurious response.

Interference rejection

The interference could be wide-band NR signal (e.g. in-band deployment) or CW signal (e.g. from unsynchronized reader/CW node) or D2R signal from other devices. Whether interference rejection to be defined needs to be discussed in WI.

6.7.4.2 Device 2a

6.7.4.2.1 TX

Transmitter power

For device 2a, the transmitter architecture is based on backscattering technology, and the backscattering power or power backscattering loss is potentially needed to be defined to ensure the coverage target. The maximum output power is potentially not needed.

The minimum backscattered modulated power can be specified for device 2a. The backscatter power is the amount of power reflected back from the device 2a when a successful command sequence is completed at the device at a specific sensitivity level.

The backscatter loss due to the modulation can be defined as scattering cross-section or modulation efficiency in equation (1). It can be observed that the modulation efficiency is defined as the ratio of the power radiated by the device to the power density incident on its antenna.

The backscatter loss or cross section of the device 2a can be defined as

$$\sigma = \frac{P_{backscatter}}{S} = \frac{R_a^2}{|Z_a + Z_{c1,c2}|^2} \cdot \frac{\lambda^2 \cdot G_d^2}{\pi} \quad (1)$$

The received backscattered signal at the test device:

$$P_{received} = \frac{P_{backscatter} \cdot G_r}{4\pi r^2} = \frac{P_t \cdot G_t}{4\pi r^2} \cdot \frac{G_r}{4\pi r^2} \cdot \sigma \quad (2)$$

Where $P_{backscatter}$ is the power backscattered back in the direction of the transmitter defined in clause 6.7.4.1.1, and S is the power density of the incident radio wave on the device antenna defined in clause 6.7.4.1.1.

Output power dynamic

For device 2a, power control is not designed for backscattering transmission, and RAN4 concluded that all the output power dynamic requirements are not relevant for backscattering type devices.

- **Transmit OFF power:** when there is no incoming CW signal, the device 2a does not emit any high-power level. The requirement of transmit OFF power is potentially not needed.

- **Transmit ON/OFF time mask:** The time mask requirement is to test the ON/OFF time period for RF chain. For device 2a, this time period is not crucial and it is potentially not needed.
- **Power control:** Power control is not designed for backscattering transmission, the power control requirement is potentially not needed for device 2a.

Output RF spectrum emissions

- **Occupied bandwidth:** The occupied bandwidth is the width of a frequency band containing 99 % of the total integrated mean power of the transmitted spectrum on the assigned channel. Occupied bandwidth is potentially needed for Device 2a. The specific value is dependent on the physical layer design of D2R, which has not yet been finalized at this stage.
- **ACLR:** According to the co-existence simulation, the ACLR requirement is potentially not needed for device 2a.

6.7.4.2.2 RX

Reference sensitivity

The reference sensitivity is the minimum mean power applied to device antenna port to guarantee the receive performance. The requirement of reference sensitivity for device 2a is potentially needed.

Adjacent channel selectivity (ACS)

For device 2a, the RF BPF is not assumed in RF architecture for the reason of cost and size consideration, So the ACS requirement which relies on digital filter is potentially not needed.

Adjacent Subcarrier selectivity (ASCS)

Same as ACS, potentially, no need to define ASCS.

In-band blocking

Same as ACS, potentially, no need to define in-band blocking requirement.

Out-of-band blocking

Same as ACS, potentially, no need to define out-of-band blocking requirement.

Spurious emissions

The spurious emissions power is the power of emissions generated or amplified in a receiver that appear at the UE antenna connector. The received power comes from R2D signal. For device 2a, LNA is not assumed in RF architecture. Hence, there is potentially no need to define spurious emissions.

Spurious response

Same as ACS, there is potentially no need to define spurious response.

Intermodulation Characteristic

Same as ACS, there is potentially no need to define intermodulation characteristic.

6.7.4.3 Device 2b

6.7.4.3.1 TX

Maximum output power

In the transmission direction (D2R), for device 2b, as it can generate the signal by itself, the device function can be more similar to legacy IoT devices than backscattering-type devices. Therefore, its maximum output power and the corresponding power class can also be specified similarly to legacy IoT devices,

However, due to the nature of energy harvesting, its transmission and reception duty cycle may be limited by the availability of the harvested energy. Moreover, the power constraint on the AIoT devices may further impact the choice of components, e.g., oscillator, amplifier, etc.; thus, a significantly lower performance/capability can be expected.

Transmit OFF power

As device type 2b, assuming the device is fully charged, it can be foreseen that it will work similarly to legacy IoT devices. Therefore, the transmit OFF power should be potentially specified for type 2b to ensure the power level when the RF part is turned off will not exceed a certain range.

Transmit ON/OFF time mask

The time mask requirement is to test the time period between the RF part turned on and off. For A-IoT devices, this time period is not a crucial requirement and hence it is potentially not needed.

Minimum output power

As device type 2b, assuming the device is fully charged, it can be foreseen that it will work similarly to legacy IoT devices. Therefore, the minimum output power should be potentially specified for type 2b to ensure the lower limit of the output power for coverage.

Power control

As device type 2b, assuming the device is fully charged, it can be foreseen that it will work similarly to legacy IoT devices. Therefore, the power control should be potentially specified for type 2b to ensure the output power range.

Frequency error

As device type 2b, assuming the device is fully charged, it can be foreseen that it will work similarly to legacy IoT devices and the LO is assumed to be used. Hence the frequency error is potentially required.

Occupied bandwidth

The transmission bandwidth configuration defines the bandwidth of the R2D/D2R signal in baseband. For the R2D channel, one or multiple PRBs are under consideration in RAN1. For D2R channel, the transmission bandwidth of 15kHz or more is being discussed.

6.7.4.3.2 RX

Reference sensitivity

The receiver sensitivity is to test the receiver capability to certain performance metrics in a given lowest RX power level. For device 2b, this requirement will be potentially left to be further defined in WI phase to consider the RF architecture and the exact RF component capability.

Maximum input power

In NR, the maximum input power is to guarantee the UE can still work normally when it is close to BS. Similarly, this requirement is still potentially needed for AIoT device to ensure the device have reasonable dynamic range.

ACS

The ACS requirements shall be derived based on co-existence simulation, and guard band/guard RB may also be defined. As the BB LPF is proposed, the ACS requirement is potentially supposed to be essential to verify its RF performance

It's expected that the AIoT system will be deployed within an NR channel from the same operator. The AIoT device should be able to receive in the presence of a wideband NR interferer. Additionally, the device might also suffer from CW interference from a nearby reader/CW node, if it's not synchronised with the serving reader.

ASCS

The ASCS requirement is supposed to define the selectivity of the in-channel receiver. However as the channel bandwidth of the device is limited to only 1RB, the ASCS is potentially not needed.

Spurious response

Spurious response is a measure of the ability of the receiver to receive a wanted signal on its assigned channel frequency without exceeding a given degradation due to the presence of an unwanted CW interfering signal at any other frequency for which a response is obtained. In the A-IoT deployment, the CW signal is set at a certain frequency for energy harvesting and reflection. Hence the scenario for this test is potentially not needed.

6.7.4.4 Testability

6.7.4.4.1 Testability of device

Based on the architecture described in clause 5, with such highly integrated architectures for device 1/2a/2b, conducted antenna connectors are assumed not to be available. Therefore, OTA testing is considered as the baseline testing methodology for device 1, device 2a and device 2b RF verification. Far-field measurement system in an anechoic chamber could be considered as the baseline setup.

To be fully functional, A-IoT devices require a dedicated energy source for harvesting to be provided during the test. According to the SI description, the devices under consideration are either battery-less or with limited energy storage capability (i.e., using a capacitor) and the energy is provided through the harvesting of radio waves, light, motion, heat, or any other power source that could be seen suitable. Note that the following description on the energy harvesting is just an example to illustrate the testability issues for Energy Harvesting. It doesn't mean to define the Energy Harvesting procedure. Device goes from OFF state to ON state when it detects energy and has been able to accumulate sufficient amount of energy for the parts to function. This Energy Harvesting Sensitivity (EHS) is the value when device starts to harvest energy. Device behaviour principle with non-constant energy availability is shown in Figure 6.7.4.4.2-1.

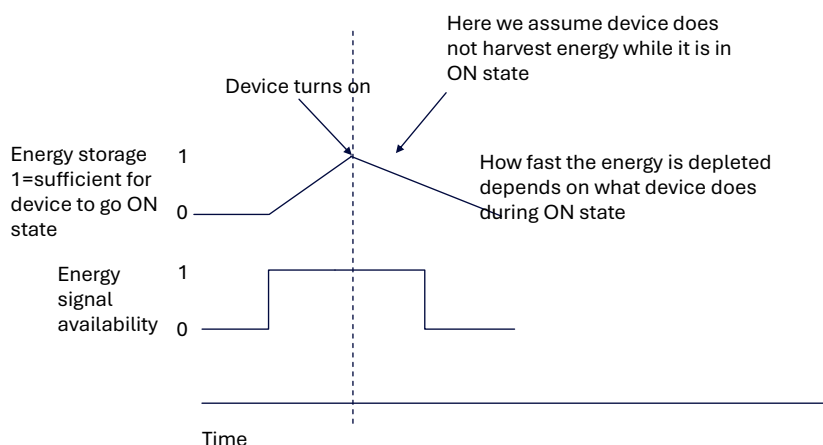


Figure 6.7.4.4.2-1: An example of energy harvesting principle

Energizing signal is a trigger for device to turn on. This is specifically for device type 1 and 2a. After device turns on, it needs to monitor the A-IoT R2D signal and device should respond to network in test environment once the signal from network is detected successfully. Device may also decide to sleep or continue monitoring.

In testing phase, for a device that harvests RF energy, tester must provide sufficient RF waveform electrical field density to the device to ensure it is turned on and time gate the RF testing to the ON periods of the A-IoT device. An external antenna may be needed to provide the CW and also provide RF energy, while a separate testing antenna is also needed, which are illustrated in Figure 6.7.4.1.2-2.

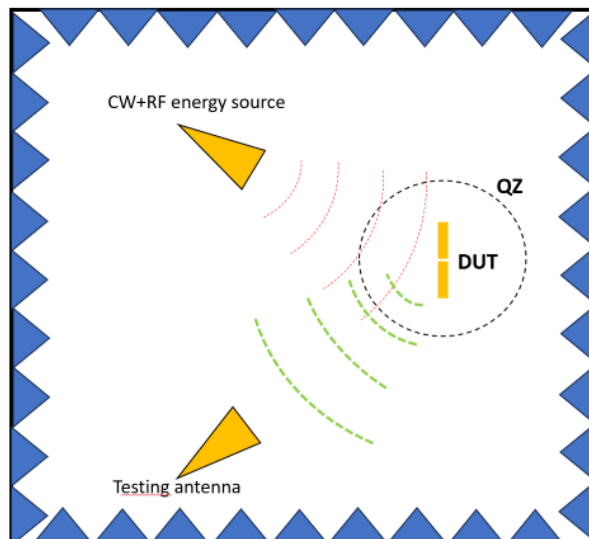


Figure 6.7.4.1.2-2: A possible OTA test setup with CW and RF energy source

Therefore, RAN4 needs to define a list of testable requirements in the WI phase and also the time domain pattern for the tests.

6.7.4.4.2 Summary of testability issues

RAN4 take OTA as the baseline testing methodology for device 1, device 2a and device 2b RF verification. The following test setup issues can be discussed and decided in the WI phase: power level of RF energy source and CW signal, duration time of harvesting, ON-OFF testing pattern, isolation between the CW antenna and the testing antenna, etc.

6.7.5 CW interference cancelation study

6.7.5.1 General

The CW cancellation can be divided into 3 stages sequentially: (1) Spatial isolation for separate TX/RX antenna configuration; (2) Analog cancellation from antenna to ADC and (3) Digital cancellation at baseband.

Spatial isolation for separate TX/RX antenna configuration

For separate TX/RX antenna configuration, the spatial isolation can be achieved by antenna isolation, or coupling loss between CW transmitter and D2R receiver.

For example, if the CW transmission source is separately deployed from the D2R receiver, the spatial isolation relies on the distance and antenna sidelobe suppression between the CW node and A-IOT node.

Analog cancellation

Interference cancellation in the RF front-end can be applied to reduce the desensitization of the receiver. In principle, it can be implemented by reconstructing a local CW with the same amplitude but opposite phase of a CW interferer received from antenna at the input of the LNA, and combine this cancellation signal with the received signal. The cancellation performance depends on the phase and amplitude error between the reconstructed and actual received CW interference.

Additionally in the case of integrated carrier-wave transmitter and D2R receiver in sharing antenna case, the interference suppression could be provided by a circulator/directional coupler.

Digital cancellation

In the digital baseband, suppression can be achieved by filtering, as the received single-tone carrier-wave is shifted to direct current (DC) at baseband. Digital interference cancellation can also be achieved by using a low complexity self-interference active cancellation technique.

6.7.5.2 Interference cancelation analysis

6.7.5.2.1 Ericsson

The min received power could be -74 dBm or lower, the signal with such low level received power should be decoded in the presence of the CW transmission which is either 33 dBm or 23 dBm for monostatic antenna and reference at antenna. The CW transmission can leak to the receiver chain via either reflection from environment or the imperfection of the isolation between transmitter and receiver when CW is transmitted by A-IoT BS or UE as reader. As the receiver has limited dynamic range and linearity, this CW leakage needs to be attenuated/isolated to mitigate the desensitization of the receiver. The CW signal frequency is used to modulate the backscatter signal and the frequency separation of the CW signal and modulated signal is around 100kHz for example as illustrated in Figure 6.7.5.2.1-1 according to the current D2R waveform design. Using an RF filter before the LNA to attenuate the CW signal is not possible. As such, a high-level CW leakage signal will be present together with a low wanted backscattered signal before LNA.

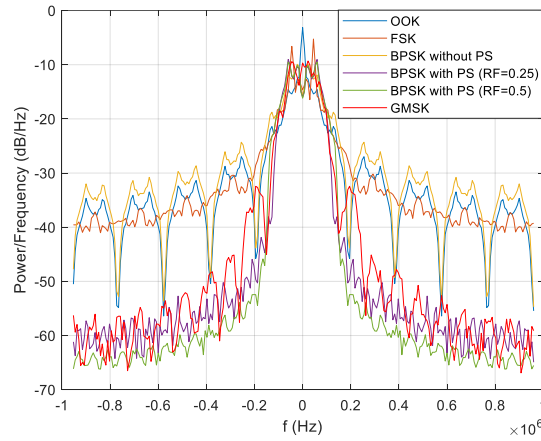


Figure 6.7.5.2.1-1: D2R waveform in frequency domain

To analysis the impact of the receiver desensitization, the backscattered signal is modelled as below [2]:

$$S_b(t) = A_b \cos(\omega_b \cdot t + \varphi_b) \cdot \cos(\omega_c \cdot t + \varphi_c)$$

with ω_c representing the CW frequency and ω_b is the message frequency.

And the leakage CW signal is modelled as below:

$$S_I(t) = A_I \cdot \cos(\omega_c \cdot t + \varphi_I)$$

Then the receiver nonlinearity is modelled with a power series below assuming the second-order nonlinear product can be filtered:

$$y_o(t) = a_1 \cdot x(t) + a_3 \cdot x(t)^3$$

Where a_1 is the small signal gain of the receiver and $a_3 < 0$ represents the nonlinear behaviour of the system.

The input signal before LNA is then:

$$x(t) = S_b(t) + S_I(t)$$

Assuming the $\varphi_b = \varphi_c = \varphi_I = 0$, at output of the LNA, only the terms associated with fundamental frequency $\cos(\omega_b t) \cos(\omega_c t)$ are of interest and they are below:

$$y_{ol}(t) = \left(a_1 A_b + \frac{9}{16} a_3 A_b^3 + \frac{9}{4} a_3 A_b A_I^2 \right) \cos(\omega_b \cdot t) \cdot \cos(\omega_c \cdot t)$$

As the CW signal is much larger than backscattered signal, the term $\frac{9}{16} \cdot a_3 \cdot A_b^3$ can be dropped and further simplified with:

$$y_{of}(t) = \left(a_1 A_b + \frac{9}{4} a_3 A_b A_I^2 \right) \cos(\omega_b \cdot t) \cdot \cos(\omega_c \cdot t)$$

The voltage of the fundamental signal at output of LNA can be reformulated as:

$$v(t) = a_1 \cdot A_b \cdot \left(1 + \frac{9}{4} \cdot \frac{a_3}{a_1} \cdot A_I^2 \right)$$

Since $a_3 < 0$, we can write:

$$v(t) = a_1 \cdot A_b \cdot \left(1 - \frac{9}{4} \cdot \frac{|a_3|}{|a_1|} \cdot A_I^2 \right)$$

It can be observed from the above that the wanted signal level is degraded with scaling factor which directly reduces the SNR, therefore the desensitization factor is:

$$SNR_D = 20 \log_{10} \left(1 - \frac{9}{4} \cdot \frac{|a_3|}{|a_1|} \cdot A_I^2 \right)$$

Further the IIP3 of LNA can be expressed with:

$$A_{IIP3_LNA}^2 = \frac{|a_1|}{|a_3|} \cdot \frac{4}{3}$$

Therefore, the SNR degradation can be formulated below:

$$SNR_D = 20 \log_{10} \left(1 - 3 \cdot \frac{A_I^2}{A_{IIP3_LNA}^2} \right) = 20 \log_{10} \left(1 - 3 \cdot \frac{P_{CWleakage}}{P_{IIP3_LNA}} \right)$$

Assuming CW signal is transmitted by A-IoT BS or UE as reader and monostatic antenna configuration, with the return loss of 15 dB, the CW leakage power will be 18 dBm. If the SNR degradation is allowed with 3 dB, the IIP3 of LNA should be 28 dBm. Such high IIP3 is demanding for design. For other combination value for SNR degradation and CW leakage power, it is illustrated in Figure 6.7.5.2.1-2 and the IIP3 value is listed in Table 6.7.5.2.1-1. Referring to TR 38.858, the reported IIP3 for WA BS is from the range of 10 dBm to -27.6 dBm, this in turn demands some CW cancellation capability before LNA.

Table 6.7.5.2.1-1: IIP3 of LNA with different SNR degradation and CW leakage power

		SNR degradation (dB)		
		-1	-3	-10
CW leakage power (dBm)	18	32	28	24
	8	22	18	14
	0	14	10	6
	-10	4	0	-4

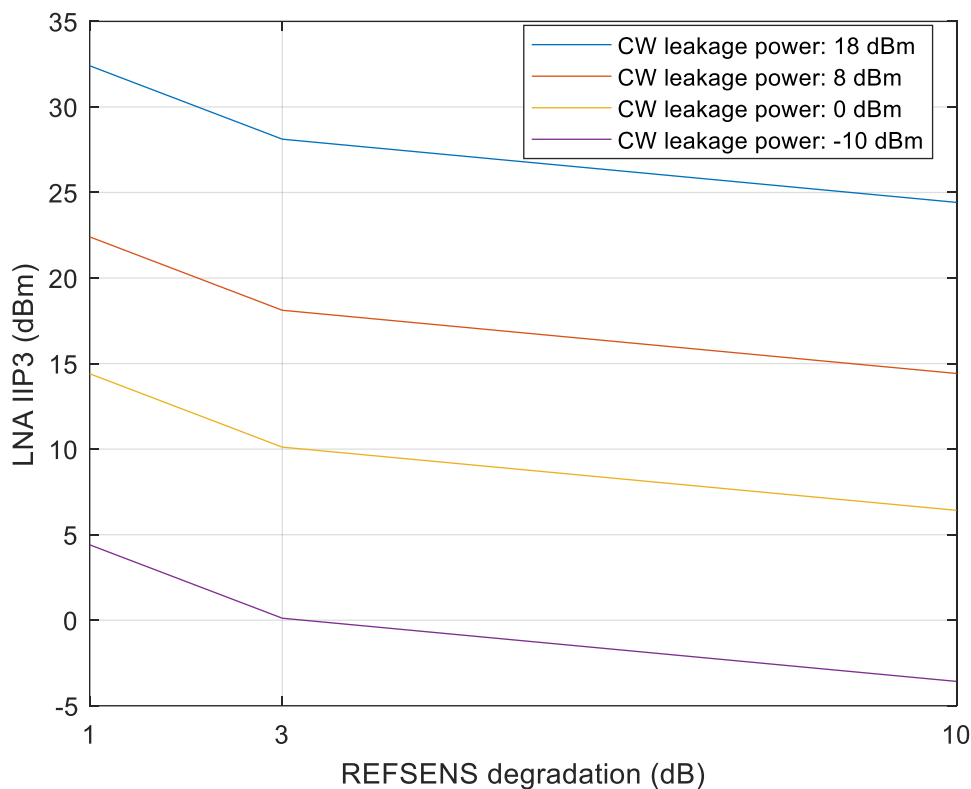


Figure 6.7.5.2.1-2: IIP3 of LNA with different SNR degradation and CW leakage power

For CW cancellation, there are passive or active cancellation schemes existing for RFID reader front-end. In both schemes, a copy of the CW leakage signal is created with same amplitude but opposite phase. The cancellation scheme is illustrated in Figure 6.7.5.2.1-3. For the passive cancellation scheme, the cancellation path needs to be calibrated and for active cancellation scheme, the cancellation path can be tuned to adapt to the leakage CW signal. However, the cancellation is limited by either the hardware imperfection or the detecting signal amplitude/phase accuracy. In [6], it is mentioned that for 40 dB CW cancellation, the amplitude error should be 0.09 dB and phase error to be 0.5 degree. This is quite demanding in design considering the current RRM measurement accuracy for RSRP is +/- 2 dB in normal condition.

When the CW and A-IoT signal are transmitted from different hardware, the active CW cancellation scheme should be used in this case. This applies to the case where the CW node is co-located with A-IoT BS for D1T1-A2 or D2T2-A2.

To enable the CW cancellation function, reader needs some time to train the circuit to generate the reverse CW leakage with the same amplitude and the opposite phase. The cancellation training time is implementation specific and also relates to the targeted cancellation performance. Considering this time and also the transient time to transmit the CW signal, the timing relation between the R2D, CW and D2R can be illustrated in Figure 6.7.5.1-4.

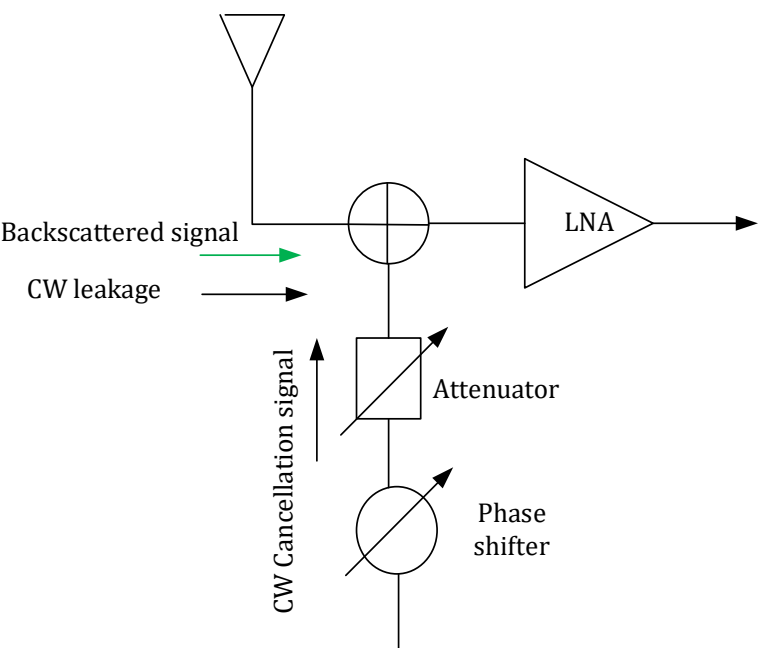


Figure 6.7.5.1-3: CW cancellation scheme

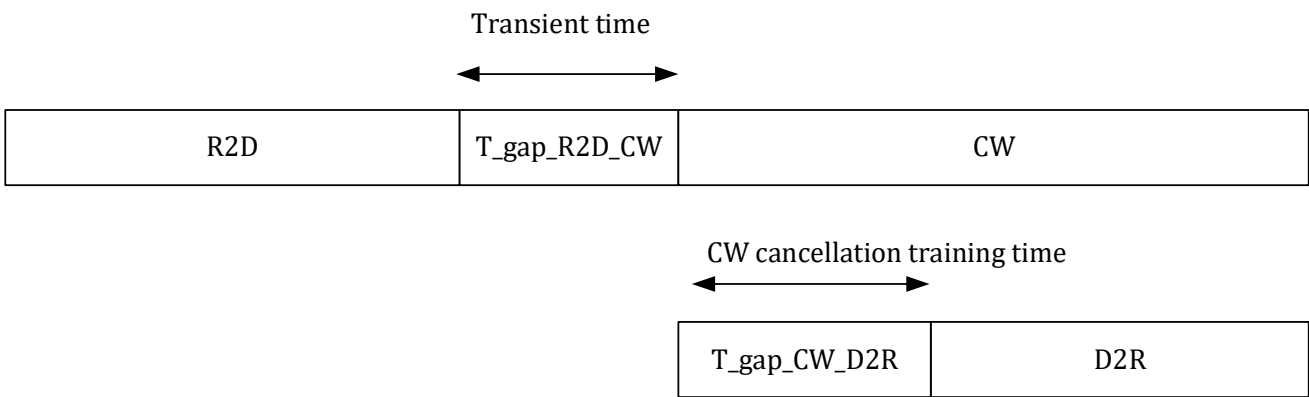


Figure 6.7.5.1-4: Transient time for CW and R2D transmission and CW cancellation training time before D2R reception

6.7.5.2.2 Qualcomm

6.7.5.2.2.1 Feasible values CW cancellation from implementation impact

In principle, the opposite phased CW can be injected to the receiver and then two signal will cancel each other out. The problem is how to know the phase of the CW in the receiver and how to adjust the cancelling signal perfectly. This is especially difficult in the intermediate node since it has smaller dimensions and shared antennas. The circuitry and relevant aspects are shown in Figure 6.7.5.2.2.1-1.

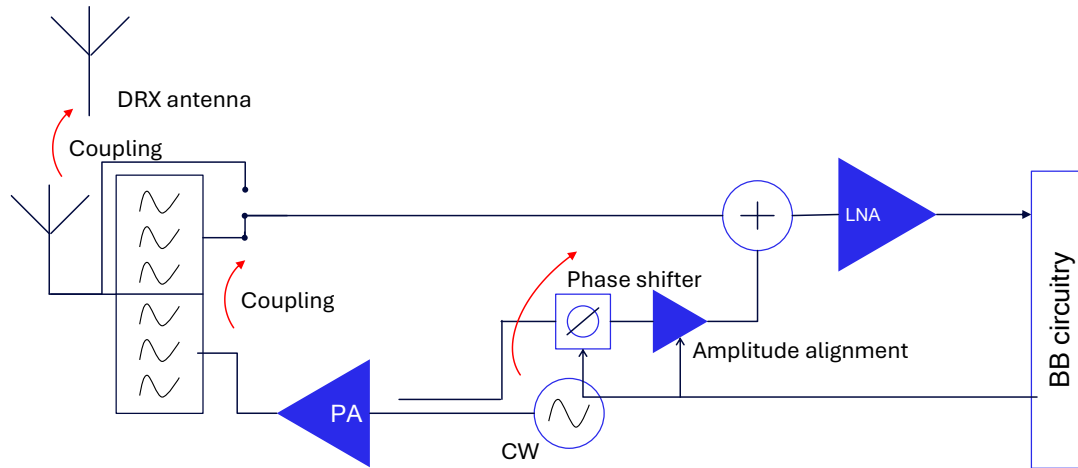


Figure 6.7.5.2.2.1-1. CW cancellation in an Intermediate node

From Figure 6.7.5.2.2.1-1 we can see that CW has many leakage paths and conclude that the cancellation circuitry has to cancel very high-level CW compared to the backscattered signal level.

RAN4 has not discussed techniques for active cancellation but it should be noted that the processing accuracy in a handset receiver typically results in the order of 1.5-4 % Rx EVM [226]. This is the level Intermediate node baseband can provide information to the phase and amplitude cancellation circuits for the analogue cancellation. In the LS [226], the typical UE Rx EVM is 2-4 % for a UE that supports 1024QAM. UE fulfilling minimum requirements and not support 1024QAM would have even smaller minimum implied assumption for Rx EVM. The analogue domain phase and amplitude controller also have finite resolution and 1% accuracy would mean 1.8 degrees phase step assuming signal can be inverted separately. This level of granularity maybe challenging in practice. Intermediate node CW cancellation circuitry processing accuracy is in the order of 1.5-4 %.

Translating this to CW cancellation in dB, we need to split this in to phase accuracy and amplitude accuracy. With a simple mathematical analysis, if we assume that waveform with an amplitude of Y is cancelled with:

$$\text{Cancellation waveform} = Y \cdot (1 - \text{amplitude_error}) \cdot \sin(\omega t + \text{phase_error})$$

And calculate the resulting CW as

$$Z = Y - \text{Cancellation waveform},$$

We can get cancellation value of Z compared to Y as in Figure 3 below. The EVM % is calculated from the *amplitude_error* and *phase_error*.

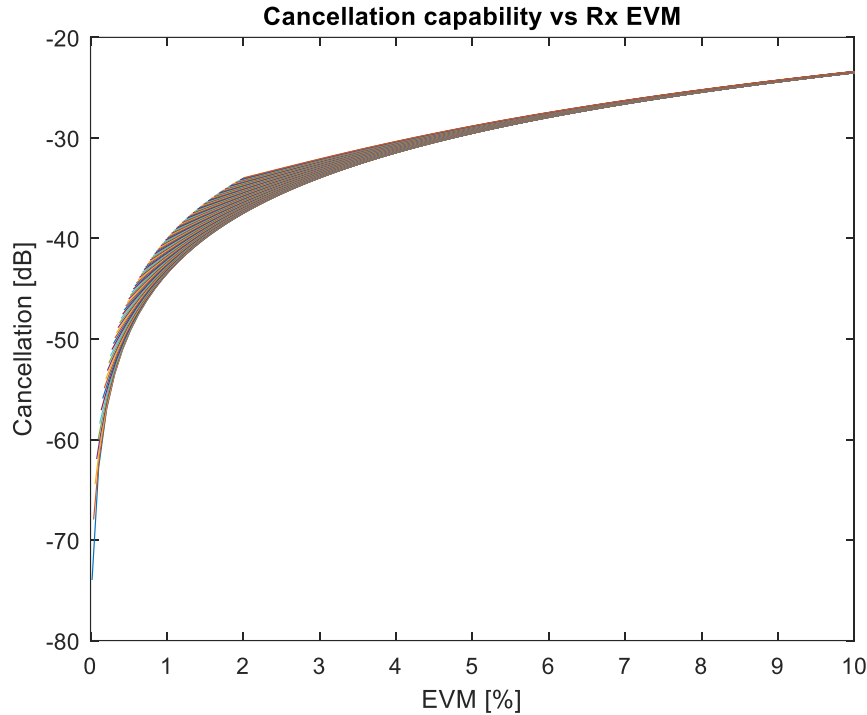


Figure 6.7.5.2.2.1-2: Cancellation as a function of EVM in the cancellation circuitry.

The curve in Figure 6.7.5.2.2.1-2 is not explicitly defined since the contribution of phase error to the EVM % is not same as its contribution to the cancellation. In other words, cancellation is more sensitive to the amplitude error than it is for phase error. In Figure 6.7.5.2.2.1-3 we show the dependency as surface plot for reference.

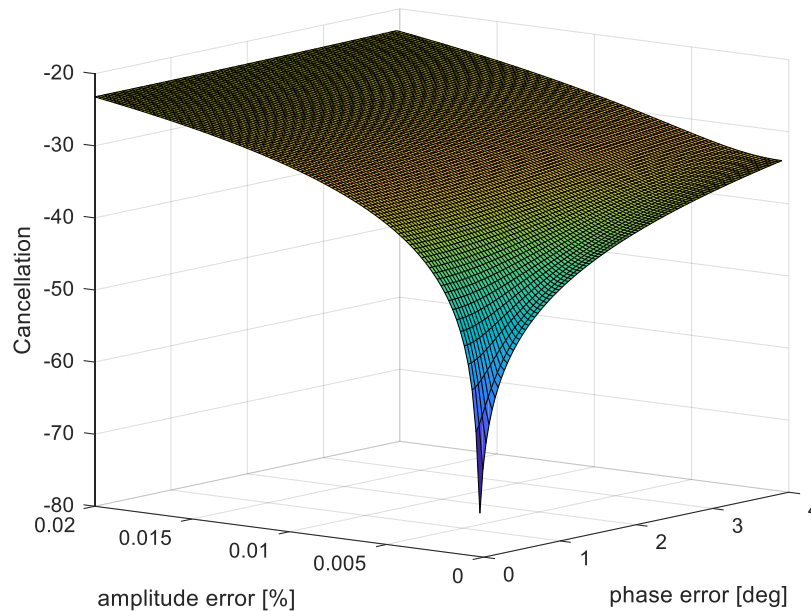


Figure 6.7.5.2.2.1-3: Cancellation sensitivity as a function of amplitude and phase error

We can see that the amplitude error and phase have different impact to cancellation and Intermediate node realistic analogue domain CW cancellation capability is at most 50 dB and more likely typical value is closer to less than 40 dB.

Coming back to the incoming signal characteristics, this means that the UE can cope with the CW of the level:

$CW_to_signal = \text{Analogue CW cancellation} + \text{Digital domain cancellation}.$

The digital domain cancellation is same process as DC offset compensation. Normally, the DC offset to be compensated is small since it is a result of non-perfections in the circuitry. In this case, the CW is much larger than the receiver modulated signal so the DC offset cancellation circuitry needs some tuning to work in this new situation. We can assume here that the digital domain cancellation can cope at least with the CW to signal ratio where LNA remain linear enough not to distort the signal.

6.7.5.2.2.2 Needed CW cancellation when CW from reader

In the first phase, to make the analysis a bit more concrete, we can assume that current handheld UE requirements are taken as baseline. Here we consider the maximum input level of -25 dBm and further assume that digital path dynamic range is defined by SNR for 256 QAM and ACS resulting in to 68 dB (12 bits) but in practice some more dynamic range is needed to accommodate some range for the digital AGC in practice. The output power is the one used in co-ex work, 23 dBm, and antenna coupling is 10 dB thus same what is used in earlier and in all other work such as MSD work in RAN4 for handhelds. Additionally, we can assume that the needed D2R signal SNR for successful decoding is 10 dB.

Table 6.7.5.2.2.2-1. Assumptions for the CW cancellation for intermediate node

Parameter	Value	Note
Max input level	-25 dBm	
Output power of CW	23 dBm	
Antenna coupling	-10 dB	Also reflection loss
Digital path dynamic range	12 bit (14 bit)	68 dB (74 dB)
SN(I)R for D2R signal	10 dB	At receiver

With the above numbers, we can calculate that the maximum CW cancellation in digital domain is $68 - 10 = 58$ dB. This will depend on number of bits used in the ACD, and for effective 14 bit ADC it would be 74 dB.

Reflected power from the antenna is 10 dB lower than the transmitted power, this means LNA without analogue domain CW cancellation will see 13 dBm CW but since the maximum input level is -25 dBm, the analogue domain cancellation has to be 38 dB for 23 dB output power and to ensure LNA max input level is not exceeded. Even if handheld device uses different antennas, the coupling can be 10 dB and the result is the same conclusion. Analogue domain CW cancellation must be 38 dB for 23 dBm output power and -25 dBm maximum input level for the receiver.

The analysis in previous section we discussed that this is feasible from the control point of view. With 38 dB in analogue domain, the total CW cancellation maximum is 112 dB.

Looking more closely, the range of the reader can be calculated from the above numbers following the logic:

Received D2R power = Max CW power – 2x pathloss – backscattering loss

Minimum received power at lowest SNR = max input level + SNR_{D2R} – DNR_{ADC}

When we make both equations equal:

Max CW power – 2x pathloss – backscattering loss $\geq$ max input level + SNR_{D2R} – DNR_{ADC}

Pathloss $\leq (-\text{max input level} - \text{SNR}_{\text{D2R}} + \text{DNR}_{\text{ADC}} + \text{Max CW power} - \text{backscattering loss})/2$

In numbers, assuming 10 dB backscattering loss:

Pathloss $\leq (-(-25 \text{ dBm}) - 10 \text{ dB} + 84 \text{ dB} + 23 \text{ dBm} - 10 \text{ dB})/2 = 56 \text{ dB}$, which is about 14 meters at 1 GHz with 14 bit ADC.

And this is assuming the analogue domain CW cancellation is as above.

To understand the CW cancellation role to the range, we can first look at the analogue domain cancellation. If the cancellation is less than 38 dB but the maximum input level can not be exceeded, the output power has to be reduced and this reduces the range accordingly. If the CW cancellation is better than the 38 dB, the output power can be increased and this increases the range.

Looking at the digital domain cancellation, firstly we state that the digital domain can cancellation can be very large if a long training period is allowed. When intermediate node is the reader, it can choose when to turn CW or and train its own receiver thus time can be long.

For a fixed output power, the digital domain dynamic range defines the range since more bits there are available, more bits can be used for the CW cancellation with 2 effective bits needed to cover 10 dB SNR needed for the successful decoding, only the training time may need to be increased. However, this analysis did not look in to the noise contribution. With the above values, the received D2R signal power is close to -100 dBm. Without knowing what is the bandwidth of the D2R signal, it is difficult to say exactly what is the noise contribution but at -100 dBm level and e.g. 200 kHz BW when $kTB = -120$ dBm, there will be some noise contribution to the SINR.

6.7.5.2.2.3 Needed CW cancellation when "CW from outside of the topology"

Case is different when CW comes from outside the topology. The level of the CW at the receiver is lower compared to the D2R signal level but the CW phase noise plays a role in the range since the down conversion will be done with a different signal source with non-correlating phase noise.

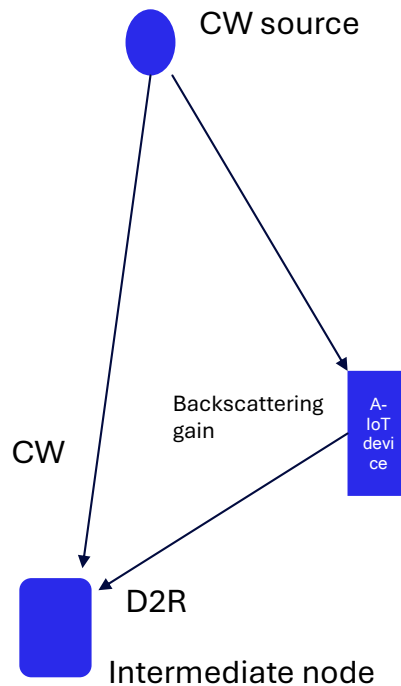


Figure 6.7.5.2.2.3-1: CW not in the reader "CW outside topology"

The way the CW nodes are distributed in the aged topologies result almost constant -39 dBm received CW in the reader. The assumption is to always use closest CW source. Looking at the statistical D2R received power distribution in Figure 2, for 90% of the connections see power higher than -101 dBm for D1T1 and -98 dBm for D2T2.

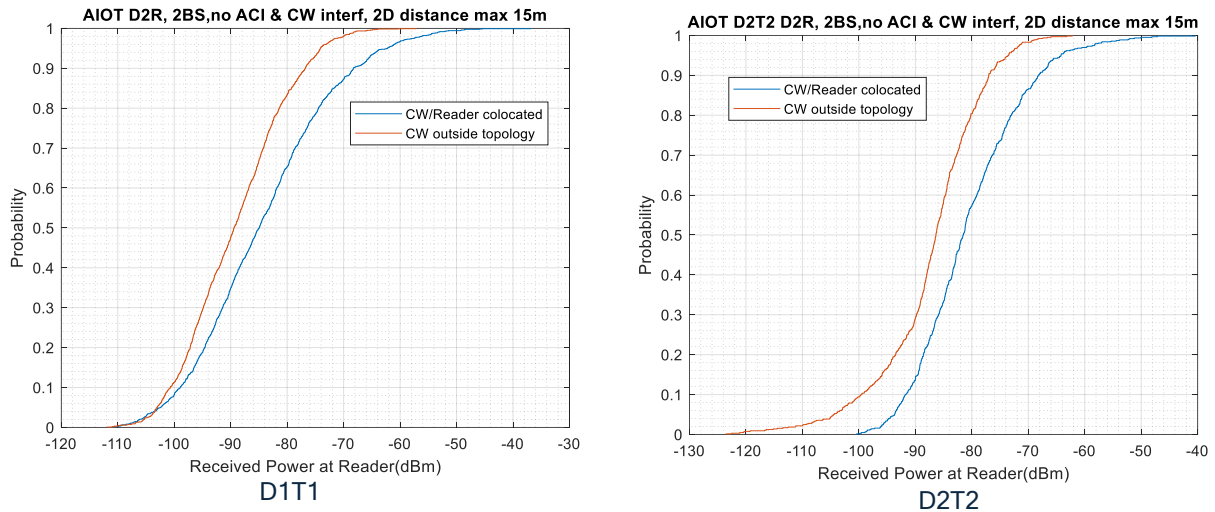


Figure 6.7.5.2.3-2: D2R received power distribution in the two topologies. CW level is almost constant at the receiver

Calculating the ratios yields that CW to D2R signal levels for 90 % probability are 62 and 62 dBc in D1T1 and D2T2 when CW comes outside the topology.

In this topology, it is difficult to relate the CW cancellation and range since it depends on how the CW source and reader are positioned. From statistical data in Figure we can see that the cancellation needs to be much lower level than when CW source is in the reader and up to the point that maybe analogue domain cancellation is not needed.

6.7.5.2.2.4 Phase noise impact when CW is outside topology

If the reader will be using a different CW to down convert the D2R signal than the CW used in the backscattering, the phase noise of the incoming CW plays a critical role in system performance.

From the same conclusion, that the CW to D2R power ratio at the reader is approximately 62 dBc, we can conclude that the integrated phase noise over the D2R bandwidth can not be bigger than -72 dBc, to allow for 10 dB SNIR. However, this assumes phase noise is the only noise or interference source and more comfortable specification could be -80 dBc below the CW. Integrated phase noise for the CW coming outside the topology over the D2R signal BW ($B_{\text{tx,D2R}}$) must be better than -80 dBc.

We ran system simulation and impact to SNR because of multiple phase noise values and in Figure 6.7.5.2.2.4-1 present the results. Here we chose 15 kHz D2R bandwidth and assumed it is offset from the CW sufficiently to avoid increase in integrated noise because of high flicker noise close to the CW. In that way, the simulated results show most optimised case. If the bandwidth is large, the integrated noise is also larger.

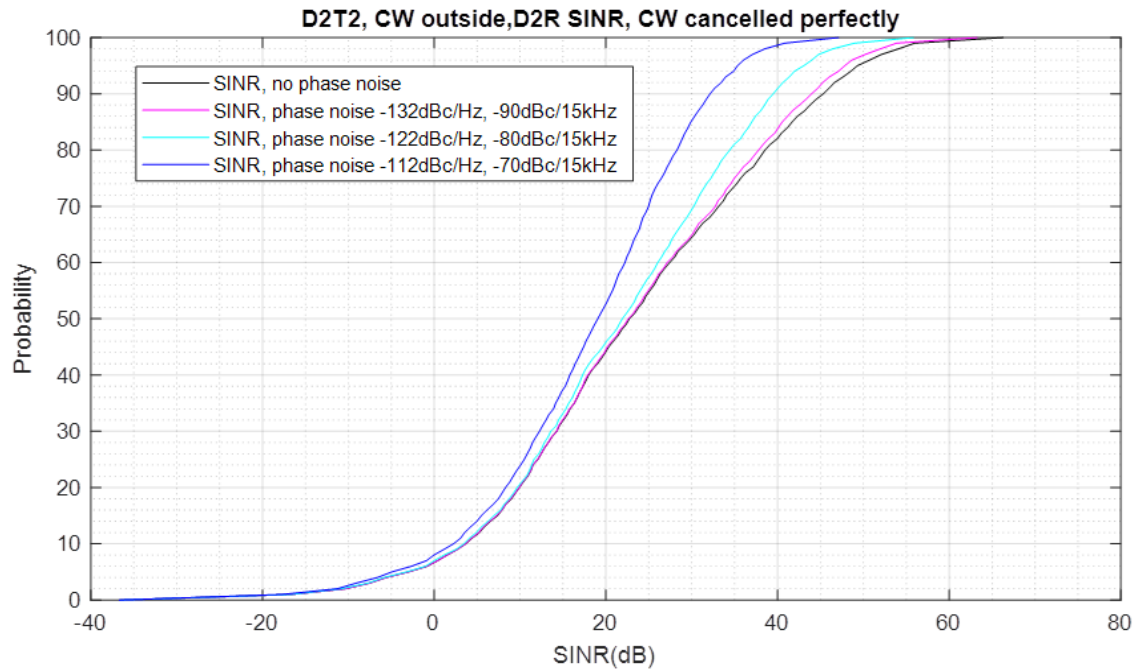


Figure 6.7.5.2.4-1: Impact of phase noise to SINR.

Typically the PLLs and oscillators are designed to meet wide BWs and in handheld device also settling time needs to be accounted for and they may not be optimised for close in noise where backscattering signals in A-IoT case are assumed to be positioned with relatively narrow bandwidth.

In practical design, the phase noise is a result of many components, such as VCO, PLL loop and $1/f$ noise. In Figure 6.7.5.2.4-2 these are summarize and the resulting phase noise is the darker red curve.

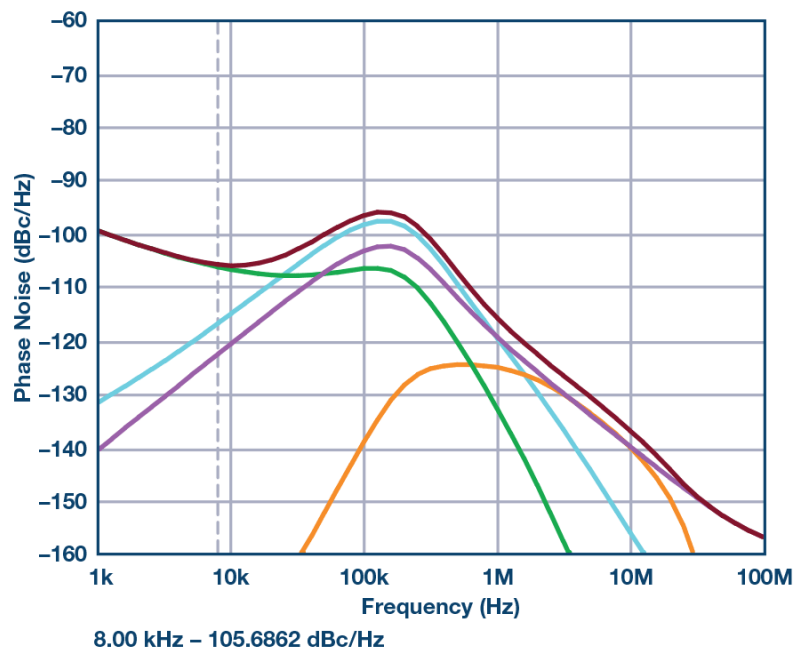


Figure 6.7.5.2.4-2: Example of a phase noise characterises and contributing sources ([159])

From Figure 6.7.5.2.4-2 we can see that -80 dBc integrated phase noise over 100 kHz specification may require dedicated high-performance design.

RAN4 then should specify this for the component which provides the phase noise to the system but this issue should be firstly understood and documented in the study item and then what kind of specification will be written then discussed in work item.

Understanding phase noise of the CW source may also be important for the radio system design. RFID uses an offset from CW in the D2R link and this helps with the phase noise since as we can observe from Figure 3, offsetting the placement of $B_{\text{Tx,D2R}}$ further away might be useful in relaxing the CW source phase noise requirements.

Placing $B_{\text{Tx,D2R}}$ further away from CW in frequency is beneficial for the system throughput and range.

6.7.5.2.3 Huawei

2 aspects are evaluated: (1) Blocking of D2R receiver; (2) Carrier-wave interference cancellation

Blocking of D2R receiver

If the carrier-wave source is separately deployed from the D2R receiver (e.g., D1T1-A1 and D1T1-B scenarios), the spatial isolation relies on the distance between the CW node and A-IOT node. For example, the path loss is 57 dB for an isolation distance of 20 meters, assuming free-space propagation. Normally the CW node and A-IOT node deploy at the same height, the antenna gain is about -5 dB at 90° offset from the boresight for both TX antenna and RX antenna. For D1T1-A1 and D1T1-B scenarios the received power at the antenna connector is $23-57-10 = -44$ dBm and $23-51-10 = -38$ dBm for 20 meters and 10 meters distance respectively, which is close to the narrowband blocking requirements for small cell NR base station. Hence there is no blocking issue for these scenarios.

In the case of integrated carrier-wave transmitter and D2R receiver in D1T1-A2, analog interference cancellation in the RF front-end can be applied when the interference power exceeds the blocking threshold of receiver. The performance heavily depends on the similarity between the reconstructed and actual received interference. A suppression of 10-30 dB can usually be expected for RF interference cancellation. Assuming the same reference clock between the carrier-wave transmitter and D2R receiver (e.g., in D1T1-A1/A2 scenario, or D1T1-B scenario with wired connection between the base station and the carrier-wave node), higher interference suppression can be reached, such as 20 dB or higher.

Overall carrier-wave interference

In the case of separated carrier-wave transmitter and D2R receiver in D1T1-A1 or D1T1-B, through spatial isolation, RF interference cancellation and high-pass filtering, the overall suppression to the carrier-wave interference can reach about $(61+10+80) = 151$ dB at 10 meters distance. For larger spatial isolation corresponding to longer distance between the carrier-wave transmitter and D2R receiver (e.g., ~60 dB @ 25 m), the carrier-wave interference can be even higher (e.g., $70+10+80 = 160$ dB).

In the case of integrated carrier-wave transmitter and D2R receiver in D1T1-A2, with ~40 dB interference suppression provided by a circulator/directional coupler and/or the antenna with high isolation between Tx and Rx, the carrier-wave interference suppression can reach about $(40+20+80) = 140$ dB.

6.7.5.2.4 vivo

Spatial domain

For A1/B scenario, the receiver and transmitter are separated, which means spatial domain isolation for the CW interference can be obtained through this deployment. The amount of spatial isolation can be achieved by pathloss isolation.

For the D1T1-A1/B scenario, the pathloss isolation can reach 61 dB, which assumes the distance between CW emitter and receiver is 20 meters for InH-DH-NLOS path loss model. Meanwhile, for the D2T2-A1/B scenario, the pathloss isolation is 54 dB when the CW emitter and receiver distance is 10 meters and the pathloss model is InH-Office-NLOS.

For A2 scenario, the D2R receiver and CW transmitter are located at same node, so the pathloss can't provide suppression in this case, and the spatial domain suppression capability mainly from Tx/Rx antenna isolation. For D1T1 we assume the antenna isolation is 15 dB, and for D2T2 we assume 10 dB which is a typical value for NR UE.

The suppression capability in spatial domain is summarized below:

Scenario	Part1 Spatial domain suppression
D1T1-A1/B	61 dB
D1T1-A2	15 dB
D2T2-A1/B	54 dB
D2T2-A2	10 dB

Analog domain

This part is the suppression before the LNA. A typical self-interference cancellation architecture is show in Figure 6.7.5.2.4-1

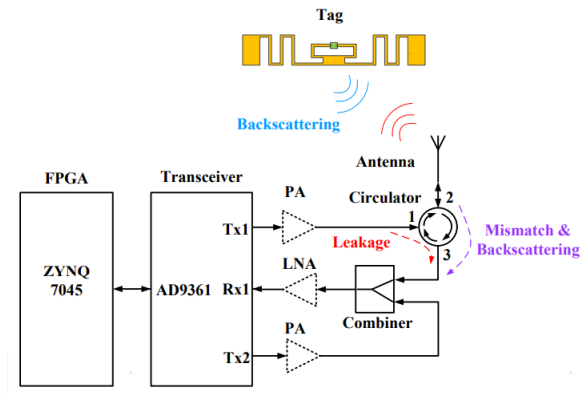


Figure 6.7.5.2.4-1: A typical self-interference cancellation architecture [1]

The basic rationale is to generate a CW that have -180° phase difference compared to the incoming interference and combine them together. According to [1][2], the RF-IC suppression capability between 30dB~ 60dB can be achieved generally, and we assume 45dB here.

For A1/B scenario, since the CW transmitter is separated from D2R receiver, the method above actually hard to be used at receiver side, so we assume 0 dB suppression capability as worst case.

The suppression capability in analog domain is summarized below:

Scenario	Part2 Analog domain suppression
D1T1-A1/B	0 dB
D1T1-A2	45 dB
D2T2-A1/B	0 dB
D2T2-A2	45 dB

IM3 product

After LNA, the CW and D2R will generate intermodulation product, and the IM3 product will overlap with D2R signal in frequency domain, and the BB filter actually cannot help on this. The traditional IM3 calculation procedure is described in the table below

Table 6.7.5.2.4-1: The model of CW total cancellation capability

System configuration									
0A	Scenarios	D2T2-A2	D2T2-B	D2T2-A2	D2T2-B	D1T1-A2	D1T1-B	D1T1-A2	D1T1-B
		InH-LOS	InH-LOS	InH-LOS	InH-LOS	InF-DH-NLOS	InF-DH-NLOS	InF-DH-NLOS	InF-DH-NLOS

0A1	CW case								
0B	Device type	Device1		Device2a		Device1		Device2a	
0C	Center frequency (GHz)	0.9				0.9			
Transmitter									
1E	CW Tx power(dBm)	23				33			
1F	D2R bandwidth (Hz)	160000				160000			
1G	Tx antenna gain(dBi)	0				2			
1N	Cable/connector loss	1				3			
Receiver									
2B	bandwidth used for the evaluated channel (Hz)	160000				160000			
2C	Receiver antenna gain(dBi)	0				2			
2D	Receiver Noise Figure(dB)	7				5			
2E	Thermal Noise(dBm/Hz)	-174				-174			
2F	Noise Power(dBm)	-114.9588002				-116.9588002			
2G	Tx/Rx spatial isolation capability	10	0	10	0	15	0	15	0
2F	RX cable, connector loss	1				3			
2J	Rx RF-IC suppression capability; For outside topology, this item is pathloss suppression interference	45	54.46	45	54.46	45	61.18	45	61.18
	CW cancellation (dB)	55	54.46	55	54.46	60	61.18	60	61.18
2K1	Remaining CW interference(dB)	-34	-33.46	-34	-33.46	-29	-30.18	-29	-30.18
2K2	D2R power (dBm), for D1T1 scenario, assuming 10m distance between reader and tag; for D2T2 scenario, assuming 5m distance between reader and tag	-72.15	-72.15	-62.15	-62.15	-88.17	-88.17	-78.17	-78.17

2K3	Rx IIP3 capability(dBm)	-6	-6	-6	-6	-10	-10	-10	-10
2K4_1	Rx IM3#1 = $2K1+2*2K2-2*2K3$	- 166.31	- 165.77	- 146.31	- 145.77	- 185.34	- 186.52	- 165.34	- 166.52
2K4_2	Rx IM3#2 = $2K2+2*2K1-2*2K3$	- 128.15	- 127.08	- 118.15	- 117.08	- 126.17	- 128.52	- 116.17	- 118.52
2K4	Rx IM3 = $2K4_1+2K4_2$	- 128.15	- 127.07	- 118.15	- 117.07	- 126.17	- 128.52	- 116.17	- 118.52
2K5	Total CW cancellation capability= $1E-2K4$	151.15	150.07	141.15	140.07	159.17	161.52	149.17	151.52

Digital domain

From our side, it is not clear how digital domain process can provide benefit on the interference suppression. Since if there is no blocking or non-linear distortion, the single tone CW interference is converted to DC in baseband, which has no impact to detection performance as long as the ratio between desired signal and residual interference is within receiver dynamic range. If blocking happens, it is also hard to overcome it in digital baseband.

6.7.5.3 Summary

CW cancellation performance is implementation specific. Spatial isolation, Analog cancellation and Digital cancellation are studied in clause 6.7.5.2. For RF CW cancellation performance, it is trade-off among the allowed receiver desensitization, receiver linearity and the CW leakage power received at antenna. To enable the CW cancellation, CW cancellation training time should be considered in the implementation. During the CW cancellation training time, D2R reception should be avoided.

6.8 Characteristics of carrier-wave waveform

6.8.1 CW transmission

For the case that D2R backscattering is transmitted in the same carrier as CW for D2R backscattering, and for topology 1, the following cases for CW transmission are studied:

Case 1-1: CW is transmitted from inside the topology, transmitted in DL spectrum

Case 1-2: CW is transmitted from inside the topology, transmitted in UL spectrum

Case 1-4: CW is transmitted from outside the topology, transmitted in UL spectrum

The observations shown in Table 6.8.1-1 are made regarding these cases.

Table 6.8.1-1: Observations on CW transmission cases for topology 1

CW Transmission case	Observations
Case 1-1	- No need for BS to support full-duplex capability in DL spectrum for scenario A1 - Spatial isolation is possible for scenario A1, reducing the received CW interference power at BS side. - Cross-link interference handling for CW at BS side for scenario A1. - BS needs to support full-duplex capability (including self-interference suppression for CW) in DL spectrum for scenario A2. - Higher CW transmission power can be assumed in the DL spectrum than that of in the UL spectrum.
Case 1-2	- No need for BS to support full-duplex capability in UL spectrum for scenario A1 - Spatial isolation is possible for scenario A1, reducing the received CW interference power at BS side. - Cross-link interference handling for CW at BS side for scenario A1. - BS needs to support full-duplex capability (including self-interference suppression for CW) in UL spectrum for scenario A2. - Lower CW transmission power can be assumed in the UL spectrum than that of in the DL spectrum.
Case 1-4	- No need for BS to support full-duplex capability in UL spectrum - Spatial isolation is possible, reducing the received CW interference power at BS side. - Cross-link interference handling for CW at BS side. - Estimation of CW interference may be needed for successful D2R reception (i.e., at BS). - Lower CW transmission power can be assumed in the UL spectrum than that of in the DL spectrum.

For the case that D2R backscattering is transmitted in the same carrier as CW for D2R backscattering, and for topology 2, the following cases for CW transmission are studied:

Case 2-2: CW is transmitted from inside the topology (i.e., intermediate UE), transmitted in UL spectrum

Case 2-3: CW is transmitted from outside the topology, transmitted in DL spectrum

Case 2-4: CW is transmitted from outside the topology, transmitted in UL spectrum

The observations shown in Table 6.8.1-2 are made regarding these cases.

Table 6.8.1-2: Observations on CW transmission cases for topology 2

CW Transmission case	Observations
Case 2-2	- No need for intermediate UE to support full-duplex capability in UL spectrum for scenario A1 - Spatial isolation is possible for scenario A1, reducing the received CW interference power at intermediate UE side. - Cross-link interference handling for CW at intermediate UE side for scenario A1. - Intermediate UE needs to support full-duplex capability (including self-interference suppression for CW) in UL spectrum for scenario A2. - Lower CW transmission power can be assumed in the UL spectrum than that of in the DL spectrum.
Case 2-3	- No need for intermediate UE to support full-duplex capability in DL spectrum - Spatial isolation is possible, reducing the received CW interference power at intermediate UE side. - Cross-link interference handling for CW at intermediate UE side. - Estimation of CW interference may be needed for successful D2R reception (i.e., at intermediate UE). - Higher CW transmission power can be assumed in the DL spectrum than that of in the UL spectrum.
Case 2-4	- No need for intermediate UE to support full-duplex capability in UL spectrum - Spatial isolation is possible, reducing the received CW interference power at intermediate UE side. - Cross-link interference handling for CW at intermediate UE side. - Estimation of CW interference may be needed for successful D2R reception (i.e., at intermediate UE). - Lower CW transmission power can be assumed in the UL spectrum than that of in the DL spectrum.

If large frequency shift is supported by device 2a, the D2R backscattering can be transmitted in a different carrier as CW for D2R backscattering (e.g., CW in DL spectrum and D2R in UL spectrum, or CW in UL spectrum and D2R in DL spectrum), and the followings are observed.

- In-band full-duplex capability is not needed for CW transmission and D2R reception at D2R receiver side, if the D2R receiver only supports device 2a with large frequency shift capability.
- No interference from CW to the corresponding D2R reception at D2R receiver side.
- For CW in DL spectrum and D2R in UL spectrum, higher CW transmission power can be assumed in the DL spectrum than that of in the UL spectrum.

6.8.2 CW characteristics

Candidates for the CW for D2R backscattering are waveforms consisting of:

Waveform 1: A single-tone unmodulated sinusoid, also referred to as 'a single tone'.

Waveform 2: Two single tones.

Table 6.8.2-1 captures observations on the above CW waveform candidates.

Table 6.8.2-1: Observations and/or comparisons of CW waveform candidates

CW waveform characteristics	Waveform 1 compared to Waveform 2 NOTE 1: Waveform 1 without frequency hopping NOTE 2: Waveform 2 with both tones from the same CW node	Waveform 1 with frequency hopping (2-hops) compared to waveform 1 without frequency hopping
D2R reception performance	Waveform 2 provides [0, 8] dB frequency diversity gain compared to waveform 1 at 1% or 10% BLER in a fading channel for a 1Rx receiver and a 1Tx CW transmitter, at least depending on the gap between the two tones and the channel's coherence bandwidth. In a TDL-A fading channel with 30ns delay spread - For the gap between [75KHz, 900KHz], the frequency diversity gains at 1% BLER target observed by 6 sources are within [0, 1.5] dB, and the frequency diversity gains at 10% BLER target observed by 3 sources are almost 0dB. - For the gap between [1.08MHz, 4.2MHz], the frequency diversity gains at 1% BLER target observed by 6 sources are within [3, 5.8] dB, and the frequency diversity gains at 10% BLER target observed by 3 sources are within [0.4, 2.5] dB. - For the gap between [5MHz, 10MHz], the frequency diversity gains at 1% BLER target observed by 5 sources are within [5, 8] dB, and the frequency diversity gains at 10% BLER target observed by 5 sources are within [1.3, 4] dB. In a TDL-D fading channel with 30ns delay spread - For 10MHz gap, 1 source [160] observed 0.7 dB@1%BLER and -0.2dB@10%BLER frequency diversity gain. (Note: loss due to the power split in TDL-D); and 1 source [161] observed 1dB@1%BLER and 0.5dB@10%BLER frequency diversity gain In a TDL-A fading channel with 150ns delay spread - For the gap is 180KHz, the frequency diversity gains at 1% BLER target observed by 2 sources are within [1, 3] dB, and the frequency diversity gains at 10% BLER target observed by 2 sources are within [0, 2.5] dB. - For the gap is 2.16MHz, the frequency diversity gains at 1% BLER target observed by 2 sources are within [7, 8] dB, and the frequency diversity gains at 10% BLER target observed by 2 sources are within [2.5, 5.5] dB. - For the gap is 5MHz, the frequency diversity gains at 1% BLER target observed by 2 sources are within [7, 8] dB, and the frequency diversity gains at 10% BLER target observed by 2	Observations on frequency diversity: - 6 sources observed that waveform 1 with frequency hopping should be used together with bit-level repetition or TB-level repetition. - 3 sources further observed that the unaligned timing (due to SFO at A-IoT device) between reader and devices makes it hard to align the boundary of each hop of the external carrier-wave with a corresponding bit or block repetition of the D2R transmission. - 2 sources observed that for milli-second level CW duration for each frequency hop/block repetition, the diversity gain loss caused by mis-alignment of a few micro-seconds level is marginal, TB level repetition can tolerate some time mis-alignment. - 1 source [162] observed that waveform 1 with frequency hopping achieves frequency diversity gain by using polynomial sweeping-based CC encoding (proposed by [162]) instead of repetitions When bit-level repetition, TB-level repetition or polynomial sweeping-based CC encoding is not used, - Waveform 1 with frequency hopping provides [0, 0.5] dB frequency diversity gain compared to waveform 1 without frequency hopping at 1% or 10% BLER in a fading channel for a 1Rx receiver and a 1Tx CW transmitter. - In a TDL-A fading channel with 30ns delay spread - For 10MHz gap between two hops, 1 source observed 0 dB@10%BLER frequency diversity gain. - In a TDL-A fading channel with 150ns delay spread - For 1.65MHz gap between two hops, 1 source observed 0.5 dB@10%BLER and 0 dB@1%BLER frequency diversity gain. When bit-level repetition, TB-level repetition is used, - Waveform 1 with frequency hopping provides [0.5, 8] dB frequency diversity gain compared to waveform 1 without frequency hopping at 1% or 10% BLER in a fading channel for a 1Rx receiver and a 1Tx CW transmitter, at least depending on the gap between the two hops and the channel's coherence bandwidth. - In a TDL-A fading channel with 30ns delay spread - For the gap of 2.16MHz, 1 source observed 4dB@1% BLER and 2dB@10% BLER frequency diversity gains.

	sources are within [2.5, 3] dB. The following observations were made by one source [161]: - Waveform 1 with antenna hopping (AH) provides [1, 5.75] dB spatial diversity gain compared to Waveform 1 with no AH at 1% or 10% BLER in a fading channel for a 1Rx receiver and a 2Tx CW transmitter. - In a TDL-D fading channel with 30ns delay spread, the spatial diversity gain is 1.6 dB at 1% BLER and 1 dB at 10% BLER. - In a TDL-A fading channel with 30ns delay spread, for the gap of 2.16MHz, the spatial diversity gain is 5.75dB at 1% BLER target and 2.5dB at 10% BLER target. For the gap of 10MHz, the spatial diversity gain is 2.2dB at 1% BLER target and 1.5dB at 10% BLER target. - Waveform 2 with AH provides [1.4, 11.5] dB spatial diversity and frequency diversity gain compared to Waveform 1 with no AH at 1% or 10% BLER in a fading channel for a 1Rx receiver and a 2Tx CW transmitter, at least depending on the gap between the two tones and the channel's coherence bandwidth. - In a TDL-D fading channel with 30ns delay spread, for the gap of 10MHz, the spatial diversity and frequency diversity gain is 2.9 dB at 1% BLER and 1.4 dB at 10% BLER. - In a TDL-A fading channel with 30ns delay spread, for the gap of 2.16MHz, the spatial diversity and frequency diversity gain is 8.25dB at 1% BLER and 5dB at 10% BLER. For the gap of 10MHz, the spatial diversity and frequency diversity gain is 11.5dB at 1% BLER and 7dB at 10% BLER. Note: The total transmission power is assumed the same for both waveforms.	- For the gap between [5MHz, 10MHz], the frequency diversity gains at 1% BLER target observed by 2 sources are within [5.8, 8] dB, and the frequency diversity gains at 10% BLER target observed by 3 sources are within [1.2, 6] dB. The 8 dB gain is achieved under the assumption of ideal channel estimation. - In a TDL-D fading channel with 30ns delay spread - For 10MHz gap, 1 source observed 1 dB@1%BLER and 0.5dB@10%BLER frequency diversity gain. - In a TDL-A fading channel with 150ns delay spread - For the gap between [1.65MHz, 2.16MHz], the frequency diversity gains at 1% BLER target observed by 2 sources are within [5.5, 8] dB, and the frequency diversity gain at 10% BLER target observed by 2 sources is 5 dB. The 8 dB gain is achieved under the assumption of ideal channel estimation. When polynomial sweeping-based CC encoding [162] is used or assumed - In a TDL-A fading channel with 30ns delay spread, and for 10MHz gap between two hops, 1 source [162] observed that waveform 1 with frequency hopping provides 4 dB frequency diversity gain compared to waveform 1 without frequency hopping at 1% BLER for a 1Rx receiver and a 1Tx CW transmitter. The following observations were made by one source ([161]): - Waveform 1 with frequency hopping and antenna hopping provides [1, 9] dB spatial diversity and frequency diversity compared to Waveform 1 with no frequency hopping and no antenna hopping at 1% or 10% BLER in a fading channel for a 1Rx receiver and a 2Tx CW transmitter, at least depending on the gap between the two tones and the channel's coherence bandwidth and assuming the ideal channel estimation. - In a TDL-D fading channel with 30ns delay spread, for the gap of 10MHz, the gain is 2.6 dB at 1% BLER and 1 dB at 10% BLER. - In a TDL-A fading channel with 30ns delay spread, for the gap of 2.16MHz, the gain is 7dB at 1% BLER target and 3.5dB at 10% BLER target. For the gap of 10MHz, the gain is 9dB at 1% BLER target and 5dB at 10% BLER target. Note: The total transmission power is assumed the same for both waveforms. Note: The above evaluations assume the same time domain resources overhead for waveform 1 with or without frequency hopping.
Spectrum utilization of backscattered signal	For the D2R transmission bandwidth corresponding to the CW waveforms, waveform 2 requires twice the frequency domain resources for D2R transmission of waveform 1, if the frequency gap between the two tones is no smaller than the transmission bandwidth of the	4 sources observed that for the D2R transmission BW corresponding to the CW waveforms, compared to waveform 1 without frequency hopping, waveform 1 with frequency hopping utilizes the same amount of frequency domain resources for D2R transmission at a certain time.

corresponding to the CW waveforms	corresponding D2R transmission.	
CW interference suppression at D2R receiver	Waveform 2 requires additional complexity if RF interference cancellation is used at least with CW waveform reconstruction, and requires individual cancellation for each of the tones, e.g. two RF or IF narrow-band bandpass filters. Note: RF interference cancellation is needed when the received CW interference power exceeds the blocking threshold of the receiver.	5 sources observed that waveform 1 with frequency hopping requires additional complexity for CW interference suppression if RF interference cancellation is used. 4 sources observed that waveform 1 with frequency hopping requires individual cancellation for each hop in different time, e.g. one tuneable RF or IF narrow-band bandpass filter, or two RF or IF narrow-band bandpass filters. 2 sources observed that for the RF interference cancellation at the D2R receiver, the frequency hopping of external carrier-wave requires fast re-detection of the parameters of the received carrier-wave interference, which at least doubled the implementation complexity of D2R receiver and may not be supported by the intermediate UE in D2T2. Note: RF interference cancellation is needed when the received CW interference power exceeds the blocking threshold of the receiver.
Relative complexity of CW generation	Waveform 2 leads to higher PAPR of the generated CW, which impacts the implementation of the power amplifier in the CW node.	2 sources observed that waveform 1 with frequency hopping requires additional complexity to construct the single-tone unmodulated sinusoid waveform at different center frequencies in a TDMed manner. 2 sources observed that waveform 1 with frequency hopping doesn't lead to higher PAPR of the generated CW. 1 source observed that waveform 1 with frequency hopping does not require additional complexity of CW generate compared to waveform 1 without frequency hopping.

For the gap between two tones or two frequency hops of a single tone to be able to leverage frequency diversity gain, the bandwidth and spectrum characteristics of the D2R transmission, and the channel coherence bandwidth, should be taken into account.

The following CW waveform characteristics which would need control of the CW node(s) are identified:

- When CW is transmitted or not transmitted
- Transmission Power
- Frequency resources, e.g. frequency location(s) for waveform (e.g. waveform 1, waveform 2)

Other CW waveform characteristics which would need control of the CW node(s), if any, can be further identified in a later stage.

6.9 Locating ambient IoT devices

6.9.1 General

A-IoT device location information may be used for the following purposes:

- (a) improving the A-IoT operation itself, e.g., by A-IoT CN sending a Command to one or more readers (e.g., the last reader(s)) associated to the device rather than sending it blindly.
- (b) providing location information to the consumer of the A-IoT service.

Locating an Ambient IoT device at "reader ID granularity" is useful for both purposes and is to be supported.

NOTE: "reader ID granularity" is likely not sufficient for purpose (b) and may be refined in the future.

6.9.2 Topology 1

For support of A-IoT device location, the A-IoT RAN node may report the A-IoT RAN node ID to the A-IoT CN in the Inventory Report or Command Response messages.

The A-IoT RAN node may additionally report a Reader ID to the A-IoT CN in the Inventory Report and/or the Command Response messages, which corresponds to the reader to which the A-IoT device responded.

NOTE 1: Whether multiple Reader ID(s) can be included by the A-IoT RAN node in the Inventory Report/Command Response message needs further discussion.

NOTE 2: Definition and format of the A-IoT RAN node ID and Reader ID needs further discussion.

NOTE 3: Whether and how to have awareness of Reader location in A-IoT CN needs further discussion.

6.9.3 Topology 2

For support of A-IoT device location, the A-IoT CN may receive in the Inventory Report and/or Command Response messages the identifier of the A-IoT enabled UE to which the A-IoT device responded.

The A-IoT CN may obtain the A-IoT enabled UE's location via ULI in NGAP messages. The A-IoT CN may also obtain the A-IoT enabled UE's location via legacy LCS mechanisms.

NOTE: Definition and format of the identifier of the A-IoT enabled UE needs further discussion.

6.9.4 Proximity determination

Proximity determination is feasible with either of the two following solutions. Potential specification impact or not will not be determined in this study item.

- Solution 1: If the reader successfully receives D2R transmission from the device in response to R2D transmission, then the device is determined as near to the reader.
- Solution 2: If the reader successfully receives D2R transmission from the device in response to R2D transmission, then the device is determined as near to the reader based on measurements at the reader side.

Proximity determination based on device-side measurements is not considered.

6.10 DO-A assessment

The study included an assessment of whether the harmonized air interface design can address the DO-A use case, only to identify which part(s) of the harmonized air interface design is/are not sufficient for the DO-A use case.

From the RAN1 perspective, at least the following aspect of the air interface for DO-DTT and DT traffic types is not sufficient for the DO-A traffic type: For DO-DTT and DT traffic types, the D2R resource(s) for D2R transmission is/are indicated in a R2D transmission, but this is not applicable at least for the first D2R transmission for DO-A traffic.

From RAN2 perspective, the DO-A traffic type/use case cannot be supported with the current design in the study item. It is assessed that, from RAN2 perspective, at least the A-IoT paging is an aspect/part of the current design which is not sufficient for the DO-A use case.

7 Evaluations

7.1 Coverage evaluations

For an evaluation scenario:

- For each link i ,
 - Step 1: Obtain the required SINR for the physical channels under target scenarios and service/reliability requirements if Budget-Alt2 is used for this link i .
 - Step 2: Obtain the receiver sensitivity using the method Budget-Alt1 (if a predefined threshold is assumed to derive the receiver sensitivity) or Budget-Alt2 (if no predefined threshold is assumed to derive the receiver sensitivity). See Clause 4.3.1 for the Budget definition.
 - Step 3: Obtain the coverage performance for link i based on the receiver sensitivity from step 2 and link budget template.
- The coverage results for each link are provided.

RAN1 has evaluated the coverage for InF-DH NLOS scenarios for D1T1 and InH-Office LOS/InF-DL NLOS scenarios for D2T2, across all devices 1/2a/2b. Coverage evaluation results comparing various R2D Tx power/EIRP value, R2D Rx sensitivity, BLER 1%/10% target, data rate ~1 kbps/~5-7 kbps, D2R coherent receiver/non-coherent receiver, R2D budget-Alt1 and Alt2, CW cancellation, backscatter loss and modulation schemes, D2R with/without channel coding schemes, etc, have been provided. The observations for coverage evaluation results can be found in Clauses 7.1.1.1.2, 7.1.1.2.2, 7.1.1.3.2, 7.1.2.1.2, 7.1.2.2.2, and 7.1.2.3.2.

All the evaluation results are contained in the attached spreadsheet file [189].

7.1.1 Link budget for D1T1

See associated file.

7.1.2 Link budget for D2T2

See associated file.

7.1.3 Link performance for D2R

See associated file.

7.1.4 Link performance for R2D

See associated file.

7.2 Latency evaluations

7.2.1 Single device latency

According to the definitions of clause 4.1, the following are the assessment results for single-device latency.

Table 7.2.1-1: Summary for single-device latency for "inventory-only" case

Source	Evaluation assumptions				Latency [ms]
	Data rate [kbps]	Re-attempt probability (X)	Message size [bits]	Gap [ms]	
[163]	5	10%	Paging: 22 Msg 1: 16 Msg 2: 18 Msg 3: 96	Gap A = 0.2 Gap B1 = 0.1	30.90
[164]	7	10%	Paging: 22 QueryRep: 4 Msg 1/2: 16 Msg 3: 96	Gap A = 0.0875 Gap B1 = 0.0875	24.09
[166]	1	10%	Paging: 128 Msg 3: 96	Gap A = 0.24 Gap B1 = 0.24	249.16
			Paging: 288 Msg 3: 256		604.71
	7		Paging: 128 Msg 3: 96		35.82
			Paging: 288 Msg 3: 256		86.62
[167]	7	10%	Paging: 20 Msg 1/2: 16 Msg 3: 96	Gap A = 0.2 Gap B1 = 0.1	24.30
[169]	7	10%	Paging: 24 Msg 1: 16 Msg 2: 24 Msg 3: 96	Gap A = 1.5 Gap B1 = 1.5	32.07
[186]	1	10%	Paging: 62 Msg 1: 16 Msg 2: 18 Msg 3: 96	Gap A = 10 Gap B1 = 20	246.70
	7			Gap A = 1.4 Gap B1 = 2.9	35.20
[171]	7	10%	Paging: 20 Msg 1/2: 16 Msg 3: 96	Gap A = 1.4 Gap B1 = 1.4	28.30
[172]	1	10%	Paging: 16 Msg 3: 96	Gap A = 0.2	124.70
			Paging: 24 Msg 3: 96		133.50
			Paging: 32 Msg 3: 96		142.40
	7		Paging: 16 Msg 3: 96		18.00
			Paging: 24 Msg 3: 96		19.27
			Paging: 32 Msg 3: 96		20.53
[173]	1	10%	Paging: 20 QueryRep = 20 Msg 1/2: 20 Msg 3: 96	Gap A = 0.25 Gap B1 = 0.25	173.97
	7				24.40
[188]	1	10%	Paging: 20 Msg 1/2: 20 Msg 3: 96	Gap A = 0.2 Gap B1 = 0.2	174.00
			Paging: 20 Msg 3: 96	Gap A = 0.2	129.10

	7		Paging: 20 Msg 1/2: 20 Msg 3: 96	Gap A = 0.2 Gap B1 = 0.2	25.42
			Paging: 20 Msg 3: 96	Gap A = 0.2	18.63
[178]	1	10%	Paging: 22 Msg 1: 16 Msg 2: 18 Msg 3: 96	Gap A = 0.2 Gap B1 = 0.1	169.44
	7				24.68
[180]	1	10%	Paging: 22 Msg 1/2: 16 Msg 3: 96	Gap A = 0.1 Gap B1 = 0.1	166.90
	7				24.00
[181]	7	10%	Paging: 30 Msg 1/2: 24 Msg 3: 200	Gap A = 0.5 Gap B1 = 0.5	45.72
	56				7.18
	640				2.14
[213]	1	10%	Paging: 16 Msg 1/2: 16 Msg 3: 96	Gap A = 0.2 Gap B1 = 0.2	160.56
			Paging: 24 Msg 1/2: 16 Msg 3: 96		169.44
			Paging: 32 Msg 1/2: 16 Msg 3: 96		178.33
	7		Paging: 16 Msg 1/2: 16 Msg 3: 96		23.41
			Paging: 24 Msg 1/2: 16 Msg 3: 96		24.68
			Paging: 32 Msg 1/2: 16 Msg 3: 96		25.94
[194]	1	10%	Paging: 20 Msg 1/2: 16 Msg 3: 96	Gap A = 0.2 Gap B1 = 0.3	167
[198]	1	10%	Paging: 22 Msg 1/2: 16 Msg 3: 96	Gap A = 0.025 Gap B1 = 0.025	166.75
[199]	1	10%	Paging: 20 Msg 1/2: 16 Msg 3: 96	Gap A = 10 Gap B1 = 10	197.8
[204]	7	10%	Paging: 26 Msg 1/2: 16 Msg 3: 96	Gap A = 0.2 Gap B1 = 0.2	25.11
[212]	7	10%	Paging: 26 Msg 1/2: 16 Msg 3: 96	Gap A = 1 Gap B1 = 1	25.7

Table 7.2.1-2: Summary for single-device latency for "inventory and command" case

Source	Evaluation assumptions				Latency [ms]
	Data rate [kbps]	Re-attempt probability (X)	Message size [bits]	Gap [ms]	
[164]	7	10%	Paging: 22 QueryRep: 4 Msg 1/2: 16 Msg 3: 96 Step C R2D/Step D D2R: 16	Gap B = 0.075 Gap C = 0.0875	29.35

[166]	1	10%	Paging: 128 Msg 3: 96 Step C R2D: 16 Step D D2R: 800	Gap A = 0.24 Gap B1 = 0.24 Gap C = 0.24	1156.64
			Paging: 288 Msg 3: 256 Step C R2D: 16 Step D D2R: 800		1512.20
	7		Paging: 128 Msg 3: 96 Step C R2D: 16 Step D D2R: 800		166.17
			Paging: 288 Msg 3: 256 Step C R2D: 16 Step D D2R: 800		216.96
[167]	7	10%	Paging: 20 Msg 1/2: 16 Msg 3: 96 Step C R2D: 400	Gap A = 0.2 Gap B1 = 0.1	87.90
[169]	7	10%	Paging: 24 Msg 1: 16 Msg 2: 24 Msg 3: 96 Step C R2D: 48 Step D D2R: 96	Gap A = 1.5 Gap B1 = 1.5 Gap B = 3 Gap C = 1.5	59.93
[186]	1	10%	Paging: 62 Msg 1: 16 Msg 2: 18 Msg 3: 96 Step C R2D: 60 Step D D2R: 33	Gap A = 10 Gap B1 = 20 Gap B = 5 Gap C = 20	377.80
	7			Gap A = 1.4 Gap B1 = 2.9 Gap B = 5 Gap C = 20	77.80
[171]	7	10%	Paging: 20 Msg 1/2: 16 Msg 3: 96 Step C R2D: 128 Step D D2R: 128	Gap A = 1.4 Gap B1 = 1.4 Gap B = 1.4 Gap C = 1.4	72.10
[172]	1	10%	Paging: 24 Msg 3: 96 Step C R2D: 64	Gap A = 0.2 Gap B = 0.1	204.80
			Paging: 24 Msg 3: 96 Step C R2D: 256		418.10
			Paging: 24 Msg 3: 96 Step C R2D: 400		578.10
	7		Paging: 24 Msg 3: 96 Step C R2D: 64		40.34
			Paging: 24 Msg 3: 96 Step C R2D: 256		59.71
			Paging: 24 Msg 3: 96		82.57

			Step C R2D: 400		
[173]	1	10%	Paging: 20 QueryRep = 20 Msg 1/2: 20 Msg 3: 96 Step C R2D: 20	Gap A = 0.25 Gap B1 = 0.25 Gap B = 0.075	196.28
	7		28.66		
[180]	1	10%	Paging: 22 Msg 1/2: 16 Msg 3: 96 Step C R2D: 96 Step D D2R: 16	Gap A = 0.1 Gap B1 = 0.1 Gap B = 0.1 Gap C = 0.1	291.60
	7		42.00		
[181]	7	10%	Paging: 30 Msg 1/2: 24 Msg 3: 200 Step C R2D: 200 Step D D2R: 10	Gap A = 0.5 Gap B1 = 0.5 Gap B = 0.5 Gap C = 0.5	80.28
	56				12.46
	640				3.62
[213]	1	10%	Paging: 24 Msg 1/2: 16 Msg 3: 96 Step C R2D: 64 Step D: 4	Gap A = 0.2 Gap B1 = 0.2 Gap B = 0.1 Gap C = 0.2	245.33
			Paging: 24 Msg 1/2: 16 Msg 3: 96 Step C R2D: 256 Step D: 4		458.67
			Paging: 24 Msg 1/2: 16 Msg 3: 96 Step C R2D: 400 Step D: 4		618.67
	7		Paging: 24 Msg 1/2: 16 Msg 3: 96 Step C R2D: 64 Step D: 4		46.61
			Paging: 24 Msg 1/2: 16 Msg 3: 96 Step C R2D: 256 Step D: 4		65.98
			Paging: 24 Msg 1/2: 16 Msg 3: 96 Step C R2D: 400 Step D: 4		88.83
[214]	1	10%	Paging: 20 Msg 1/2: 20 Msg 3: 96 Step C R2D: 96	Gap A = 0.2 Gap B1 = 0.2 Gap B = 0.2	280.89
			Paging: 20 Msg 3: 96 Step C R2D: 96		236
	7		Paging: 20 Msg 1/2: 20 Msg 3: 96 Step C R2D: 96		40.88
			Paging: 20 Msg 3: 96 Step C R2D: 96		34.1

[194]	1	10%	Paging: 20 Msg 1/2: 16 Msg 3: 96 Step C R2D: 400	Gap A = 0.2 Gap B1 = 0.3 Gap B = 0.1	612
[198]	1	10%	Paging: 22 Msg 1/2: 16 Msg 3: 96 Step C R2D: 20 Step D D2R: 16	Gap A = 0.025 Gap B1 = 0.025 Gap B = 0.025 Gap C = 0.025	206.81
[199]	1	10%	Paging: 20 Msg 1/2: 16 Msg 3: 96 Step C R2D: 128 Step D D2R: 128	Gap A = 10 Gap B1 = 10 Gap B = 10 Gap C = 10	504.4
[204]	7	10%	Paging: 26 Msg 1/2: 16 Msg 3: 96 Step C R2D: 96 Step D D2R: 16	Gap A = 0.2 Gap B1 = 0.2 Gap B = 0.4 Gap C = 0.2	43.56
[207]	1	10%	Paging: 22 Msg 1/2: 16 Msg 3: 96 Step C R2D: 96	Gap A = 0.1 Gap B1 = 0.1 Gap B = 0.1 Gap C = 0.1	273.7
	7				39.4
	1		Paging: 22 Msg 1/2: 16 Msg 3: 96 Step C R2D: 256 Step D D2R: 16		469.3
	7				67.4
	1				451.4
	7				64.8
[212]	7	10%	Paging: 26 Msg 1/2: 16 Msg 3: 96 Step C R2D: 96 Step D D2R: 96	Gap A = 1 Gap B1 = 1 Gap B = 1	42.1

Observations:

For the "inventory-only" case, considering 10% re-attempt probability:

- Assuming Msg3 size (Device ID size) of 96 bits
 - For data rate of 1 kbps, 10 sources [166], [186], [172], [173], [188], [178], [180], [194], [198], [199] provide results, and the single device latency is [124.70 ms ~ 249.16 ms].
 - For data rate of 5~7 kbps, 14 sources [163], [164], [166], [167], [169], [186], [171], [172], [173], [188], [178], [180], [204], [212] provide results, the single device latency is [18.00 ms ~ 35.82 ms].
- Assuming Msg3 size (Device ID size) of larger than 200 bits
 - For data rate of 1 kbps, 1 source [166] provides results, the single device latency is 604.71 ms.
 - For data rate of 7 kbps, 2 source [166], [181] provide results, the single device latency is [45.72 ms ~ 86.62 ms].
 - For data rate of 56 kbps, 1 source [181] provides results, the single device latency is 7.18 ms.
 - For data rate of 640 kbps, 1 source [181] provides results, the single device latency is 2.14 ms.

For the "inventory and command" case, considering 10% re-attempt probability:

- 11 sources provide results including Step D D2R transmission
 - Assuming Msg3 size (Device ID) of 96 bits and Step D D2R message size small than or equal to 400 bits,
 - For data rate of 1 kbps, 6 sources [186], [180], [213], [198], [199], [207] provide results, and the single device latency is [206.81 ms ~ 618.67 ms].
 - For data rate of 7 kbps, 9 sources [164], [169], [186], [171], [180], [213], [204], [207], [212] provide results, the single device latency is [29.35 ms ~ 88.83 ms].
 - Assuming Msg3 size (Device ID) of 96 bits and Step D D2R message size larger than 400 bits,
 - For data rate of 1 kbps, 1 source [166] provides results, and the single device latency is 1156.64 ms.
 - For data rate of 7 kbps, 1 source [166] provides results, and the single device latency is 166.17 ms.
 - Assuming Msg3 size (Device ID) larger than 200 bits and Step D D2R message size smaller than or equal to 400 bits,
 - For data rate of 56 kbps, 1 source [181] provide results, the single device latency is 12.46 ms.
 - For data rate of 640 kbps, 1 source [181] provide results, the single device latency is 3.62 ms.
 - Assuming Msg3 size (Device ID) larger than 200 bits and Step D D2R message size larger than 400 bits
 - For data rate of 1 kbps, 1 source [166] provides results, and the single device latency is 1512.20 ms.
 - For data rate of 7 kbps, 1 source [166] provides results, and the single device latency is 216.96 ms.
- 5 sources provide results not including Step D D2R transmission
 - Assuming Msg3 size (Device ID) of 96 bits and Step C R2D message size small than or equal to 400 bits,
 - For data rate of 1 kbps, 5 sources [172], [173], [214], [207], [167] provide results, and the single device latency is [196.28 ms ~ 612 ms].
 - For data rate of 7 kbps, 5 sources [167], [172], [173], [214], [207] provide results, the single device latency is [28.66 ms ~ 87.90 ms].

7.2.2 Inventory completion time for multiple devices

For the inventory-only use case, the, inventory completion time for multiple A-IoT devices is defined as the time a reader successfully completed the inventory process for at least 99% of all A-IoT devices within the coverage of the reader, assuming device density of 1.5 devices per m².

Note: The study does not define a target for this inventory completion time.

Table 7.2.2-1: Summary of inventory completion time for multiple devices

Source	Evaluation assumption For data rate column, value "X" means same data rate X for both R2D and D2R, value "(X,Y)" means data rate X for R2D and data rate Y for D2R.							Inventory completion time [s]
	No. of devices	Success rate	Random access scheme	Multiplexing resources (TDMA/FDMA/CDMA)	Data rate [kbps]	Energy harvesting consideration	Device type	
[163]	600	90%	Slotted ALOHA	1	5	No	-	17.4 (multi-round) 25.6 (single round)
				2 (FDMA)				14.5
				4 (FDMA)				9.5
				8 (FDMA)				5.3
				12 (FDMA)				3.6
		95%		1				18.6 (multi-round) 37.6 (single round)
				2 (FDMA)				15.5
				4 (FDMA)				10.2
				8 (FDMA)				5.7
		99%		1				19.6 (multi-round) 129.4 (single round)
				2 (FDMA)				16.3
				4 (FDMA)				10.7
				8 (FDMA)				6.0
				12 (FDMA)				4.0
[164]	600	99%	Slotted ALOHA	1	7	No	-	191.1
	6000			30 (FDMA)				15.61
				60(FDMA)				11.56
				1				1910.99
				30 (FDMA)				156
				60 (FDMA)				115.2
[166]	600	100%	Slotted ALOHA	1	(35, 5)	No	-	16.5959
					(35, 16)			6.3974
					(35, 50)			3.2451
					(126, 540)			1.5563

					Distribution based			12.6481
				2 (FDMA)	(35, 5)			12.9208
					(35, 16)			4.8847
					(35, 50)			2.4009
					(126, 540)			1.1014
					Distribution based			9.8101
				4 (FDMA)	(35, 5)			8.6877
					(35, 16)			3.2456
					(35, 50)			1.5635
					(126, 540)			0.6885
					Distribution based			6.5811
				8 (FDMA)	(35, 5)			4.9540
					(35, 16)			1.8434
					(35, 50)			0.8820
					(126, 540)			0.3792
					Distribution based			3.7499
				1	Distribution based			Direction 1
[167]	600	99%	Slotted ALOHA	1	1	No	-	90
					5			18.4
					60			2.2
				2	1			69
					5			14
					60			1.5
				4	1			46.5
					5			9.4
					60			1
				6	1			33
					5			7.1
					60			0.7
				8	1			25
					5			5.5
					60			0.57
[168]	44	99%	Slotted ALOHA	12 (FDMA)	5-7	No	-	1.09~1.46
	275							7.5~9.05
	1111							29.48~35.39
[186]	600	99%	Slotted ALOHA	2 (FDMA)	(56, 160)	Direction 1	1	36.40 (paging interval = 2 s)
				4 (FDMA)				46.40 (paging interval = 4 s)
				2 (FDMA)	(14, 40)			36.40 (paging interval = 2 s)
				4 (FDMA)				62.40 (paging interval = 4 s)
								36.40 (paging interval = 2 s)

								58.40 (paging interval = 4 s)					
				8 (FDMA)				36.40 (paging interval = 2 s) 46.40 (paging interval = 4 s)					
		100%		2 (FDMA)	(56, 160)			39.42 (paging interval = 2 s) 64.02 (paging interval = 4 s)					
				4 (FDMA)				39.17 (paging interval = 2 s) 59.64 (paging interval = 4 s)					
				2 (FDMA)				(14, 40)			40.34 (paging interval = 2 s) 69.63 (paging interval = 4 s)		
				4 (FDMA)	39.83 (paging interval = 2 s) 67.03 (paging interval = 4 s)								
				8 (FDMA)	39.43 (paging interval = 2 s) 63.96 (paging interval = 4 s)								
				99%							2 (FDMA)	(56, 160)	
								4 (FDMA)	35.80 (paging interval = 20 ms) 37.16 (paging interval = 40 ms)				
		2 (FDMA)						(14, 40)			36.44 (paging interval = 20 ms) 58.36 (paging interval = 40 ms)		
		4 (FDMA)									35.90 (paging interval = 20 ms) 55.32 (paging interval = 40 ms)		
		8 (FDMA)									35.84 (paging interval = 20 ms) 49.32 (paging interval = 40 ms)		
		100%					2 (FDMA)				(56, 160)		
				4 (FDMA)	35.99 (paging interval = 20 ms) 53.55 (paging interval = 40 ms)								
				2 (FDMA)	(14, 40)			55.57 (paging interval = 20 ms) 72.66 (paging interval = 40 ms)					
				4 (FDMA)				51.18 (paging interval = 20 ms) 62.32 (paging interval = 40 ms)					
				8 (FDMA)				46.35 (paging interval = 20 ms) 57.34 (paging interval = 40 ms)					
				99%				Non-Slotted ALOHA based solution	2 (FDMA)	(56, 160)			43.08
		4 (FDMA)			41.45								
		2 (FDMA)			(14, 40)				43.07				
		4 (FDMA)							41.45				
		8 (FDMA)	39.95										
		100%			2 (FDMA)		(56, 160)			43.95			
				4 (FDMA)	42.07								
				2 (FDMA)	(14, 40)					43.99			
				4 (FDMA)					42.07				
				8 (FDMA)					41.27				

		99%	Slotted ALOHA based solution	2 (FDMA)	(56, 160)	Direction 2	2	43.68		
				4 (FDMA)				39.84		
				2 (FDMA)	(14, 40)			75.24		
				4 (FDMA)				50.07		
				8 (FDMA)	(56, 160)			42.21		
				2 (FDMA)				44.01		
		100%	Slotted ALOHA based solution	4 (FDMA)	(14, 40)			39.92		
				2 (FDMA)				76.96		
				4 (FDMA)	(14, 40)			51.36		
				8 (FDMA)				43.68		
				99%	Non-Slotted ALOHA based solution			2 (FDMA)	(56, 160)	43.08
								4 (FDMA)		41.44
		2 (FDMA)	(14, 40)					59.97		
		4 (FDMA)						43.09		
		8 (FDMA)	(56, 160)					41.46		
		2 (FDMA)						43.96		
		100%	Non-Slotted ALOHA based solution	4 (FDMA)	(14, 40)			42.09		
				2 (FDMA)				67.11		
				4 (FDMA)	(14, 40)			43.69		
				8 (FDMA)				42.10		
				99%	Slotted ALOHA based solution			2 (FDMA)	(56, 160)	43.68
								4 (FDMA)		39.84
		2 (FDMA)	(14, 40)					107.68		
		4 (FDMA)						76.96		
		8 (FDMA)	(56, 160)					61.60		
		2 (FDMA)						44.06		
		100%	Slotted ALOHA based solution	4 (FDMA)	(14, 40)			41.05		
				2 (FDMA)				107.68		
				4 (FDMA)	(14, 40)			87.73		
				8 (FDMA)				71.87		
[170]	600			99%	Slotted ALOHA	1	1	No	-	33.992 (2-step for step B)
						2 (FDMA)				91.306 (3-step for step B)
		3 (FDMA)	26.627 (2-step for step B)							
		4 (FDMA)	85.966 (3-step for step B)							
			24.183 (2-step for step B)							
[171]	389	99%	Slotted ALOHA	1	50	No	-	81.539 (3-step for step B)		
				2 (FDMA)				22.951 (2-step for step B)		
								80.347 (3-step for step B)		
								4.29 (initial TDMA slots = 64)		
								3.24 (initial TDMA slots = 128)		
								2.31 (initial TDMA slots = 32)		
								1.46 (initial TDMA slots = 64)		

								2.04 (initial TDMA slots = 128)
				4 (FDMA)				1.19 (initial TDMA slots = 32) 1.22 (initial TDMA slots = 64) 1.71 (initial TDMA slots = 128)
				8 (FDMA)				1.10 (initial TDMA slots = 32) 1.16 (initial TDMA slots = 64) 1.55 (initial TDMA slots = 128)
[173]	600	99%	Slotted ALOHA	1	1	No	-	132.3
				2 (TDMA+FDMA)				92.0
				4 (TDMA+FDMA)				72.8
				6 (TDMA+FDMA)				65.7
				8 (TDMA+FDMA)				60.0
				1	7			19.7
				2 (TDMA+FDMA)				13.9
				4 (TDMA+FDMA)				10.2
				6 (TDMA+FDMA)				9.3
				8 (TDMA+FDMA)				9.1
				1	50			3.6
				2 (TDMA+FDMA)				2.4
				4 (TDMA+FDMA)				1.7
				6 (TDMA+FDMA)				1.6
				8 (TDMA+FDMA)				1.5
[175]	943	99%	Slotted ALOHA	1	(56, 56)	Direction 1	2	6.32 (22 rounds) 8.16 (5 rounds)
					(56, 160)			4.34 (22 rounds) 5.71 (5 rounds)
					(252, 960)			1.13 (22 rounds) 1.49 (5 rounds)
					(252, 4000)			0.99 (22 rounds) 1.32 (5 rounds)
	11781				(56, 56)			74.07 (22 rounds) 101.87 (5 rounds)
					(56, 160)			50.62 (22 rounds) 71.24 (5 rounds)
					(252, 960)			13.12 (22 rounds) 18.54 (5 rounds)
					(252, 4000)			11.52 (22 rounds) 16.45 (5 rounds)
[178]	600	99%	Slotted ALOHA	1	7	No	-	13.1
[180]	500	99%	Slotted ALOHA	TDMA for R2D and D2R	7	No	-	11.3 (One paging associated with multiple D2R slots)

								13 (Each paging providing a single D2R slot)
				TDMA for R2D, TDMA + FDMA for D2R				2.9 (One paging associated with multiple D2R slots) 3.9 (Each paging providing a single D2R slot)
				TDMA + FDMA for R2D and D2R				1.8 (One paging associated with multiple D2R slots) 2.5 (Each paging providing a single D2R slot)
	10000			TDMA for R2D and D2R				229 (One paging associated with multiple D2R slots) 263 (Each paging providing a single D2R slot)
				TDMA for R2D, TDMA + FDMA for D2R				43 (One paging associated with multiple D2R slots) 50 (Each paging providing a single D2R slot)
				TDMA + FDMA for R2D and D2R				22 (One paging associated with multiple D2R slots) 31 (Each paging providing a single D2R slot)
[181] [209]	600	100%	Slotted ALOHA	8 (4 TDM resources, 2 FDM resources)	(28, 70)	Direction 1: No DCM	1	20
						Direction 2: Case 1 DCM (periodic query)		5
						Direction 2: Case 1 DCM (aperiodic query)		2.5
						Direction 2: Case 2 DCM (periodic query)		11 (4 groups)
	600	100%	Slotted ALOHA	16 (4 TDM resources, 4 FDM resources)	(28, 70)	Direction 1: No DCM	2	200
						Direction 2: Case 1 DCM (periodic query)		20
						Direction 2: Case 1 DCM (aperiodic query)		20

						Direction 2: Case 2 DCM (periodic query)		10 (8 groups)
[181]	20	100%	Slotted ALOHA	8 (2 TDMA resources, 4 FDMA resources)	(28, 70)	Direction 1, RF incident = -28 dBm	1	1.3
						Direction 1, RF incident = -32 dBm		5
						Direction 2, Case 1 DCM, RF incident = - 28 dBm		1.35
						Direction 2, Case 1 DCM, RF incident = - 32 dBm		5.05
						Direction 2, Case 2 DCM, RF incident = - 28 dBm		0.2
						Direction 2, Case 2 DCM, RF incident = - 32 dBm		0.6
				16 (4 TDMA resources, 4 FDMA resources)		Direction 1, RF incident = -28 dBm	2	>10
						Direction 1, RF incident = -32 dBm		>10
						Direction 2, Case 1 DCM, RF incident = - 28 dBm		0.4
						Direction 2, Case 1 DCM, RF incident = - 32 dBm		>10
						Direction 2, Case 2 DCM, RF incident = - 28 dBm		0.4
						Direction 2, Case 2 DCM,		1.7

						RF incident = - 32 dBm		
[191]	600	99%	Slotted ALOHA Collision based re- attempts by re- paging	1	7	No	-	23.90 (rounds=35, TDMA slots per access round=128)
				2(FDMA+TDMA)				11.77 (cycles =9 TDMA slots per access round=128)
				4 (FDMA+TDMA)				6.47 (rounds=5, TDMA slots per access round=128)
				8 (FDMA+TDMA)				3.92 (rounds=3, TDMA slots per access round=128)
			Slotted ALOHA Collision based re- attempts by re- paging	1				17.43 (rounds=5, TDMA slots per access round=512)
				2(FDMA+TDMA)				11.38 (rounds=3 TDMA slots per access round=512)
				4 (FDMA+TDMA)				8.37 (rounds=3, TDMA slots per access round=512)
				8 (FDMA+TDMA)				5.75 (rounds=2, TDMA slots per access round=512)
	6000		Slotted ALOHA Collision based re- attempts by re- paging	1				237.97 (rounds=35, TDMA slots per access round=1280)
				2(FDMA+TDMA)				118.71 (rounds=9 TDMA slots per access round=1280)
				4 (FDMA+TDMA)				65.10 (rounds=5, TDMA slots per access round=1280)
				8 (FDMA+TDMA)				38.59 (rounds=3, TDMA slots per access round=1280)
			Slotted ALOHA Collision based re- attempts by re- paging	1				177.27 (rounds=5, TDMA slots per access round=5120)
				2(FDMA+TDMA)				111.98 (rounds=3 TDMA slots per access round=5120)
				4 (FDMA+TDMA)				85.65 (rounds=3, TDMA slots per access round=5120)
				8 (FDMA+TDMA)				57.10 (rounds=2, TDMA slots per access round=5120)
[213]	600	90%	Slotted ALOHA	1	5	No	-	16.14
		99%				17.53		
		90%		4 (FDMA)		No		7.58

		99%	Slotted ALOHA					8.23
		90%	Slotted ALOHA	4 (CDMA)		No		7.58
		99%	Slotted ALOHA with anti-collision solution	1		No		8.23
		90%						14.61
		99%						15.87
		90%	Slotted ALOHA	1		Direction 1 ON/OFF duration = 10ms/100ms		16.35
		99%						17.94
		90%	Slotted ALOHA	1		Direction 1 ON/OFF duration = 100ms/1s		16.36
		99%						18.92
		90%	Slotted ALOHA	1		Direction 1 ON/OFF duration = 500ms/5s		16.36
		99%						19.93
		[195]	600	100%		Slotted ALOHA		1 (TDMA)
1 (TDMA)	Direction 1: OFF when energy <90% Rx power = - 28dBm				16.034 (paging period 3ms)			
1 (TDMA)	Direction 2: SLEEP w R2D control Rx power = - 28dBm				11.186 (paging period 3ms)			
1 (TDMA)	Direction 1: OFF when fully discharged Rx power = - 28dBm				13.546 (paging period 4ms)			
1 (TDMA)	Direction 1: OFF when energy <90% Rx power = - 28dBm				9.676 (paging period 4ms)			

				1 (TDMA)		Direction 2: SLEEP w R2D control Rx power = - 28dBm		5.634 (paging period 4ms)
				1 (TDMA)		Direction 1: OFF when fully discharged Rx power = - 28dBm		10.688 (paging period 6.5ms)
				1 (TDMA)		Direction 1: OFF when energy <90% Rx power = - 28dBm		7.864 (paging period 6.5ms)
				1 (TDMA)		Direction 2: SLEEP w R2D control Rx power = - 28dBm		2.71 (paging period 6.5ms)
				1 (TDMA)		Direction 1: OFF when fully discharged Rx power = - 28dBm		69.206 (paging period 10ms)
				1 (TDMA)		Direction 1: OFF when energy <90% Rx power = - 28dBm		53.481 (paging period 10ms)
				1 (TDMA)		Direction 2: SLEEP w R2D control Rx power = - 28dBm		16.893 (paging period 10ms)
				1 (TDMA)		Direction 1: OFF when fully discharged Rx power = - 28dBm		114.45 (paging period 16ms)
				1 (TDMA)		Direction 1: OFF when energy <90% Rx power = - 28dBm		84.909 (paging period 16ms)

				1 (TDMA)		Direction 2: SLEEP w R2D control Rx power = - 28dBm		27.027 (paging period 16ms)
				2 (TDMA)		Direction 1: OFF when fully discharged Rx power = - 28dBm		54.048 (paging period 16ms)
				2 (TDMA)		Direction 1: OFF when energy <90% Rx power = - 28dBm		44.627 (paging period 16ms)
				2 (TDMA)		Direction 2: SLEEP w R2D control Rx power = - 28dBm		12.884 (paging period 16ms)
				4 (TDMA)		Direction 1: OFF when fully discharged Rx power = - 28dBm		25.686 (paging period 16ms)
				4 (TDMA)		Direction 1: OFF when energy <90% Rx power = - 28dBm		22.393 (paging period 16ms)
				4 (TDMA)		Direction 2: SLEEP w R2D control Rx power = - 28dBm		6.662 (paging period 16ms)
[196]	600	99%	Slotted ALOHA	1	-	No	-	18.42
				2 (FDMA)				16.91
				4 (FDMA)				16.24
				8 (FDMA)				15.91
[199]	389	99%	Slotted ALOHA	1	50	No	-	>10s (Initial TDMA slots=32) 4.35s (Initial TDMA slots=64) 3.43s (Initial TDMA slots=128)
				2 (TDMA+FDMA)				1.91s (Initial TDMA slots=32) 1.10s (Initial TDMA slots=64) 1.77s (Initial TDMA slots=128)

				4 (TDMA+FDMA)				0.62s (Initial TDMA slots=32) 0.67s (Initial TDMA slots=64) 1.18s (Initial TDMA slots=128)
				8 (TDMA+FDMA)				0.44s (Initial TDMA slots=32) 0.50s (Initial TDMA slots=64) 0.89s (Initial TDMA slots=128)
[209]	20	100%	Slotted ALOHA	8 (TDMA+FDMA)	Case 1: (7, 7) Case 2: (28, 70)	No DCM	1	20
						Case 1 DCM (periodic Query)		5
						Case 1 DCM (aperiodic Query)		2.5
						Case 2 DCM (periodic Query)		11 (4 groups)
				16 (TDMA+FDMA)		No DCM	2	200
						Case 1 DCM (periodic Query)		20
						Case 1 DCM (aperiodic Query)		20
						Case 2 DCM (periodic Query)		10 (8 groups)
[212]	600	90	Slotted ALOHA	1	7	No	-	54.43
				2 (FDMA)				27.21
				4 (FDMA)				13.62
		99		1				105.43
				2 (FDMA)				52.72
				4 (FDMA)				26.36

8 Conclusions and recommendations

It is feasible to support a harmonized air interface design, with minimized differences (where necessary) for Ambient IoT, to enable the devices described in Clause 3 for addressing the objectives stated in [2]. The extent to which maximum distance targets, differentiated per device and deployment/topology scenario, as described in Clause 4.1, can be met under different sets of evaluation assumptions is reported in Clause 7.1. A latency target for inventory of a single device in TR 38.848, can be met under different evaluation assumptions according to evaluations reported in Clause 7.2.1. The addressed subset of design targets described in TR 38.848 can be met or taken into account in feasible designs, and the design targets left to WGs in Clause 5 of TR 38.848 have been refined. Evaluations have also reported the inventory completion time for multiple devices. Corresponding evaluation scenarios, assumptions, and methodologies have been developed.

The basic blocks of A-IoT device architectures based on RF-ED receiver for device 1, 2a, and 2b have been identified, and also for receiver architectures based on IF-ED or ZIF for device 2b. For device 2a, the characteristics of blocks representing reflection amplifier(s) and large frequency shifter have been studied. Aspects to consider further regarding whether the large frequency shifter block is feasible, and its necessity or effectiveness have also been identified.

Purposes for which clocks(s) may be used by the different A-IoT devices 1/2a/2b have been identified, together with descriptions of the corresponding power consumption, and clock accuracy characteristics.

Regarding the potential impact of device (un)availability due to energy harvesting, two 'Directions' were found, depending on whether the reader does not, or can, provide information to a device based on which the device may become available/unavailable. Company solutions are reported in Clause 6.2, grouped according to the Direction they apply to.

The physical channels studied are PRDCH in R2D, and PDRCH in D2R, for both of which the generation process has been described at high level. Feasible candidates for the methods involved in the processes have also been documented, together with evaluations and pros/cons reported by companies. Information related to R2D reception and D2R scheduling have been listed together with ways they could be conveyed.

The physical signals studied have potential purposes involving start indication, timing acquisition/tracking, calibration/frequency synchronization at the device and the reader, channel or interference estimation at the reader, and indicating the end of a R2D/D2R transmission. Some of those purposes could be provided by information in a PRDCH.

Pros and cons of multiplexing/multiple access options for R2D and D2R links have been studied by company reports. Options and relationships among the resource allocations for messages of the random access procedures have been identified. The needed timing relationships between corresponding transmissions have been captured, for example the minimum time between a R2D transmission and the corresponding D2R transmission following it; and others.

Carrier-wave waveforms and their characteristics have been identified, together with cases for how CW transmission could be deployed in terms of the CW node and the spectrum used. Two CW waveforms were studied: a single-tone unmodulated sinusoid; or two unmodulated single-tone sinusoids. Their characteristics were studied and compared in terms of D2R backscattering reception performance and CW interference suppression at the reader, spectrum utilization of the backscattered signal, and relative complexity of CW generation. The effects of frequency hopping and antenna hopping were included.

For Topology 2, RAN1 studied resource allocation and/or control of the intermediate UE via the Uu interface via higher-layer signaling, or L1 and higher-layer signaling.

The harmonized air interface design focuses on DT and DO-DTT use cases. The study also assessed whether the harmonized air interface design can address the DO-A use case, and concluded that it cannot. The part(s) which is/are not sufficient for the DO-A use case are summarized in Clause 6.10.

RAN1 recommends that in a normative phase, the design should start from the descriptions in this TR.

(1) The following techniques are recommended in the first normative phase:

- PRDCH and PDRCH, as respectively the single physical channels for R2D and D2R
- Modulations of OOK-1 and OOK-4 in R2D

- Multiplexing in time domain for R2D
 - Convolutional FEC in D2R and no FEC in R2D
 - R2D and D2R signal(s)
- (2) At least the following aspects may be considered for inclusion or non-inclusion in a given first or subsequent Release:
- Device 1 and/or Device 2a (with or without large frequency shift) and/or Device 2b (with RF-ED and/or IF-ED and/or ZIF receiver architecture)
 - D1T1 and/or D2T2
 - Scenario A1 and/or A2 and/or B and/or C
 - Line code(s) Manchester/PIE for R2D, and Manchester / Miller / no line code for D2R
 - Multiple access by TDMA and/or FDMA and/or CDMA for D2R; FDMA for device 2b (IF-ED/ZIF receiver) in R2D
 - OOK and/or BPSK and/or MSK modulation for D2R
 - Carrier-wave waveform 1 and/or 2, and transmission case(s)
 - Device (un)availability direction 1 and/or 2
 - Proximity determination solution 1 and/or 2

From RAN2 perspective, RAN2 has studied Topology 1 and Topology 2, and has concluded they are feasible. In detail:

- General AS procedures, A-IoT paging, A-IoT random access, A-IoT data transmission and Topology 2 aspects have been studied:
 - See clause 6.3.1 for the overall AS procedure and the information useful to be visible to the reader from CN, etc.
 - See clause 6.3.3 for the A-IoT paging function, including paging message content, paging monitoring behavior, multiple (subsequent) A-IoT paging messages for the same service, etc.
 - See clause 6.3.4) for the A-IoT random access, including the contention-free access and contention-based random access procedure, the message contents, the re-access solutions, etc.
 - See clause 6.3.2 and 6.3.5 for protocol stack for A-IoT radio interface between A-IoT device and reader, required functionalities, and A-IoT data transmission functionalities, etc.
 - See clause 6.3.6 for the Topology 2 specific aspects, including the architecture options and management of radio resources in A-IoT radio interface, etc.
- From RAN2 perspective, the descriptions under clause 6.3 and 6.10 are considered as the baseline conclusions.
- Following is further concluded:
 - At least following features are recommended for the normative phase in the A-IoT MAC layer:
 - A-IoT paging
 - A-IoT random access:
 - For contention-based random access, further down-selection between *Solution 1* (3step-only) and *Solution 3* (unified solution) in clause 6.3.4.
 - A-IoT data transmission
 - For Topology 2, at least the following aspects are considered:
 - Forwarding of A-IoT upper layer data and A-IoT radio resource management.

- No further down-selection for Topology 2 architecture options (which are feasible) from RAN2 perspective (the final decision depends on TSGs/WGs outcome).

Logical system architectures have been studied, as documented in clause 6.4.

Impacts on CN-RAN interface, signaling and procedures have been studied to support A-IoT Inventory operation and Command operation for topology 1 and topology 2 respectively, as documented in clause 6.5.

Solutions for locating an Ambient IoT device with minimal specification impact are identified, as documented in clause 6.9.

In RAN3, no evaluation and down selection for T2 solutions have been performed which requires coordination among WGs.

RAN4 has performed adjacent channel co-existence studies between ambient IOT and NR for the scenario that NR macro BS outdoor and ambient IOT system indoor. RAN4 evaluated a total of 9 deployment scenarios as described in Table 6.6.2-1. For each deployment scenario, a total of 4 cases were performed as described in Table 6.6.2-1. The co-existence cases are described in Table 6.6.2-2. The performance metrics for NR system impacted by ambient IOT are throughput loss at the cell edge and cell average. The performance metric for ambient IOT system impacted by NR system is SINR degradation for inter-system interference impact analysis. Meanwhile, in addition to SINR degradation, ambient IOT intra-system performance evaluated by link level simulation is also one of the key aspects affecting ambient IOT system performance, which can be further analyzed in WI phase. The summary of co-existence evaluation can be found in clause 6.6.5.

RAN4 has studied the RF requirements impact for ambient IOT, including system parameters, ambient IoT BS, intermediate UE, devices (device 1, 2a and 2b) and CW node, which can be found in clause 6.7.1, 6.7.2, 6.7.3 and 6.7.4.

RAN4 has also studied the feasibility of CW interference cancellation for both ambient IoT BS and intermediate UE, and the feasible interference cancellation values can be found in clause 6.7.5.

Moreover, RAN4 also studied the testability issue for ambient IOT devices, which can be found in clause 6.7.4.4.

Annex A: Device receiver sensitivity for *Budget-Alt1*

The receiver sensitivity for device 1, device 2a, and device 2b assuming *Budget-Alt1* is used in the coverage evaluation is summarized as follows.

Table A-1: Device 1 receiver sensitivity (*Budget-Alt1*)

Company	-40 dBm	-36 dBm	-30 dBm	Reference
[197]		×		
[199]	×	×		
[215]			×	
[200]		×		
[193]		×		
[207]		×		
[191]	×	×	×	[216], [217], [218]
[201]	×	×	×	
[203]		×		
[192]		×		
[208]			×	[219]
[212]		×	×	
[211]			×	
[204]			×	
[210]	×	×		
[190]			×	
[202]	×	×	×	
[209]		×	×	
[214]			×	
[185]			×	
[194]		×		
[195]		×		
[198]			×	
[213]		×		[216]
[205]		×		
[196]			×	

Table A-2: Device 2a receiver sensitivity (*Budget-Alt1*)

Company	-50 dBm	-45 dBm	-40 dBm	Reference
[197]		×		
[199]			×	
[215]			×	
[200]		×		
[193]		×		
[207]			×	
[191]	×	×	×	[216], [217], [218]
[201]			×	
[203]		×		
[192]		×		
[208]			×	[220], [221]
[212]			×	
[211]			×	
[204]			×	
[210]		×		
[190]		×		
[202]		×	×	
[209]		×	×	
[185]			×	
[194]		×	×	
[195]			×	
[198]		×		
[213]		×		[220], [222]
[205]			×	
[196]		×		
[214]			×	

Table A-3: Device 2b receiver sensitivity (*Budget-Alt 1*)

Company	-85 dBm	-60 dBm	-55 dBm	-50 dBm	-45 dBm	-40 dBm	Reference
[199]					×		
[215]					×		
[193]					×		
[191]				×			[216], [217], [218]
[201]						×	
[203]					×		
[192]					×		[220], [221]
[212]					×		
[211]					×		
[210]					×		
[190]					×		
[202]					×	×	
[209]					×	×	
[195]		×					

[198]					×		[223], [224], [225]
[213]			×				
[196]	×						
[214]						×	

Annex B (informative): Change history

Change history							
Date	Meeting	TDoc	CR	Rev	Cat	Subject/Comment	New version
2024-02	RAN1#116	R1-2401795				TR skeleton	0.0.1
2024-08	RAN1#118	R1-2407489				Inclusion of agreements up to RAN1#117	0.1.0
2024-09	RAN#105	RP-241784				First version for information to RAN#105	1.0.0
2024-10	RAN#118b	R1-2409993				Addition of the decisions made in RAN1#118bis. Endorsed as v1.1.0 in post-meeting email discussion [Post-118bis-AIoT-01]	1.1.0
2024-11						MCC clean-up – mainly fixing formatting issues	1.1.1
2024-12	RAN#106	RP-242466				For approval to RAN#106	2.0.0
2024-12	RAN#106	RP-243319				Replacement of one FFS and one Editor's Note during RAN#106	2.0.1
2024-12	RAN#106					RAN#106 approval - TR under change control as part of Release 19	19.0.0